

JEE MAIN 2027 EDITION · SESSION 1: 22-30 JAN 2027 (TENTATIVE)

JEE MAIN

99+

THE COMPLETE PERSONALIZED PREPARATION
SYSTEM

*A Personalized Strategy, Formula, PYQ, Practice, Revision &
Mock-Test Handbook*

● Physics

● Chemistry

● Mathematics

● Strategy · PYQs · Mocks

PREPARED FOR

Gayathri
Devu

MASTER BOOK

Strategy · Physics · Chemistry · Maths · Formulas · PYQs · Mocks · Plans
· Trackers

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PART I

Start Here



Welcome, verified exam facts, the 99+ roadmap and percentile strategy, your personal profile, the diagnostic test and the plan selector.

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Welcome, Gayathri

This book is a working system, not a reading book. Every page is here for one job: moving your JEE Main 2027 score up by the next few marks, again and again, until you are at the level where 99+ percentile becomes a realistic outcome.

It combines five things that are usually sold separately:

1 · A coaching module set

Concept summaries, formulas, traps, shortcuts and original practice questions for every chapter in the current NTA syllabus: 20 Physics units, 20 Chemistry units and 14 Mathematics units.

2 · A formula & fact handbook

Every formula, constant, reaction, reagent and NCERT fact you need, compressed for 1-day, 3-day, 7-day and final-24-hour revision.

3 · A strategy manual

Question selection, answer-choice techniques, speed methods, exam-hall timing and negative-marking control, built from verified exam rules.

4 · A mock-test companion

A full-length original mock test, subject mini-tests, a mock-analysis sheet and a percentile-planning section based on historical data.

5 · A personal planner

30- to 180-day plans at 6, 8 and 10 hours a day, final 30/14/7/3-day plans, timetables, weekly scorecards and printable trackers.

6 · A mistake-elimination system

The Mistake Book: eight mistake types, how to diagnose each one, and a fix routine that stops you making the same mistake twice.

Your timeline, in one line

✓ VERIFIED FACT Verified: JEE Main 2027 Session 1 dates

NTA's tentative examination calendar (public notice dated 16 September 2026) schedules **JEE Main 2027 Session 1 on 22, 23, 24, 28, 29 and 30 January 2027**, with 31 January as a buffer day. NTA has said these dates are tentative. The detailed Information Bulletin, with eligibility, pattern and syllabus confirmation, had **not** been released when this book was compiled (28 September 2026). Session 2 is expected in April 2027 but had not been announced. Sources: [NTA-CAL27], [NTA-WEB].

Counting from 28 September 2026, you have about **116 days to Session 1** and about **185–190 days to an early-April Session 2**. That is why this book's **120-day plan** is built to peak in late January and its **180-day plan** carries you through to April. If you are reading this later, the Study-Plan Selector (page [29](#)) picks the right plan for the days you have left.

Three promises this book makes, and one it doesn't

What this book promises

- **Accuracy over hype.** Exam rules are taken from NTA documents and labelled *Verified*. Trend numbers from coaching analyses are labelled *Third-party analysis*. Our own advice is labelled *Recommendation*.
- **Complete coverage.** Every unit of the official syllabus has a module. Topics that NTA removed in 2024 are flagged so you don't waste time on them.
- **Original material.** Every practice question here was written for this book. PYQs are referenced and classified, not copied.

What it will not promise

No book, coach or strategy can *guarantee* a percentile, rank or college. Percentile depends on your raw score **and** on how everyone else in your shift performs. What we can do is show you, with historical ranges, what score band usually corresponds to 99 percentile, and then give you a system for reaching the top of that band with a safety margin.

How to start today (40 minutes)

1. Read **How to Use This Book** (page 6): 5 minutes.
2. Fill in your **Personalization Profile** (page 19): 10 minutes. Leave blanks where you're unsure.
3. Take the **Current-Level Diagnostic Test** (page 21): 60 minutes, today or tomorrow.
4. Use the **Study-Plan Selector** (page 29) to choose your plan, print that plan's pages, and start Day 1.

★ **IMPORTANT** The one habit that matters most

Every question you get wrong goes into **The Mistake Book** (page ?) *the same day*, with its mistake type. Students who keep this habit reliably get more out of the same number of study hours, because they stop losing marks to the same errors.

Prepared with care for **Gayathri Devu**, target: **JEE Main 2027, 99+ percentile**.

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How to Use This Book

The visual code

Every box in this book has one job, and its colour and label tell you what that job is. After a week you will scan pages by colour.

FORMULA Formula

Formulas and standard results you must be able to write from memory. Example: $v^2 = u^2 + 2as$.

SHORTCUT Shortcut

A legitimate faster method, with *when to use it* and *when not to use it*.

IMPORTANT Important

A concept that questions are built on. If you skip this, later questions in the chapter stop making sense.

MISTAKE ALERT Mistake Alert

A trap that has cost students marks: sign errors, unit slips, misread conditions, NCERT exceptions.

PYQ PATTERN PYQ Pattern

How this idea is typically asked in JEE Main: the recurring question *structure*, not a copied question.

REVISION Revision

What to re-read on Day 2, Day 7 and Day 30. Tick the boxes as you go.

PRACTICE Practice

Original questions with answer, full solution, shortcut, concept tested, difficulty, target time and common trap.

MOCK TEST Mock Test

Timed test material and instructions: sit it like the real exam.

WARNING Warning

Rules where a mistake costs you directly, such as negative marking or exam-day regulations.

Three kinds of statement, three labels

Label	Meaning	Example
 VERIFIED	Taken from an official NTA/NCERT/JoSAA document or an official announcement reported by a reputable outlet	"Each wrong answer carries –1 mark."
 THIRD-PARTY ANALYSIS	From published analyses by coaching institutes or education portals; useful but not official	"Current Electricity usually has 2–3 questions per shift."
 RECOMMENDATION	This book's strategy advice, derived from the rules, the trend data and learning research	"Attempt Chemistry first if it is your fastest subject."

Difficulty and question tags

Tag	Meaning
Easy	Easy: direct formula or fact. Target: under 1.5 min.
Medium	Medium: two steps, or one idea plus careful calculation. Target: 2–3 min.
Hard	Hard: multi-concept, long algebra, or an unusual twist. Target: 3–5 min, attempt only in Round 2.
NV	Numerical-value answer (Section B style). There are no options to check against, so be careful: a wrong answer here also costs –1.
Speed	Time-pressure item: solve it in the target time or skip it.
Tricky	Contains a deliberate trap. Read the <i>Common trap</i> line after solving.

Navigating the PDF

On a laptop or tablet

- The **Contents** pages are clickable: tap any line to jump there.
- Every page has **quick-jump tabs** in the footer: *Contents · Physics · Chemistry · Maths · Formulas · Plans · Trackers*.
- Open your PDF reader's **bookmarks/outline panel** for a collapsible map of the whole book.
- "Page 123"-style references in the text are clickable too.
- The **progress bar** at the top of each page shows how far through the book you are, coloured by subject.

On paper

- Print the **Study Planner** and **Mock Analysis Template** PDFs separately; they're designed for A4 printing.
- Print the **Quick Revision Book** double-sided and keep it in your bag.
- The **Exam Day Command Sheet** is a single page. Print two copies: one for your desk, one for exam morning.
- Chapter modules print well in grayscale: every box also has a text label.

NOTE About interactivity, honestly

A standard PDF cannot play real animations reliably across devices, so this book doesn't try to. What it does have: clickable contents, internal links, bookmarks, quick-jump tabs, QR codes to official resources, and **fillable checkboxes and text fields**. These work in Adobe Acrobat Reader, Xodo, Foxit and most desktop browsers. On some phone viewers, fields may show but not save. In that case, print the trackers.

The daily loop this book is designed around

- 1 Learn**
Read the module's concept summary and formula box (40–60 min per sub-topic)
- 2 Practise**
Solve the module's practice set, timed, then check solutions
- 3 Log**
Every error goes into The Mistake Book with its type
- 4 Revise**
Same-day 10-min recap → Day 2 → Day 7 → Day 14 → Day 30
- 5 Test**
Chapter test → mixed tests → full mocks, analysed with the Mock Analysis Sheet

What to read when

Situation	Go to
Starting preparation	Roadmap (page 14) → Profile (page 19) → Diagnostic (page 21) → Plan Selector (page 29)
Learning a chapter	Its chapter module in Part II, III or IV
A mock went badly	Mock Analysis Sheet (page ?) → The Mistake Book (page ?) → Mental game (page ?)
Final month	Final 30/14/7/3-day plans (page ?) → Formula Handbook (page ?)
Night before and exam morning	Exam-Eve Plan (page ?) → Exam Day Command Sheet (page ?) → Last-Minute Revision Book (page ?)

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Know Your Exam: JEE Main 2027

⚠ WARNING Status as of 28 September 2026

The **JEE (Main) 2027 Information Bulletin has not yet been released**. The only official 2027-specific announcement so far is NTA's tentative calendar for Session 1. Everything else below comes from the **JEE (Main) 2026 Information Bulletin** and NTA notices, and is marked as such. Pattern and syllabus have been stable since 2025 and 2024 respectively, but **re-check every row against the 2027 bulletin when it is published**. The checklist at the end of this chapter is there for exactly that.

2027 dates

Item	Detail	Status
Session 1 exam days	22, 23, 24, 28, 29, 30 January 2027 (31 January buffer)	✓ VERIFIED tentative, NTA calendar dated 16 Sep 2026 [NTA-CAL27]
Information Bulletin & registration	Not released as of 28 Sep 2026. Portals expect late October 2026	THIRD-PARTY ANALYSIS expected, not official
Session 2	Not announced. In 2026 it ran 2–8 April; 2025: 2–8 April	✓ VERIFIED past years only [R-DATES26], [R-DATES25]
Official websites	jeemain.nta.nic.in · nta.ac.in	✓ VERIFIED



JEE Main official portal
jeemain.nta.nic.in



NTA (public notices,
calendar)
nta.ac.in



NCERT textbooks (free PDFs)
ncert.nic.in/textbook.php



JoSAA (counselling, seat
matrix)
josaa.nic.in

Paper 1 (B.E./B.Tech.) pattern

VERIFIED FACT Verified from the JEE (Main) 2026 Information Bulletin [NTA-IB26]

	Physics	Chemistry	Mathematics	Total
Section A: MCQs (4 options, one correct)	20	20	20	60
Section B: Numerical-value questions	5	5	5	15
Questions to attempt	25 (all compulsory)	25	25	75
Marks	100	100	100	300

- **Mode:** Computer Based Test (CBT). **Duration:** 3 hours. In 2026 the shifts ran 9:00–12:00 and 15:00–18:00.
- **Section B has no internal choice.** Since 2025 all five numerical questions are compulsory. The old 10-given-attempt-5 format is gone.
- **Languages:** 13 (English, Hindi, Assamese, Bengali, Gujarati, Kannada, Malayalam, Marathi, Odia, Punjabi, Tamil, Telugu, Urdu).

Marking scheme

WARNING Verified: the numbers that decide your strategy [NTA-IB26]

Response	Section A (MCQ)	Section B (Numerical)
Correct	+4	+4
Incorrect	−1	−1
Unanswered / Marked for Review without an answer	0	0

Negative marking applies to numerical questions too. Older books that say "no negative marking in Section B" describe a pattern that no longer exists.

☆ IMPORTANT What −1 really means

A random guess on a 4-option MCQ has expected value $\frac{1}{4}(+4) + \frac{3}{4}(-1) = +0.25$ marks. That looks positive, but it's tiny, and the variance is large. If you can **eliminate two options with certainty**, the expected value rises to $\frac{1}{2}(4) + \frac{1}{2}(-1) = +1.5$. The honest rule, used throughout this book: **attempt when you have eliminated at least two options through real reasoning. Never attempt a pure blind guess.** On numerical questions a guess is almost always wrong, so answer only after solving.

Calculators, tables, rough work

- **No calculator of any kind.** NTA's November 2025 clarification states that the on-screen calculator mentioned in the first 2026 bulletin was a typographical error, and that calculators are not allowed in JEE Main. **✓ VERIFIED** [NTA-CALC], [R-CALC]
- **Numerical answers:** entered with the on-screen numeric keypad. In recent papers the Section B instructions have asked for answers rounded to the nearest integer. Read the instruction on your screen on the day.

||| **THIRD-PARTY ANALYSIS**

- Rough sheets are provided at the centre and must be returned. Don't carry your own paper. ✓ VERIFIED [NTA-IB26]

What this means for you: practise all mental arithmetic, approximation and long calculation **by hand from Day 1**. The Speed Manual (page ?) has drills for this.

How your percentile is calculated

✓ **VERIFIED FACT** Verified: NTA normalisation [NTA-IB26], [R-NORM]

Each shift's raw scores are converted into **NTA percentile scores** separately:

$$\text{Percentile} = \frac{100 \times \text{number of candidates in that shift with raw score} \leq \text{yours}}{\text{total candidates who appeared in that shift}}$$

- The topper of every shift gets 100 percentile. Percentiles are reported to 7 decimal places.
- Percentiles are also computed **subject-wise** (Physics, Chemistry, Maths). Your total percentile is computed from your total raw score. It is **not** the average of the subject percentiles.
- If you take both sessions, **the better of your two NTA scores** is used for the merit list/rank.

Why the same marks give different percentiles in different shifts: in a harder shift, fewer candidates score high, so fewer marks are needed to beat 99% of that shift. The 99+ Strategy chapter (page 15) shows the historical ranges.

Tie-breaking (for rank, when percentiles are equal)

✓ **VERIFIED** as per the 2026 bulletin, reported in [R-TIE]. Ties are resolved in this order:

1. Higher NTA score in **Mathematics** → 2. **Physics** → 3. **Chemistry** → 4. Lower ratio of incorrect to correct answers (all subjects) → 5. same ratio in Maths → 6. Physics → 7. Chemistry → 8. Otherwise the same rank.

Strategy note: Maths is the first tie-breaker, and a low wrong-to-right ratio is the next. That is two more reasons, beyond the marks themselves, to protect accuracy.

Eligibility essentials

Rule	Detail	Status
Age limit	None for JEE Main (institutes may have their own)	✓ VERIFIED 2026 bulletin
Year of passing Class 12	2026 exam: passed in 2024 or 2025, or appearing in 2026. For 2027 the same pattern would mean 2025, 2026 or appearing in 2027. Confirm in the 2027 bulletin.	✓ VERIFIED (2026); 2027 pending
75% criterion (for NIT/IIIT/CFTI admission via JoSAA)	75% aggregate in Class 12 (65% for SC/ST), or top-20 percentile of your board	✓ VERIFIED 2026 bulletin / JoSAA
JEE Advanced qualifying cut-off (General)	2026: 93.4123549 · 2025: 93.1023262 · 2024: 93.2362181 · 2023: 90.7788642	✓ VERIFIED as reported [R-CUTOFF]

Facts that matter specifically for you

Female-only seat pool (JoSAA) ✓ VERIFIED In JoSAA counselling, IITs and the NIT+ system institutes run a *Female-only (including supernumerary)* pool alongside the *Gender-neutral* pool. The supernumerary seats exist to raise female enrolment to at least about 20%. A female candidate is considered for **both** pools, so the ranks you need can be meaningfully different from gender-neutral closing ranks. Check the official JoSAA past opening/closing ranks for your target branches. [JOSAA]

Application fee ✓ VERIFIED (2026 bulletin, reported [R-FEE]) For one paper at an Indian centre, the fee was **₹800 for female candidates of all categories**, against ₹1,000 for General-category male candidates. Check the 2027 bulletin for the current figure.

Exam-day rules (from 2026 guidelines) ✓ VERIFIED Report early: in 2026, gates closed 30 minutes before the exam started. Carry your admit card, a valid photo ID and the photographs the admit card asks for. Metallic items and jewellery are restricted, and face authentication/biometric verification happens at entry. Candidates in customary or religious attire should report earlier for frisking. **The admit card's instructions override anything written here.**

Practical planning ⚙ RECOMMENDATION

- Visit or map your centre route 1–2 days before. Plan to travel with a family member for early-morning shifts.
- If your exam dates may coincide with your period, plan comfort and medication with your doctor in advance, and take at least two full mocks at the same time of day to rehearse. Check the admit card for permitted items.
- Keep the exam-morning routine (page ?) identical to your mock-day routine.

Recent official changes worth knowing

Year	Change	Status
2024	Syllabus reduced and aligned to the rationalised NCERT (e.g., Communication Systems, States of Matter, Surface Chemistry, s-Block, Hydrogen, Polymers, Mathematical Reasoning and Induction removed)	✓ VERIFIED [NTA-SYL]
2025	Section B optional questions scrapped: 5 compulsory numerical questions per subject	✓ VERIFIED
2026	Aadhaar/DigiLocker-based identity details at registration; face authentication at centres; calculator clarification	✓ VERIFIED [NTA-IB26], [NTA-CALC]
2026–27	Ministry-announced multi-level (four-tier) scrutiny of question papers to reduce errors. No change to difficulty, pattern or syllabus announced	📊 THIRD-PARTY ANALYSIS reported [R-REFORM]

Verify-on-release checklist (when the 2027 bulletin comes out)

- ☐ Number of questions per section (20 MCQ + 5 NV?) and whether all are compulsory
- ☐ Marking: +4 / –1 in both sections?
- ☐ Syllabus PDF: any unit or topic change from the 2024–2026 syllabus?
- ☐ Eligibility: permitted years of passing Class 12
- ☐ Session 2 dates and registration window
- ☐ Shift timings, reporting time, gate-closing time
- ☐ Calculator/rough-sheet rules
- ☐ Tie-breaking order
- ☐ Fee for female candidates
- ☐ Exam-city choices near you (choose four with care)

- ☐ Photo/signature and document specifications
- ☐ Update page **9** of this book by hand where anything changed

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The 99+ Percentile Roadmap

The roadmap has five phases. Each phase ends with a **gate**: a measurable condition you must meet before the next phase gets most of your time. The phase lengths below assume the 180-day plan. Shorter plans compress them (see the Plan Selector, page 29).

- 1 **Phase 0 · Diagnose**
Days 1–3: profile, diagnostic test, pick plan, set up Mistake Book
- 2 **Phase 1 · Build**
Concepts + chapter practice for every Must-do and High chapter; weekly mixed tests
- 3 **Phase 2 · Strengthen**
Chapter-wise PYQs, second pass on weak chapters, formula mastery
- 4 **Phase 3 · Mock & Fix**
2–3 full mocks a week, each followed by analysis and targeted repair
- 5 **Phase 4 · Peak**
Final 14 days: revision-heavy, mocks in the exam time slot, sleep locked

Phase	Share of time (180-day plan)	Gate to pass before moving on	Status
0 · Diagnose	3 days	Profile filled, diagnostic scored, plan chosen, Mistake Book started	⌚ RECOMMENDATION
1 · Build	~45% (Days 4–85)	Every Must-do chapter: module read, practice set $\geq 75\%$ correct, chapter test taken	⌚ RECOMMENDATION
2 · Strengthen	~25% (Days 86–130)	PYQs for all Must-do + High chapters done; formula recall $\geq 90\%$ on a blank-sheet test	⌚ RECOMMENDATION
3 · Mock & Fix	~20% (Days 131–166)	Last 5 full mocks averaging your target score band with ≤ 8 negative marks	⌚ RECOMMENDATION
4 · Peak	Final 14 days	Revision cycle complete; exam-slot mocks stable; sleep 8 h	⌚ RECOMMENDATION

☆ IMPORTANT The central idea

99 percentile isn't reached by knowing "more chapters". It's reached by **converting what you know into marks at high accuracy within 180 minutes**. That's why this book puts as much weight on selection, speed, accuracy and mistake elimination as on content.

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The 99+ Percentile Strategy

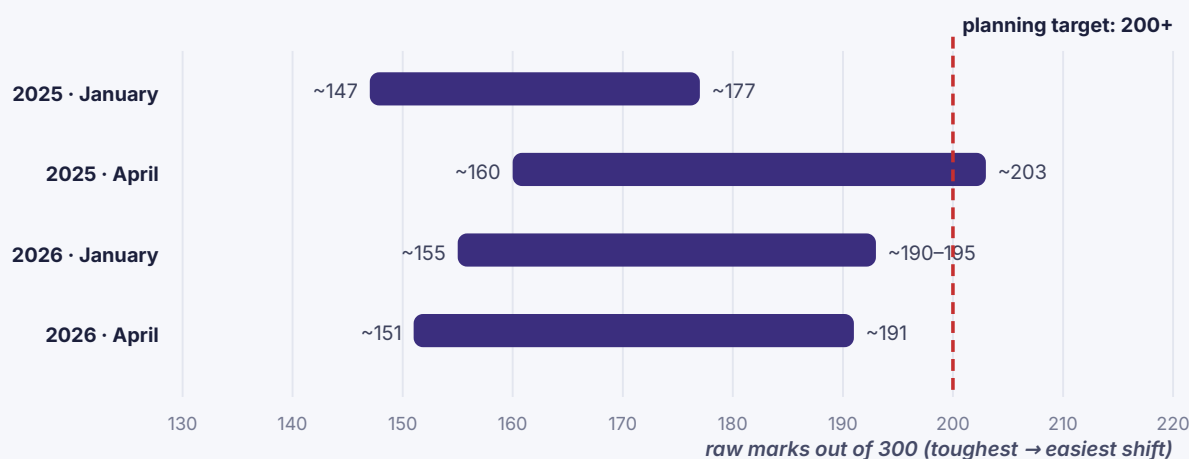
What score has 99 percentile needed? (historical, approximate)

VERIFIED FACT Third-party analyses of past results, labelled approximate

Marks needed for about the 99th percentile varied widely **by shift**. Ranges compiled from published shift-wise marks-vs-percentile analyses:

Session	Approx. raw marks for ~99 percentile (toughest → easiest shift)	Sources
2025 · January	~147 → ~177	[T-VED-25]
2025 · April	~160 → ~203	[T-VED-25]
2026 · January	~155 → ~190–195	[T-VED-26], [T-COMP]
2026 · April	~151 → ~191	[T-PW-S2]

Published tables for other percentile bands (95, 99.5, 99.9) **disagree with one another by large margins**, so this book doesn't reproduce them. Use the ranges above for 99 percentile only, and treat them as approximate.



Reading the chart: in a tough shift, 150-odd marks reached 99 percentile. In an easy shift, even 190–200 wasn't always enough. You won't know in advance which kind of shift you'll get.

STRATEGY The planning target this book uses

Plan for the **easy-shift case**. That means a target of **200+ raw marks in the actual exam**, which in turn means consistently scoring **200–215 in full-length mocks of JEE-Main standard** in the final month, with **≤ 8 negative marks**. If you hit this, you're in the 99+ zone in almost every historical shift. If your shift turns out tough, the same preparation lands you well above 99. **This is a target band, not a guarantee.**

Percentile → approximate rank

$$\text{Approx. rank} \approx \frac{(100 - \text{percentile}) \times N}{100}$$

where N is the number of unique candidates. In 2026, $N \approx 15.38$ lakh appeared **✓ VERIFIED** [R-2026STAT]. So 99 percentile $\approx$ rank 15,000 and 99.5 percentile $\approx$ rank 7,700 (common rank list, before category and female-pool effects). This is a rough approximation: NTA's actual rank list merges sessions and applies tie-breaks.

Why percentile varies by shift

1. **Paper difficulty differs.** NTA sets a different paper for each shift. A lengthy Maths section in one shift pulls that shift's whole score distribution down.
2. **Normalisation is within the shift.** Your percentile compares you only with candidates in your shift (page 9).
3. **Candidate pool differs slightly by shift and session.** More repeaters and stronger candidates often appear in Session 2 (April), when many are attempting a second time. This is one reason April cut-offs have often been higher. THIRD-PARTY ANALYSIS

What you control: your raw score's safety margin above the band. Build the margin through accuracy and selection, not through more attempts.

Accuracy targets

Metric	99+ zone target	Why
Overall accuracy (correct ÷ attempted)	$\geq 90\%$	At 90% accuracy on 58 attempts: $52 \times 4 - 6 = 202$
Negative marks per paper	≤ 8	Each wrong answer costs 5 marks of swing versus leaving it: the +4 you didn't get and the -1 you did
Section B (NV) accuracy	$\geq 90\%$	No options to verify against, so every NV answer must be checked once
Silly-mistake losses	≤ 2 questions	Measured in the Mock Analysis Sheet
Unattempted "known" questions	0	Skipping a question you could solve is a selection mistake

Minimum safe preparation milestones RECOMMENDATION

- ☐ **Coverage:** 100% of Must-do and High-priority chapters (PYQ Priority Matrix, page ?); at least 90% of the full syllabus
- ☐ **Formulas:** blank-sheet recall of each subject's formula list at $\geq 90\%$
- ☐ **PYQs:** all 2024–2026 PYQs (post-syllabus-change papers) for Must-do chapters; High-priority chapters from 2022 onward
- ☐ **Practice volume:** 60–80 timed questions a day on normal days (6–8 h plan)
- ☐ **Mocks:** at least 20 full-length mocks before Session 1 (at least 30 before Session 2), each analysed
- ☐ **Stability:** last 5 mocks within ± 15 marks of each other, averaging 200+
- ☐ **Mistake Book:** no mistake *type* accounts for more than 30% of lost marks in the last 3 mocks

Attempt strategy: six scenarios

All six aim at roughly 200+. Pick the one closest to your profile after 3–4 mocks, then refine it. **Time allocations are for the 180-minute paper**, with the subject order you choose (page ?).

A · High-Accuracy Strategy (*for careful solvers, slower speed*)

	Phy	Chem	Maths	Total
Attempt	19	21	16	56
Correct	18	20	15	53
Wrong	1	1	1	3
Marks	71	79	59	209
Time (min)	60	45	70	+5 review

Leave 19 questions. Accept only questions you can finish in time. Accuracy $\approx 95\%$.

B · Balanced Attempt Strategy (*the default*)

	Phy	Chem	Maths	Total
Attempt	20	22	18	60
Correct	18	20	16	54
Wrong	2	2	2	6
Marks	70	78	62	210
Time (min)	60	45	65	+10 review

Accuracy 90%. The most robust plan for most students.

C · Speed Strategy (*fast and accurate students only*)

	Phy	Chem	Maths	Total
Attempt	22	23	21	66
Correct	20	21	18	59
Wrong	2	2	3	7
Marks	78	82	69	229
Time (min)	55	40	75	+10 review

Use only if your mocks show $\geq 88\%$ accuracy at this attempt level. More attempts at lower accuracy lose marks.

D · Weak-Maths Strategy

	Phy	Chem	Maths	Total
Attempt	21	23	13	57
Correct	20	22	12	54
Wrong	1	1	1	3
Marks	79	87	47	213
Time (min)	60	45	65	+10 review

Maths: attempt only Must-do models (Matrices, Vectors & 3D, Sequences, Binomial, P&C, definite integrals, the conic basics). Every mark you gain here comes from Physics and Chemistry accuracy.

E · Weak-Physics Strategy

	Phy	Chem	Maths	Total
Attempt	15	23	20	58
Correct	14	22	18	54
Wrong	1	1	2	4
Marks	55	87	70	212
Time (min)	55	45	70	+10 review

Physics: bank Modern Physics, Semiconductors, Units & Errors, EM Waves, Ray and Wave Optics, KTG, and the experiment-based questions: formula-direct questions first.

F · Weak-Chemistry Strategy

	Phy	Chem	Maths	Total
Attempt	21	17	20	58
Correct	20	15	18	53
Wrong	1	2	2	5
Marks	79	58	70	207
Time (min)	60	45	65	+10 review

Chemistry is the most *fixable* weak subject: NCERT-line Inorganic, Biomolecules, Practical Chemistry and formula-based Physical Chemistry can add 20+ marks in 6 weeks (Chemistry strategy, page [159](#)).

**MISTAKE ALERT** The most common strategy error

Students raise the number of attempts when a mock goes badly. The data from your own Mock Analysis Sheets will usually show the opposite: **scores fall when attempts rise above your accuracy-safe level**. Adjust attempts only from evidence, and in steps of about 2 questions per subject.

Section timing checkpoints (default Balanced plan, order Chem → Phy → Maths)

Clock (from start)	You should be...
0:00–0:05	Reading instructions, calm breathing, first scan of Chemistry
0:45	Chemistry Round 1 done ($\approx$ 20 attempted); move to Physics
1:45	Physics Round 1 done; move to Maths
2:40	Maths Round 1 done; start Round 2 (marked questions, all subjects)
2:50	Stop new questions. Review marked answers and numerical entries
3:00	Submit

Full exam-hall protocol: page [?](#).

PART I · START HERE

Student Personalization Profile

Fill this in on Day 1, in pen or directly in the PDF (the fields are fillable). Update it after every 5th mock. The rest of the book refers back to these answers.

Field	Your details	If unknown, this book assumes...
Student name	Gayathri Devu	—
Current class / year		Class 12 in 2026–27, or a Class 12 pass-out taking a drop year
Current preparation level	(Beginner / Intermediate / Advanced)	Intermediate: Class 11 done once, Class 12 partially done
Target percentile	99+	—
Target score (raw, /300)		200–215 in mocks → 200+ in the exam (page 15)
Strongest subject		Unknown: decided by the Diagnostic Test
Weakest subject		Unknown: decided by the Diagnostic Test
Average daily study time	hours	8 hours on school days, 10 on holidays
Exam attempt	(1st / 2nd / 3rd)	First attempt in Session 1, Jan 2027; second in Session 2, Apr 2027
Target college / branch (optional)		—
School / college schedule		School 8:00–14:00 on weekdays (Class 12 student)
Current mock score	/300	Not yet measured: take a baseline mock in Week 1
Exam date		Session 1: 22–30 Jan 2027 (tentative, NTA calendar)

NOTE About the assumptions column

Where you leave a field blank, the plans in this book use the assumption shown. Every assumption is marked; none is presented as fact about you. When you fill in the real detail, use the adjustment table below.

Personal adjustment table

If your profile says...	Then adjust like this	Go to
Level = Beginner	Use the next-longer plan if possible. Phase 1 is +25% time; start every chapter with its "Core theory" and Basic questions only	Plan Selector, page 29
Level = Advanced	Skim Core theory; start at the PYQ Pattern box; do only M/H practice questions; mocks begin 4 weeks earlier	Mock system, page ?
Weakest = Maths	Scenario D; 40% of daily problem-solving time to Maths; drill the Must-do models first	page 15, ?
Weakest = Physics	Scenario E; formula-first approach; daily 20-min "direct formula" drill	page 32
Weakest = Chemistry	Scenario F; NCERT reading block every day; Inorganic flash-review nightly	page 159
Study time $\leq$ 6 h/day (school days)	Use the 6-hour variants; move mocks to Sunday mornings; skip Low-frequency topics until Phase 3	Plans, page ?

If your profile says...	Then adjust like this	Go to
Mock score < 120	The priority is concept gaps, not speed. Phase 1 continues until chapter tests reach $\geq 70\%$	Roadmap, page 14
Mock score 120–170	The priority is accuracy and selection. Mistake Book twice daily; attempt-limit drills	The Mistake Book, page ?
Mock score 170–200	The priority is speed and Section B accuracy. Speed Manual drills; Round-1/Round-2 discipline	Speed Manual, page ?

My why, in one sentence (optional)

Baseline numbers (fill after the Diagnostic Test and the first mock)

	Physics	Chemistry	Maths	Total
Diagnostic score (/40 each)				
Baseline mock (/100 each)				
Accuracy %				
Target by Session 1				

PART I · START HERE

Current-Level Diagnostic Test

MOCK TEST Instructions

30 questions · 60 minutes · +4 / -1 (same marking as JEE Main, NV questions included). Use a timer, no calculator, no notes. Mark the time at which you finish each subject. The purpose is **measurement**, not a good score. Don't prepare for it, and don't look at the solutions until you've finished.

Diagnostic · Physics

1 Easy 1 min Basic

DP-01

The dimensional formula of Planck's constant h is:

- (A) $[ML^2T^{-2}]$ (B) $[ML^2T^{-1}]$ (C) $[MLT^{-1}]$ (D) $[ML^{-1}T^{-2}]$

2 Easy 1 min Basic

DP-02

A ball is thrown vertically upward at 20 m s^{-1} ($g = 10 \text{ m s}^{-2}$). It returns to the thrower's hand after:

- (A) 2 s (B) 4 s (C) 6 s (D) 8 s

3 Easy 1.5 min Concept

DP-03

A 2 kg block **sliding** on a rough horizontal floor ($\mu_k = 0.3$, $g = 10 \text{ m s}^{-2}$) is pulled by a horizontal force of 10 N in the direction of motion. Its acceleration is:

- (A) 1 m s^{-2} (B) 2 m s^{-2} (C) 3 m s^{-2} (D) 5 m s^{-2}

4 Medium 1.5 min Concept

DP-04

A solid sphere rolls without slipping. The fraction of its total kinetic energy that is rotational is:

- (A) $2/5$ (B) $2/7$ (C) $5/7$ (D) $1/2$

5 Medium 1.5 min Concept

DP-05

The escape speed from Earth is 11.2 km s^{-1} . For a planet of the same density but twice Earth's radius it is:

- (A) 11.2 km s^{-1} (B) 15.8 km s^{-1} (C) 22.4 km s^{-1} (D) 44.8 km s^{-1}

6 Medium 1.5 min Concept

DP-06

An ideal monatomic gas is heated at constant pressure. The fraction of the heat supplied that is converted into work is:

- (A) $2/5$ (B) $3/5$ (C) $2/3$ (D) $1/3$

7 Easy 1.5 min Basic

DP-07

Capacitors of $2 \mu\text{F}$ and $3 \mu\text{F}$ are connected in series across a 10 V battery. The charge on each capacitor is:

- (A) $12 \mu\text{C}$ (B) $20 \mu\text{C}$ (C) $30 \mu\text{C}$ (D) $50 \mu\text{C}$

8 Easy 1.5 min NV

DP-08

A uniform wire of resistance $32\ \Omega$ is bent into a circle. Find the resistance (in Ω) between two diametrically opposite points.

Your answer (numerical value):

9 Easy 1.5 min Basic

DP-09

An object is placed 30 cm in front of a convex lens of focal length 20 cm. The image is:

- (A) 60 cm behind the lens, real, inverted, twice the size
 (B) 12 cm behind the lens, virtual, erect
 (C) 60 cm in front of the lens, virtual, erect
 (D) 12 cm behind the lens, real, diminished

10 Easy 1 min NV

DP-10

Photons of energy 6.0 eV fall on a metal of work function 2.0 eV. Find the stopping potential (in volts).

Your answer (numerical value):

Diagnostic · Chemistry

1 Easy 1 min Basic

DC-01

The number of moles of oxygen **atoms** in 88 g of CO_2 is:

- (A) 2 (B) 4 (C) 6 (D) 8

2 Easy 1 min Basic

DC-02

The number of radial nodes in a 3p orbital is:

- (A) 0 (B) 1 (C) 2 (D) 3

3 Easy 1 min Basic

DC-03

Which molecule has zero dipole moment?

- (A) NH_3 (B) H_2O (C) BF_3 (D) CHCl_3

4 Medium 1.5 min Concept

DC-04

For a reaction, $\Delta H = -100\ \text{kJ mol}^{-1}$ and $\Delta S = -200\ \text{J K}^{-1}\text{mol}^{-1}$ (assume both are temperature independent). The reaction is spontaneous:

- (A) above 500 K (B) below 500 K
 (C) at all temperatures (D) at no temperature

5 Easy 1 min Basic

DC-05

The pH of $10^{-3}\ \text{M}$ NaOH at $25\ ^\circ\text{C}$ is:

- (A) 3 (B) 10 (C) 11 (D) 12

6 Easy 1 min NV

DC-06

A first-order reaction has a half-life of 10 min. Find the time (in minutes) needed for 87.5% completion.

Your answer (numerical value):

7 Medium 1.5 min Concept

DC-07

The correct order of first ionisation enthalpy is:

- (A) $B < C < N < O$ (B) $B < C < O < N$ (C) $C < B < O < N$ (D) $O < N < C < B$

8 Medium 1.5 min Concept

DC-08

The spin-only magnetic moment of $[\text{Fe}(\text{CN})_6]^{3-}$ is closest to:

- (A) 1.73 BM (B) 3.87 BM (C) 5.92 BM (D) 0 BM

9 Easy 1 min Basic

DC-09

The most stable carbocation among the following is:

- (A) CH_3^+ (B) CH_3CH_2^+ (C) $(\text{CH}_3)_2\text{CH}^+$ (D) $(\text{CH}_3)_3\text{C}^+$

10 Medium 1.5 min NV

DC-10

How many structurally isomeric **alcohols** have the formula $\text{C}_4\text{H}_{10}\text{O}$?Your answer (numerical value):

Diagnostic · Mathematics

1 Easy 1 min Basic

DM-01

If $f(x) = 2x + 3$ and $g(x) = x^2$, then $(f \circ g)(2)$ equals:

- (A) 11 (B) 49 (C) 14 (D) 7

2 Easy 1 min Basic

DM-02

 $\left| \frac{3+4i}{1-i} \right|$ equals:

- (A) $5/\sqrt{2}$ (B) $5\sqrt{2}$ (C) $5/2$ (D) $7/\sqrt{2}$

3 Easy 1 min Basic

DM-03

If α, β are the roots of $x^2 - 5x + 6 = 0$, then $\alpha^2 + \beta^2$ is:

- (A) 13 (B) 25 (C) 37 (D) 11

4 Medium 1 min Concept

DM-04

If A is a 3×3 matrix with $|A| = 4$, then $|\text{adj } A|$ is:

- (A) 4 (B) 16 (C) 64 (D) 8

5 Easy 1.5 min NV

DM-05

Find the number of distinct arrangements of the letters of the word LEVEL.

Your answer (numerical value):

6 Easy 1 min Basic

DM-06

The sum of the first 20 terms of the AP 3, 7, 11, ... is:

- (A) 820 (B) 800 (C) 780 (D) 840

7 Easy 1 min Basic

DM-07

 $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$ equals:

- (A) 1 (B) 2 (C) 4 (D) 1/2

8 Easy 1 min Basic

DM-08

 $\int_0^{\pi/2} \sin^2 x \, dx$ equals:

- (A) $\pi/2$ (B) $\pi/4$ (C) 1 (D) $\pi/8$

9 Easy 1 min NV

DM-09

Find the perpendicular distance of the point (1, 2) from the line $3x + 4y - 1 = 0$.Your answer (numerical value):

10 Easy 1 min Basic

DM-10

Two fair dice are thrown. The probability that the sum is 7 is:

- (A) 1/6 (B) 1/12 (C) 5/36 (D) 7/36

Diagnostic — Answers & Solutions

Diagnostic · Physics

1 (B) 2 (B) 3 (B) 4 (B) 5 (C) 6 (A) 7 (A) 8 8 9 (A) 10 4

Diagnostic · Chemistry

1 (B) 2 (B) 3 (C) 4 (B) 5 (C) 6 30 7 (B) 8 (A) 9 (D) 10 4

Diagnostic · Mathematics

1 (A) 2 (A) 3 (A) 4 (B) 5 30 6 (A) 7 (B) 8 (B) 9 2 10 (A)

DIAGNOSTIC · PHYSICS

1 Answer: (B) Dimensions of Planck's constant Easy 1 min Basic

DP-01

SOLUTION $E = h\nu \Rightarrow h = E/\nu = [ML^2T^{-2}]/[T^{-1}] = [ML^2T^{-1}]$.SHORTCUT Same dimensions as angular momentum ($L = nh/2\pi$ in Bohr's model).

COMMON TRAP Confusing with the dimensions of energy, which is option (A).

2 Answer: (B) Vertical projectile time of flight Easy 1 min Basic

DP-02

SOLUTION Time to top $= u/g = 2$ s; the descent takes the same time. Total $= 2u/g = 4$ s.SHORTCUT $T = 2u/g$ for return to the launch level.

COMMON TRAP Answering the time to the top (2 s).

3 Answer: (B) Kinetic friction Easy 1.5 min Concept

DP-03

SOLUTION Friction $f = \mu_k mg = 0.3 \times 20 = 6$ N opposing motion. $a = (10 - 6)/2 = 2 \text{ m s}^{-2}$.SHORTCUT Net force $= F - \mu mg$.

COMMON TRAP Ignoring friction gives 5.

4 Answer: (B) Rolling kinetic energy split Medium 1.5 min Concept

DP-04

SOLUTION $K_{\text{rot}}/K_{\text{tot}} = \frac{\frac{1}{2}I\omega^2}{\frac{1}{2}mv^2(1 + k^2/R^2)} = \frac{k^2/R^2}{1 + k^2/R^2} = \frac{2/5}{7/5} = \frac{2}{7}$.SHORTCUT Fraction $= \frac{k^2/R^2}{1 + k^2/R^2}$; for a solid sphere $k^2/R^2 = 2/5$.COMMON TRAP Answering $k^2/R^2 = 2/5$ itself.

5 Answer: (C) *Escape velocity scaling* **Medium** ⌚ 1.5 min **Concept**

DP-05

SOLUTION $v_e = \sqrt{2GM/R}$ and $M = \frac{4}{3}\pi R^3 \rho$, so $v_e = R\sqrt{\frac{8}{3}\pi G \rho} \propto R$ at fixed ρ . Doubling R doubles v_e .

SHORTCUT Fixed density: $v_e \propto R$. Fixed mass: $v_e \propto 1/\sqrt{R}$.

COMMON TRAP Using $v_e \propto 1/\sqrt{R}$ (fixed-mass case) gives the wrong direction.

6 Answer: (A) *First law, isobaric process* **Medium** ⌚ 1.5 min **Concept**

DP-06

SOLUTION $W = nR\Delta T$, $Q = nC_p\Delta T = \frac{5}{2}nR\Delta T$, so $W/Q = 2/5$.

SHORTCUT $W/Q = R/C_p = 1 - 1/\gamma$.

COMMON TRAP Using C_v instead of C_p gives 2/3.

7 Answer: (A) *Series capacitors* **Easy** ⌚ 1.5 min **Basic**

DP-07

SOLUTION $C_{eq} = \frac{2 \times 3}{2 + 3} = 1.2 \mu\text{F}$; $Q = C_{eq}V = 12 \mu\text{C}$ (the same on both in series).

SHORTCUT Series: same Q ; $C_{eq} = C_1C_2/(C_1 + C_2)$.

COMMON TRAP Adding capacitances as if in parallel gives 50.

8 Answer: 8 *Resistance of a ring* **Easy** ⌚ 1.5 min **NV**

DP-08

SOLUTION Each semicircle is 16Ω ; they're in parallel: $16 \parallel 16 = 8 \Omega$.

SHORTCUT Two equal halves in parallel: $R/4$.

COMMON TRAP Answering 16 (one half only).

9 Answer: (A) *Thin lens formula* **Easy** ⌚ 1.5 min **Basic**

DP-09

SOLUTION $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ with $u = -30$: $\frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60}$, so $v = +60$ cm. $m = v/u = -2$: real, inverted, magnified.

SHORTCUT Object between f and $2f \rightarrow$ image beyond $2f$, real, magnified.

COMMON TRAP Sign error with u gives 12 cm.

10 Answer: 4 *Photoelectric equation* **Easy** ⌚ 1 min **NV**

DP-10

SOLUTION $K_{max} = h\nu - \phi = 4.0$ eV, so $eV_0 = 4.0$ eV and $V_0 = 4$ V.

SHORTCUT In eV units, V_0 (volts) = $E_{photon} - \phi$ numerically.

COMMON TRAP Adding instead of subtracting the work function.

DIAGNOSTIC · CHEMISTRY**1 Answer: (B)** *Mole concept* **Easy** ⌚ 1 min **Basic**

DC-01

SOLUTION $88/44 = 2$ mol CO_2 ; each has 2 O atoms, so 4 mol O atoms.

SHORTCUT Moles of atoms = moles of molecules $\times$ atomicity.

COMMON TRAP Stopping at 2 (moles of molecules).

2 Answer: (B) *Nodes of orbitals* **Easy** ⌚ 1 min **Basic**

DC-02

SOLUTION Radial nodes = $n - l - 1 = 3 - 1 - 1 = 1$. (Angular nodes = $l = 1$.)

SHORTCUT Radial = $n - l - 1$; angular = l ; total = $n - 1$.

COMMON TRAP Giving the total nodes (2).

3 Answer: (C) *Dipole moment & shape* **Easy** ⌚ 1 min **Basic**

DC-03

SOLUTION BF_3 is trigonal planar and symmetric, so the bond dipoles cancel. The others are pyramidal, bent or tetrahedral with unequal substituents.

SHORTCUT Symmetric shapes (linear AX_2 , trigonal planar AX_3 , tetrahedral AX_4) with identical atoms $\rightarrow \mu = 0$.

COMMON TRAP Polar bonds don't imply a polar molecule.

4 Answer: (B) *Gibbs energy & spontaneity* **Medium** ⌚ 1.5 min **Concept**

DC-04

SOLUTION $\Delta G = \Delta H - T\Delta S = -100 + 0.2T$ (kJ). $\Delta G < 0$ when $T < 500$ K.

SHORTCUT Both negative $\rightarrow$ enthalpy-driven $\rightarrow$ spontaneous at low T , below $T = \Delta H/\Delta S$.

COMMON TRAP Unit mismatch: J vs kJ gives 0.5 K or 500 000 K.

5 Answer: (C) *pH of strong base* Easy 1 min Basic

DC-05

SOLUTION $[\text{OH}^-] = 10^{-3}$, so $\text{pOH} = 3$ and $\text{pH} = 14 - 3 = 11$.**SHORTCUT** $\text{pH} + \text{pOH} = 14$ at 25°C .**COMMON TRAP** Reporting pOH as pH .**6 Answer: 30** *First-order half-life* Easy 1 min NV

DC-06

SOLUTION 87.5% complete means 12.5% = $1/8$ remains, which is 3 half-lives: $3 \times 10 = 30$ min.**SHORTCUT** Remaining fraction = $(1/2)^n$ for first order.**COMMON TRAP** Using linear (zero-order) thinking.**7 Answer: (B)** *Ionisation enthalpy trend* Medium 1.5 min Concept

DC-07

SOLUTION Increases across the period, with one exception: N ($2p^3$, half-filled) $>$ O ($2p^4$). So $\text{B} < \text{C} < \text{O} < \text{N}$.**SHORTCUT** Period-2 exceptions: $\text{Be} > \text{B}$ and $\text{N} > \text{O}$.**COMMON TRAP** Forgetting the $\text{N} > \text{O}$ exception gives (A).**8 Answer: (A)** *Low-spin complexes* Medium 1.5 min Concept

DC-08

SOLUTION Fe^{3+} is d^5 . CN^- is a strong-field ligand, so the complex is low spin: $t_{2g}^5 e_g^0$, one unpaired electron. $\mu = \sqrt{n(n+2)} = \sqrt{3} \approx 1.73$ BM.**SHORTCUT** Strong-field ligands (CN^- , CO) pair up d^4 – d^7 in octahedral complexes.**COMMON TRAP** Treating it as high spin gives 5.92 BM.**9 Answer: (D)** *Carbocation stability* Easy 1 min Basic

DC-09

SOLUTION Stability grows with hyperconjugation and +I donation: $3^\circ > 2^\circ > 1^\circ >$ methyl.**SHORTCUT** More α -hydrogens (hyperconjugation) $\rightarrow$ more stable alkyl carbocation.**COMMON TRAP** None here; this is a speed item.**10 Answer: 4** *Structural isomers of alcohols* Medium 1.5 min NV

DC-10

SOLUTION Butan-1-ol, butan-2-ol, 2-methylpropan-1-ol, 2-methylpropan-2-ol. (The 3 ethers are not alcohols. Counting stereoisomers is not asked.)**SHORTCUT** Place $-\text{OH}$ on each distinct carbon of each C_4 skeleton: n-butane gives 2, isobutane gives 2.**COMMON TRAP** Including ethers gives 7.**DIAGNOSTIC · MATHEMATICS****1 Answer: (A)** *Composition of functions* Easy 1 min Basic

DM-01

SOLUTION $(f \circ g)(2) = f(g(2)) = f(4) = 11$.**SHORTCUT** Apply the inner function first.**COMMON TRAP** Computing $g(f(2)) = 49$.**2 Answer: (A)** *Modulus of a quotient* Easy 1 min Basic

DM-02

SOLUTION $|z_1/z_2| = |z_1|/|z_2| = 5/\sqrt{2}$.**SHORTCUT** Never rationalise when only the modulus is asked.**COMMON TRAP** Rationalising wastes time and invites arithmetic slips.**3 Answer: (A)** *Symmetric functions of roots* Easy 1 min Basic

DM-03

SOLUTION $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 25 - 12 = 13$.**SHORTCUT** Use Vieta's relations; don't solve for the roots.**COMMON TRAP** Forgetting the $-2\alpha\beta$ term.**4 Answer: (B)** *Determinant of adjoint* Medium 1 min Concept

DM-04

SOLUTION $|\text{adj } A| = |A|^{n-1} = 4^2 = 16$ for $n = 3$.**SHORTCUT** $|\text{adj } A| = |A|^{n-1}$ and $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$.**COMMON TRAP** Using $|A|^n = 64$.

5 Answer: 30 *Permutations with repetition* **Easy** ⌚ 1.5 min **NV**

DM-05

SOLUTION 5 letters with L repeated twice and E repeated twice: $\frac{5!}{2!2!} = 30$.

SHORTCUT Divide by the factorial of each repeated letter's count.

COMMON TRAP Dividing by 2! only once gives 60.

6 Answer: (A) *AP sum* **Easy** ⌚ 1 min **Basic**

DM-06

SOLUTION $S_{20} = \frac{20}{2}[2(3) + 19(4)] = 10 \times 82 = 820$.

SHORTCUT $S_n = \frac{n}{2}(\text{first} + \text{last})$; the last term is $3 + 76 = 79$, so 10×82 .

COMMON TRAP Using $20d$ instead of $19d$ gives 840.

7 Answer: (B) *Standard trigonometric limit* **Easy** ⌚ 1 min **Basic**

DM-07

SOLUTION $1 - \cos 2x = 2 \sin^2 x$, so the limit is $2 \left(\frac{\sin x}{x} \right)^2 \rightarrow 2$.

SHORTCUT $1 - \cos kx \approx \frac{k^2 x^2}{2}$ near 0.

COMMON TRAP Forgetting to square k (giving 1) or dropping the $\frac{1}{2}$ (giving 4).

8 Answer: (B) *Definite integral* **Easy** ⌚ 1 min **Basic**

DM-08

SOLUTION By King's property, $\int_0^{\pi/2} \sin^2 x \, dx = \int_0^{\pi/2} \cos^2 x \, dx$. Their sum is $\int_0^{\pi/2} 1 \, dx = \pi/2$, so each is $\pi/4$.

SHORTCUT Average value of $\sin^2$ over a quarter period is $\frac{1}{2}$: $\frac{1}{2} \cdot \frac{\pi}{2}$.

COMMON TRAP Answering $\pi/2$ by forgetting the halving.

9 Answer: 2 *Distance of point from line* **Easy** ⌚ 1 min **NV**

DM-09

SOLUTION $d = \frac{|3(1) + 4(2) - 1|}{\sqrt{3^2 + 4^2}} = \frac{10}{5} = 2$.

SHORTCUT Substitute the point, take the modulus, divide by $\sqrt{a^2 + b^2}$.

COMMON TRAP Forgetting the constant term.

10 Answer: (A) *Classical probability* **Easy** ⌚ 1 min **Basic**

DM-10

SOLUTION Favourable pairs: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1), which is 6 of 36 outcomes. $P = 1/6$.

SHORTCUT Sum 7 is the most likely sum, with 6 outcomes.

COMMON TRAP Counting unordered pairs gives 3.

Interpreting your score

Score each subject out of 40 (+4 correct, -1 wrong, 0 blank).

Subject score (/40)	Starting level for that subject	What it tells you
32–40	Advanced start	Foundations are solid. Your gains will come from speed, PYQs and hard questions
20–31	Intermediate start	The core is there. Gaps are chapter-specific: use the self-audit below
below 20	Beginner start	Rebuild concepts first. Don't start mocks until the chapter tests reach 70%
Total (/120)		Overall level to enter in your profile
90+		Advanced
55–89		Intermediate
below 55		Beginner

Also note:

- Your **strongest subject** is the one with the highest score *per minute*, not just the highest score.
- If you didn't finish in 60 minutes, you have a **speed gap**: add a daily Speed Drill (page ?).
- Two or more wrong answers in a subject *where you felt confident* means an **accuracy gap**: start the Mistake Book routine immediately.

Syllabus self-audit

Rate each unit honestly: **0** = not studied · **1** = read, weak · **2** = can solve standard questions · **3** = can solve JEE-level questions under time.

Physics

Unit	0	1	2	3
Units & Measurements	☐	☐	☐	☐
Kinematics	☐	☐	☐	☐
Laws of Motion	☐	☐	☐	☐
Work, Energy, Power	☐	☐	☐	☐
Rotational Motion	☐	☐	☐	☐
Gravitation	☐	☐	☐	☐
Solids & Liquids	☐	☐	☐	☐
Thermodynamics	☐	☐	☐	☐
Kinetic Theory	☐	☐	☐	☐
Oscillations & Waves	☐	☐	☐	☐
Electrostatics	☐	☐	☐	☐
Current Electricity	☐	☐	☐	☐
Magnetism	☐	☐	☐	☐
EMI & AC	☐	☐	☐	☐
EM Waves	☐	☐	☐	☐
Optics	☐	☐	☐	☐
Dual Nature	☐	☐	☐	☐
Atoms & Nuclei	☐	☐	☐	☐
Electronic Devices	☐	☐	☐	☐
Experimental Skills	☐	☐	☐	☐

Chemistry

Unit	0	1	2	3
Basic Concepts	☐	☐	☐	☐
Atomic Structure	☐	☐	☐	☐
Chemical Bonding	☐	☐	☐	☐
Thermodynamics	☐	☐	☐	☐
Solutions	☐	☐	☐	☐
Equilibrium	☐	☐	☐	☐
Redox & Electrochem	☐	☐	☐	☐
Chemical Kinetics	☐	☐	☐	☐
Periodicity	☐	☐	☐	☐
p-Block	☐	☐	☐	☐
d- & f-Block	☐	☐	☐	☐
Coordination Cpd	☐	☐	☐	☐
Purification & Analysis	☐	☐	☐	☐
GOC	☐	☐	☐	☐
Hydrocarbons	☐	☐	☐	☐
Haloalkanes/arenes	☐	☐	☐	☐
Oxygen compounds	☐	☐	☐	☐
Nitrogen compounds	☐	☐	☐	☐
Biomolecules	☐	☐	☐	☐
Practical Chemistry	☐	☐	☐	☐

Mathematics

Unit	0	1	2	3
Sets, Relations, Functions	☐	☐	☐	☐
Complex & Quadratics	☐	☐	☐	☐
Matrices & Determinants	☐	☐	☐	☐
Permutations & Comb.	☐	☐	☐	☐
Binomial Theorem	☐	☐	☐	☐
Sequences & Series	☐	☐	☐	☐
Limits, Cont., Diff.	☐	☐	☐	☐
Integral Calculus	☐	☐	☐	☐
Differential Equations	☐	☐	☐	☐
Coordinate Geometry	☐	☐	☐	☐
3D Geometry	☐	☐	☐	☐
Vector Algebra	☐	☐	☐	☐
Statistics & Probability	☐	☐	☐	☐
Trigonometry	☐	☐	☐	☐

PART I · START HERE

Study-Plan Selector

Answer three questions and you'll have your plan. Each plan in Part X comes in **6 h, 8 h and 10 h per day** variants.

- 1 Q1 · How many days to your exam?**
Count from today to the first day of your session
- 2 Q2 · How many hours can you study per day?**
Honest average, including school days
- 3 Q3 · What is your level?**
From the Diagnostic Test (page [21](#))
- 4 Your plan**
Read it off the tables below and print those pages

Step 1: days → plan

Days left	Plan	Where
170 or more	180-Day Plan	page ?
110–169	120-Day Plan	page ?
80–109	90-Day Plan	page ?
50–79	60-Day Plan	page ?
25–49	30-Day Plan	page ?
14–24	Final 14 Days	page ?
fewer than 14	Final 7 / 3 Days + Exam-Eve	page ?

VERIFIED FACT For Gayathri, starting on 28 September 2026 (calculation)

- **Session 1 (from 22 Jan 2027):** 116 days → **120-Day Plan**, starting from its Day 5. Days 1–4 are the diagnostic and set-up days, done in the first weekend.
- **Session 2 (expected early April 2027, not yet announced):** about 185 days → **180-Day Plan**, with Session 1 used as a full-length "real mock" at its Day ~116.
- Recommended: **follow the 120-Day Plan to Session 1, then the "Session 1 → Session 2 bridge"** (the last block of the 180-Day Plan) up to April.

Step 2: hours → variant

Honest daily average	Variant	Notes
up to 6.5 h	6 h	School-going Class 12 on weekdays; mocks on Sundays
6.5 – 9 h	8 h	Most common; school + evening study, or dropper with coaching
9 h or more	10 h	Dropper or holiday period. Not more than 10 h: beyond that, retention and sleep suffer
School weekdays + free weekends	Hybrid: 6 h Mon–Fri, 10 h Sat–Sun	Averages about 7.1 h/day. The plans mark which tasks move to the weekend

⚠ WARNING Sleep isn't negotiable

Every variant keeps **8 hours of sleep** (7.5 minimum on the occasional late night). For 13–18-year-olds, the American Academy of Sleep Medicine recommends 8–10 hours [LS-AASM]. Memory consolidation happens during sleep: cutting sleep to add study hours reliably *reduces* what you retain.

Step 3: level → modifications

Level	Modification
Beginner	Choose the <i>next longer</i> plan if your days allow. Otherwise drop Low-frequency topics (page ?) until Phase 3 and double the Basic-question practice in Phase 1
Intermediate	Use the plan as written
Advanced	Compress Phase 1 by 25%. Use the saved days for extra full mocks and Hard-question sets

Step 4: weak-subject weighting

Default split of *problem-solving time* is Physics 34% · Chemistry 30% · Maths 36%. Adjust by 10 percentage points towards your weakest subject:

Weakest subject	Physics	Chemistry	Maths
none / balanced	34%	30%	36%
Maths	30%	25%	45%
Physics	44%	25%	31%
Chemistry	30%	40%	30%

My selection

Days left today	
Plan chosen	
Hours variant	
Level modification	
Weak-subject weighting	
Start date	
Re-check date (after 5th mock)	

Physics

Strategy for maximising Physics marks, then 20 unit modules: concepts, formulas, graphs, PYQ patterns, traps, shortcuts, practice sets and chapter tests.

•	Physics: The Complete Strategy	32
01	Units & Measurements	35
02	Kinematics	42
03	Laws of Motion	49
04	Work, Energy & Power	56
05	Rotational Motion & Centre of Mass	62
06	Gravitation	69
07	Properties of Solids & Liquids (incl. Thermal Properties)	75
08	Thermodynamics	83
09	Kinetic Theory of Gases	89
10	Oscillations & Waves	94

11	Electrostatics & Capacitors	100
12	Current Electricity	107
13	Magnetic Effects of Current & Magnetism	113
14	Electromagnetic Induction & Alternating Current	120
15	Electromagnetic Waves	126
16	Optics: Ray & Wave	130
17	Dual Nature of Matter & Radiation	137
18	Atoms & Nuclei	142
19	Electronic Devices	148
20	Experimental Skills	153

PART II · PHYSICS

Physics: The Complete Strategy

What Physics in JEE Main actually tests

JEE Main Physics is mostly "**one idea + clean calculation**". A typical question identifies one principle (conservation of energy, Gauss's law, lens formula, Bohr's model), sets up a standard situation, and checks whether you can apply it quickly and without sign or unit errors. Long multi-concept problems do appear, but a minority of each paper is truly hard. The rest rewards **formula fluency, standard-model recognition and calculation discipline**.

||| **THIRD-PARTY ANALYSIS** [T-ANALYSIS26]

✓ VERIFIED FACT Syllabus structure (official)

The official Physics syllabus has **20 units**. Units 1–19 are theory. **Unit 20, Experimental Skills**, lists 18 standard experiments (Vernier callipers, screw gauge, metre bridge, resonance tube, and others). The syllabus document states that the theory part carries 80% weightage and the experimental part 20% [NTA-SYL]. In practice, directly lab-based questions have been fewer than this per shift, but the experiments' underlying physics (errors, least count, graphs) is tested often. ||| **THIRD-PARTY ANALYSIS**

Unit-wise weight and priority

"Typical questions per shift" is a range compiled from published PYQ analyses of 2024–2026 papers

||| **THIRD-PARTY ANALYSIS** [T-WT-VED], [T-WT-CD], [T-PHY26]. **Priority** is this book's recommendation

🕒 **RECOMMENDATION**, balancing frequency, difficulty and the time needed to master the unit.

#	Unit	Typical Q / shift	Difficulty	Priority	Study hours (Intermediate)
1	Units & Measurements (+ errors)	1–2	Easy	Must-do	6
2	Kinematics	1	Medium	High	10
3	Laws of Motion	1	Medium	High	10
4	Work, Energy & Power (+ collisions)	1–2	Medium	High	8
5	Rotational Motion (+ centre of mass)	1–2	Hard	Must-do	16
6	Gravitation	1–2	Easy–Med	Must-do	7
7	Properties of Solids & Liquids (+ thermal properties)	2–3	Medium	Must-do	14
8	Thermodynamics	1–2	Medium	Must-do	9
9	Kinetic Theory of Gases	1	Easy	Must-do	5
10	Oscillations & Waves	1–2	Medium	High	12
11	Electrostatics (+ capacitors)	2–3	Medium	Must-do	16
12	Current Electricity	2–3	Easy–Med	Must-do	12
13	Magnetic Effects & Magnetism	2–3	Medium	Must-do	14
14	EMI & AC	1–2	Medium	High	12

#	Unit	Typical Q / shift	Difficulty	Priority	Study hours (Intermediate)
15	Electromagnetic Waves	1	Easy	Must-do	3
16	Optics (ray + wave)	2–3	Medium	Must-do	16
17	Dual Nature of Matter & Radiation	1	Easy	Must-do	4
18	Atoms & Nuclei	1–2	Easy	Must-do	6
19	Electronic Devices	1–2	Easy	Must-do	5
20	Experimental Skills	0–2 direct	Easy	High	6

☆ **IMPORTANT** The "easy block"

Units **1, 9, 15, 17, 18, 19** (plus the direct parts of 6 and 12) together usually give **5–8 questions a shift** that are formula-direct and fast. This is the highest-return study time in Physics: about 30 hours of work for roughly 20–30 marks. Master it first if Physics is your weak subject.

How to study a Physics chapter (the 5-pass method) **RECOMMENDATION**

- Pass 1 · Understand (NCERT + module core theory).** Read NCERT for the chapter, then this module's *Concept summary*. Write the formula box by hand once.
- Pass 2 · Model bank.** List the chapter's standard models (block on incline, rolling body, charged sphere, LCR circuit). For each, write *the one equation you start from*.
- Pass 3 · Practice.** Do the module's practice set timed, then 30–60 questions from a standard question bank, then PYQs.
- Pass 4 · Error harvest.** Every wrong question goes into the Mistake Book with its type.
- Pass 5 · Compress.** Reduce the chapter to its half-page Formula Handbook entry and its trap list. Revise from those only.

Using NCERT for Physics

NCERT Physics isn't sufficient on its own for JEE-level problem solving, but it's **necessary** for four things:

- **Definitions and statements** (Lenz's law, Huygens' principle, Brewster's law, Kepler's laws): statement-type questions come from here.
- **Semiconductors, EM waves, Dual Nature, Atoms, Nuclei:** NCERT depth is close to JEE Main depth in these chapters.
- **Units & Measurements** (significant figures, error rules) and the **Lab Manual**, which covers the Unit 20 experiments.
- **Graphs and tables** printed in NCERT (binding-energy curve, stress–strain curve, I–V characteristics, EM spectrum table).

Physics: how to maximise marks

Strong student (Physics $\geq$ 75/100 in mocks)

- Aim for 22–23 attempts at 92%+ accuracy.
- Your losses are silly: signs, unit conversions, reading "radius" as "diameter". Keep a 10-item personal trap list and read it before every mock.
- Practise hard Rotation, EMI and Electrostatics *mixed* problems. These separate 90 from 75.
- Do Section B (NV) Physics first. NV questions are often direct, and strong students can bank them.

Average student (50–75)

- Master the easy block first (above): it's worth 20–30 marks.
- For every medium chapter, learn the **three most-asked models** only. Don't collect tricks.
- In the exam, do Round 1 on "formula $\rightarrow$ numbers" questions only. Leave all multi-concept questions for Round 2.
- Weekly target: 150 Physics questions, with 40 of them PYQs.

Weak student (< 50)

- Physics is where the formula handbook pays off most. Daily 20-minute blank-sheet formula recall.
- Chapter order: Units $\rightarrow$ Modern (17, 18, 19) $\rightarrow$ EM Waves $\rightarrow$ KTG $\rightarrow$ Current $\rightarrow$ Electrostatics $\rightarrow$ Optics $\rightarrow$ Thermodynamics $\rightarrow$ Gravitation, then Mechanics.
- Attempt 15–17 questions, all Easy/Medium, at 90%+ accuracy. That alone is 55–65 marks.
- Use dimensional analysis and limiting cases (Speed Manual) to verify answers.

Physics exam-hall rules RECOMMENDATION

- **Read the last line first.** Is it asking for speed or velocity, total or net, radius or diameter, in cm or m?
- **Write the "given" in SI units** before any algebra.
- **Estimate before you compute.** A rough order of magnitude eliminates 1–2 options in most numerical MCQs.
- **Two-minute rule.** No progress in 2 minutes means mark for review and move on.
- **Section B:** recompute once, using a different path where possible (energy vs force, for example), before entering the number.

PART II · PHYSICS · UNIT 01

Units & Measurements

01

1–2 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 1

NCERT

~6 hours

STUDY TIME

Concept summary

- Every physical quantity = **numerical value** × **unit**. The SI system has **7 base units**. Everything else is derived.
- Dimensions** express a quantity in terms of $[M]$, $[L]$, $[T]$, $[A]$, $[K]$, $[\text{mol}]$, $[\text{cd}]$. They are used to check equations, derive relations and convert units.
- Significant figures** record how precisely something was measured. **Errors** measure how far a measurement may be from the true value.
- JEE Main asks almost only four things here: *dimensions of an unfamiliar constant*, *error propagation*, *significant figures / rounding*, and *instrument readings* (Vernier, screw gauge; see Unit 20).

Core theory

SI base units

Quantity	Unit	Symbol	Dimension
Length	metre	m	$[L]$
Mass	kilogram	kg	$[M]$
Time	second	s	$[T]$
Electric current	ampere	A	$[A]$
Thermodynamic temperature	kelvin	K	$[K]$
Amount of substance	mole	mol	$[\text{mol}]$
Luminous intensity	candela	cd	$[\text{cd}]$

Plane angle (radian) and solid angle (steradian) are **dimensionless**, but they do have units. Since the 2019 SI revision, base units are defined by fixing exact values of constants: h , e , k_B , N_A , c and others.

Dimensions you must know by heart

Quantity	Dimension	Quantity	Dimension
Force	$[MLT^{-2}]$	Pressure, stress, modulus, energy density	$[ML^{-1}T^{-2}]$
Work, energy, torque	$[ML^2T^{-2}]$	Power	$[ML^2T^{-3}]$
Momentum, impulse	$[MLT^{-1}]$	Angular momentum, h	$[ML^2T^{-1}]$
Gravitational constant G	$[M^{-1}L^3T^{-2}]$	Surface tension, spring constant	$[MT^{-2}]$
Coefficient of viscosity η	$[ML^{-1}T^{-1}]$	Moment of inertia	$[ML^2]$
Charge	$[AT]$	Potential (voltage)	$[ML^2T^{-3}A^{-1}]$
Resistance	$[ML^2T^{-3}A^{-2}]$	Capacitance	$[M^{-1}L^{-2}T^4A^2]$
Inductance	$[ML^2T^{-2}A^{-2}]$	Magnetic field B	$[MT^{-2}A^{-1}]$
Permittivity ϵ_0	$[M^{-1}L^{-3}T^4A^2]$	Permeability μ_0	$[MLT^{-2}A^{-2}]$
Electric field	$[MLT^{-3}A^{-1}]$	Magnetic flux	$[ML^2T^{-2}A^{-1}]$
Specific heat	$[L^2T^{-2}K^{-1}]$	Latent heat	$[L^2T^{-2}]$
Thermal conductivity	$[MLT^{-3}K^{-1}]$	Boltzmann constant	$[ML^2T^{-2}K^{-1}]$

⚡ **SHORTCUT** Build dimensions from a defining equation, never from memory alone

Unknown constant? Write the simplest equation it appears in and solve for it. ϵ_0 : from $F = \frac{q^2}{4\pi\epsilon_0 r^2}$ we get $[\epsilon_0] = \frac{[A^2 T^2]}{[MLT^{-2}][L^2]} = [M^{-1} L^{-3} T^4 A^2]$. B : from $F = qvB$. L : from $\epsilon = L dI/dt$. η : from $F = 6\pi\eta r v$. **Time saved:** 30–60 s per question, with far fewer memory errors.

Uses of dimensional analysis

1. **Checking** an equation: every added or subtracted term must have the same dimensions (principle of homogeneity).
2. **Deriving** a relation when the quantity depends on 3 others as a product of powers.
3. **Converting units:** $n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$ for a quantity of dimension $[M^a L^b T^c]$.

⚠ **MISTAKE ALERT** Limitations examiners test

Dimensional analysis **cannot**: find dimensionless constants (2π , $\frac{1}{2}$); derive relations that involve sums ($s = ut + \frac{1}{2}at^2$ as a whole) or trig/log/exp functions; tell apart quantities with the same dimensions (work vs torque). Arguments of $\sin$, $\log$, e^x **must be dimensionless**. That's often the key to the whole question.

Significant figures

- All non-zero digits count. Zeros **between** non-zeros count. **Leading** zeros never count (0.00520 has 3). **Trailing zeros after a decimal point** count (2.50 has 3).
- Trailing zeros in an integer without a decimal point are ambiguous. Use scientific notation: 4.50×10^3 has 3.
- **Multiplication/division:** the result has as many s.f. as the *least* precise factor.
- **Addition/subtraction:** the result has as many *decimal places* as the term with the fewest decimal places.
- **Rounding:** if the digit to drop is > 5 , round up; if < 5 , leave it; if exactly 5 (followed by nothing), make the preceding digit **even**.

Errors

Σ **FORMULA** Error formulas

Operation	Maximum error
$Z = A \pm B$	$\Delta Z = \Delta A + \Delta B$
$Z = AB$ or A/B	$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$
$Z = \frac{A^p B^q}{C^r}$	$\frac{\Delta Z}{Z} = p \frac{\Delta A}{A} + q \frac{\Delta B}{B} + r \frac{\Delta C}{C}$
n readings	mean $\bar{a} = \frac{\sum a_i}{n}$, mean absolute error $\overline{\Delta a} = \frac{\sum a_i - \bar{a} }{n}$
Relative / percentage error	$\delta a = \frac{\overline{\Delta a}}{\bar{a}}$, percentage = $\delta a \times 100\%$

Errors always **add**, even in subtraction and division. That's the maximum-error convention used in JEE.

⚡ **SHORTCUT** Percentage error of a formula in one line

$g = 4\pi^2 L/T^2$: %error in g = % L + 2×% T . Multiply each percentage by the **absolute value of its power** and add. Constants ($4\pi^2$) contribute nothing. **When NOT to use:** when a quantity appears in a sum, e.g. $Z = \frac{A}{A+B}$. There you must differentiate properly.

Instruments (details in Unit 20, page 153)

- **Vernier callipers:** least count = 1 MSD – 1 VSD. If N VSD = $(N - 1)$ MSD, then LC = MSD/ N .

- **Screw gauge:** least count = $\frac{\text{pitch}}{\text{number of circular-scale divisions}}$.
- **Zero error:** true reading = observed reading – (zero error), with the zero error's sign included.

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures (2024–2026 papers)

1. "Find the dimensions of a, b in $X = \dots$ " Van der Waals-type equations, $y = A \sin(\omega t - kx)$, $P = \frac{a - t^2}{bx}$. Use homogeneity term by term.
2. **Combination checks:** which of these has the dimensions of speed / time / energy? ($1/\sqrt{\mu_0 \epsilon_0}$, RC , L/R , $\sqrt{LC}$, hc/λ , $\epsilon_0 E^2$, B^2/μ_0)
3. **Error in a derived quantity** (density of a sphere, g from a pendulum, Young's modulus): the power rule.
4. **Significant-figure arithmetic** and rounding.
5. **Vernier/screw-gauge reading with zero error**, sometimes combined with an error calculation.
6. **New unit system:** "If the units of mass, length and time are changed to..., find the new value of 1 J."

Question classification

Type	Difficulty	Approach	Target time
Dimensions of a named quantity	E	Defining equation	45 s
Dimensions of constants in an unfamiliar formula	E–M	Homogeneity, term by term	1.5 min
Percentage error, power rule	E	One line	45 s
Error with a sum/difference inside	M	Differentiate / add absolute errors	2 min
Instrument reading + zero error	E–M	LC first, then sign of zero error	1.5 min
Dimensional derivation (3 variables)	M	Solve 3 equations in M, L, T	2 min

Selection rule: almost always Round-1 questions. Skip only if the unfamiliar formula is very long.

Common mistakes

MISTAKE ALERT Mistake alerts

- Treating an angle as having a dimension. It's dimensionless, even though it has a unit (rad).
- Subtracting errors in $A - B$. **Errors add.**
- Forgetting to multiply by the power: %error in r^3 is $3 \times \%r$.
- Getting the zero-error sign backwards. With the jaws closed, a reading *above* zero is a **positive** zero error, and you subtract it.
- Counting leading zeros as significant.
- Rounding intermediate results. Keep one extra digit and round at the end.

Practice questions

P01 · Practice

1 Easy 1 min Basic

P01-01

The dimensions of $\frac{1}{2}\epsilon_0 E^2$ (ϵ_0 = permittivity of free space, E = electric field) are:

- (A) $[MLT^{-2}]$ (B) $[ML^{-1}T^{-2}]$ (C) $[ML^2T^{-2}]$ (D) $[ML^{-2}T^{-1}]$

2 Medium 1.5 min Concept

P01-02

In $\left(P + \frac{a}{V^2}\right)(V - b) = RT$, the dimensions of a/b are those of:

- (A) pressure (B) energy (C) force (D) momentum

3 Easy 0.75 min Speed

P01-03

In $g = 4\pi^2 L/T^2$, the percentage errors in L and T are 1% and 2%. The maximum percentage error in g is:

- (A) 3% (B) 4% (C) 5% (D) 1%

4 Easy 1 min Basic

P01-04

The product 12.5×1.25 , reported to the correct number of significant figures, is:

- (A) 15.625 (B) 15.63 (C) 15.6 (D) 16

5 Medium 2 min JEE

P01-05

A Vernier calliper has 1 MSD = 1 mm and 10 VSD = 9 MSD. With the jaws closed, the Vernier zero lies to the **right** of the main-scale zero and the 3rd Vernier division coincides with a main-scale division. When a rod is measured, the main-scale reading is 32 mm and the 6th Vernier division coincides. The corrected diameter is:

- (A) 3.29 cm (B) 3.26 cm (C) 3.23 cm (D) 3.20 cm

6 Medium 2 min JEE

P01-06

A screw gauge has pitch 0.5 mm and 50 circular-scale divisions. With the studs closed, the reference line reads +4 divisions on the circular scale. Measuring a wire, the main-scale reading is 2.5 mm and the circular-scale reading is 36. The diameter of the wire is:

- (A) 2.86 mm (B) 2.82 mm (C) 2.90 mm (D) 2.54 mm

7 Medium 1.5 min NV

P01-07

In a new system, the unit of mass is 5 kg, the unit of length is 2 m and the unit of time is 2 s. What is the numerical value of 100 J in this system?

Your answer (numerical value):

8 Medium 2 min Concept

P01-08

The time period T of oscillation of a small liquid drop depends on its density ρ , radius r and surface tension σ . Dimensional analysis gives $T \propto$:

- (A) $\sqrt{\rho r^3/\sigma}$ (B) $\sqrt{\sigma/(\rho r^3)}$ (C) $\sqrt{\rho r/\sigma}$ (D) $\rho r^3/\sigma$

9 Medium 2 min Calc

P01-09

Five measurements of a time period are 1.52 s, 1.48 s, 1.55 s, 1.45 s and 1.50 s. The period, with its mean absolute error, is:

- (A) (1.50 ± 0.03) s (B) (1.50 ± 0.05) s (C) (1.51 ± 0.02) s (D) (1.50 ± 0.10) s

P01 · Chapter Test

1 Easy 0.75 min Speed

P01-T1

The dimensions of $\mu_0 \epsilon_0$ are:

- (A) $[L^2 T^{-2}]$ (B) $[L^{-2} T^2]$ (C) $[L T^{-1}]$ (D) dimensionless

2 Easy 0.5 min Basic

P01-T2

Which pair has the same dimensions?

- (A) torque and work (B) momentum and impulse per unit time
(C) stress and strain (D) power and energy

3 Easy 1 min Basic

P01-T3

 $V = (100 \pm 5) \text{ V}$ and $I = (10 \pm 0.2) \text{ A}$. The percentage error in $R = V/I$ is:

- (A) 3% (B) 5% (C) 7% (D) 2.5%

4 Easy 0.5 min NV

P01-T4

How many significant figures are there in 0.007020?

Your answer (numerical value):

5 Medium 1.5 min Concept

P01-T5

 E = energy, J = angular momentum, M = mass, G = gravitational constant. The quantity $\frac{EJ^2}{M^5G^2}$ has the dimensions of:

- (A) length (B) time
(C) mass (D) an angle (dimensionless)

Answers & Solutions

P01 · Practice

1 (B) 2 (B) 3 (C) 4 (C) 5 (C) 6 (B) 7 20 8 (A) 9 (A)

P01 · Chapter Test

1 (B) 2 (A) 3 (C) 4 4 5 (D)

P01 · PRACTICE

1 Answer: (B) Dimensions of energy density Easy 1 min Basic

P01-01

SOLUTION $\frac{1}{2}\epsilon_0 E^2$ is the **energy per unit volume** of an electric field: $[ML^2T^{-2}]/[L^3] = [ML^{-1}T^{-2}]$.**SHORTCUT** Energy density has the dimensions of pressure. Same for $B^2/2\mu_0$.**COMMON TRAP** Answering the dimension of energy.

2 Answer: (B) Homogeneity in van der Waals equation Medium 1.5 min Concept

P01-02

SOLUTION a/V^2 must be a pressure, so $[a] = [PV^2]$. b must be a volume, so $[b] = [V]$. Then $[a/b] = [PV]$, which is energy (PV has units $\text{N m}^{-2} \times \text{m}^3 = \text{J}$).**SHORTCUT** $a/b \sim PV \rightarrow \text{energy}$.**COMMON TRAP** Stopping at " a has dimensions of PV^2 " and choosing pressure.

3 Answer: (C) Percentage error with powers Easy 0.75 min Speed

P01-03

SOLUTION $\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2\frac{\Delta T}{T} = 1\% + 4\% = 5\%$.**SHORTCUT** Power rule: add $|\text{power}| \times \%\text{error}$.**COMMON TRAP** Using -2 and subtracting gives 3% .

4 Answer: (C) Significant figures in multiplication Easy 1 min Basic

P01-04

SOLUTION Both factors have 3 s.f., so the result keeps 3 s.f.: $15.625 \rightarrow 15.6$.**SHORTCUT** Multiplication: keep the least number of s.f.**COMMON TRAP** Applying the decimal-place rule, which is for addition only.

5 Answer: (C) Vernier reading with zero error **Medium** ⌚ 2 min JEE P01-05

SOLUTION $LC = 1 \text{ MSD} - 1 \text{ VSD} = 1 - 0.9 = 0.1 \text{ mm}$. Zero error = $+3 \times 0.1 = +0.3 \text{ mm}$ (zero to the right means positive).
Observed = $32 + 6(0.1) = 32.6 \text{ mm}$. Corrected = $32.6 - 0.3 = 32.3 \text{ mm} = 3.23 \text{ cm}$.

SHORTCUT Corrected = observed – (signed zero error).

COMMON TRAP Adding the positive zero error gives 3.29 cm.

6 Answer: (B) Screw gauge with zero error **Medium** ⌚ 2 min JEE P01-06

SOLUTION $LC = 0.5/50 = 0.01 \text{ mm}$. Zero error = $+4 \times 0.01 = +0.04 \text{ mm}$. Observed = $2.5 + 36(0.01) = 2.86 \text{ mm}$. Corrected = $2.86 - 0.04 = 2.82 \text{ mm}$.

SHORTCUT Observed – zero error.

COMMON TRAP Forgetting the correction gives 2.86.

7 Answer: 20 Conversion to a new unit system **Medium** ⌚ 1.5 min NV P01-07

SOLUTION New energy unit = $(5 \text{ kg})(2 \text{ m})^2/(2 \text{ s})^2 = 5 \text{ J}$. So $100 \text{ J} = 100/5 = 20$ new units.

SHORTCUT Compute the size of the new unit, then divide.

COMMON TRAP Inverting the ratio gives 500.

8 Answer: (A) Dimensional derivation **Medium** ⌚ 2 min Concept P01-08

SOLUTION Let $T = k\rho^a r^b \sigma^c$. $[\sigma] = [MT^{-2}]$. M: $a + c = 0$. L: $-3a + b = 0$. T: $-2c = 1$. So $c = -\frac{1}{2}$, $a = \frac{1}{2}$, $b = \frac{3}{2}$, giving $T \propto \sqrt{\rho r^3/\sigma}$.

SHORTCUT The time dimension comes only from σ , so σ must sit under a square root in the denominator. That leaves (A) or (C), and only (A) balances M and L .

COMMON TRAP Taking $[\sigma] = [MLT^{-2}]$ (the dimensions of force).

9 Answer: (A) Mean absolute error **Medium** ⌚ 2 min Calc P01-09

SOLUTION Mean = $7.50/5 = 1.50 \text{ s}$. Deviations: 0.02, 0.02, 0.05, 0.05, 0.00. Mean absolute error = $0.14/5 = 0.028 \approx 0.03 \text{ s}$.

SHORTCUT Round the error to the same decimal place as the mean.

COMMON TRAP Using the largest deviation (0.05) instead of the mean deviation.

P01 · CHAPTER TEST

1 Answer: (B) Speed of light from constants **Easy** ⌚ 0.75 min Speed P01-T1

SOLUTION $c = 1/\sqrt{\mu_0 \epsilon_0}$, so $\mu_0 \epsilon_0 = 1/c^2$, which has dimensions $[L^{-2}T^2]$.

SHORTCUT Memorise: $1/\sqrt{\mu_0 \epsilon_0} = \text{speed}$.

COMMON TRAP Answering the dimension of c^2 .

2 Answer: (A) Same dimensions **Easy** ⌚ 0.5 min Basic P01-T2

SOLUTION Torque ($r \times F$) and work ($F \cdot d$) are both $[ML^2T^{-2}]$. Impulse per unit time is force. Strain is dimensionless. Power is energy per unit time.

SHORTCUT Same dimensions don't mean the same physical quantity: torque is a vector, work a scalar.

COMMON TRAP Thinking momentum = impulse per unit time. Momentum = impulse.

3 Answer: (C) Error in a quotient **Easy** ⌚ 1 min Basic P01-T3

SOLUTION $\frac{\Delta R}{R} = \frac{5}{100} + \frac{0.2}{10} = 5\% + 2\% = 7\%$.

SHORTCUT Quotient: add the relative errors.

COMMON TRAP Subtracting (3%).

4 Answer: 4 Counting significant figures **Easy** ⌚ 0.5 min NV P01-T4

SOLUTION Leading zeros don't count. 7, 0 (between non-zeros), 2 and the trailing 0 after the decimal point do: 4 s.f.

SHORTCUT Start counting from the first non-zero digit.

COMMON TRAP Excluding the final zero gives 3.

5 Answer: (D) Dimensionless combination **Medium** ⌚ 1.5 min Concept P01-T5

SOLUTION $[EJ^2] = [ML^2T^{-2}][M^2L^4T^{-2}] = [M^3L^6T^{-4}]$. $[M^5G^2] = [M^5][M^{-2}L^6T^{-4}] = [M^3L^6T^{-4}]$. The ratio is dimensionless.

SHORTCUT Count powers of T first: numerator T^{-4} , denominator T^{-4} .

COMMON TRAP Arithmetic slip in the powers of G .

**REVISION** Revision tracker · Units & Measurements

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Kinematics

02

~1 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

High

PRIORITY

Class 11 · Ch 2, 3

NCERT

~10 hours

STUDY TIME

Concept summary

- Kinematics** describes motion (position, velocity, acceleration) without asking what causes it.
- Everything comes from three definitions: $v = \frac{dx}{dt}$, $a = \frac{dv}{dt} = v \frac{dv}{dx}$, and the **slope/area** rules for graphs.
- For constant acceleration, the equations of motion are exact. For variable acceleration, **integrate**.
- 2D motion (projectiles, relative velocity, circular motion) = two independent 1D motions plus vector addition.

Core theory & definitions

Term	Meaning	Key point
Distance	Total path length (scalar)	Never decreases; $\geq$ displacement
Displacement	Change in position vector	Can be zero for non-zero distance
Average velocity	$\Delta \vec{r} / \Delta t$	Average speed = distance/time $\geq$ average velocity
Instantaneous velocity	$d\vec{r}/dt$	Always tangent to the path
Acceleration	$d\vec{v}/dt$	Can be non-zero at an instant where $v = 0$ (top of a vertical throw)

Formulas

Σ FORMULA Uniformly accelerated motion (1D)

$$v = u + at \quad s = ut + \frac{1}{2}at^2 \quad v^2 = u^2 + 2as \quad s = \frac{(u+v)}{2}t$$

$$s_{n^{\text{th}}} = u + \frac{a}{2}(2n - 1) \quad (\text{displacement in the } n^{\text{th}} \text{ second})$$

Free fall from rest: $h = \frac{1}{2}gt^2$, $v = \sqrt{2gh}$. **Thrown up with u :** $H_{\text{max}} = \frac{u^2}{2g}$, time to top = u/g , return time = $2u/g$.

From rest, distances in successive equal intervals are in the ratio 1 : 3 : 5 : 7 : ...

Σ FORMULA Variable acceleration

$$v = \int a \, dt \quad x = \int v \, dt \quad a = v \frac{dv}{dx} \Rightarrow \int v \, dv = \int a \, dx$$

Use $a = v \, dv/dx$ whenever a or v is given **as a function of position**.

Σ FORMULA Projectile motion (ground to ground, launch speed u at angle θ)

$$T = \frac{2u \sin \theta}{g} \quad H = \frac{u^2 \sin^2 \theta}{2g} \quad R = \frac{u^2 \sin 2\theta}{g} \quad R_{\text{max}} = \frac{u^2}{g} \quad (\theta = 45^\circ)$$

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta} = x \tan \theta \left(1 - \frac{x}{R}\right)$$

- Complementary angles (θ and $90^\circ - \theta$) give the **same range**. $H_1/H_2 = \tan^2 \theta$ for the pair, and $H_1 H_2 = R^2/16$.
- $\frac{H}{R} = \frac{\tan \theta}{4}$ and $\frac{H}{T^2} = \frac{g}{8}$.
- Horizontal launch** from height h at speed u : $t = \sqrt{2h/g}$, range = $u\sqrt{2h/g}$.
- At the highest point: speed = $u \cos \theta$, acceleration = g (downward), and velocity $\perp$ acceleration.

Σ FORMULA Relative velocity

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$$

River crossing (width d , boat speed v relative to water, river speed u):

- Minimum time: head straight across. $t_{\min} = d/v$, drift $= ud/v$.
- Zero drift (shortest path, needs $v > u$): head upstream at $\sin \alpha = u/v$ from the perpendicular. Then $t = \frac{d}{\sqrt{v^2 - u^2}}$.

Rain-person problems: $\vec{v}_{\text{rain, rel. person}} = \vec{v}_{\text{rain}} - \vec{v}_{\text{person}}$. Hold the umbrella along the **relative** velocity.

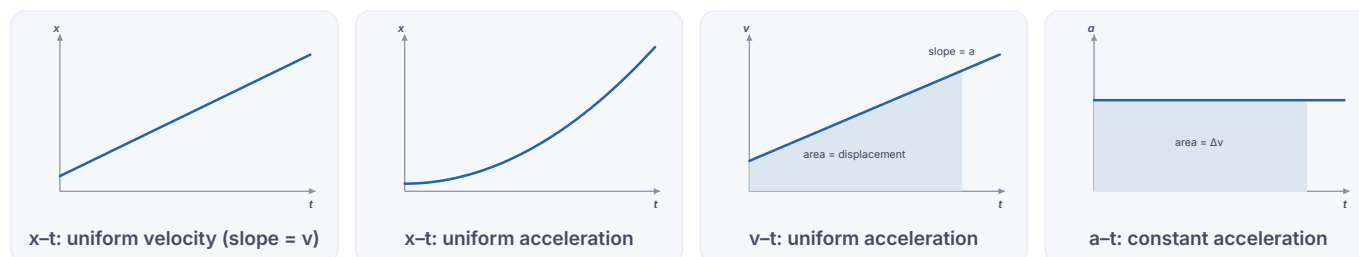
Σ FORMULA Circular motion

$$v = \omega r \quad a_c = \frac{v^2}{r} = \omega^2 r \quad a_t = \frac{dv}{dt} = \alpha r \quad a_{\text{net}} = \sqrt{a_c^2 + a_t^2}$$

$\omega = 2\pi/T = 2\pi f$. In uniform circular motion, speed is constant but velocity is not, so the particle accelerates towards the centre.

Σ FORMULA Vectors (used everywhere)

$|\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}$; direction $\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$ from $\vec{A}$. $\vec{A} \cdot \vec{B} = AB \cos \theta$; $|\vec{A} \times \vec{B}| = AB \sin \theta$; component of $\vec{A}$ along $\vec{B} = \vec{A} \cdot \hat{B}$. $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}| \iff \vec{A} \perp \vec{B}$.

Graphs and how to read them

Graph	Slope gives	Area gives
$x-t$	velocity	—
$v-t$	acceleration	displacement (signed); distance = total unsigned area
$a-t$	jerk (not tested)	change in velocity
$v-x$	dv/dx , so $a = v \times \text{slope}$	—

⚠ MISTAKE ALERT Graph traps

- A straight $x-t$ line means **constant velocity**, not constant acceleration.
- A curved $x-t$ graph bending upward means positive acceleration.
- On a $v-t$ graph, the area **below** the time axis is negative displacement. Distance counts it as positive.
- A $v-t$ graph can never be vertical (that would need infinite acceleration). An $x-t$ graph can never go backwards in time.

Standard models & assumptions

- Air resistance neglected; g constant (take 10 m s^{-2} unless told otherwise).
- Projectiles: horizontal velocity constant, vertical motion is free fall.
- "Particle" means size is ignored and there is no rotation.

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **Variable acceleration:** $x(t)$ or $v(t)$ given as a polynomial, asked for the distance (not displacement) in an interval. Find where $v = 0$ first.
2. **Projectile relations:** H vs R , complementary angles, equation of trajectory $\rightarrow u$ and θ .
3. **River-boat / rain-umbrella** relative velocity.
4. **Graph conversion:** given an $a-t$ or $v-t$ graph, find displacement or the shape of another graph.
5. n^{th} -**second distance** and ratio questions.
6. **Circular motion:** net acceleration with a tangential component.

Shortcuts

SHORTCUT Distance vs displacement for polynomial motion

If $v(t)$ changes sign inside the interval, split the interval at the root of $v = 0$. Compute $|x(t_1) - x(0)| + |x(t_2) - x(t_1)|$. **Time saved:** avoids integrating $|v|$. **Common mistake:** forgetting that the particle may reverse direction.

SHORTCUT Trajectory equation $\rightarrow$ everything

Write $y = ax - bx^2$. Then $\tan \theta = a$, $R = a/b$, $H = a^2/(4b)$, and $u^2 = \frac{g}{2b \cos^2 \theta}$. **When NOT to use:** when the launch point isn't at the origin.

Question classification

Type	Difficulty	Target time
Equations of motion, free fall	E	1 min
Projectile formulas / ratios	E-M	1.5 min
Variable acceleration with calculus	M	2-3 min
Relative motion (river, rain, two bodies)	M	2 min
Graph interpretation	E-M	1.5 min
Projectile on an incline, or two-body chase	H	3-4 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Using equations of motion when acceleration is **not** constant.
- Sign convention: pick "up = +" and **stay with it**. With up positive, g enters as -10 .
- In river problems, confusing "shortest time" (head straight across) with "shortest path" (head upstream).
- Treating speed at the top of a projectile's flight as zero. It's $u \cos \theta$.

Practice questions

P02 · Practice

1 Easy 1 min Speed

P02-01

A particle starts from rest with uniform acceleration and covers 15 m in the 3rd second. The distance it covers in the 5th second is:

- (A) 21 m (B) 25 m (C) 27 m (D) 30 m

2 Medium 1.5 min Concept

P02-02

A projectile's maximum height equals its horizontal range. The angle of projection is:

- (A) $\tan^{-1} 4$ (B) $\tan^{-1} 2$ (C) 45° (D) $\tan^{-1}(1/4)$

3 Medium 1.5 min Concept

P02-03

Rain falls vertically at 12 m s^{-1} . A woman walks east at 5 m s^{-1} . To protect herself she should hold her umbrella:

- (A) vertically
(B) tilted towards the east at $\tan^{-1}(5/12)$ from the vertical
(C) tilted towards the west at $\tan^{-1}(5/12)$ from the vertical
(D) tilted towards the east at $\tan^{-1}(12/5)$ from the vertical

4 Medium 1.5 min NV

P02-04

A river 100 m wide flows at 3 m s^{-1} . A boat moves at 5 m s^{-1} relative to the water. Find the time (in s) to cross along the shortest path, landing directly opposite.

Your answer (numerical value):

5 Medium 2 min Tricky

P02-05

A particle moves along the x -axis with $v = 3t^2 - 6t$ (SI units), starting at $x = 0$. The distance it travels in the first 3 s is:

- (A) 0 m (B) 4 m (C) 8 m (D) 12 m

6 Easy 1 min Concept

P02-06

A particle moves so that $v = 2\sqrt{x}$ (SI units). Its acceleration is:

- (A) 1 m s^{-2} , constant (B) 2 m s^{-2} , constant
(C) 4 m s^{-2} , constant (D) proportional to $\sqrt{x}$

7 Easy 1.5 min NV

P02-07

A ball is thrown horizontally at 20 m s^{-1} from the top of a 45 m tower ($g = 10 \text{ m s}^{-2}$). Find the horizontal distance (in m) from the foot of the tower where it lands.

Your answer (numerical value):

8 Medium 1.5 min JEE

P02-08

A particle moves on a circle of radius 2 m. Its speed increases at a constant 3 m s^{-2} . When its speed is 4 m s^{-1} , the magnitude of its acceleration is:

- (A) $\sqrt{73} \text{ m s}^{-2}$ (B) 11 m s^{-2} (C) 5 m s^{-2} (D) 8 m s^{-2}

9 Easy 0.5 min Speed

P02-09

If $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$ for non-zero vectors, the angle between them is:

- (A) 0° (B) 45° (C) 90° (D) 180°

P02 · Chapter Test

1 Easy 1 min Basic

P02-T1

A stone dropped from height h reaches the ground in time T . At time $T/2$, its height above the ground is:

- (A) $h/2$ (B) $h/4$ (C) $3h/4$ (D) $h/8$

2 Easy 1 min Basic

P02-T2

Two projectiles are launched at the same speed at 30° and 60° . The ratio of their maximum heights $H_{30} : H_{60}$ is:

- (A) 1 : 1 (B) 1 : 3 (C) 3 : 1 (D) $1 : \sqrt{3}$

3 Medium 1.5 min JEE

P02-T3

A projectile's trajectory is $y = \sqrt{3}x - \frac{x^2}{20}$ (SI units, $g = 10 \text{ m s}^{-2}$). Its speed of projection is:

- (A) 10 m s^{-1} (B) 20 m s^{-1} (C) $20\sqrt{3} \text{ m s}^{-1}$ (D) 40 m s^{-1}

4 Easy 1 min NV

P02-T4

A car moving at 20 m s^{-1} brakes uniformly and stops in 40 m. Find the magnitude of its deceleration (in m s^{-2}).

Your answer (numerical value):

5 Easy 0.5 min Speed

P02-T5

The angular speed of the minute hand of a clock is:

- (A) $\pi/30 \text{ rad s}^{-1}$ (B) $\pi/1800 \text{ rad s}^{-1}$ (C) $\pi/3600 \text{ rad s}^{-1}$ (D) $\pi/21600 \text{ rad s}^{-1}$

Answers & Solutions

P02 · Practice

1 (C) 2 (A) 3 (B) 4 25 5 (C) 6 (B) 7 60 8 (A) 9 (C)

P02 · Chapter Test

1 (C) 2 (B) 3 (B) 4 5 5 (B)

P02 · PRACTICE

1 Answer: (C) Distance in n th second Easy 1 min Speed

P02-01

SOLUTION $s_n = \frac{a}{2}(2n-1)$ for $u = 0$. $s_3 = \frac{5a}{2} = 15 \Rightarrow a = 6 \text{ m s}^{-2}$. $s_5 = \frac{9a}{2} = 27 \text{ m}$.

SHORTCUT $s_5 : s_3 = 9 : 5$, so $15 \times 9/5 = 27$.

COMMON TRAP Using $s = \frac{1}{2}at^2$ for the whole 5 s.

2 Answer: (A) $H = R$ condition Medium 1.5 min Concept

P02-02

SOLUTION $\frac{H}{R} = \frac{\tan \theta}{4} = 1 \Rightarrow \tan \theta = 4$.

SHORTCUT Memorise $H/R = \tan \theta/4$.

COMMON TRAP Guessing 45° (which gives the maximum range, not $H = R$).

3 Answer: (B) Rain-person relative velocity Medium 1.5 min Concept

P02-03

SOLUTION $\vec{v}_{\text{rain, woman}} = \vec{v}_{\text{rain}} - \vec{v}_{\text{woman}} = (-5\hat{i} - 12\hat{j}) \text{ m s}^{-1}$. The rain appears to come from the east, falling towards the west. The umbrella must face the incoming rain, so tilt it **forward (east)** at $\tan^{-1}(5/12)$ from the vertical. Relative speed = 13 m s^{-1} .

SHORTCUT Tilt the umbrella in the direction you are walking.

COMMON TRAP Measuring the angle from the horizontal gives (D).

4 Answer: 25 River crossing: zero drift **Medium** ⌚ 1.5 min **NV**

P02-04

SOLUTION For zero drift the boat heads upstream so that its upstream component cancels the river: $5 \sin \alpha = 3$. The across-component is $\sqrt{25 - 9} = 4 \text{ m s}^{-1}$, so $t = 100/4 = 25 \text{ s}$.

SHORTCUT $t = d/\sqrt{v^2 - u^2}$.

COMMON TRAP Answering $100/5 = 20 \text{ s}$ (that's the minimum-time case, which has drift).

5 Answer: (C) Distance for velocity with sign change **Medium** ⌚ 2 min **Tricky**

P02-05

SOLUTION $x = t^3 - 3t^2$. $v = 0$ at $t = 0$ and $t = 2 \text{ s}$. $x(2) = -4$, $x(3) = 0$. Distance = $|-4 - 0| + |0 - (-4)| = 8 \text{ m}$. (Displacement = 0.)

SHORTCUT Split at the root of $v = 0$.

COMMON TRAP Answering the displacement (0 m).

6 Answer: (B) Acceleration from $v(x)$ **Easy** ⌚ 1 min **Concept**

P02-06

SOLUTION $a = v \frac{dv}{dx} = 2\sqrt{x} \cdot \frac{1}{\sqrt{x}} = 2 \text{ m s}^{-2}$.

SHORTCUT $v^2 = 4x$; compare with $v^2 = 2ax$, giving $a = 2$.

COMMON TRAP Differentiating v with respect to x and stopping there.

7 Answer: 60 Horizontal projection **Easy** ⌚ 1.5 min **NV**

P02-07

SOLUTION Fall time $t = \sqrt{2h/g} = \sqrt{9} = 3 \text{ s}$. $x = ut = 60 \text{ m}$.

SHORTCUT Fall time depends only on height.

COMMON TRAP Using the projectile range formula $u^2 \sin 2\theta/g$ (which is for ground-to-ground launches).

8 Answer: (A) Net acceleration in non-uniform circular motion **Medium** ⌚ 1.5 min **JEE**

P02-08

SOLUTION $a_c = v^2/r = 16/2 = 8$; $a_t = 3$. $a = \sqrt{64 + 9} = \sqrt{73} \approx 8.5 \text{ m s}^{-2}$.

SHORTCUT The two components are perpendicular: add in quadrature.

COMMON TRAP Adding them directly (11).

9 Answer: (C) Equal sum and difference **Easy** ⌚ 0.5 min **Speed**

P02-09

SOLUTION Squaring: $A^2 + B^2 + 2\vec{A} \cdot \vec{B} = A^2 + B^2 - 2\vec{A} \cdot \vec{B} \Rightarrow \vec{A} \cdot \vec{B} = 0$.

SHORTCUT Diagonals of a parallelogram are equal only for a rectangle.

COMMON TRAP None; this is a pure speed item.

P02 · CHAPTER TEST**1 Answer: (C)** Free fall at half time **Easy** ⌚ 1 min **Basic**

P02-T1

SOLUTION Distance fallen $\propto t^2$: in $T/2$ it falls $h/4$. Height above ground = $3h/4$.

SHORTCUT Ratio 1 : 3 in two equal halves of the time.

COMMON TRAP Answering the distance fallen ($h/4$).

2 Answer: (B) Complementary angles **Easy** ⌚ 1 min **Basic**

P02-T2

SOLUTION $H \propto \sin^2 \theta$: $\sin^2 30^\circ : \sin^2 60^\circ = \frac{1}{4} : \frac{3}{4} = 1 : 3$. (The ranges are equal.)

SHORTCUT $H_1/H_2 = \tan^2 \theta_1$ for complementary angles: $\tan^2 30^\circ = 1/3$.

COMMON TRAP Confusing heights with ranges (1:1).

3 Answer: (B) Speed from trajectory equation **Medium** ⌚ 1.5 min **JEE**

P02-T3

SOLUTION $\tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ$, $\cos^2 \theta = \frac{1}{4}$. $\frac{g}{2u^2 \cos^2 \theta} = \frac{1}{20} \Rightarrow \frac{10}{u^2/2} = \frac{1}{20} \Rightarrow u^2 = 400$, so $u = 20 \text{ m s}^{-1}$.

SHORTCUT $u^2 = \frac{g}{2b \cos^2 \theta}$ with $b = 1/20$.

COMMON TRAP Forgetting $\cos^2 \theta$.

4 Answer: 5 Braking distance **Easy** ⌚ 1 min **NV**

P02-T4

SOLUTION $0 = u^2 - 2as \Rightarrow a = \frac{400}{80} = 5 \text{ m s}^{-2}$.

SHORTCUT Stopping distance = $u^2/2a$.

COMMON TRAP Forgetting the factor 2.

5 Answer: (B) Angular speed of a clock hand **Easy** ⌚ 0.5 min **Speed**

P02-T5

SOLUTION $\omega = 2\pi/T = 2\pi/3600 = \pi/1800 \text{ rad s}^{-1}$.

SHORTCUT Minute hand: $T = 1 \text{ h}$. Hour hand: $T = 12 \text{ h}$.

COMMON TRAP Using 60 s (the second hand).

 **REVISION** Revision tracker · Kinematics

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 03

Laws of Motion

03

~1 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

High

PRIORITY

Class 11 · Ch 4

NCERT

~10 hours

STUDY TIME

Concept summary

- **Newton I:** a body stays at rest or in uniform motion unless a net external force acts. It defines inertial frames.
- **Newton II:** $\vec{F}_{net} = \frac{d\vec{p}}{dt}$, which is $m\vec{a}$ for constant mass. **Newton III:** forces come in equal and opposite pairs acting on *different* bodies.
- **Method for every problem:** isolate each body → draw a **free-body diagram (FBD)** → choose axes → write $\sum F = ma$ per axis → add the **constraint** relations (string length, contact) → solve.
- **Friction** adjusts itself up to $\mu_s N$ (static) and is $\mu_k N$ once sliding. **Circular motion** needs a net inward force mv^2/r supplied by real forces.

Formulas

Σ FORMULA Momentum & impulse

$$\vec{p} = m\vec{v} \quad \vec{J} = \int \vec{F} dt = \Delta\vec{p} \quad \text{No external force} \Rightarrow \vec{p}_{total} = \text{constant}$$

Recoil: $m_{bullet}v_{bullet} = M_{gun}V_{gun}$. For a rebound from a wall: $|\Delta p| = m(v_1 + v_2)$.

Σ FORMULA Standard systems

System	Acceleration	Tension / normal force
Atwood machine ($m_2 > m_1$)	$a = \frac{(m_2 - m_1)g}{m_1 + m_2}$	$T = \frac{2m_1m_2g}{m_1 + m_2}$
Block m_1 on smooth table pulled by hanging m_2	$a = \frac{m_2g}{m_1 + m_2}$	$T = \frac{m_1m_2g}{m_1 + m_2}$
Lift accelerating up at a	—	apparent weight $N = m(g + a)$
Lift accelerating down at a	—	$N = m(g - a)$; free fall: $N = 0$
Smooth incline	$a = g \sin \theta$	$N = mg \cos \theta$
Rough incline, sliding down	$a = g(\sin \theta - \mu_k \cos \theta)$	—
Pendulum in a car accelerating at a	string at $\tan \theta = a/g$ from the vertical	$T = m\sqrt{g^2 + a^2}$

Σ FORMULA Friction

$$f_s \leq \mu_s N \quad f_k = \mu_k N \quad \mu_k < \mu_s \quad \text{angle of repose: } \tan \theta_r = \mu_s \quad \text{angle of friction: } \tan \lambda = \mu_s$$

- Minimum force to move a block on a horizontal surface (pull at angle ϕ above the horizontal): $F_{min} = \frac{\mu mg}{\sqrt{1 + \mu^2}}$ at $\tan \phi = \mu$.
- Two blocks (top m on bottom M , force on the bottom block): they move together as long as $F \leq \mu_s(m + M)g$.

Σ FORMULA Circular dynamics

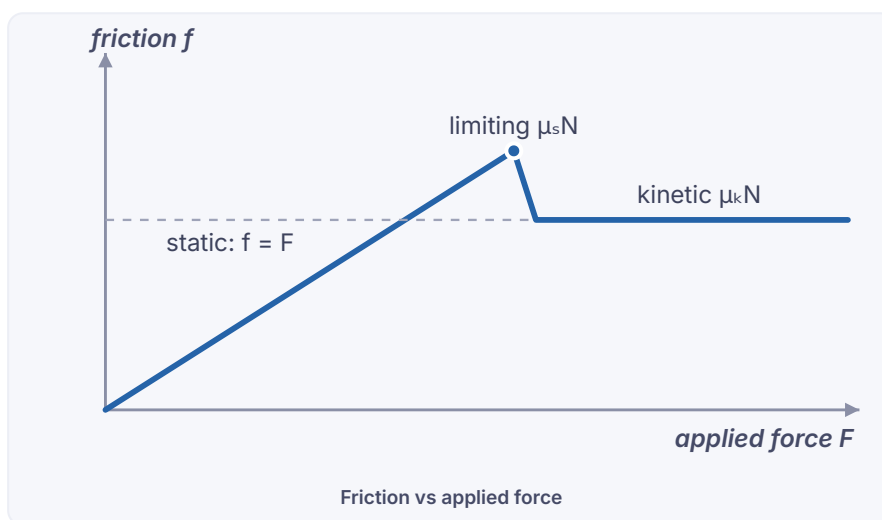
Level road: $v_{max} = \sqrt{\mu rg}$ Banked road, no friction: $v = \sqrt{rg \tan \theta}$

Banked road with friction: $v_{max} = \sqrt{rg \frac{\mu + \tan \theta}{1 - \mu \tan \theta}}$ $v_{min} = \sqrt{rg \frac{\tan \theta - \mu}{1 + \mu \tan \theta}}$

Conical pendulum (string length L , angle θ): $\tan \theta = \frac{v^2}{rg}$, $T = 2\pi \sqrt{\frac{L \cos \theta}{g}}$.

Σ FORMULA Equilibrium of concurrent forces

$\sum \vec{F} = 0$. **Lami's theorem** for three forces: $\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$, where each angle is the one **opposite** that force (between the other two).

**☆ IMPORTANT Friction is a self-adjusting force**

Below the limit, static friction equals exactly what's needed to prevent slipping. It is **not** $\mu_s N$. The value $\mu_s N$ is only the maximum. Always first check whether the body slips: compare the applied force (or the force needed for common motion) with $\mu_s N$.

Standard models & assumptions

- Strings are massless and inextensible (same tension throughout, same acceleration magnitude at both ends). Pulleys are massless and frictionless unless stated otherwise.
- "Smooth" means no friction. "Rough" means friction, with μ given.
- **Pseudo force** $-m\vec{a}_0$ acts on every body when you analyse from a frame accelerating at $\vec{a}_0$.

Frequently tested concepts & question patterns**▮ PYQ PATTERN Recurring structures**

1. **Two-block systems** with friction between blocks: find the maximum force for common motion, or the accelerations when slipping occurs.
2. **Incline + friction**: acceleration, minimum/maximum force to hold a block, time to slide down.
3. **Banked roads** and conical pendulums.
4. **Connected bodies over pulleys**: tension and acceleration, sometimes with a movable pulley ($a_1 = 2a_2$ constraint).
5. **Impulse from a force–time graph**: area under the $F-t$ graph.
6. **Equilibrium of a mass hung by two strings**: Lami's theorem or components.

Shortcuts

⚡ SHORTCUT System approach for acceleration

For connected bodies moving with the same acceleration magnitude: $a = \frac{(\text{net driving force})}{(\text{total mass})}$. Then find any tension from **one** body's FBD. **When NOT to use:** when bodies can slip relative to each other. Check the friction limit first. **Time saved:** about 1 minute per pulley problem.

⚡ SHORTCUT Pulley constraint in 5 seconds

For a movable pulley carrying block B, with the string's free end attached to A: $x_A + 2x_B = \text{const} \Rightarrow a_A = 2a_B$. Count the strings supporting the pulley.

Question classification

Type	Difficulty	Target time
Lift, recoil, impulse	E	1 min
Atwood / pulley system	E–M	1.5 min
Incline with friction	M	2 min
Banked road / conical pendulum	M	2 min
Two-block friction (slip check)	M–H	3 min
Variable force + pseudo force combined	H	3–4 min (Round 2)

Common mistakes

⚠ MISTAKE ALERT Mistake alerts

- Writing friction = $\mu_s N$ when the body doesn't move.
- Putting a Newton-III pair on the **same** FBD.
- Forgetting that $N \neq mg$ on an incline, or when a force is applied at an angle.
- Using the pseudo force in an inertial frame too, which counts it twice.
- Impulse on a rebound: $m(v_1 + v_2)$, **not** $m(v_1 - v_2)$.

Practice questions

P03 · Practice

1 Easy 1 min Basic

P03-01

Masses of 3 kg and 5 kg hang from the two ends of a light string over a frictionless pulley ($g = 10 \text{ m s}^{-2}$). The tension in the string is:

- (A) 30 N (B) 37.5 N (C) 40 N (D) 50 N

2 Easy 0.75 min Speed

P03-02

A 60 kg person stands on a scale in a lift that accelerates **downward** at 2 m s^{-2} ($g = 10$). The scale reads:

- (A) 720 N (B) 600 N (C) 480 N (D) 120 N

3 Medium 1.5 min Concept

P03-03

A block slides down a 30° incline with $\mu_k = \frac{1}{2\sqrt{3}}$ ($g = 10 \text{ m s}^{-2}$). Its acceleration is:

- (A) 5 m s^{-2} (B) 2.5 m s^{-2} (C) 7.5 m s^{-2} (D) 1.25 m s^{-2}

4 Easy 1 min NV

P03-04

A road of radius 50 m is banked at an angle θ with $\tan \theta = 0.2$. Find the speed (in m s^{-1}) at which a car needs no friction ($g = 10 \text{ m s}^{-2}$).

Your answer (numerical value):

5 Medium 2 min NV

P03-05

A 2 kg block rests on a 4 kg block that lies on a frictionless floor. The coefficient of static friction between the blocks is 0.3. Find the maximum horizontal force (in N) that can be applied to the **lower** block so that the blocks move together ($g = 10 \text{ m s}^{-2}$).

Your answer (numerical value):

6 Easy 1 min Basic

P03-06

A 0.2 kg ball hits a wall perpendicularly at 10 m s^{-1} and rebounds at 8 m s^{-1} . The magnitude of the impulse on the ball is:

- (A) 0.4 N s (B) 2.0 N s (C) 3.6 N s (D) 1.8 N s

7 Medium 1 min Concept

P03-07

A small bob hangs from the roof of a car accelerating horizontally at $g/\sqrt{3}$. In equilibrium relative to the car, the string makes an angle with the vertical of:

- (A) 30° , towards the back of the car (B) 30° , towards the front of the car
(C) 60° , towards the back of the car (D) 45° , towards the back of the car

8 Medium 2 min JEE

P03-08

A 10 kg mass hangs from two strings making angles of 30° and 60° with the horizontal ceiling ($g = 10 \text{ m s}^{-2}$). The tension in the string at 60° to the horizontal is:

- (A) 50 N (B) $50\sqrt{3}$ N (C) 100 N (D) $100/\sqrt{3}$ N

9 Easy 0.75 min Speed

P03-09

A 5 kg gun fires a 50 g bullet at 400 m s^{-1} . The recoil speed of the gun is:

- (A) 0.4 m s^{-1} (B) 4 m s^{-1} (C) 40 m s^{-1} (D) 2 m s^{-1}

P03 · Chapter Test

1 Medium 1 min Tricky

P03-T1

A 5 kg block rests on a horizontal floor with $\mu_s = 0.5$ ($g = 10$). A horizontal force of 20 N is applied. The friction force on the block is:

- (A) 25 N (B) 20 N (C) 5 N (D) 0 N

2 Easy 0.5 min Speed

P03-T2

If $\mu_s = 1/\sqrt{3}$ between a block and an incline, the angle of repose is:

- (A) 30° (B) 45° (C) 60° (D) 15°

3 Easy 1 min Basic

P03-T3

The maximum safe speed on a level circular road of radius 40 m with $\mu = 0.4$ ($g = 10$) is:

- (A) $4\sqrt{10} \text{ m s}^{-1}$ (B) 16 m s^{-1} (C) 8 m s^{-1} (D) 40 m s^{-1}

4 Medium 1.5 min NV

P03-T4

A force $F = 6t \text{ N}$ acts on a 2 kg body initially at rest. Find its speed (in m s^{-1}) at $t = 2 \text{ s}$.

Your answer (numerical value):

5 Easy 1 min Basic

P03-T5

A 2 kg block on a smooth table is connected by a string over a pulley to a hanging 3 kg block ($g = 10$). The tension in the string is:

- (A) 12 N (B) 20 N (C) 30 N (D) 18 N

Answers & Solutions

P03 · Practice

1 (B) 2 (C) 3 (B) 4 10 5 18 6 (C) 7 (A) 8 (B) 9 (B)

P03 · Chapter Test

1 (B) 2 (A) 3 (A) 4 6 5 (A)

P03 · PRACTICE

1 Answer: (B) Atwood machine tension Easy 1 min Basic

P03-01

SOLUTION $T = \frac{2m_1m_2g}{m_1 + m_2} = \frac{2(3)(5)(10)}{8} = 37.5 \text{ N}$.

SHORTCUT T lies between the two weights (30 N and 50 N). It's their harmonic mean.

COMMON TRAP Taking T equal to the lighter weight (30 N).

2 Answer: (C) Apparent weight Easy 0.75 min Speed

P03-02

SOLUTION $N = m(g - a) = 60 \times 8 = 480 \text{ N}$.

SHORTCUT Accelerating down means you feel lighter.

COMMON TRAP Adding a gives 720 N.

3 Answer: (B) Rough incline acceleration Medium 1.5 min Concept

P03-03

SOLUTION $a = g(\sin \theta - \mu_k \cos \theta) = 10 \left(\frac{1}{2} - \frac{1}{2\sqrt{3}} \cdot \frac{\sqrt{3}}{2} \right) = 10(0.5 - 0.25) = 2.5 \text{ m s}^{-2}$.

SHORTCUT Friction removes $\mu \cot \theta$ of the smooth-incline value: $\mu \cot 30^\circ = 0.5$, so half of $g \sin \theta = 5$.

COMMON TRAP Using $\sin \theta$ in the friction term.

4 Answer: 10 Banking speed Easy 1 min NV

P03-04

SOLUTION $v = \sqrt{rg \tan \theta} = \sqrt{50 \times 10 \times 0.2} = \sqrt{100} = 10 \text{ m s}^{-1}$.

SHORTCUT Design speed of a banked road: $v^2 = rg \tan \theta$.

COMMON TRAP Using $\sin \theta$ instead of $\tan \theta$.

5 Answer: 18 Two-block common motion **Medium** ⌚ 2 min **NV** P03-05

SOLUTION The upper block is accelerated only by friction, so $a_{max} = \mu_s g = 3 \text{ m s}^{-2}$. For common motion $F_{max} = (2 + 4)(3) = 18 \text{ N}$.

SHORTCUT $F_{max} = \mu_s g (m + M)$ when pushing the lower block.

COMMON TRAP Using the lower block's mass alone (12 N), or μ times the total weight used differently.

6 Answer: (C) Impulse on rebound **Easy** ⌚ 1 min **Basic** P03-06

SOLUTION $|\Delta p| = m(v_1 + v_2) = 0.2 \times 18 = 3.6 \text{ N s}$ (the velocity reverses).

SHORTCUT Rebound: add the speeds.

COMMON TRAP Subtracting the speeds gives 0.4 N s.

7 Answer: (A) Pendulum in accelerating car **Medium** ⌚ 1 min **Concept** P03-07

SOLUTION In the car's frame, a pseudo force ma acts backward. $\tan \theta = a/g = 1/\sqrt{3} \Rightarrow \theta = 30^\circ$, with the bob hanging towards the **back**.

SHORTCUT The bob lags behind the acceleration.

COMMON TRAP Choosing "front".

8 Answer: (B) Equilibrium with two strings **Medium** ⌚ 2 min **JEE** P03-08

SOLUTION Horizontal: $T_1 \cos 30^\circ = T_2 \cos 60^\circ \Rightarrow T_2 = \sqrt{3} T_1$. Vertical: $T_1 \sin 30^\circ + T_2 \sin 60^\circ = 100 \Rightarrow \frac{T_1}{2} + \frac{3T_1}{2} = 100 \Rightarrow T_1 = 50 \text{ N}$, $T_2 = 50\sqrt{3} \text{ N}$.

SHORTCUT The strings are perpendicular to each other ($30^\circ + 60^\circ = 90^\circ$), so $T_2 = W \cos 30^\circ$ and $T_1 = W \sin 30^\circ$.

COMMON TRAP Swapping which string carries more tension. The steeper string carries more.

9 Answer: (B) Recoil of a gun **Easy** ⌚ 0.75 min **Speed** P03-09

SOLUTION $V = \frac{mv}{M} = \frac{0.05 \times 400}{5} = 4 \text{ m s}^{-1}$.

SHORTCUT Momentum conservation from rest.

COMMON TRAP Using 50 kg for 50 g.

P03 · CHAPTER TEST

1 Answer: (B) Static friction below the limit **Medium** ⌚ 1 min **Tricky** P03-T1

SOLUTION Limiting friction $= \mu_s N = 25 \text{ N} > 20 \text{ N}$, so the block doesn't move. Static friction balances the applied force: 20 N.

SHORTCUT Friction = applied force whenever there is no slipping.

COMMON TRAP Answering $\mu_s N$ (25 N).

2 Answer: (A) Angle of repose **Easy** ⌚ 0.5 min **Speed** P03-T2

SOLUTION $\tan \theta_r = \mu_s = 1/\sqrt{3} \Rightarrow \theta_r = 30^\circ$.

SHORTCUT Angle of repose = angle of friction.

COMMON TRAP None.

3 Answer: (A) Level-road maximum speed **Easy** ⌚ 1 min **Basic** P03-T3

SOLUTION $v = \sqrt{\mu r g} = \sqrt{0.4 \times 40 \times 10} = \sqrt{160} = 4\sqrt{10} \approx 12.6 \text{ m s}^{-1}$.

SHORTCUT $\sqrt{160}$ lies between 12 and 13.

COMMON TRAP Forgetting the square root.

4 Answer: 6 Impulse of variable force **Medium** ⌚ 1.5 min **NV** P03-T4

SOLUTION $J = \int_0^2 6t \, dt = 3t^2 \Big|_0^2 = 12 \text{ N s}$. $v = J/m = 6 \text{ m s}^{-1}$.

SHORTCUT Impulse = area under the F - t graph (a triangle: $\frac{1}{2} \times 2 \times 12$).

COMMON TRAP Using F at $t = 2$ with $v = at$ (constant acceleration assumed): 12 m s^{-1} .

5 Answer: (A) Table-pulley system **Easy** ⌚ 1 min **Basic** P03-T5

SOLUTION $a = \frac{m_2 g}{m_1 + m_2} = \frac{30}{5} = 6 \text{ m s}^{-2}$. $T = m_1 a = 12 \text{ N}$.

SHORTCUT $T = \frac{m_1 m_2 g}{m_1 + m_2} = \frac{60}{5}$.

COMMON TRAP Taking T = the hanging weight (30 N).

**REVISION** Revision tracker · Laws of Motion

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 04

Work, Energy & Power

04

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

High

PRIORITY

Class 11 · Ch 5

NCERT

~8 hours

STUDY TIME

Concept summary

- **Work-energy theorem:** $W_{net} = \Delta K$. This is the single most powerful problem-solving tool in mechanics. It skips accelerations and time entirely.
- **Conservative forces** (gravity, spring, electrostatic) store potential energy: $F = -\frac{dU}{dx}$. If only conservative forces do work, $K + U$ is constant.
- **Power** is the rate of doing work: $P = \vec{F} \cdot \vec{v}$.
- **Collisions:** momentum is always conserved (no external impulse). Kinetic energy is conserved only in elastic collisions.

Formulas

Σ FORMULA Work & energy

$$W = \vec{F} \cdot \vec{s} = F s \cos \theta \quad W = \int \vec{F} \cdot d\vec{r} \text{ (area under } F-x \text{)} \quad K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

$$U_{gravity} = mgh \quad U_{spring} = \frac{1}{2}kx^2 \quad W_{spring} = \frac{1}{2}k(x_1^2 - x_2^2)$$

$$W_{all \text{ forces}} = \Delta K \quad W_{non-conservative} = \Delta(K + U) \quad P = \frac{dW}{dt} = \vec{F} \cdot \vec{v}$$

Units: 1 kWh = 3.6×10^6 J; 1 hp = 746 W; 1 eV = 1.6×10^{-19} J.

Σ FORMULA Potential energy curves

$F = -dU/dx$. Equilibrium where $dU/dx = 0$: **stable** if $d^2U/dx^2 > 0$ (minimum of U), **unstable** if < 0 (maximum), **neutral** if U is flat. A particle with total energy E is confined to where $U(x) \leq E$ (turning points at $U = E$).

Σ FORMULA Vertical circle (string / light rod, radius R)

Case	Minimum speed at the lowest point	Minimum speed at the top
String (or inside a track)	$\sqrt{5gR}$	$\sqrt{gR}$
Light rod (or tube)	$\sqrt{4gR}$	0

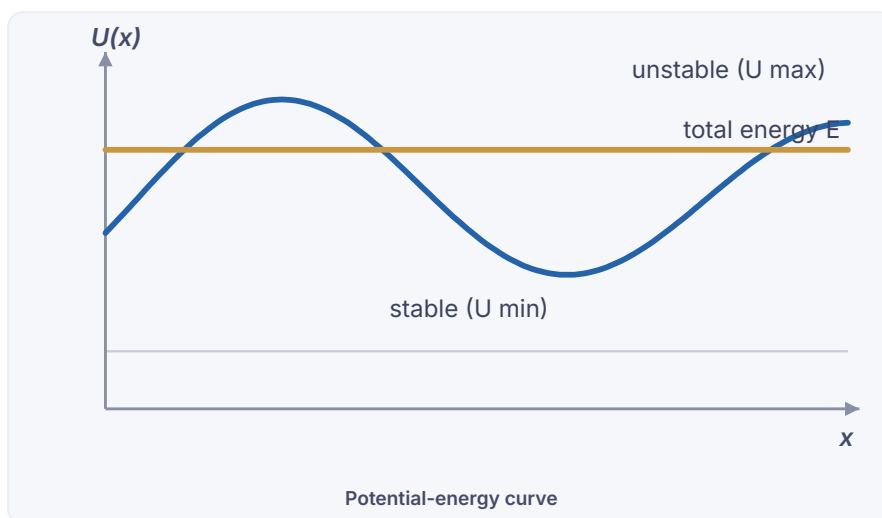
$T_{bottom} - T_{top} = 6mg$ (string, full circle). If $\sqrt{2gR} < u < \sqrt{5gR}$ at the bottom, the string goes slack somewhere above the horizontal diameter. If $u \leq \sqrt{2gR}$, the bob only oscillates below the horizontal diameter.

Σ FORMULA Collisions

Coefficient of restitution: $e = \frac{\text{velocity of separation}}{\text{velocity of approach}}$ (along the line of impact). Elastic: $e = 1$. Perfectly inelastic: $e = 0$. **1D elastic, m_2 initially at rest:**

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2} u_1 \quad v_2 = \frac{2m_1}{m_1 + m_2} u_1$$

- Equal masses exchange velocities. A heavy body hitting a light one at rest: the light one leaves at about $2u_1$.
- **KE lost in any 1D collision:** $\Delta K = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (u_1 - u_2)^2 (1 - e^2)$.
- **Perfectly inelastic, m_2 at rest:** fraction of KE lost = $\frac{m_2}{m_1 + m_2}$.
- **Bouncing ball:** $h_n = e^{2n} h_0$, $e = \sqrt{h_1/h_0}$.
- **Oblique elastic collision of equal masses (one at rest):** they move off at 90° to each other.



Standard models & assumptions

- Springs are massless and obey Hooke's law. "Smooth" surfaces do no work (the normal force $\perp$ displacement).
- The normal force and tension in a vertical circle do no work. Only gravity changes the speed.
- In collisions, impulsive forces dominate: ignore gravity and friction *during* the brief impact.

Frequently tested concepts & question patterns

▮ PYQ PATTERN Recurring structures

1. **Work by a variable force** $F(x)$ or from an F - x graph (area).
2. **Spring-block** energy: maximum compression, speed at a given compression.
3. **Vertical circle:** minimum speed, tension at a point, where the string slackens.
4. **Collisions:** final velocities, KE loss, coefficient of restitution from bounce heights.
5. **Power:** vehicle on an incline at constant speed; pumping water; $P = Fv$ with variable force.
6. $U(x) \rightarrow$ **force and equilibrium type**.
7. **Kinetic energy-momentum relation:** percentage changes, $p = \sqrt{2mK}$.

Shortcuts

SHORTCUT Energy method first

If the question asks for a **speed, height or compression** (not a time), use energy conservation or the work–energy theorem straight away. **Time saved:** 1–2 minutes compared with a force/kinematics approach. **When NOT to use:** when the question asks for time or acceleration, or when the path matters for a non-conservative force whose work you can't easily compute.

SHORTCUT Percentage changes

$K \propto p^2$. If p rises by $x\%$ (small), K rises by about $2x\%$. For large changes use exact ratios: $K \times 4 \Rightarrow p \times 2$ (a 300% rise in K is a 100% rise in p).

Question classification

Type	Difficulty	Target time
Work from $F(x)$ or graph area	E	1 min
Spring–block energy	E	1 min
Power on an incline	E–M	1.5 min
1D collision velocities / KE loss	M	2 min
Vertical circle tension/slack point	M–H	2.5 min
2D collision, or collision + spring combined	H	3–4 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Taking work by friction as $\mu mg \times d$ on an incline. Use $\mu mg \cos \theta \times d$.
- Sign errors in W_{spring} when the spring is stretched from x_1 to x_2 .
- Conserving KE in an inelastic collision.
- For a rod in a vertical circle, using the string condition $\sqrt{5gR}$.
- Forgetting that $P = Fv$ uses the **force exerted by the engine**, not the net force.

Practice questions

P04 · Practice

1 Easy ⌚ 1 min NV

P04-01

A force $F = 3x^2$ N acts along the x -axis. Find the work done (in J) as a body moves from $x = 0$ to $x = 2$ m.

Your answer (numerical value):

2 Easy ⌚ 1 min Basic

P04-02

A 2 kg block moving at 4 m s^{-1} on a smooth floor hits a spring of constant 200 N m^{-1} . The maximum compression is:

- (A) 0.2 m (B) 0.4 m (C) 0.8 m (D) 0.04 m

3 Medium 1.5 min JEE

P04-03

A 1000 kg car climbs an incline with $\sin \theta = 1/20$ at a constant 10 m s^{-1} against a resistive force of 500 N ($g = 10$). The engine's power is:

- (A) 5 kW (B) 10 kW (C) 15 kW (D) 20 kW

4 Medium 1 min Basic

P04-04

A bob on a light string of length 0.9 m is to complete a vertical circle ($g = 10$). The minimum speed needed at the lowest point is:

- (A) 3 m s^{-1} (B) $3\sqrt{5} \text{ m s}^{-1}$ (C) $3\sqrt{2} \text{ m s}^{-1}$ (D) 6 m s^{-1}

5 Easy 1 min Basic

P04-05

A 2 kg ball moving at 6 m s^{-1} collides elastically head-on with a stationary 4 kg ball. The velocity of the 4 kg ball afterwards is:

- (A) 2 m s^{-1} (B) 3 m s^{-1} (C) 4 m s^{-1} (D) 6 m s^{-1}

6 Medium 1.5 min NV

P04-06

A 1 kg block moving at 10 m s^{-1} collides with and sticks to a stationary 4 kg block. Find the percentage of the initial kinetic energy lost.

Your answer (numerical value):

7 Easy 1 min Concept

P04-07

A particle has potential energy $U(x) = x^2 - 4x$ (SI units). Its equilibrium position and type are:

- (A) $x = 2 \text{ m}$, stable (B) $x = 2 \text{ m}$, unstable (C) $x = 4 \text{ m}$, stable (D) $x = 0$, neutral

8 Easy 1 min Basic

P04-08

A ball dropped from 10 m rebounds to 6.4 m. The coefficient of restitution is:

- (A) 0.64 (B) 0.8 (C) 0.36 (D) 0.6

9 Easy 1 min Basic

P04-09

A 1 kg block slides on a floor with $\mu_k = 0.2$ from an initial speed of 6 m s^{-1} ($g = 10$). It stops after:

- (A) 3 m (B) 6 m (C) 9 m (D) 18 m

P04 · Chapter Test

1 Easy 0.5 min Speed

P04-T1

The work done by Earth's gravity on a satellite in a circular orbit during one full revolution is:

- (A) positive (B) negative (C) zero (D) $2\pi r \times mg$

2 Easy 1 min Basic

P04-T2

If the kinetic energy of a body increases by 300%, its momentum increases by:

- (A) 100% (B) 200% (C) 300% (D) 50%

3 Medium 1 min Calc

P04-T3

A machine gun fires 10 bullets per second, each of mass 50 g, at 1000 m s^{-1} . The power delivered to the bullets is:

- (A) 25 kW (B) 250 kW (C) 500 kW (D) 2.5 kW

4 Easy 0.75 min NV

P04-T4

The potential energy of a particle is $U = 5x^3 \text{ J}$ (x in m). Find the magnitude of the force (in N) on it at $x = 1 \text{ m}$.

Your answer (numerical value):

5 Easy 0.5 min Tricky

P04-T5

A 2 kg body is lifted 5 m vertically at **constant speed** ($g = 10$). The net work done on the body is:

- (A) 100 J (B) -100 J (C) 0 (D) 200 J

Answers & Solutions

P04 · Practice

1 8 2 (B) 3 (B) 4 (B) 5 (C) 6 80 7 (A) 8 (B) 9 (C)

P04 · Chapter Test

1 (C) 2 (A) 3 (B) 4 15 5 (C)

P04 · PRACTICE

1 Answer: 8 Work by variable force Easy 1 min NV

P04-01

SOLUTION $W = \int_0^2 3x^2 dx = x^3 \Big|_0^2 = 8 \text{ J}$.

SHORTCUT Integrate: the antiderivative of $3x^2$ is x^3 .

COMMON TRAP Using $F(2) \times 2 = 24 \text{ J}$.

2 Answer: (B) Spring maximum compression Easy 1 min Basic

P04-02

SOLUTION $\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow x = v\sqrt{m/k} = 4\sqrt{0.01} = 0.4 \text{ m}$.

SHORTCUT $x = v\sqrt{m/k}$.

COMMON TRAP Forgetting the square root gives 0.04.

3 Answer: (B) Power on an incline Medium 1.5 min JEE

P04-03

SOLUTION At constant speed, the engine force $= mg \sin \theta + f = 500 + 500 = 1000 \text{ N}$. $P = Fv = 10 \text{ kW}$.

SHORTCUT Constant speed means driving force = sum of resisting forces.

COMMON TRAP Omitting the gravity component (5 kW).

4 Answer: (B) Vertical circle, minimum speed Medium 1 min Basic

P04-04

SOLUTION $u_{\min} = \sqrt{5gR} = \sqrt{45} = 3\sqrt{5} \approx 6.7 \text{ m s}^{-1}$.

SHORTCUT String: $\sqrt{5gR}$ at the bottom. Rod: $\sqrt{4gR}$.

COMMON TRAP Using the rod condition gives 6.

5 Answer: (C) 1D elastic collision Easy 1 min Basic

P04-05

SOLUTION $v_2 = \frac{2m_1}{m_1 + m_2}u_1 = \frac{4}{6} \times 6 = 4 \text{ m s}^{-1}$. (The 2 kg ball rebounds at 2 m s^{-1} .)

SHORTCUT Check: relative speed of separation = relative speed of approach: $4 - (-2) = 6$.

COMMON TRAP Assuming the balls exchange velocities (true only for equal masses).

- 6 Answer: 80** KE loss in perfectly inelastic collision **Medium** ⌚ 1.5 min **NV** P04-06
- SOLUTION** $v = \frac{1 \times 10}{5} = 2 \text{ m s}^{-1}$. $K_i = 50 \text{ J}$, $K_f = \frac{1}{2}(5)(4) = 10 \text{ J}$. Loss = $40/50 = 80\%$.
- SHORTCUT** Fraction lost = $\frac{m_2}{m_1 + m_2} = \frac{4}{5}$.
- COMMON TRAP** Reporting the percentage retained (20).

- 7 Answer: (A)** Equilibrium from $U(x)$ **Easy** ⌚ 1 min **Concept** P04-07
- SOLUTION** $dU/dx = 2x - 4 = 0 \Rightarrow x = 2$. $d^2U/dx^2 = 2 > 0$, a minimum, so stable.
- SHORTCUT** U is an upward parabola, so its vertex is a stable equilibrium.
- COMMON TRAP** Taking the root of $U = 0$ ($x = 4$) as the equilibrium.

- 8 Answer: (B)** Restitution from bounce height **Easy** ⌚ 1 min **Basic** P04-08
- SOLUTION** $e = \sqrt{h_1/h_0} = \sqrt{0.64} = 0.8$.
- SHORTCUT** Speed $\propto \sqrt{h}$.
- COMMON TRAP** Using h_1/h_0 directly (0.64).

- 9 Answer: (C)** Stopping distance with friction **Easy** ⌚ 1 min **Basic** P04-09
- SOLUTION** $\mu mg d = \frac{1}{2}mv^2 \Rightarrow d = \frac{v^2}{2\mu g} = \frac{36}{4} = 9 \text{ m}$.
- SHORTCUT** Mass cancels.
- COMMON TRAP** Forgetting the 2 gives 18 m.

P04 · CHAPTER TEST

- 1 Answer: (C)** Work by centripetal force **Easy** ⌚ 0.5 min **Speed** P04-T1
- SOLUTION** Gravity is always perpendicular to the velocity in a circular orbit, so it does no work at any instant.
- SHORTCUT** Force $\perp$ velocity $\rightarrow$ no work.
- COMMON TRAP** Multiplying force by the circumference.
- 2 Answer: (A)** KE-momentum relation **Easy** ⌚ 1 min **Basic** P04-T2
- SOLUTION** $K' = 4K \Rightarrow p' = \sqrt{2mK'} = 2p$, an increase of 100%.
- SHORTCUT** $p \propto \sqrt{K}$.
- COMMON TRAP** Reading "+300%" as $\times 3$ instead of $\times 4$.
- 3 Answer: (B)** Power of a machine gun **Medium** ⌚ 1 min **Calc** P04-T3
- SOLUTION** KE per bullet = $\frac{1}{2}(0.05)(10^6) = 2.5 \times 10^4 \text{ J}$. Power = $10 \times 2.5 \times 10^4 = 2.5 \times 10^5 \text{ W} = 250 \text{ kW}$.
- SHORTCUT** $P = \frac{1}{2}nmv^2$.
- COMMON TRAP** Unit slip: 50 g as 0.5 kg.
- 4 Answer: 15** Force from potential energy **Easy** ⌚ 0.75 min **NV** P04-T4
- SOLUTION** $F = -dU/dx = -15x^2$; at $x = 1$, $|F| = 15 \text{ N}$.
- SHORTCUT** Force = -slope of U .
- COMMON TRAP** Reporting $U(1) = 5$.
- 5 Answer: (C)** Net work at constant speed **Easy** ⌚ 0.5 min **Tricky** P04-T5
- SOLUTION** $W_{\text{net}} = \Delta K = 0$ (constant speed). The applied force does +100 J, and gravity does -100 J.
- SHORTCUT** Constant speed means zero net work.
- COMMON TRAP** Answering the work done by the lifting force alone.

 REVISION Revision tracker · Work, Energy & Power

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Rotational Motion & Centre of Mass

05

1–2 per shift

TYPICAL QUESTIONS

Medium–Hard

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 6

NCERT

~16 hours

STUDY TIME

Concept summary

- The **centre of mass (COM)** moves as if all the mass and all the external forces were concentrated there: $M\vec{a}_{cm} = \vec{F}_{ext}$.
- Rotation** has an exact analogue for every linear quantity: $\theta \leftrightarrow x, \omega \leftrightarrow v, \alpha \leftrightarrow a, I \leftrightarrow m, \tau \leftrightarrow F, L \leftrightarrow p$.
- Moment of inertia** depends on the mass *and* how far it lies from the axis. The parallel and perpendicular axis theorems give I about new axes.
- Rolling without slipping:** $v_{cm} = \omega R$. Rolling KE splits into translational and rotational parts in a fixed ratio.
- Angular momentum is conserved** when there is no external torque: $I_1\omega_1 = I_2\omega_2$.

Formulas

Σ FORMULA Centre of mass

$$\vec{r}_{cm} = \frac{\sum m_i \vec{r}_i}{\sum m_i} \quad x_{cm} = \frac{\int x dm}{\int dm} \quad \vec{v}_{cm} = \frac{\sum m_i \vec{v}_i}{M}$$

Body	COM location
Semicircular ring (radius R)	$\frac{2R}{\pi}$ from the centre
Semicircular disc	$\frac{4R}{3\pi}$ from the centre
Solid hemisphere	$\frac{3R}{8}$ from the centre of the flat face
Hollow hemisphere	$\frac{R}{2}$ from the centre
Solid cone (height h)	$\frac{h}{4}$ from the base
Triangle (lamina)	centroid

Removed-part trick: $x_{cm} = \frac{m_1 x_1 - m_2 x_2}{m_1 - m_2}$ (treat the hole as negative mass).

Σ FORMULA Rotational kinematics & dynamics

$$\omega = \omega_0 + \alpha t \quad \theta = \omega_0 t + \frac{1}{2} \alpha t^2 \quad \omega^2 = \omega_0^2 + 2\alpha\theta$$

$$\vec{\tau} = \vec{r} \times \vec{F} \quad \tau = I\alpha \quad \vec{L} = \vec{r} \times \vec{p} = I\vec{\omega} \quad \vec{\tau}_{ext} = \frac{d\vec{L}}{dt}$$

$$K_{rot} = \frac{1}{2} I \omega^2 = \frac{L^2}{2I} \quad W = \int \tau d\theta \quad P = \tau \omega \quad \text{angular impulse} = \int \tau dt = \Delta L$$

Σ FORMULA Moments of inertia

Body (mass M)	Axis	I
Thin ring, radius R	central, $\perp$ plane	MR^2
Thin ring	diameter	$\frac{1}{2}MR^2$
Disc / solid cylinder	central axis	$\frac{1}{2}MR^2$
Disc	diameter	$\frac{1}{4}MR^2$
Hollow cylinder (thin)	central axis	MR^2
Solid sphere	diameter	$\frac{2}{5}MR^2$
Hollow sphere (thin shell)	diameter	$\frac{2}{3}MR^2$
Thin rod, length L	$\perp$ through centre	$\frac{1}{12}ML^2$
Thin rod	$\perp$ through one end	$\frac{1}{3}ML^2$
Rectangular plate $a \times b$	$\perp$ plane through centre	$\frac{1}{12}M(a^2 + b^2)$
Solid cylinder (length l)	$\perp$ axis through centre	$M \left(\frac{R^2}{4} + \frac{l^2}{12} \right)$

Parallel axis: $I = I_{cm} + Md^2$ (any body). **Perpendicular axis:** $I_z = I_x + I_y$ (planar bodies only). **Radius of gyration:** $k = \sqrt{I/M}$.

Σ FORMULA Rolling without slipping

$$v_{cm} = \omega R \quad K = \frac{1}{2}mv^2 \left(1 + \frac{k^2}{R^2} \right) \quad a_{incline} = \frac{g \sin \theta}{1 + k^2/R^2} \quad \mu_{min} = \frac{\tan \theta}{1 + R^2/k^2}$$

Body	k^2/R^2	K_{rot}/K_{total}	a on incline
Ring / hollow cylinder	1	1/2	$\frac{1}{2}g \sin \theta$
Disc / solid cylinder	1/2	1/3	$\frac{2}{3}g \sin \theta$
Hollow sphere	2/3	2/5	$\frac{3}{5}g \sin \theta$
Solid sphere	2/5	2/7	$\frac{5}{7}g \sin \theta$

Race down an incline (smallest k^2/R^2 wins): solid sphere > disc > hollow sphere > ring. The race result is independent of mass and radius. **Sliding $\rightarrow$ rolling on a rough floor** (initial speed v_0 , $\omega_0 = 0$): the final rolling speed is $\frac{v_0}{1 + k^2/R^2}$. For a solid sphere, $\frac{5}{7}v_0$.

Σ FORMULA Equilibrium of a rigid body

$\sum \vec{F} = 0$ and $\sum \vec{\tau} = 0$ about *any* point. Take torques about the point where the most unknown forces act.

Ladder (uniform, smooth wall, rough floor, at angle θ with the floor): on the verge of slipping, $\mu = \frac{1}{2 \tan \theta}$.

Linear $\leftrightarrow$ rotational comparison

Linear	Rotational
x, v, a	θ, ω, α
m	I
$F = ma$	$\tau = I\alpha$
$p = mv$	$L = I\omega$
$K = \frac{1}{2}mv^2$	$K = \frac{1}{2}I\omega^2$
$P = Fv$	$P = \tau\omega$

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **MI about a shifted axis:** parallel/perpendicular axis on a disc, ring, rod or plate; MI of composite bodies (rod + spheres, square of rods).
2. **Rolling on an incline:** acceleration, speed at the bottom, which body reaches first, minimum friction.
3. **Angular momentum conservation:** a person walking on a turntable, a skater, two discs coupled (KE lost).
4. **COM of composite or cut-out bodies**, and COM motion after an explosion.
5. **Torque and angular momentum as cross products** in vector form.
6. **Rolling KE ratios**, and the fraction of KE that is rotational.

Shortcuts

SHORTCUT The k^2/R^2 key

Memorise one number per body: ring 1, disc 1/2, hollow sphere 2/3, solid sphere 2/5. Every rolling formula in the table above is generated from it. **Time saved:** 1–2 min per rolling question. **Common mistake:** using $I/(MR^2)$ about the wrong axis, such as a disc about its diameter (1/4) when it rolls about its central axis (1/2).

SHORTCUT Coupled rotating bodies

Two coaxial bodies stick together: $\omega_f = \frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$ and $\Delta K = \frac{I_1 I_2 (\omega_1 - \omega_2)^2}{2(I_1 + I_2)}$. This is the rotational version of a perfectly inelastic collision.

Question classification

Type	Difficulty	Target time
MI from the table + one theorem	E	1 min
Rolling acceleration / KE ratios	E–M	1.5 min
Angular momentum conservation	M	2 min
COM of a cut-out body	M	2 min
Torque/L as a cross product	E	1 min
Rolling + slipping, or impulse on a rod	H	3–4 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Applying the perpendicular axis theorem to a 3D body (a sphere or cylinder). It works for **planar** bodies only.
- Using the parallel axis theorem between two axes, neither of which passes through the COM. Always go through I_{cm} .
- Assuming friction does work in pure rolling. Static friction at the contact point does **no** work.
- Using $v = \omega R$ when the body is still slipping.
- Conserving KE when two rotating bodies couple. Only angular momentum is conserved.

Practice questions

P05 · Practice

1 Medium 1.5 min Concept

P05-01

The moment of inertia of a uniform disc (mass M , radius R) about a **tangent lying in its plane** is:

- (A) $\frac{5}{4}MR^2$ (B) $\frac{3}{2}MR^2$ (C) $\frac{1}{2}MR^2$ (D) $\frac{3}{4}MR^2$

2 Easy 1 min Basic

P05-02

A solid cylinder rolls without slipping down a 30° incline ($g = 10$). Its acceleration is:

- (A) 5 m s^{-2} (B) $10/3 \text{ m s}^{-2}$ (C) 2.5 m s^{-2} (D) $25/7 \text{ m s}^{-2}$

3 Easy 1 min Concept

P05-03

A skater spins with MI 4 kg m^2 at 2 rev/s . She pulls in her arms, reducing her MI to 2 kg m^2 . The ratio (new KE)/(old KE) is:

- (A) $1/2$ (B) 1 (C) 2 (D) 4

4 Easy 0.75 min Speed

P05-04

The centre of mass of a thin uniform wire bent into a semicircle of radius R is at a distance from the centre of:

- (A) $2R/\pi$ (B) $4R/3\pi$ (C) $R/2$ (D) $\pi R/4$

5 Easy 1 min NV

P05-05

A force $\vec{F} = (2\hat{i} + 3\hat{j}) \text{ N}$ acts at the point $\vec{r} = (\hat{i} - \hat{j}) \text{ m}$. Find the magnitude of the torque (in N m) about the origin.

Your answer (numerical value):

6 Easy 0.75 min Speed

P05-06

A solid sphere, a disc and a ring (all of different masses and radii) are released together from rest at the top of the same incline and roll without slipping. The order in which they reach the bottom is:

- (A) ring, disc, sphere (B) sphere, disc, ring
(C) disc, sphere, ring (D) all together

7 Easy 1 min NV

P05-07

A 2 kg disc rolls without slipping at 2 m s^{-1} . Find its total kinetic energy (in J).

Your answer (numerical value):

8 Hard 2.5 min JEE

P05-08

A uniform ladder rests against a smooth vertical wall, making 60° with a rough horizontal floor. The minimum coefficient of friction at the floor for equilibrium is:

- (A) $\frac{1}{2\sqrt{3}}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

9 Medium 1.5 min Concept

P05-09

A solid sphere is set sliding (not rotating) at speed v_0 on a rough horizontal floor. When it starts rolling without slipping, its speed is:

- (A) v_0 (B) $\frac{5}{7}v_0$ (C) $\frac{2}{7}v_0$ (D) $\frac{2}{3}v_0$

10

Easy

⌚ 0.75 min

Speed

P05-10

A wheel starts from rest with a constant angular acceleration of 2 rad s^{-2} . The angle it turns through in the first 5 s is:

(A) 10 rad

(B) 25 rad

(C) 50 rad

(D) 5 rad



P05 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

P05-T1

The radius of gyration of a solid sphere of radius R about a diameter is:

(A) $\sqrt{2/5} R$ (B) $\frac{2}{5} R$ (C) $\sqrt{2/3} R$ (D) $R/\sqrt{2}$

2

Easy

⌚ 0.75 min

Basic

P05-T2

The moment of inertia of a thin ring (mass M , radius R) about a diameter is:

(A) MR^2 (B) $\frac{1}{2}MR^2$ (C) $\frac{1}{4}MR^2$ (D) $2MR^2$

3

Easy

⌚ 1 min

NV

P05-T3

A constant torque of 10 N m acts on a flywheel of $MI \ 2 \text{ kg m}^2$ that starts from rest. Find its angular speed (in rad/s) after 4 s.

Your answer (numerical value):

4

Medium

⌚ 1 min

Concept

P05-T4

A disc rotates freely at ω_0 . An identical disc, initially at rest, is gently placed on it coaxially, and they rotate together. The final angular speed is:

(A) ω_0 (B) $\omega_0/2$ (C) $\omega_0/\sqrt{2}$ (D) $2\omega_0$

5

Medium

⌚ 1 min

Concept

P05-T5

A hollow sphere and a solid sphere of equal mass and radius roll without slipping at the same speed. The ratio of their kinetic energies (hollow : solid) is:

(A) 25 : 21

(B) 21 : 25

(C) 5 : 3

(D) 1 : 1

Answers & Solutions

P05 · Practice

1 (A) 2 (B) 3 (C) 4 (A) 5 5 6 (B) 7 6 8 (A) 9 (B) 10 (B)

P05 · Chapter Test

1 (A) 2 (B) 3 20 4 (B) 5 (A)

P05 · PRACTICE

1

Answer: (A) Parallel + perpendicular axis

Medium

⌚ 1.5 min

Concept

P05-01

SOLUTION About a diameter: $\frac{1}{4}MR^2$ (perpendicular axis theorem: $I_x + I_y = \frac{1}{2}MR^2$). A tangent in the plane is parallel to a diameter at distance R : $\frac{1}{4}MR^2 + MR^2 = \frac{5}{4}MR^2$.

SHORTCUT In-plane tangent: $\frac{5}{4}$. Tangent $\perp$ plane: $\frac{3}{2}$.

COMMON TRAP Starting from the central-axis value gives $\frac{3}{2}MR^2$ (the tangent perpendicular to the plane).

2 Answer: (B) Rolling acceleration **Easy** ⌚ 1 min Basic

P05-02

SOLUTION $a = \frac{g \sin \theta}{1 + k^2/R^2} = \frac{5}{3/2} = \frac{10}{3} \text{ m s}^{-2}.$

SHORTCUT Disc/cylinder: $\frac{2}{3}g \sin \theta.$

COMMON TRAP Using the solid sphere's k^2/R^2 gives 25/7.

3 Answer: (C) Angular momentum conservation **Easy** ⌚ 1 min Concept

P05-03

SOLUTION L is conserved: $\omega_2 = 4 \text{ rev/s}$. $K = L^2/2I$ with L fixed, so $K \propto 1/I$ and the ratio is 2. The extra energy comes from the work she does pulling her arms in.

SHORTCUT Constant L : $K \propto 1/I$.

COMMON TRAP Assuming KE is conserved (ratio 1).

4 Answer: (A) COM of a semicircular wire **Easy** ⌚ 0.75 min Speed

P05-04

SOLUTION For a semicircular ring, $y_{cm} = \frac{\int R \sin \theta \lambda R d\theta}{\lambda \pi R} = \frac{2R}{\pi}.$

SHORTCUT Ring: $2R/\pi$. Disc (lamina): $4R/3\pi$.

COMMON TRAP Mixing up the wire and lamina results.

5 Answer: 5 Torque as a cross product **Easy** ⌚ 1 min NV

P05-05

SOLUTION $\vec{\tau} = \vec{r} \times \vec{F} = (1)(3) - (-1)(2) = 5$ along $\hat{k}$. Magnitude 5 N m.

SHORTCUT In 2D, $\tau_z = xF_y - yF_x$.

COMMON TRAP Computing $\vec{F} \times \vec{r}$ changes the sign only, not the magnitude. The real trap is arithmetic with the negative y .

6 Answer: (B) Rolling race **Easy** ⌚ 0.75 min Speed

P05-06

SOLUTION $a \propto 1/(1 + k^2/R^2)$: sphere (2/5) > disc (1/2) > ring (1). Mass and radius don't matter.

SHORTCUT Smaller k^2/R^2 reaches first.

COMMON TRAP Thinking the heavier body wins.

7 Answer: 6 Rolling kinetic energy **Easy** ⌚ 1 min NV

P05-07

SOLUTION $K = \frac{1}{2}mv^2(1 + \frac{1}{2}) = \frac{1}{2}(2)(4)(1.5) = 6 \text{ J}.$

SHORTCUT Disc: $K = \frac{3}{4}mv^2$.

COMMON TRAP Counting only the translational KE (4 J).

8 Answer: (A) Ladder friction **Hard** ⌚ 2.5 min JEE

P05-08

SOLUTION Forces: weight W at the middle, floor normal $N_1 = W$, friction f , wall normal $N_2 = f$. Torques about the foot:

$$N_2 L \sin 60^\circ = W \frac{L}{2} \cos 60^\circ \Rightarrow N_2 = \frac{W}{2 \tan 60^\circ}. \text{ Then } \mu \geq f/N_1 = \frac{1}{2 \tan 60^\circ} = \frac{1}{2\sqrt{3}}.$$

SHORTCUT $\mu_{min} = \frac{1}{2 \tan \theta}$ (θ measured from the floor).

COMMON TRAP Measuring θ from the wall, which gives $\frac{\tan \theta}{2}$ with the wrong angle.

9 Answer: (B) Sliding to rolling **Medium** ⌚ 1.5 min Concept

P05-09

SOLUTION Angular momentum about the contact point is conserved (friction acts through that point): $mv_0 R = mvR +$

$$\frac{2}{5}mR^2(v/R) \Rightarrow v = \frac{5}{7}v_0.$$

SHORTCUT $v = \frac{v_0}{1 + k^2/R^2}.$

COMMON TRAP Conserving kinetic energy, which friction dissipates during slipping.

10 Answer: (B) Rotational kinematics **Easy** ⌚ 0.75 min Speed

P05-10

SOLUTION $\theta = \frac{1}{2}at^2 = \frac{1}{2}(2)(25) = 25 \text{ rad}.$

SHORTCUT Same form as $s = \frac{1}{2}at^2$.

COMMON TRAP Using $\theta = \alpha t$ (10).

P05 · CHAPTER TEST

1 Answer: (A) Radius of gyration **Easy** ⌚ 0.5 min **Speed**

P05-T1

SOLUTION $k = \sqrt{I/M} = \sqrt{\frac{2}{5}}R$.

SHORTCUT $k^2/R^2 =$ the fraction in $I = (\cdot)MR^2$.

COMMON TRAP Forgetting the square root.

2 Answer: (B) Perpendicular axis for a ring **Easy** ⌚ 0.75 min **Basic**

P05-T2

SOLUTION $I_z = I_x + I_y = 2I_{dia} = MR^2 \Rightarrow I_{dia} = \frac{1}{2}MR^2$.

SHORTCUT Diameter MI = half the central-axis MI (for any symmetric planar body).

COMMON TRAP Using the disc value (1/4).

3 Answer: 20 Torque and angular speed **Easy** ⌚ 1 min **NV**

P05-T3

SOLUTION $\alpha = \tau/I = 5 \text{ rad s}^{-2}$; $\omega = \alpha t = 20 \text{ rad/s}$.

SHORTCUT Angular impulse $\tau t = I\omega$: $40 = 2\omega$.

COMMON TRAP None; speed item.

4 Answer: (B) Coupling identical discs **Medium** ⌚ 1 min **Concept**

P05-T4

SOLUTION $I\omega_0 = 2I\omega \Rightarrow \omega = \omega_0/2$. (Half of the KE is lost.)

SHORTCUT Rotational perfectly inelastic collision.

COMMON TRAP Conserving KE gives $\omega_0/\sqrt{2}$.

5 Answer: (A) Rolling KE ratio **Medium** ⌚ 1 min **Concept**

P05-T5

SOLUTION $K \propto 1 + k^2/R^2$: $(1 + \frac{2}{3}) : (1 + \frac{2}{5}) = \frac{5}{3} : \frac{7}{5} = 25 : 21$.

SHORTCUT Compare the $(1 + k^2/R^2)$ factors.

COMMON TRAP Inverting the ratio.



REVISION Revision tracker · Rotational Motion

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Gravitation

06

1-2 per shift

TYPICAL QUESTIONS

Easy-Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 7

NCERT

~7 hours

STUDY TIME

Concept summary

- **Newton's law:** every pair of masses attracts with $F = Gm_1m_2/r^2$ along the line joining them. Spherically symmetric bodies act as point masses for external points.
- g falls with height and with depth. It is maximum at the surface.
- **Potential energy** is negative, $U = -GMm/r$, with zero at infinity. Bound orbits have negative total energy.
- **Satellites:** gravity provides the centripetal force. Everything (speed, period, energy) follows from $\frac{GMm}{r^2} = \frac{mv^2}{r}$.
- **Kepler's laws** summarise planetary motion. The second law is angular momentum conservation.

Formulas

Σ FORMULA Acceleration due to gravity

$$g = \frac{GM}{R^2} \quad g_h = \frac{g}{(1+h/R)^2} \approx g \left(1 - \frac{2h}{R}\right) \quad (h \ll R) \quad g_d = g \left(1 - \frac{d}{R}\right)$$

- g is zero at the centre of the Earth.
- For the same small fractional decrease in g : $d = 2h$.
- Effect of rotation at latitude λ : $g' = g - \omega^2 R \cos^2 \lambda$. g is minimum at the equator.

Σ FORMULA Field and potential

Source	Field E_g	Potential V
Point mass / outside a sphere ($r \geq R$)	$\frac{GM}{r^2}$	$-\frac{GM}{r}$
Inside a thin shell ($r < R$)	0	$-\frac{GM}{R}$ (constant)
Inside a solid sphere ($r < R$)	$\frac{GMr}{R^3}$	$-\frac{GM(3R^2 - r^2)}{2R^3}$

At the centre of a solid sphere, $V = -\frac{3}{2} \frac{GM}{R}$. $E_g = -dV/dr$.

Σ FORMULA Energy, escape and orbits

$$U = -\frac{GMm}{r} \quad v_{esc} = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \approx 11.2 \text{ km s}^{-1}$$

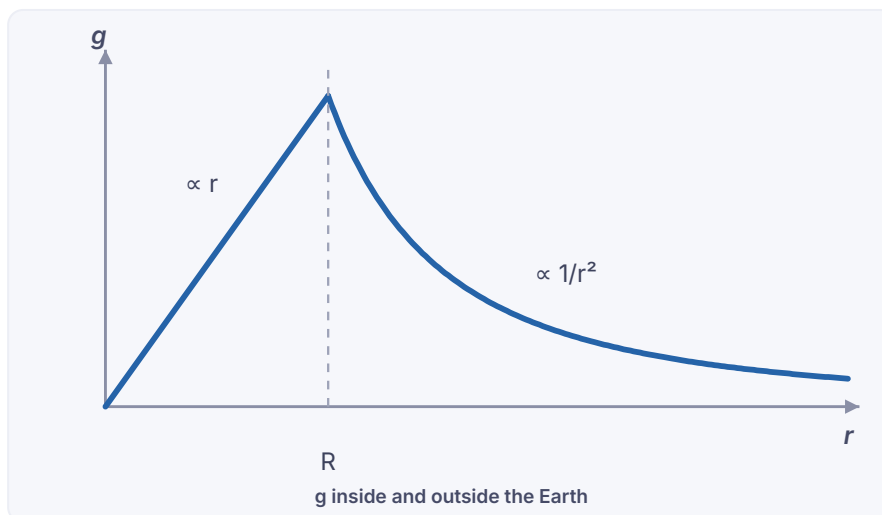
$$v_{orb} = \sqrt{\frac{GM}{r}} \quad T = 2\pi\sqrt{\frac{r^3}{GM}} \quad \text{near the surface: } v_o = \sqrt{gR} \approx 7.9 \text{ km s}^{-1}, T \approx 84 \text{ min}$$

$$K = \frac{GMm}{2r} \quad U = -\frac{GMm}{r} \quad E = -\frac{GMm}{2r} \quad K = -E = -\frac{1}{2}U$$

- $v_{esc} = \sqrt{2} v_{orb}$ at the same point. Binding energy $= +\frac{GMm}{2r}$.
- Energy to move a satellite from r_1 to r_2 : $\Delta E = \frac{GMm}{2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$.
- **Geostationary orbit:** $T = 24$ h, equatorial, same sense as Earth's rotation, $r \approx 42\,000$ km from the centre ($h \approx 36\,000$ km).
- Escape speed does **not** depend on the mass of the body or the direction of launch.

Σ FORMULA Kepler's laws

1. Orbits are ellipses with the Sun at one focus.
2. The radius vector sweeps equal areas in equal times: $\frac{dA}{dt} = \frac{L}{2m} = \text{constant}$, so speed is maximum at perihelion. $v_p r_p = v_a r_a$.
3. $T^2 \propto a^3$, where a is the semi-major axis: $T^2 = \frac{4\pi^2}{GM} a^3$.



Standard models & assumptions

- Earth is a uniform sphere, $R = 6400 \text{ km}$, $g = 9.8$ (or 10) m s^{-2} , $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{kg}^{-2}$.
- Satellites are in circular orbits unless an ellipse is stated. The satellite's own mass doesn't affect its orbit.

Frequently tested concepts & question patterns

▮ PYQ PATTERN Recurring structures

1. **g vs height and depth:** ratios, "at what height/depth does g become $x\%$ ", comparisons.
2. **Satellite energies:** KE/PE/total relations, energy to shift orbits, binding energy.
3. **Escape speed** for planets with changed M , R or density; escape vs orbital speed.
4. **Kepler's third law:** period ratios; areal velocity.
5. **Field/potential inside shells and spheres** (graph-based).
6. **Null point** between two masses.

Shortcuts

⚡ SHORTCUT Scaling instead of computing

Most gravitation MCQs are ratio questions. Write the dependence and scale it: $g \propto M/R^2 \propto \rho R$, $v_{esc} \propto \sqrt{M/R} \propto R\sqrt{\rho}$, $T \propto r^{3/2}$, $v_{orb} \propto r^{-1/2}$. **Time saved:** 1 minute. **Common mistake:** holding the wrong variable fixed (mass vs density).

⚡ SHORTCUT Null point between masses m_1, m_2 a distance d apart

From m_1 : $x = \frac{d\sqrt{m_1}}{\sqrt{m_1} + \sqrt{m_2}}$.

Question classification

Type	Difficulty	Target time
g at height/depth	E	1 min
Kepler III ratio	E	45 s
Satellite energy relations	E-M	1 min
Escape/orbital speed scaling	E	1 min
Field/potential inside bodies	M	2 min
Elliptical orbits (angular momentum + energy)	M-H	3 min

Common mistakes



MISTAKE ALERT Mistake alerts

- Using h (height above the surface) where $r = R + h$ is needed.
- Applying $g(1 - 2h/R)$ when h is not small compared with R .
- Treating PE as positive, or forgetting that total energy is negative.
- Claiming g inside a **shell** is non-zero. The field is zero, but the potential is not.
- Taking $T^2 \propto r^3$ with r measured from the surface.

Practice questions

P06 · Practice

1

Easy

⌚ 0.75 min

Speed

P06-01

The acceleration due to gravity at a height equal to Earth's radius, above the surface, is:

- (A) $g/2$ (B) $g/4$ (C) $g/\sqrt{2}$ (D) $g/8$

2

Easy

⌚ 1 min

Basic

P06-02

At what height above Earth's surface does g decrease by 1%? ($R = 6400$ km; assume $h \ll R$.)

- (A) 64 km (B) 32 km (C) 16 km (D) 128 km

3

Easy

⌚ 0.75 min

Speed

P06-03

A planet's orbital radius is 4 times Earth's. Its orbital period is:

- (A) 2 years (B) 4 years (C) 8 years (D) 16 years

4

Easy

⌚ 0.75 min

Basic

P06-04

A satellite in a circular orbit has kinetic energy K . Its total mechanical energy is:

- (A) K (B) $-K$ (C) $-2K$ (D) $2K$

5

Medium

⌚ 1 min

Concept

P06-05

A satellite orbits just above Earth's surface ($v_o \approx 7.9$ km s⁻¹). The minimum *additional* speed it needs to escape is about:

- (A) 11.2 km/s (B) 3.3 km/s (C) 7.9 km/s (D) 1.4 km/s

6 Medium 1.5 min JEE

P06-06

The energy needed to move a satellite of mass m from a circular orbit of radius $2R$ to one of radius $3R$ ($R =$ Earth's radius) is:

- (A) $mgR/12$ (B) $mgR/6$ (C) $mgR/3$ (D) $mgR/2$

7 Easy 0.75 min Concept

P06-07

Inside a uniform thin spherical shell of mass M and radius R :

- (A) field and potential are both zero
 (B) the field is zero and the potential is $-GM/R$ everywhere
 (C) the field is GM/R^2 and the potential is zero
 (D) the field increases linearly with r

8 Easy 1 min NV

P06-08

Masses of 1 kg and 4 kg are 3 m apart. Find the distance (in m) from the 1 kg mass of the point between them where the net gravitational field is zero.

Your answer (numerical value):

9 Easy 0.75 min Speed

P06-09

A planet has twice Earth's mass and twice its radius. Its surface gravity is:

- (A) g (B) $2g$ (C) $g/2$ (D) $g/4$

P06 · Chapter Test

1 Easy 0.5 min Concept

P06-T1

The escape speed from a planet depends on:

- (A) the mass of the object being launched (B) the angle of launch
 (C) the mass and radius of the planet (D) all of the above

2 Easy 0.5 min Speed

P06-T2

The weight of a body at the centre of the Earth is:

- (A) the same as at the surface (B) half of that at the surface
 (C) zero (D) infinite

3 Easy 1 min Basic

P06-T3

If the orbital speed just above Earth's surface is 7.9 km/s, the orbital speed at a height R above the surface is about:

- (A) 3.95 km/s (B) 5.6 km/s (C) 7.9 km/s (D) 11.2 km/s

4 Easy 0.5 min NV

P06-T4

The orbital radius of a satellite is made 4 times larger. By what factor does its period increase?

Your answer (numerical value):

5 Easy 0.5 min Basic

P06-T5

The binding energy of a satellite of mass m in a circular orbit of radius r around a planet of mass M is:

- (A) GMm/r (B) $GMm/2r$ (C) $-GMm/2r$ (D) $2GMm/r$

Answers & Solutions

P06 · Practice

1 (B) 2 (B) 3 (C) 4 (B) 5 (B) 6 (A) 7 (B) 8 1 9 (C)

P06 · Chapter Test

1 (C) 2 (C) 3 (B) 4 8 5 (B)

P06 · PRACTICE

1 Answer: (B) g at height R Easy 0.75 min Speed

P06-01

SOLUTION $g_h = g \left(\frac{R}{R+h} \right)^2 = g \left(\frac{1}{2} \right)^2 = g/4$.

SHORTCUT r doubles, so g falls by a factor of 4.

COMMON TRAP Using $g(1 - 2h/R)$, which gives $-g$ and is invalid for large h .

2 Answer: (B) Small-height decrease Easy 1 min Basic

P06-02

SOLUTION $\frac{\Delta g}{g} = \frac{2h}{R} = 0.01 \Rightarrow h = \frac{0.01 \times 6400}{2} = 32 \text{ km}$.

SHORTCUT Depth for the same 1% decrease: 64 km, which is $2h$.

COMMON TRAP Forgetting the 2 gives 64 km.

3 Answer: (C) Kepler's third law Easy 0.75 min Speed

P06-03

SOLUTION $T \propto r^{3/2}$: $4^{3/2} = 8$, so 8 years.

SHORTCUT $\sqrt{4^3} = 8$.

COMMON TRAP Using $T \propto r^2$ gives 16.

4 Answer: (B) Satellite energy relations Easy 0.75 min Basic

P06-04

SOLUTION $E = -\frac{GMm}{2r} = -K$ (and $U = -2K$).

SHORTCUT $K : U : E = 1 : -2 : -1$.

COMMON TRAP Confusing U with E .

5 Answer: (B) Extra speed to escape Medium 1 min Concept

P06-05

SOLUTION $v_e = \sqrt{2} v_o \approx 11.2 \text{ km/s}$. Additional $= (\sqrt{2} - 1)v_o \approx 0.414 \times 7.9 \approx 3.3 \text{ km/s}$.

SHORTCUT $\Delta v = (\sqrt{2} - 1)v_o$.

COMMON TRAP Answering v_e itself.

6 Answer: (A) Energy to shift orbits Medium 1.5 min JEE

P06-06

SOLUTION $\Delta E = \frac{GMm}{2} \left(\frac{1}{2R} - \frac{1}{3R} \right) = \frac{GMm}{12R} = \frac{mgR}{12}$, using $GM = gR^2$.

SHORTCUT Take the difference of total energies $-GMm/2r$.

COMMON TRAP Using the difference of potential energies gives $mgR/6$.

7 Answer: (B) Field inside a shell Easy 0.75 min Concept

P06-07

SOLUTION By the shell theorem the field inside is zero, so the potential is constant and equal to its surface value $-GM/R$.

SHORTCUT Zero field $\Rightarrow$ constant (not zero) potential.

COMMON TRAP Assuming zero potential.

8 Answer: 1 Null point between two masses **Easy** ⌚ 1 min **NV**

P06-08

SOLUTION $\frac{G(1)}{x^2} = \frac{G(4)}{(3-x)^2} \Rightarrow 3-x = 2x \Rightarrow x = 1 \text{ m.}$

SHORTCUT $x = \frac{d\sqrt{m_1}}{\sqrt{m_1} + \sqrt{m_2}} = \frac{3}{3}.$

COMMON TRAP Measuring from the heavier mass (2 m).

9 Answer: (C) Surface gravity of another planet **Easy** ⌚ 0.75 min **Speed**

P06-09

SOLUTION $g \propto M/R^2 = 2/4 = 1/2.$

SHORTCUT Ratio scaling.

COMMON TRAP Using M/R gives g .

P06 · CHAPTER TEST

1 Answer: (C) What escape speed depends on **Easy** ⌚ 0.5 min **Concept**

P06-T1

SOLUTION $v_e = \sqrt{2GM/R}$ is independent of the object's mass and of the launch direction (energy is a scalar).

SHORTCUT Energy is a scalar, so direction doesn't matter.

COMMON TRAP Choosing (B).

2 Answer: (C) Weight at Earth's centre **Easy** ⌚ 0.5 min **Speed**

P06-T2

SOLUTION $g_d = g(1 - d/R) = 0$ at $d = R.$

SHORTCUT g grows linearly from 0 at the centre to the surface value.

COMMON TRAP Thinking gravity is strongest at the centre.

3 Answer: (B) Orbital speed at height R **Easy** ⌚ 1 min **Basic**

P06-T3

SOLUTION $v \propto 1/\sqrt{r}$; r doubles, so $v = 7.9/\sqrt{2} \approx 5.6 \text{ km/s.}$

SHORTCUT Divide by $\sqrt{2}.$

COMMON TRAP Halving the speed (3.95).

4 Answer: 8 Period scaling **Easy** ⌚ 0.5 min **NV**

P06-T4

SOLUTION $T \propto r^{3/2} = 4^{3/2} = 8.$

COMMON TRAP Answering 16 (r^2).

5 Answer: (B) Binding energy **Easy** ⌚ 0.5 min **Basic**

P06-T5

SOLUTION Binding energy $= -E = +\frac{GMm}{2r}$: the energy needed to free the satellite completely.

SHORTCUT Binding energy is positive.

COMMON TRAP Choosing the (negative) total energy itself.

REVISION Revision tracker · Gravitation

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Properties of Solids & Liquids (incl. Thermal Properties)

07

2–3 per shift
TYPICAL QUESTIONS

Medium
DIFFICULTY

Must-do
PRIORITY

**Class 11 · Ch 8, 9,
10**
NCERT

~14 hours
STUDY TIME

Concept summary

This unit bundles four mini-chapters, and each regularly gets a question:

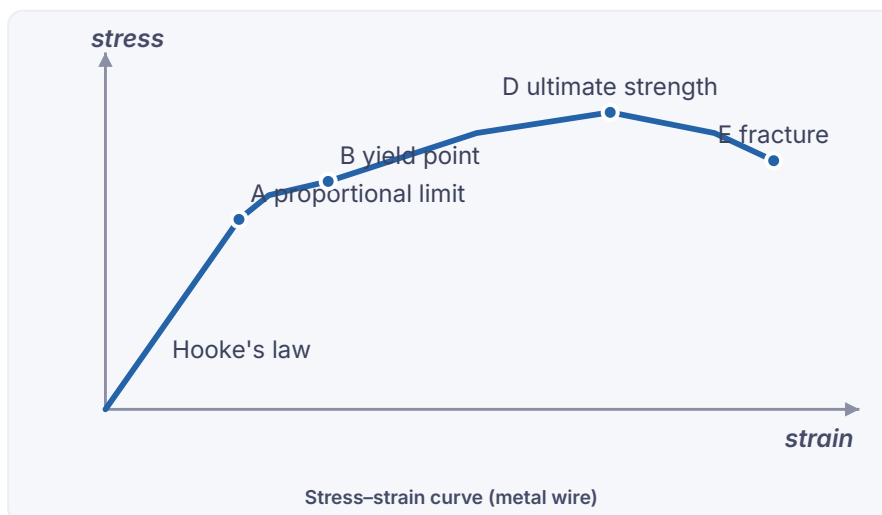
1. **Elasticity:** stress, strain, the three moduli, stress–strain curve, elastic energy.
2. **Fluids:** pressure, Pascal's law, buoyancy, continuity, Bernoulli, viscosity (Stokes, terminal velocity).
3. **Surface tension:** surface energy, excess pressure, capillary rise, angle of contact.
4. **Thermal properties:** expansion, calorimetry, latent heat, conduction, radiation, Newton's law of cooling.

Formulas: Elasticity

Σ FORMULA Moduli & energy

$$\text{Stress} = \frac{F}{A} \quad \text{Strain} = \frac{\Delta L}{L} \quad Y = \frac{FL}{A \Delta L} \quad B = -\frac{\Delta P}{\Delta V/V} \quad \eta_{\text{rigidity}} = \frac{F/A}{\theta}$$

- **Compressibility** $= 1/B$. **Poisson's ratio** $\sigma = -\frac{\text{lateral strain}}{\text{longitudinal strain}}$ (between 0 and 0.5 for real materials).
- **Elastic energy:** $U = \frac{1}{2} F \Delta L$; energy density $u = \frac{1}{2} \text{stress} \times \text{strain} = \frac{\text{stress}^2}{2Y}$.
- **Thermal stress** in a rod clamped at both ends: $Y \propto \Delta T$.
- **Wire under its own weight** (mass M , hanging): $\Delta L = \frac{MgL}{2AY}$.
- For the same material and load: $\Delta L \propto \frac{L}{r^2}$.



☆ IMPORTANT Stress–strain curve vocabulary

Proportional limit (Hooke's law holds up to here) → elastic limit (full recovery up to here) → yield point (plastic flow begins) → ultimate tensile strength → fracture. **Ductile** materials (copper) have a long plastic region. **Brittle** materials (glass) fracture soon after the elastic limit. **Elastomers** (rubber) have large elastic strain but no Hooke's-law region.

Formulas: Fluids

Σ FORMULA Statics

$$P = P_0 + \rho gh \quad \text{Pascal (hydraulic lift): } \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \text{Buoyancy: } F_B = \rho_{fluid} V_{sub} g$$

Floating body: fraction submerged = $\rho_{body} / \rho_{fluid}$. Gauge pressure = $P - P_0$.

Σ FORMULA Dynamics

$$A_1 v_1 = A_2 v_2 \quad P + \frac{1}{2} \rho v^2 + \rho gh = \text{constant} \quad \text{Torricelli: } v = \sqrt{2gh}$$

- Hole at depth h below the surface in a tank of height H (on the ground): horizontal range = $2\sqrt{h(H-h)}$, maximum at $h = H/2$.
- Venturi meter:** $v_1 = A_2 \sqrt{\frac{2\Delta P}{\rho(A_1^2 - A_2^2)}}$.
- Lift on a wing / roof blown off: faster flow means lower pressure (Bernoulli).

Σ FORMULA Viscosity

$$F = -\eta A \frac{dv}{dx} \quad \text{Stokes: } F = 6\pi\eta r v \quad \text{Terminal velocity: } v_t = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

ρ = density of the sphere, σ = density of the fluid. **Reynolds number** $R_e = \frac{\rho v D}{\eta}$: flow is streamline for $R_e \lesssim 1000$ and turbulent for $R_e \gtrsim 2000$. Critical velocity $v_c = \frac{R_e \eta}{\rho D}$. The viscosity of liquids falls with temperature, while that of gases rises.

Formulas: Surface tension

Σ FORMULA Surface tension

$$T = \frac{F}{L} \quad W = T \Delta A \quad \text{Excess pressure: drop } \frac{2T}{R}, \text{ soap bubble } \frac{4T}{R}, \text{ air bubble in liquid } \frac{2T}{R}$$

$$\text{Capillary rise: } h = \frac{2T \cos \theta}{\rho g r}$$

- $\theta < 90^\circ$ (water-glass): the liquid rises and the meniscus is concave. $\theta > 90^\circ$ (mercury-glass): the liquid is depressed and the meniscus is convex.
- Combining n drops** of radius r into one: $R = n^{1/3} r$; energy released = $4\pi T(nr^2 - R^2)$.
- Detergents **lower** surface tension. Surface tension falls as temperature rises.
- If a capillary tube is shorter than h , the liquid doesn't overflow. The radius of curvature of the meniscus adjusts instead.

Formulas: Thermal properties

Σ FORMULA Expansion & calorimetry

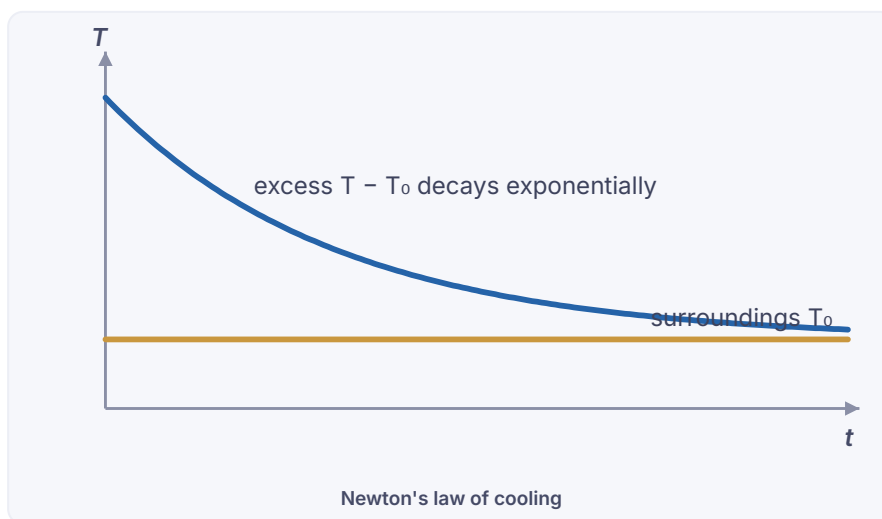
$$\Delta L = L \alpha \Delta T \quad \beta = 2\alpha \quad \gamma = 3\alpha \quad Q = mc \Delta T \quad Q = mL$$

- Principle of calorimetry:** heat lost = heat gained (no losses).
- Water: $c = 4186 \text{ J kg}^{-1} \text{K}^{-1}$ ($1 \text{ cal g}^{-1} \text{ } ^\circ\text{C}^{-1}$). $L_f = 80 \text{ cal/g}$ ($3.34 \times 10^5 \text{ J/kg}$). $L_v = 540 \text{ cal/g}$ ($2.26 \times 10^6 \text{ J/kg}$).
- Anomalous expansion of water:** it contracts from 0°C to 4°C and has maximum density at 4°C .

Σ FORMULA Heat transfer

$$\frac{dQ}{dt} = \frac{kA \Delta T}{L} \quad R_{th} = \frac{L}{kA} \quad \text{series: } R = R_1 + R_2 \quad \text{parallel: } \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

Two slabs of equal length in series: $k_{eq} = \frac{2k_1k_2}{k_1 + k_2}$. Equal area in parallel: $k_{eq} = \frac{k_1 + k_2}{2}$. **Radiation:** Stefan-Boltzmann $P = e\sigma AT^4$; net $P = e\sigma A(T^4 - T_0^4)$; $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}$. **Wien's law:** $\lambda_m T = b = 2.9 \times 10^{-3} \text{ m K}$. **Newton's law of cooling:** $\frac{dT}{dt} = -k(T - T_0)$, approximately $\frac{T_1 - T_2}{t} = k \left(\frac{T_1 + T_2}{2} - T_0 \right)$.

**Frequently tested concepts & question patterns****▮ PYQ PATTERN Recurring structures**

1. **Young's modulus / extension** comparisons between wires (same material, different L and r); elastic energy.
2. **Bernoulli + continuity:** pressure difference in a pipe, efflux speed, range of a jet.
3. **Terminal velocity** scaling (drops merging), Stokes force.
4. **Capillary rise** changes when r or θ changes; excess pressure in bubbles; energy when drops merge.
5. **Calorimetry with phase change:** ice + water mixing. Check whether all the ice melts.
6. **Conduction through composite slabs** (junction temperature), and **Newton's law of cooling** (time for successive intervals).
7. **Stefan/Wien:** power ratios for temperature changes; peak wavelength.

Shortcuts**⚡ SHORTCUT Junction temperature of two slabs in series**

$\theta_{junction} = \frac{\theta_1/R_1 + \theta_2/R_2}{1/R_1 + 1/R_2}$, the weighted mean with weights kA/L . This is the "Kirchhoff" approach used for resistors, with heat current in place of electric current.

⚡ SHORTCUT Is all the ice melted?

First compute the heat the warm water can give up by cooling to 0°C , Q_{avail} . Compare it with $m_{ice}L_f$ (plus any heat needed to warm the ice to 0°C). If Q_{avail} is smaller, the final temperature is 0°C with some ice left. **Saves** you from impossible negative answers.

Question classification

Type	Difficulty	Target time
Moduli / extension ratios	E	1 min
Bernoulli / continuity	M	2 min
Terminal velocity scaling	E	1 min
Capillarity / excess pressure	E–M	1.5 min
Calorimetry with phase change	M	2.5 min
Composite conduction / Newton cooling	M	2 min
Stefan / Wien ratios	E	1 min

Common mistakes



MISTAKE ALERT Mistake alerts

- Soap bubbles have **two** surfaces: $4T/R$, not $2T/R$.
- Stokes' law uses the **radius**, not the diameter.
- Using $^{\circ}\text{C}$ in Stefan's law. T must be in **kelvin**.
- Assuming the final temperature is above 0°C without checking that all the ice has melted.
- Forgetting that $\gamma = 3\alpha$ (volume) when a question switches from length to volume.
- In Bernoulli, mixing gauge and absolute pressure.

Practice questions

P07 · Practice

1

Easy

⌚ 1 min

Calc

P07-01

A 2 m wire of cross-section 1 mm^2 stretches by 1 mm under a 10 kg load ($g = 10$). Young's modulus is:

- (A) $2 \times 10^{10} \text{ N m}^{-2}$ (B) $2 \times 10^{11} \text{ N m}^{-2}$ (C) $1 \times 10^{11} \text{ N m}^{-2}$ (D) $2 \times 10^9 \text{ N m}^{-2}$

2

Easy

⌚ 1 min

Basic

P07-02

A wire stretched by 2 mm under a load of 50 N stores elastic energy of:

- (A) 0.05 J (B) 0.1 J (C) 0.025 J (D) 100 J

3

Easy

⌚ 0.75 min

NV

P07-03

Water flows out of a small hole 5 m below the free surface of a large open tank. Find the efflux speed (in m/s, $g = 10$).

Your answer (numerical value):

4

Easy

⌚ 0.5 min

Speed

P07-04

If a pipe's diameter halves, the speed of an incompressible fluid in it becomes:

- (A) twice (B) four times (C) half (D) one-fourth

5

Medium

⌚ 1 min

Concept

P07-05

Eight identical rain drops, each falling at terminal velocity v , merge into one drop. Its terminal velocity is:

- (A) $2v$ (B) $4v$ (C) $8v$ (D) $16v$

6 Medium 1.5 min NV

P07-06

Find the height (in cm) to which water rises in a clean glass capillary of radius 0.07 mm. Take surface tension 0.07 N/m, contact angle 0° , $\rho = 1000 \text{ kg m}^{-3}$, $g = 10$.

Your answer (numerical value):

7 Easy 0.75 min Basic

P07-07

The excess pressure inside a soap bubble of radius 1 cm (surface tension 0.03 N m^{-1}) is:

- (A) 6 Pa (B) 12 Pa (C) 3 Pa (D) 24 Pa

8 Easy 1 min NV

P07-08

100 g of water at 80°C is mixed with 200 g of water at 20°C . Find the final temperature (in $^\circ\text{C}$), neglecting heat losses.

Your answer (numerical value):

9 Easy 1 min Basic

P07-09

The heat needed to convert 10 g of ice at 0°C into water at 20°C is ($L_f = 80 \text{ cal/g}$):

- (A) 800 cal (B) 1000 cal (C) 200 cal (D) 1600 cal

10 Medium 1 min Concept

P07-10

Two rods of equal length and area, with conductivities k and $2k$, are joined end to end. The equivalent conductivity of the combination is:

- (A) $3k/2$ (B) $4k/3$ (C) $3k$ (D) $2k/3$

11 Easy 0.5 min Speed

P07-11

If the absolute temperature of a black body doubles, the power it radiates becomes:

- (A) 2 times (B) 4 times (C) 8 times (D) 16 times

12 Medium 2 min NV

P07-12

A body cools from 80°C to 70°C in 5 minutes in surroundings at 15°C . Using the average-temperature form of Newton's law of cooling, find the time (in minutes) it takes to cool from 70°C to 60°C .

Your answer (numerical value):

P07 · Chapter Test

1 Easy 0.5 min Basic

P07-T1

A wire's longitudinal strain is 2×10^{-3} and its lateral strain is -6×10^{-4} . Poisson's ratio is:

- (A) 0.3 (B) 3.3 (C) 0.03 (D) 0.12

2 Easy 0.5 min Speed

P07-T2

The pistons of a hydraulic lift have areas in the ratio 1 : 50. A 200 N force on the small piston can support a load of weight:

- (A) 4 N (B) 250 N (C) 10 000 N (D) 2 000 N

3 Easy 0.75 min Basic

P07-T3

A star's spectrum peaks at 500 nm. Its surface temperature is about ($b = 2.9 \times 10^{-3} \text{ m K}$):

- (A) 580 K (B) 5800 K (C) 58 000 K (D) 1450 K

4 Easy 0.5 min NV

P07-T4

A solid has $\alpha = 2 \times 10^{-5} \text{ K}^{-1}$. Its volume expansion coefficient is $x \times 10^{-5} \text{ K}^{-1}$. Find x .

Your answer (numerical value):

5 Easy 1 min Calc

P07-T5

A sphere of radius 1 mm moves at 0.1 m s^{-1} through a liquid of viscosity 1 Pa s. The viscous drag is about:

- (A) $1.9 \times 10^{-3} \text{ N}$ (B) $6.0 \times 10^{-4} \text{ N}$ (C) $1.9 \times 10^{-2} \text{ N}$ (D) $3.8 \times 10^{-3} \text{ N}$

Answers & Solutions

P07 · Practice

- 1 (B) 2 (A) 3 10 4 (B) 5 (B) 6 20 7 (B) 8 40 9 (B) 10 (B)
11 (D) 12 6

P07 · Chapter Test

- 1 (A) 2 (C) 3 (B) 4 6 5 (A)

P07 · PRACTICE

1 Answer: (B) Young's modulus Easy 1 min Calc

P07-01

SOLUTION $Y = \frac{FL}{A\Delta L} = \frac{100 \times 2}{10^{-6} \times 10^{-3}} = 2 \times 10^{11} \text{ N m}^{-2}$.

SHORTCUT Convert $\text{mm}^2 \rightarrow 10^{-6} \text{ m}^2$ and $\text{mm} \rightarrow 10^{-3} \text{ m}$ first.

COMMON TRAP Converting 1 mm^2 as 10^{-3} m^2 .

2 Answer: (A) Elastic energy Easy 1 min Basic

P07-02

SOLUTION $U = \frac{1}{2}F\Delta L = \frac{1}{2}(50)(2 \times 10^{-3}) = 0.05 \text{ J}$.

SHORTCUT The factor $\frac{1}{2}$ is there because the force rises linearly from 0 to F .

COMMON TRAP Forgetting the $\frac{1}{2}$ (0.1 J).

3 Answer: 10 Torricelli's theorem Easy 0.75 min NV

P07-03

SOLUTION $v = \sqrt{2gh} = \sqrt{100} = 10 \text{ m s}^{-1}$.

SHORTCUT Same as the speed of free fall through h .

COMMON TRAP Measuring h from the bottom instead of from the free surface.

4 Answer: (B) Continuity equation Easy 0.5 min Speed

P07-04

SOLUTION Av is constant and $A \propto d^2$: the area becomes $1/4$, so the speed is $\times 4$.

SHORTCUT Speed $\propto 1/d^2$.

COMMON TRAP Taking speed $\propto 1/d$.

5 Answer: (B) Terminal velocity of merged drops Medium 1 min Concept

P07-05

SOLUTION Volume $\times 8$ means $R = 2r$. $v_t \propto r^2$, so $v' = 4v$.

SHORTCUT $v' = n^{2/3}v$.

COMMON TRAP Assuming $v_t \propto \text{volume}$ ($8v$).

6 Answer: 20 Capillary rise **Medium** ⌚ 1.5 min **NV**

P07-06

$$\text{SOLUTION } h = \frac{2T}{\rho g r} = \frac{0.14}{1000 \times 10 \times 7 \times 10^{-5}} = \frac{0.14}{0.7} = 0.2 \text{ m} = 20 \text{ cm.}$$

SHORTCUT hr is constant for a given liquid: here $hr = 1.4 \times 10^{-5} \text{ m}^2$.**COMMON TRAP** Using the diameter for r gives 10 cm.**7 Answer: (B)** Excess pressure in a soap bubble **Easy** ⌚ 0.75 min **Basic**

P07-07

$$\text{SOLUTION } \Delta P = 4T/R = 0.12/0.01 = 12 \text{ Pa.}$$

SHORTCUT Two surfaces, so $4T/R$.**COMMON TRAP** Using $2T/R$ gives 6 Pa.**8 Answer: 40** Method of mixtures **Easy** ⌚ 1 min **NV**

P07-08

$$\text{SOLUTION } 100(80 - T) = 200(T - 20) \Rightarrow 8000 + 4000 = 300T \Rightarrow T = 40 \text{ }^\circ\text{C.}$$

SHORTCUT Mass-weighted mean: $\frac{100 \times 80 + 200 \times 20}{300}$.**COMMON TRAP** Taking a simple average (50).**9 Answer: (B)** Latent heat + sensible heat **Easy** ⌚ 1 min **Basic**

P07-09

$$\text{SOLUTION } \text{Melt: } 10 \times 80 = 800 \text{ cal. Warm: } 10 \times 1 \times 20 = 200 \text{ cal. Total } 1000 \text{ cal.}$$

SHORTCUT Always add the phase-change step.**COMMON TRAP** Stopping after melting (800).**10 Answer: (B)** Series conduction **Medium** ⌚ 1 min **Concept**

P07-10

$$\text{SOLUTION } \text{Series, equal lengths: } k_{eq} = \frac{2k_1k_2}{k_1 + k_2} = \frac{2(k)(2k)}{3k} = \frac{4k}{3}.$$

SHORTCUT The harmonic mean is closer to the smaller value.**COMMON TRAP** Using the arithmetic mean (the parallel formula).**11 Answer: (D)** Stefan's law **Easy** ⌚ 0.5 min **Speed**

P07-11

$$\text{SOLUTION } P \propto T^4 = 2^4 = 16.$$

COMMON TRAP Doubling a Celsius temperature is **not** doubling T .**12 Answer: 6** Newton's law of cooling **Medium** ⌚ 2 min **NV**

P07-12

$$\text{SOLUTION } \frac{10}{5} = k(75 - 15) \Rightarrow k = \frac{1}{30}. \text{ Next: } \frac{10}{t} = k(65 - 15) = \frac{50}{30} \Rightarrow t = 6 \text{ min.}$$

SHORTCUT Equal temperature drops: $t_2/t_1 = \frac{\text{mean excess}_1}{\text{mean excess}_2} = \frac{60}{50}$.**COMMON TRAP** Assuming the same 5 min, i.e. constant cooling rate.**P07 · CHAPTER TEST****1 Answer: (A)** Poisson's ratio **Easy** ⌚ 0.5 min **Basic**

P07-T1

$$\text{SOLUTION } \sigma = -\frac{-6 \times 10^{-4}}{2 \times 10^{-3}} = 0.3.$$

SHORTCUT Must lie between 0 and 0.5.**COMMON TRAP** Inverting the ratio gives 3.3.**2 Answer: (C)** Hydraulic lift **Easy** ⌚ 0.5 min **Speed**

P07-T2

$$\text{SOLUTION } F_2 = F_1 \frac{A_2}{A_1} = 200 \times 50 = 10\,000 \text{ N.}$$

SHORTCUT Pascal's law multiplies the force by the area ratio.**COMMON TRAP** Dividing by the ratio.**3 Answer: (B)** Wien's displacement law **Easy** ⌚ 0.75 min **Basic**

P07-T3

$$\text{SOLUTION } T = b/\lambda_m = \frac{2.9 \times 10^{-3}}{5 \times 10^{-7}} = 5800 \text{ K.}$$

SHORTCUT Remember: the Sun $\approx 5800 \text{ K} \approx 500 \text{ nm}$.**COMMON TRAP** Using $500 \times 10^{-6} \text{ m}$ for nm.

4 Answer: 6 Volume expansion coefficient **Easy** 0.5 min NV

P07-T4

SOLUTION $\gamma = 3\alpha = 6 \times 10^{-5} \text{ K}^{-1}$.

SHORTCUT $\alpha : \beta : \gamma = 1 : 2 : 3$.

COMMON TRAP Using $\gamma = \alpha^3$.

5 Answer: (A) Stokes' law **Easy** 1 min Calc

P07-T5

SOLUTION $F = 6\pi\eta rv = 6\pi(1)(10^{-3})(0.1) = 6\pi \times 10^{-4} \approx 1.88 \times 10^{-3} \text{ N}$.

SHORTCUT $6\pi \approx 18.85$.

COMMON TRAP Omitting the factor π gives (B).

 REVISION Revision tracker · Solids & Liquids

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 08

Thermodynamics

08

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 11

NCERT

~9 hours

STUDY TIME

Concept summary

- **Zeroth law:** thermal equilibrium is transitive, which defines temperature.
- **First law:** energy conservation for heat and work: $Q = \Delta U + W$. Here W is the work done **by** the gas (NCERT convention).
- **Internal energy** of an ideal gas depends **only on temperature**: $\Delta U = nC_v\Delta T$ for any process.
- **Work** is the area under the P - V curve. It depends on the path. Heat also depends on the path.
- **Second law:** no engine converts heat entirely into work in a cycle (Kelvin-Planck), and heat doesn't flow by itself from cold to hot (Clausius). The Carnot engine sets the maximum efficiency.

Formulas

Σ FORMULA First law & process table (ideal gas, n moles)

Process	Condition	W (by gas)	ΔU	Q
Isochoric	V const	0	$nC_v\Delta T$	$nC_v\Delta T$
Isobaric	P const	$P\Delta V = nR\Delta T$	$nC_v\Delta T$	$nC_p\Delta T$
Isothermal	T const	$nRT \ln \frac{V_2}{V_1}$	0	$= W$
Adiabatic	$Q = 0, PV^\gamma$ const	$\frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1}$	$-W$	0
Cyclic	returns to start	area enclosed	0	$= W$
Free expansion	into vacuum, insulated	0	0	0

Adiabatic relations: $PV^\gamma = \text{const}$, $TV^{\gamma-1} = \text{const}$, $P^{1-\gamma}T^\gamma = \text{const}$.

Σ FORMULA Heat capacities

$$C_p - C_v = R \quad \gamma = \frac{C_p}{C_v}$$

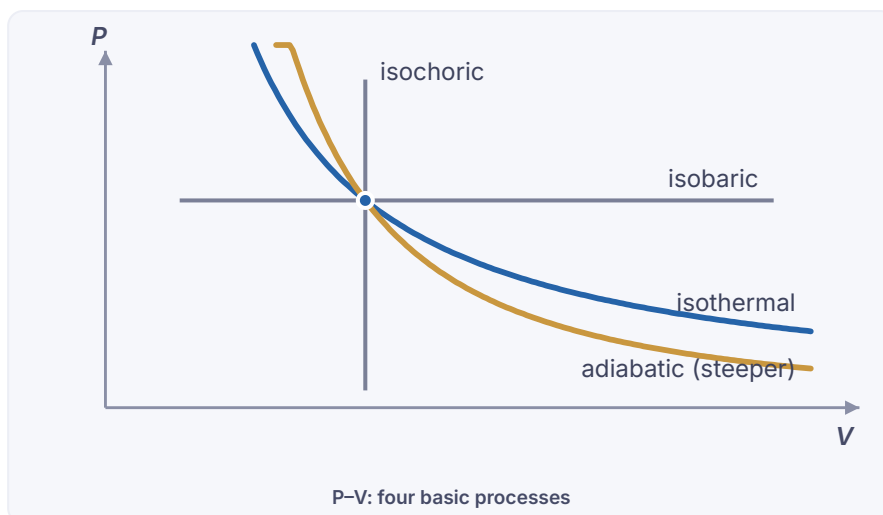
Gas	C_v	C_p	γ
Monatomic	$\frac{3}{2}R$	$\frac{5}{2}R$	$5/3 \approx 1.67$
Diatomic (rigid, room temperature)	$\frac{5}{2}R$	$\frac{7}{2}R$	$7/5 = 1.4$
Polyatomic (nonlinear, rigid)	$3R$	$4R$	$4/3 \approx 1.33$

Mixture: $C_{v,mix} = \frac{n_1C_{v1} + n_2C_{v2}}{n_1 + n_2}$. In an isobaric process, the fractions of Q are $\Delta U/Q = 1/\gamma$ and $W/Q = 1 - 1/\gamma$.

Σ FORMULA Engines & refrigerators

$$\eta = \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1} \quad \eta_{Carnot} = 1 - \frac{T_2}{T_1} \quad \text{COP}_{ref} = \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2} \quad \text{COP} = \frac{1 - \eta}{\eta}$$

Temperatures are in **kelvin**. Carnot cycle: two isothermals + two adiabatics. It is reversible, and no engine working between the same two temperatures is more efficient.



☆ IMPORTANT Reading P-V diagrams

- The **adiabatic** curve is steeper than the isothermal through the same point: slope ratio = γ .
- A clockwise cycle on a P - V diagram means net work done **by** the gas ($W > 0$), which is an engine. Anticlockwise means work is done on the gas (a refrigerator).
- On a P - T or V - T diagram, convert to P - V before finding work. Straight lines through the origin on a V - T diagram are isobaric.

Standard models & assumptions

- Ideal gas: $PV = nRT$, $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$.
- Quasi-static (slow) processes for the formulas above. Free expansion is not quasi-static.
- "Insulated/thermally isolated/sudden" suggests adiabatic. "Slow in contact with a reservoir" means isothermal.

Frequently tested concepts & question patterns

▮ PYQ PATTERN Recurring structures

1. **Process identification** from graphs (P - V , V - T , P - T) and signs of Q , W , ΔU for each leg.
2. **Adiabatic compression/expansion**: final T or P with γ .
3. **Work in a cycle** as an area; efficiency of a given cycle.
4. **Carnot efficiency / refrigerator COP**, including "what T_1 increase raises η to ...".
5. **Heat supplied in isobaric process** and the split between ΔU and W .
6. **Mixtures of gases**: γ_{mix} .

Shortcuts

⚡ SHORTCUT ΔU first, always

For any process of an ideal gas, write $\Delta U = nC_v\Delta T$ first. Then get whichever of Q or W is easier, and the other from the first law. **Time saved**: avoids integrating heat along a path.

⚡ SHORTCUT Temperature ratio in adiabatics

$T_2/T_1 = (V_1/V_2)^{\gamma-1}$. Monatomic: $\gamma - 1 = 2/3$, so compressing to $1/8$ of the volume gives $8^{2/3} = 4$, i.e. $T \times 4$.

Question classification

Type	Difficulty	Target time
First law sign bookkeeping	E	45 s
Carnot / COP	E	1 min
Adiabatic T , P , V relations	E–M	1.5 min
Cyclic process work/efficiency from graph	M	2–3 min
Isothermal work with logs	M	1.5 min
γ of mixtures, polytropic processes	M–H	2.5 min

Common mistakes

MISTAKE ALERT Mistake alerts

- Sign convention: in this book (NCERT), W is work done **by** the system. Some books use work **on** the system, which flips the sign.
- Using $^{\circ}\text{C}$ in Carnot efficiency.
- Assuming $\Delta U = 0$ in an adiabatic process. It's Q that is zero there. $\Delta U = 0$ in isothermal and cyclic processes.
- Mixing $\ln$ and $\log_{10}$: $\ln 2 = 0.693$, $\ln 10 = 2.303$.
- Using $\gamma = 5/3$ for air (diatomic, so $7/5$).

Practice questions

P08 · Practice

1 **Medium** ⌚ 1.5 min **Calc** P08-01

Two moles of an ideal gas expand isothermally at 300 K from 10 L to 20 L. The work done by the gas is about ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$, $\ln 2 = 0.693$):

- (A) 1.73 kJ (B) 3.46 kJ (C) 6.92 kJ (D) 4.99 kJ

2 **Medium** ⌚ 1 min **Concept** P08-02

A monatomic ideal gas at temperature T is compressed adiabatically to one-eighth of its volume. Its new temperature is:

- (A) $2T$ (B) $4T$ (C) $8T$ (D) $16T$

3 **Easy** ⌚ 1 min **NV** P08-03

An ideal gas goes round a rectangular cycle on a P – V diagram with corners at $V = 1 \text{ L}$ and 3 L and $P = 1 \times 10^5 \text{ Pa}$ and $3 \times 10^5 \text{ Pa}$. Find the magnitude of the net work per cycle (in J).

Your answer (numerical value):

4 **Easy** ⌚ 0.5 min **Speed** P08-04

A gas absorbs 500 J of heat and does 200 J of work. The change in its internal energy is:

- (A) 700 J (B) 300 J (C) -300 J (D) 200 J

5 Easy 0.5 min Speed

P08-05

A Carnot engine works between 500 K and 300 K. Its efficiency is:

- (A) 60% (B) 40% (C) 20% (D) 66.7%

6 Easy 0.75 min NV

P08-06

An ideal (Carnot) refrigerator keeps its interior at 250 K in a room at 300 K. Find its coefficient of performance.

Your answer (numerical value):

7 Medium 1 min Concept

P08-07

Heat is supplied to a rigid diatomic ideal gas at constant pressure. The fraction of the heat that increases internal energy is:

- (A) $2/7$ (B) $5/7$ (C) $3/5$ (D) $2/5$

8 Easy 0.75 min Concept

P08-08

At the same point on a P - V diagram, the ratio (slope of adiabatic)/(slope of isothermal) is:

- (A) 1 (B) γ (C) $1/\gamma$ (D) $\gamma - 1$

9 Easy 0.5 min Tricky

P08-09

An ideal gas expands freely into a vacuum inside an insulated container. Its temperature:

- (A) rises (B) falls
(C) stays the same (D) falls to absolute zero

P08 · Chapter Test

1 Easy 0.5 min Basic

P08-T1

Which of the following is a state function?

- (A) heat (B) work
(C) internal energy (D) the heat rejected in a cycle

2 Easy 0.5 min Speed

P08-T2

For a rigid diatomic ideal gas, γ is:

- (A) $5/3$ (B) $7/5$ (C) $4/3$ (D) $9/7$

3 Easy 0.75 min Basic

P08-T3

One mole of an ideal gas is heated by 10 K at constant pressure. The work done by the gas is ($R = 8.314$):

- (A) 8.3 J (B) 83.1 J (C) 207.9 J (D) 124.7 J

4 Easy 0.5 min NV

P08-T4

An engine absorbs 1000 J per cycle and rejects 750 J. Find its efficiency (in %).

Your answer (numerical value):

P08-T5

5 Easy 0.5 min Concept

A cyclic process on a P - V diagram is traversed anticlockwise. The net work done by the gas is:

- (A) positive
(B) negative
(C) zero
(D) equal to the heat absorbed, which is positive

Answers & Solutions

P08 · Practice

1 (B) 2 (B) 3 400 4 (B) 5 (B) 6 5 7 (B) 8 (B) 9 (C)

P08 · Chapter Test

1 (C) 2 (B) 3 (B) 4 25 5 (B)

P08 · PRACTICE

1 Answer: (B) Isothermal work Medium 1.5 min Calc

P08-01

SOLUTION $W = nRT \ln(V_2/V_1) = 2 \times 8.314 \times 300 \times 0.693 \approx 3457 \text{ J} \approx 3.46 \text{ kJ}$.

SHORTCUT $2 \times 8.314 \times 300 \approx 4988$; $\times 0.7 \approx 3.49$. Closest option: 3.46.

COMMON TRAP Using $\log_{10} 2 = 0.301$ gives 1.5 kJ.

2 Answer: (B) Adiabatic temperature change Medium 1 min Concept

P08-02

SOLUTION $TV^{\gamma-1} = \text{const}$ with $\gamma - 1 = 2/3$: $T_2 = T \cdot 8^{2/3} = 4T$.

SHORTCUT $8^{2/3} = (8^{1/3})^2 = 4$.

COMMON TRAP Using the isothermal relation (unchanged T), or γ in place of $\gamma - 1$.

3 Answer: 400 Work in a rectangular cycle Easy 1 min NV

P08-03

SOLUTION Area = $\Delta P \Delta V = (2 \times 10^5)(2 \times 10^{-3}) = 400 \text{ J}$.

SHORTCUT Work = enclosed area. $1 \text{ L} = 10^{-3} \text{ m}^3$.

COMMON TRAP Leaving litres unconverted.

4 Answer: (B) First law bookkeeping Easy 0.5 min Speed

P08-04

SOLUTION $\Delta U = Q - W = 500 - 200 = 300 \text{ J}$.

SHORTCUT $Q = \Delta U + W$.

COMMON TRAP Adding the two.

5 Answer: (B) Carnot efficiency Easy 0.5 min Speed

P08-05

SOLUTION $\eta = 1 - 300/500 = 0.4$.

COMMON TRAP Using T_2/T_1 (60%).

6 Answer: 5 Refrigerator COP Easy 0.75 min NV

P08-06

SOLUTION $\text{COP} = \frac{T_2}{T_1 - T_2} = \frac{250}{50} = 5$.

COMMON TRAP Using $T_1/(T_1 - T_2) = 6$ (the heat-pump COP).

7 Answer: (B) Isobaric heat split Medium 1 min Concept

P08-07

SOLUTION $\frac{\Delta U}{Q} = \frac{C_v}{C_p} = \frac{5/2}{7/2} = \frac{5}{7}$.

SHORTCUT $\Delta U/Q = 1/\gamma$.

COMMON TRAP Giving the work fraction (2/7).

8 Answer: (B) Slopes of adiabatic vs isothermal **Easy** ⌚ 0.75 min **Concept**

P08-08

SOLUTION Isothermal: $dP/dV = -P/V$. Adiabatic: $dP/dV = -\gamma P/V$. Ratio = γ .

SHORTCUT Adiabatic is steeper by a factor $\gamma > 1$.

COMMON TRAP Choosing $1/\gamma$.

9 Answer: (C) Free expansion **Easy** ⌚ 0.5 min **Tricky**

P08-09

SOLUTION $Q = 0$ (insulated) and $W = 0$ (no external pressure), so $\Delta U = 0$. For an ideal gas, U depends only on T , so T is unchanged.

SHORTCUT Free expansion of an ideal gas: $\Delta T = 0$, though it's **not** an isothermal quasi-static process.

COMMON TRAP Treating it as an adiabatic *reversible* expansion, which cools the gas.

P08 · CHAPTER TEST

1 Answer: (C) State functions **Easy** ⌚ 0.5 min **Basic**

P08-T1

SOLUTION Internal energy depends only on the state. Heat and work depend on the path. (Their difference $Q - W = \Delta U$ is path-independent, but each alone is not.)

SHORTCUT U, H, P, V, T are state functions; Q and W are path functions.

COMMON TRAP Treating heat as a property the system "has".

2 Answer: (B) γ for a diatomic gas **Easy** ⌚ 0.5 min **Speed**

P08-T2

SOLUTION $C_v = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$, so $\gamma = 7/5$.

SHORTCUT $\gamma = 1 + \frac{2}{f}$ with $f = 5$.

COMMON TRAP Using the monatomic value.

3 Answer: (B) Isobaric work **Easy** ⌚ 0.75 min **Basic**

P08-T3

SOLUTION $W = nR\Delta T = 8.314 \times 10 \approx 83.1$ J.

SHORTCUT $P\Delta V = nR\Delta T$.

COMMON TRAP Computing the heat $nC_p\Delta T$ instead.

4 Answer: 25 Engine efficiency **Easy** ⌚ 0.5 min **NV**

P08-T4

SOLUTION $\eta = 1 - 750/1000 = 0.25 = 25\%$.

SHORTCUT $W = 250$ J.

COMMON TRAP Answering 75.

5 Answer: (B) Anticlockwise cycle **Easy** ⌚ 0.5 min **Concept**

P08-T5

SOLUTION Anticlockwise means net work is done **on** the gas, so $W_{by} < 0$ and the gas rejects net heat ($Q = W < 0$).

SHORTCUT Clockwise = engine; anticlockwise = refrigerator.

COMMON TRAP Taking work as always positive.

REVISION Revision tracker · Thermodynamics

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Kinetic Theory of Gases

09

~1 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 12

NCERT

~5 hours

STUDY TIME

Concept summary

- A gas is a huge number of point molecules in random motion. They collide elastically, and their pressure comes from momentum transfer to the walls.
- Temperature is a measure of average translational KE:** $\langle \frac{1}{2}mv^2 \rangle = \frac{3}{2}k_B T$, independent of the gas.
- Equipartition:** each quadratic degree of freedom carries $\frac{1}{2}k_B T$ on average, and that fixes C_v and γ .
- The **mean free path** is the average distance between collisions.

Formulas

Σ FORMULA Ideal gas & pressure

$$PV = nRT = Nk_B T \quad P = \frac{1}{3}\rho v_{rms}^2 = \frac{1}{3}\frac{N}{V}m v_{rms}^2 \quad P = \frac{2}{3}E_{tr} \text{ (per unit volume)}$$

$R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$, $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$, $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$. Total translational KE of a gas $= \frac{3}{2}PV = \frac{3}{2}nRT$.

Σ FORMULA Molecular speeds (M = molar mass in kg/mol)

$$v_{rms} = \sqrt{\frac{3RT}{M}} \quad v_{avg} = \sqrt{\frac{8RT}{\pi M}} \quad v_{mp} = \sqrt{\frac{2RT}{M}}$$

$v_{rms} : v_{avg} : v_{mp} = \sqrt{3} : \sqrt{8/\pi} : \sqrt{2} \approx 1.73 : 1.60 : 1.41$. At the same T , speed $\propto 1/\sqrt{M}$. For one gas, speed $\propto \sqrt{T}$. Average translational KE per molecule: $\frac{3}{2}k_B T$ (about $6.2 \times 10^{-21} \text{ J}$ at 300 K).

Σ FORMULA Degrees of freedom & heat capacities

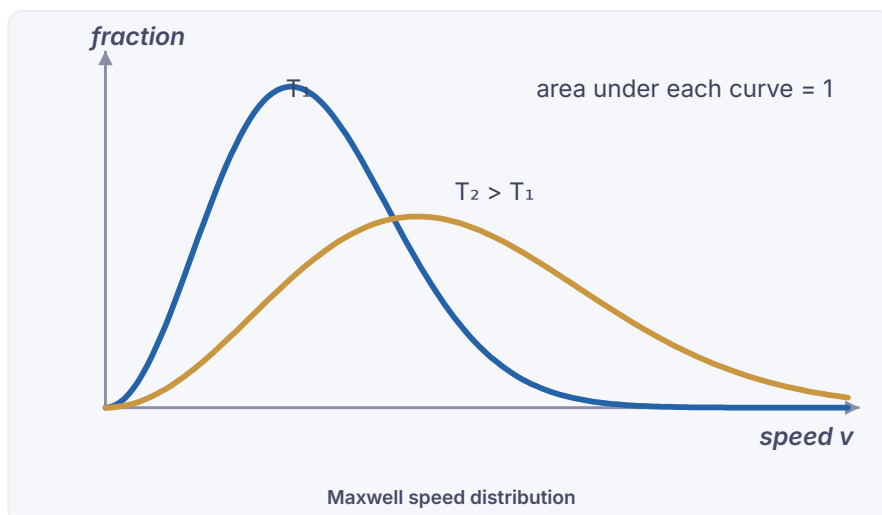
Molecule	Translational + rotational f	U per mole	C_v	$\gamma = 1 + 2/f$
Monatomic (He, Ar)	3	$\frac{3}{2}RT$	$\frac{3}{2}R$	5/3
Diatomic, rigid (O_2 , N_2 at room T)	5	$\frac{5}{2}RT$	$\frac{5}{2}R$	7/5
Diatomic with vibration (high T)	7	$\frac{7}{2}RT$	$\frac{7}{2}R$	9/7
Nonlinear polyatomic, rigid	6	$3RT$	$3R$	4/3

Each vibrational mode contributes **2** quadratic terms (kinetic + potential), i.e. $k_B T$ per molecule.

Σ FORMULA Mean free path

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 n} = \frac{k_B T}{\sqrt{2}\pi d^2 P}$$

n = number density, d = molecular diameter. At constant T , $\lambda \propto 1/P$. At constant n , λ doesn't depend on T .



Standard models & assumptions

- Molecules are point masses (negligible volume), with no intermolecular forces except during collisions, and elastic collisions.
- Large number of molecules, random motion, equally likely in all directions (so $\overline{v_x^2} = \frac{1}{3}\overline{v^2}$).

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

- rms speed** at a temperature; temperature at which two gases have equal speeds.
- Speed ratios** (rms/avg/most probable) and scaling with T , M .
- Degrees of freedom** $\rightarrow C_v, C_p, \gamma$, including mixtures.
- Pressure-KE relation**: $E = \frac{3}{2}PV$.
- Mean free path** dependence on P , T , d .
- Maxwell distribution curve**: how the peak shifts with T (the area stays constant).

Shortcuts

SHORTCUT Equal speeds of two gases

Speeds are equal when T/M is equal: $\frac{T_1}{M_1} = \frac{T_2}{M_2}$. No square roots needed.

SHORTCUT γ of a mixture

$\gamma_{mix} = \frac{\sum n_i C_{p,i}}{\sum n_i C_{v,i}}$. Never average the γ values directly.

Question classification

Type	Difficulty	Target time
rms speed / temperature scaling	E	1 min
$f \rightarrow \gamma, C_v$	E	45 s
Mixture γ	M	1.5 min
Mean free path	E	1 min
KE-pressure relation	E	1 min

Common mistakes

MISTAKE ALERT Mistake alerts

- Using M in g/mol inside $\sqrt{3RT/M}$. Convert to **kg/mol**.
- Using $^{\circ}\text{C}$ instead of K.
- Averaging γ values for mixtures.
- Forgetting that vibration adds 2 (not 1) to f per mode.
- Thinking $\frac{3}{2}k_B T$ depends on the gas's molar mass. It doesn't.

Practice questions

P09 · Practice

1 Easy 1 min Calc

P09-01

The rms speed of oxygen molecules ($M = 32$ g/mol) at 27°C is about ($R = 8.314$):

- (A) 484 m s^{-1} (B) 342 m s^{-1} (C) 1600 m s^{-1} (D) 250 m s^{-1}

2 Easy 1 min NV

P09-02

Find the temperature (in K) at which the rms speed of hydrogen equals that of oxygen at 47°C .

Your answer (numerical value):

3 Easy 0.5 min Speed

P09-03

To double the rms speed of a gas's molecules, its absolute temperature must be:

- (A) doubled (B) quadrupled (C) halved (D) increased by $\sqrt{2}$

4 Easy 0.5 min Speed

P09-04

A gas has molecules with 6 degrees of freedom. Its γ is:

- (A) $5/3$ (B) $7/5$ (C) $4/3$ (D) $9/7$

5 Medium 1.5 min Concept

P09-05

One mole of a monatomic gas is mixed with one mole of a rigid diatomic gas. γ of the mixture is:

- (A) 1.5 (B) 1.4 (C) 1.53 (D) 1.67

6 Easy 1 min Basic

P09-06

The total translational kinetic energy of the molecules of an ideal gas filling 1 m^3 at a pressure of 10^5 Pa is:

- (A) 10^5 J (B) $1.5 \times 10^5 \text{ J}$ (C) $3 \times 10^5 \text{ J}$ (D) $0.67 \times 10^5 \text{ J}$

7 Easy 0.5 min Concept

P09-07

At constant temperature, if the pressure of a gas is doubled, its mean free path:

- (A) doubles (B) halves (C) is unchanged (D) becomes 4 times

8 Easy 1 min NV

P09-08

The average translational kinetic energy of a gas molecule at 300 K is $x \times 10^{-21} \text{ J}$. Find x to the nearest integer ($k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$).

Your answer (numerical value):

P09 · Chapter Test

1 Easy 0.5 min Speed

P09-T1

The ratio $v_{rms} : v_{avg} : v_{mp}$ is:

- (A) $\sqrt{2} : \sqrt{8/\pi} : \sqrt{3}$ (B) $\sqrt{3} : \sqrt{8/\pi} : \sqrt{2}$ (C) $\sqrt{3} : \sqrt{2} : \sqrt{8/\pi}$ (D) $1 : 1 : 1$

2 Easy 0.5 min Basic

P09-T2

The internal energy of 2 mol of a rigid diatomic ideal gas at temperature T is:

- (A) $3RT$ (B) $5RT$ (C) $7RT$ (D) $\frac{5}{2}RT$

3 Easy 0.75 min Basic

P09-T3

At the same temperature, the ratio of rms speeds of H_2 and O_2 molecules is:

- (A) $1 : 4$ (B) $4 : 1$ (C) $16 : 1$ (D) $1 : 16$

4 Easy 0.5 min NV

P09-T4

How many degrees of freedom (translational + rotational) does a rigid **nonlinear** triatomic molecule have?

Your answer (numerical value):

5 Easy 0.75 min Basic

P09-T5

The pressure of a gas at constant volume doubles when it is heated from 27°C to:

- (A) 54°C (B) 327°C (C) 600°C (D) 273°C

Answers & Solutions

P09 · Practice

1 (A) 2 20 3 (B) 4 (C) 5 (A) 6 (B) 7 (B) 8 6

P09 · Chapter Test

1 (B) 2 (B) 3 (B) 4 6 5 (B)

P09 · PRACTICE

1 Answer: (A) rms speed Easy 1 min Calc

P09-01

SOLUTION $v_{rms} = \sqrt{\frac{3 \times 8.314 \times 300}{0.032}} = \sqrt{2.34 \times 10^5} \approx 484 \text{ m s}^{-1}$.

SHORTCUT $3RT \approx 7483$; $7483/0.032 \approx 233\,800$; $\sqrt{} \approx 484$.

COMMON TRAP Using $M = 32$ (g) gives about 15 m/s, which isn't among the options, so re-check the units.

2 Answer: 20 Equal rms speeds Easy 1 min NV

P09-02

SOLUTION Equal speeds need T/M equal: $\frac{T_{H_2}}{2} = \frac{320}{32} \Rightarrow T_{H_2} = 20 \text{ K}$.

SHORTCUT $T \propto M$ for equal speeds.

COMMON TRAP Using 47 instead of 320 K.

3 Answer: (B) Speed-temperature scaling Easy 0.5 min Speed

P09-03

SOLUTION $v_{rms} \propto \sqrt{T}$, so $T \rightarrow 4T$.

COMMON TRAP Doubling it.

4 Answer: (C) γ from degrees of freedom Easy 0.5 min Speed

P09-04

SOLUTION $\gamma = 1 + 2/f = 1 + 1/3 = 4/3$.

5 Answer: (A) γ of a mixture **Medium** ⌚ 1.5 min **Concept**

P09-05

SOLUTION $C_v = \frac{1.5R + 2.5R}{2} = 2R$, $C_p = C_v + R = 3R$, so $\gamma = 1.5$.

SHORTCUT Average C_v , then add R .

COMMON TRAP Averaging the γ values: $(1.67 + 1.4)/2 \approx 1.53$.

6 Answer: (B) Translational KE from pressure **Easy** ⌚ 1 min **Basic**

P09-06

SOLUTION $E = \frac{3}{2}PV = 1.5 \times 10^5 \text{ J}$.

SHORTCUT $P = \frac{2}{3}(E/V)$.

COMMON TRAP Using $P = \frac{1}{3}(E/V)$.

7 Answer: (B) Mean free path vs pressure **Easy** ⌚ 0.5 min **Concept**

P09-07

SOLUTION $\lambda = \frac{k_B T}{\sqrt{2} \pi d^2 P} \propto 1/P$.

SHORTCUT More molecules per volume means more frequent collisions.

8 Answer: 6 Average KE per molecule **Easy** ⌚ 1 min **NV**

P09-08

SOLUTION $\frac{3}{2}k_B T = 1.5 \times 1.38 \times 10^{-23} \times 300 = 6.21 \times 10^{-21} \text{ J}$, so $x \approx 6$.

SHORTCUT Worth memorising: about $6.2 \times 10^{-21} \text{ J}$ at room temperature ($\approx 0.04 \text{ eV}$).

COMMON TRAP Using $\frac{1}{2}k_B T$ per molecule.

P09 · CHAPTER TEST

1 Answer: (B) Speed ratio **Easy** ⌚ 0.5 min **Speed**

P09-T1

SOLUTION From the formulas: $\sqrt{3}$, $\sqrt{8/\pi} \approx 1.60$, $\sqrt{2}$.

SHORTCUT rms > avg > mp.

2 Answer: (B) Internal energy **Easy** ⌚ 0.5 min **Basic**

P09-T2

SOLUTION $U = n \frac{f}{2} RT = 2 \times \frac{5}{2} RT = 5RT$.

COMMON TRAP Forgetting $n = 2$.

3 Answer: (B) rms speed ratio of two gases **Easy** ⌚ 0.75 min **Basic**

P09-T3

SOLUTION $v \propto 1/\sqrt{M}$: $\sqrt{32/2} = 4$.

COMMON TRAP Inverting the ratio.

4 Answer: 6 Degrees of freedom **Easy** ⌚ 0.5 min **NV**

P09-T4

SOLUTION 3 translational + 3 rotational = 6.

SHORTCUT Linear molecules have 2 rotational degrees of freedom; nonlinear ones have 3.

COMMON TRAP Answering 5 (as for a linear molecule).

5 Answer: (B) Gay-Lussac's law **Easy** ⌚ 0.75 min **Basic**

P09-T5

SOLUTION $P \propto T$: $T_2 = 2 \times 300 = 600 \text{ K} = 327^\circ \text{C}$.

SHORTCUT Double the kelvin temperature, then convert back.

COMMON TRAP Doubling the Celsius value.

REVISION Revision tracker · Kinetic Theory

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 10

Oscillations & Waves

10

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

High

PRIORITY

Class 11 · Ch 13, 14

NCERT

~12 hours

STUDY TIME

Concept summary

- **SHM** is motion under a restoring force proportional to displacement: $a = -\omega^2 x$. Its solution is a sine or cosine, and its energy oscillates between KE and PE.
- To find the period of **any** SHM system, write the restoring force or torque as $-(\text{const}) \times \text{displacement}$ and read off ω .
- A **wave** carries energy, not matter. Its speed depends on the medium: tension/density for strings, elasticity/density for sound.
- **Superposition** gives interference, standing waves (strings, pipes) and beats.

WARNING Syllabus note

The official Unit 10 text (2024–2026) lists waves up to "**Standing waves in strings and organ pipes, fundamental mode and harmonics. Beats.**" The **Doppler effect is not listed**, so treat it as low priority [NTA-SYL].

Formulas: Oscillations

Σ FORMULA SHM kinematics & energy

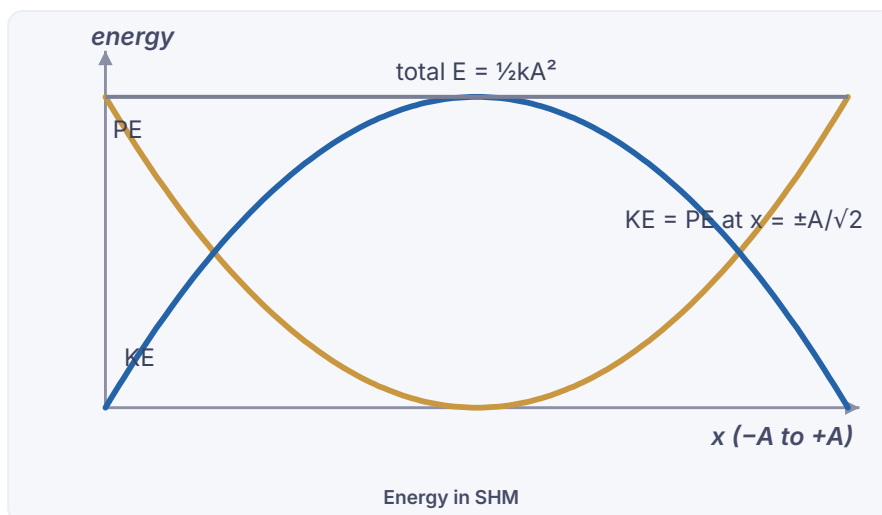
$$x = A \sin(\omega t + \phi) \quad v = \omega \sqrt{A^2 - x^2} \quad a = -\omega^2 x \quad v_{\max} = A\omega \quad a_{\max} = A\omega^2$$

$$K = \frac{1}{2}m\omega^2(A^2 - x^2) \quad U = \frac{1}{2}m\omega^2 x^2 \quad E = \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}kA^2$$

- $K = U$ at $x = A/\sqrt{2}$. KE and PE each oscillate at frequency $2f$.
- Phase difference: v leads x by $\pi/2$; a leads x by π .

Σ FORMULA Time periods of standard systems

System	Period
Spring-mass	$T = 2\pi\sqrt{m/k}$ (horizontal or vertical; the same)
Springs in series / parallel	$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2}$; $k_p = k_1 + k_2$
Spring cut into n equal parts	each part has nk
Simple pendulum (small angle)	$T = 2\pi\sqrt{L/g}$
Pendulum in a lift accelerating up / down at a	$2\pi\sqrt{L/(g \pm a)}$
Pendulum in a car accelerating horizontally at a	$2\pi\sqrt{L/\sqrt{g^2 + a^2}}$
Physical (compound) pendulum	$2\pi\sqrt{I/(mgd)}$, d = pivot-to-COM distance
Liquid in a U-tube (total column length L)	$2\pi\sqrt{L/2g}$
Floating cylinder (submerged length h)	$2\pi\sqrt{h/g}$



Formulas: Waves

Σ FORMULA Wave speed & equation

$$v = f\lambda = \frac{\omega}{k} \quad \text{string: } v = \sqrt{T/\mu} \quad \text{sound: } v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}} \text{ (Laplace)}$$

$$y = A \sin(kx - \omega t) \text{ (travelling in } +x) \quad k = \frac{2\pi}{\lambda} \quad \omega = 2\pi f \quad v_{\text{particle}} = \frac{\partial y}{\partial t} = -v \frac{\partial y}{\partial x}$$

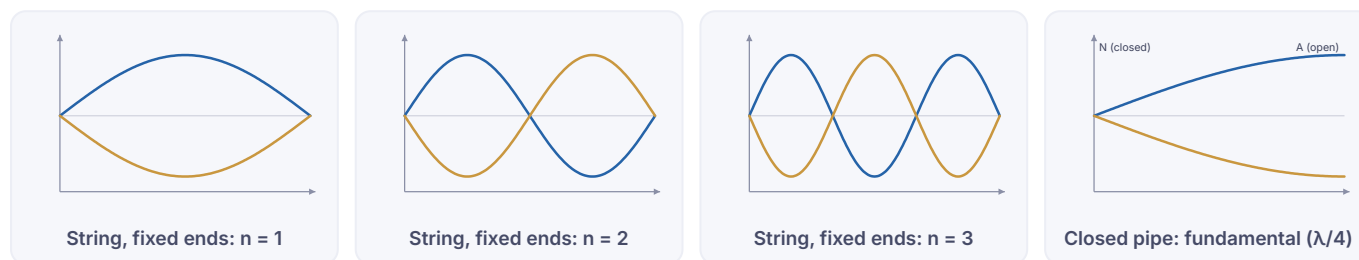
- Speed of sound $\propto \sqrt{T}$ (kelvin), independent of pressure at constant T . Newton's formula $\sqrt{P/\rho}$ was 15% too low.
- Intensity $\propto A^2 \omega^2$. Phase difference $\Delta\phi = \frac{2\pi}{\lambda} \Delta x$.
- Reflection from a fixed end: phase change π . From a free end: no phase change.

Σ FORMULA Superposition, standing waves, beats

Interference: $A_R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$; $I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$. **Standing wave** $y = 2A \sin kx \cos \omega t$: nodes $\lambda/2$ apart, node-to-antinode $\lambda/4$.

System	Allowed frequencies	Harmonics present
String fixed at both ends	$f_n = \frac{nv}{2L}$	all ($n = 1, 2, 3, \dots$)
Open pipe (both ends open)	$f_n = \frac{nv}{2L}$	all
Closed pipe (one end closed)	$f_n = \frac{(2n-1)v}{4L}$	odd only (1, 3, 5, ...)

End correction $e \approx 0.6r$: closed pipe $L \rightarrow L + e$; open pipe $L \rightarrow L + 2e$. **Resonance tube:** $\lambda = 2(L_2 - L_1)$, end correction $e = \frac{L_2 - 3L_1}{2}$. **Beats:** $f_{\text{beat}} = |f_1 - f_2|$. Loading a fork with wax **lowers** its frequency; filing its prongs **raises** it.



Nodes (N) where the envelope pinches; antinodes (A) at maximum amplitude.

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **Period of a system:** springs in combination, pendulum in an accelerating frame, liquid columns.
2. **SHM energy:** where $K = U$; speed at a given displacement; phase from given x and v .
3. **Wave equation reading:** find v , λ , f , direction and particle velocity from $y(x, t)$.
4. **Organ pipes and strings:** harmonics, overtones, open vs closed comparisons, resonance tube.
5. **Beats with wax/filing** to determine an unknown frequency.
6. **Superposition of two SHMs** in the same direction (resultant amplitude and phase).

Shortcuts

SHORTCUT Overtone vs harmonic

For a closed pipe, the **first overtone is the 3rd harmonic**. For open pipes and strings, the first overtone is the 2nd harmonic. Translate the question's word before computing.

SHORTCUT Period by "restoring constant"

Write $F = -k_{eff}x$ (or $\tau = -\kappa\theta$). Then $T = 2\pi\sqrt{m/k_{eff}}$ (or $2\pi\sqrt{I/\kappa}$). Works for liquids, floating bodies and springs with pulleys.

Question classification

Type	Difficulty	Target time
SHM formulas (v_{max} , a_{max} , energy)	E	1 min
Spring combinations / pendulum variants	E-M	1.5 min
Wave-equation reading	E	1 min
Pipes and strings harmonics	M	2 min
Beats with loading	M	1.5 min
Composite SHM (pulley + spring, rolling + spring)	H	3 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Closed pipe: only **odd** harmonics. Asked for the "second harmonic of a closed pipe"? It doesn't exist.
- Confusing μ (mass per unit **length**) with density.
- Using $^{\circ}\text{C}$ in $v \propto \sqrt{T}$.
- Wax lowers frequency. After deciding the sign of the change, check which way the beats moved.
- The spring-mass period is independent of g . The pendulum period is independent of mass.

Practice questions

P10 · Practice

1 Easy 0.75 min Basic

P10-01

A particle performs SHM with amplitude 10 cm and period 2 s. Its maximum speed is:

- (A) $0.1\pi \text{ m s}^{-1}$ (B) $0.2\pi \text{ m s}^{-1}$ (C) $0.05\pi \text{ m s}^{-1}$ (D) $\pi \text{ m s}^{-1}$

2 Easy 0.5 min Speed

P10-02

In SHM of amplitude A , kinetic energy equals potential energy at a displacement of:

- (A) $A/2$ (B) $A/\sqrt{2}$ (C) $A/4$ (D) $\sqrt{3}A/2$

3 Medium 1 min Basic

P10-03

A mass m hangs from two springs of constants k and $2k$ joined in series. The period of oscillation is:

- (A) $2\pi\sqrt{3m/2k}$ (B) $2\pi\sqrt{m/3k}$ (C) $2\pi\sqrt{2m/3k}$ (D) $2\pi\sqrt{3m/k}$

4 Medium 1 min Concept

P10-04

A simple pendulum has period T . In a lift accelerating **upward** at $g/3$, its period becomes:

- (A) $\frac{\sqrt{3}}{2}T$ (B) $\frac{2}{\sqrt{3}}T$ (C) $\frac{3}{4}T$ (D) T

5 Easy 0.75 min NV

P10-05

A string with linear mass density 0.01 kg m^{-1} is under 100 N tension. Find the transverse wave speed (in m/s).

Your answer (numerical value):

6 Medium 1.5 min Concept

P10-06

A pipe closed at one end is 25 cm long, and the speed of sound is 340 m/s (ignore end correction). The frequency of its **first overtone** is:

- (A) 680 Hz (B) 1020 Hz (C) 340 Hz (D) 1360 Hz

7 Medium 1.5 min Tricky

P10-07

Fork A (256 Hz) and fork B give 4 beats per second. When B is loaded with a little wax, the beat frequency **decreases**. The original frequency of B was:

- (A) 252 Hz (B) 260 Hz (C) 256 Hz (D) 248 Hz

8 Easy 1 min NV

P10-08

A wave is described by $y = 0.02 \sin[2\pi(10t - 0.5x)]$ (SI units). Find its speed (in m/s).

Your answer (numerical value):

9 Medium 1.5 min JEE

P10-09

In a resonance-tube experiment with a 500 Hz fork, the first and second resonances occur at 16.0 cm and 50.0 cm. The speed of sound is:

- (A) 330 m/s (B) 340 m/s (C) 350 m/s (D) 320 m/s

10 Easy 1 min Basic

P10-10

A 1 m string fixed at both ends vibrates in 3 loops at 300 Hz. The wave speed on the string is:

- (A) 100 m/s (B) 200 m/s (C) 300 m/s (D) 600 m/s

P10 · Chapter Test

1 Easy 0.5 min Speed

P10-T1

The maximum acceleration of a particle in SHM with $A = 5$ cm and $\omega = 10$ rad/s is:

- (A) 0.5 m s^{-2} (B) 5 m s^{-2} (C) 50 m s^{-2} (D) 0.05 m s^{-2}

2 Easy 0.5 min Speed

P10-T2

If the absolute temperature of air is made 4 times larger, the speed of sound in it becomes:

- (A) 4 times (B) 2 times (C) 16 times (D) unchanged

3 Medium 1 min Concept

P10-T3

A mass on a spring has period T . The spring is cut into two equal halves and the same mass is hung from one half. The new period is:

- (A) $T/2$ (B) $T/\sqrt{2}$ (C) $\sqrt{2}T$ (D) $2T$

4 Easy 0.5 min NV

P10-T4

Two sources of 510 Hz and 514 Hz sound together. Find the number of beats heard per second.

Your answer (numerical value):

5 Easy 0.75 min Basic

P10-T5

An open pipe and a closed pipe have the same length. The ratio of their fundamental frequencies (open : closed) is:

- (A) 1 : 2 (B) 2 : 1 (C) 1 : 1 (D) 4 : 1

Answers & Solutions

P10 · Practice

1 (A) 2 (B) 3 (A) 4 (A) 5 100 6 (B) 7 (B) 8 20 9 (B) 10 (B)

P10 · Chapter Test

1 (B) 2 (B) 3 (B) 4 4 5 (B)

P10 · PRACTICE

1 Answer: (A) Maximum speed in SHM Easy 0.75 min Basic

P10-01

SOLUTION $\omega = 2\pi/T = \pi \text{ rad/s}$; $v_{\max} = A\omega = 0.1\pi \approx 0.314 \text{ m s}^{-1}$.

COMMON TRAP Forgetting to convert cm to m.

2 Answer: (B) Equal kinetic and potential energy Easy 0.5 min Speed

P10-02

SOLUTION $\frac{1}{2}k(A^2 - x^2) = \frac{1}{2}kx^2 \Rightarrow x = A/\sqrt{2}$.

COMMON TRAP Answering $A/2$.

3 Answer: (A) Springs in series Medium 1 min Basic

P10-03

SOLUTION $k_s = \frac{k \cdot 2k}{3k} = \frac{2k}{3}$; $T = 2\pi\sqrt{m/k_s} = 2\pi\sqrt{3m/2k}$.

SHORTCUT Series springs behave like parallel resistors.

COMMON TRAP Adding them ($3k$), which is the parallel case.

4 Answer: (A) Pendulum in accelerating lift **Medium** ⌚ 1 min **Concept**

P10-04

SOLUTION $g_{eff} = g + g/3 = \frac{4}{3}g$. $T' = T\sqrt{g/g_{eff}} = T\sqrt{3/4} = \frac{\sqrt{3}}{2}T$.

SHORTCUT Upward acceleration raises g_{eff} , so the period gets shorter.

COMMON TRAP Using $g - a$.

5 Answer: 100 Wave speed on a string **Easy** ⌚ 0.75 min **NV**

P10-05

SOLUTION $v = \sqrt{T/\mu} = \sqrt{10^4} = 100 \text{ m s}^{-1}$.

6 Answer: (B) Overtone of a closed pipe **Medium** ⌚ 1.5 min **Concept**

P10-06

SOLUTION Fundamental = $\frac{v}{4L} = \frac{340}{1} = 340 \text{ Hz}$. The first overtone of a closed pipe is the 3rd harmonic: 1020 Hz.

SHORTCUT Closed pipe: 1, 3, 5, ... × fundamental.

COMMON TRAP Taking the 2nd harmonic (680 Hz).

7 Answer: (B) Beats with wax loading **Medium** ⌚ 1.5 min **Tricky**

P10-07

SOLUTION B is 252 or 260 Hz. Wax lowers B's frequency. If B were 252, lowering it would *increase* the beats. The beats decreased, so B = 260 Hz.

SHORTCUT Wax pulls f down. Fewer beats means f_B was above f_A .

COMMON TRAP Assuming wax raises the frequency.

8 Answer: 20 Speed from wave equation **Easy** ⌚ 1 min **NV**

P10-08

SOLUTION $\omega = 20\pi$, $k = \pi$, so $v = \omega/k = 20 \text{ m s}^{-1}$ (towards $+x$).

SHORTCUT $v = \text{coefficient of } t \div \text{coefficient of } x = 10/0.5$.

COMMON TRAP Reading $\lambda = 0.5 \text{ m}$. It is actually $1/0.5 = 2 \text{ m}$, so the wrong reading gives 5 m/s.

9 Answer: (B) Resonance tube **Medium** ⌚ 1.5 min **JEE**

P10-09

SOLUTION $\lambda/2 = L_2 - L_1 = 34 \text{ cm}$, so $\lambda = 0.68 \text{ m}$ and $v = f\lambda = 340 \text{ m s}^{-1}$. (End correction = $(L_2 - 3L_1)/2 = 1 \text{ cm}$.)

SHORTCUT The difference between successive resonances is always $\lambda/2$, so the end correction cancels.

COMMON TRAP Using $4L_1$ as λ gives 320 m/s.

10 Answer: (B) Standing wave on a string **Easy** ⌚ 1 min **Basic**

P10-10

SOLUTION 3 loops means $3\lambda/2 = 1 \text{ m}$, so $\lambda = 2/3 \text{ m}$. $v = 300 \times \frac{2}{3} = 200 \text{ m s}^{-1}$.

SHORTCUT $f_1 = 100 \text{ Hz} = v/2L$, so $v = 200$.

COMMON TRAP Taking $\lambda = L/3$.

P10 · CHAPTER TEST

1 Answer: (B) Maximum acceleration **Easy** ⌚ 0.5 min **Speed**

P10-T1

SOLUTION $a_{max} = \omega^2 A = 100 \times 0.05 = 5 \text{ m s}^{-2}$.

2 Answer: (B) Sound speed vs temperature **Easy** ⌚ 0.5 min **Speed**

P10-T2

SOLUTION $v \propto \sqrt{T}$.

3 Answer: (B) Cutting a spring **Medium** ⌚ 1 min **Concept**

P10-T3

SOLUTION Each half has spring constant $2k$, so $T' = 2\pi\sqrt{m/2k} = T/\sqrt{2}$.

SHORTCUT $k \propto 1/\text{length}$.

COMMON TRAP Thinking a shorter spring is "weaker".

4 Answer: 4 Beat frequency **Easy** ⌚ 0.5 min **NV**

P10-T4

SOLUTION $|514 - 510| = 4$.

5 Answer: (B) Open vs closed pipe **Easy** ⌚ 0.75 min **Basic**

P10-T5

SOLUTION Open: $v/2L$. Closed: $v/4L$. Ratio 2 : 1.

REVISION Revision tracker · Oscillations & Waves

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Electrostatics & Capacitors

11

2–3 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 1, 2

NCERT

~16 hours

STUDY TIME

Concept summary

- Charge is **quantised** ($q = ne$) and **conserved**. **Coulomb's law** gives the force between point charges, and superposition handles many charges.
- The **electric field** $\vec{E} = \vec{F}/q_0$ is a vector. The **potential** V is a scalar, and $\vec{E} = -\nabla V$. Scalar potentials are usually easier to add.
- Gauss's law** gives the field of symmetric distributions (line, sheet, sphere) in one line.
- Conductors** in equilibrium: $E = 0$ inside, all charge on the surface, the surface is an equipotential, and the field just outside is σ/ϵ_0 ($\perp$ to the surface).
- Capacitors** store energy $\frac{1}{2}CV^2$. Dielectrics increase C by a factor K .

Formulas

Σ FORMULA Coulomb, field, potential

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \quad k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2} \quad E = \frac{kq}{r^2} \quad V = \frac{kq}{r}$$

$$\vec{E} = -\frac{dV}{dr} \hat{r} \quad V_A - V_B = -\int_B^A \vec{E} \cdot d\vec{l} \quad U_{12} = \frac{kq_1 q_2}{r} \quad W_{ext} = q(V_f - V_i)$$

In a medium of dielectric constant K : F and E are reduced by a factor K .

Σ FORMULA Electric dipole ($\vec{p} = q\vec{d}$, from – to +)

Quantity	Axial point (distance $r \gg d$)	Equatorial point
Field	$\frac{2kp}{r^3}$ (along $\vec{p}$)	$\frac{kp}{r^3}$ (opposite to $\vec{p}$)
Potential	$\frac{kp}{r^2}$	0

General point: $V = \frac{kp \cos \theta}{r^2}$, $E = \frac{kp}{r^3} \sqrt{1 + 3 \cos^2 \theta}$. In a uniform field: torque $\vec{\tau} = \vec{p} \times \vec{E}$, energy $U = -\vec{p} \cdot \vec{E}$, and net force zero. Work to rotate from θ_1 to θ_2 is $pE(\cos \theta_1 - \cos \theta_2)$. Stable equilibrium at $\theta = 0$, unstable at $\theta = 180^\circ$.

Σ FORMULA Gauss's law results

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

Distribution	Field
Infinite line charge (λ)	$\frac{\lambda}{2\pi\epsilon_0 r}$
Infinite plane sheet (σ)	$\frac{\sigma}{2\epsilon_0}$ (independent of distance)
Near a charged conductor's surface	$\frac{\sigma}{\epsilon_0}$
Thin spherical shell (Q, R)	0 inside; $\frac{kQ}{r^2}$ outside
Uniformly charged solid sphere	$\frac{kQr}{R^3}$ inside; $\frac{kQ}{r^2}$ outside

Flux through a closed surface depends **only** on the enclosed charge. A charge q at the centre of a cube gives flux $q/6\epsilon_0$ through each face. At a corner of the cube, the flux through the whole cube is $q/8\epsilon_0$.

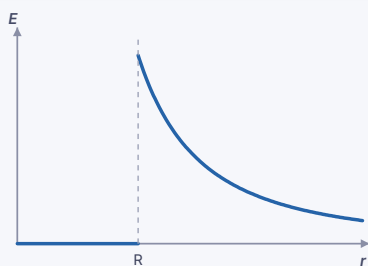
Σ FORMULA Capacitors

$$C = \frac{Q}{V} \quad \text{parallel plate: } C = \frac{\epsilon_0 A}{d} \quad \text{with dielectric filling the gap: } C = \frac{K\epsilon_0 A}{d}$$

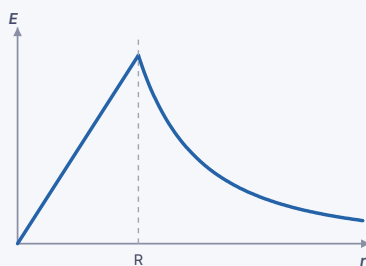
$$\text{slab of thickness } t < d: C = \frac{\epsilon_0 A}{d - t + t/K} \quad \text{metal slab: } C = \frac{\epsilon_0 A}{d - t}$$

$$\text{isolated sphere: } C = 4\pi\epsilon_0 R \quad U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV \quad u = \frac{1}{2}\epsilon_0 E^2$$

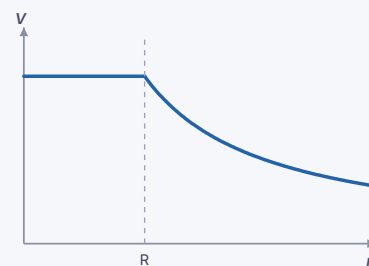
Series: $\frac{1}{C} = \sum \frac{1}{C_i}$ (same Q). **Parallel:** $C = \sum C_i$ (same V). **Sharing charge** between two capacitors: common $V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$; energy lost = $\frac{C_1 C_2 (V_1 - V_2)^2}{2(C_1 + C_2)}$. **Inserting a dielectric:** battery connected $\rightarrow V$ fixed, so Q, C, U all rise by K . Battery disconnected $\rightarrow Q$ fixed, so V and U fall by K . Force between the plates: $F = \frac{Q^2}{2\epsilon_0 A}$.



Conducting / hollow sphere



Uniformly charged solid sphere



Potential of a conducting sphere

☆ IMPORTANT Conductors vs insulators in one table

	Conductor (charged, isolated)	Insulator (uniform volume charge)
Charge location	outer surface only	throughout the volume
E inside	0	grows $\propto r$ (sphere)
V inside	constant = surface value	maximum at the centre, $\frac{3}{2}kQ/R$
Sharp points	high σ , high E (corona discharge)	—

Standard models & assumptions

- Point charges; infinite sheets and lines as idealisations; fringing ignored for parallel-plate capacitors.
- "Earthed" means $V = 0$, not $Q = 0$.

- The potential at infinity is zero.

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **Superposition:** net force or field at a point due to charges at the corners of a triangle or square; null points on a line.
2. **Gauss's law / flux:** charge inside a cube, flux through a face; field of a sphere at $r < R$ vs $r > R$.
3. **Dipoles:** torque, work to rotate, axial vs equatorial ratios.
4. **Capacitor networks:** equivalent C , charge on one capacitor, energy stored.
5. **Dielectric insertion** with and without a battery; partially filled capacitors.
6. **Potential energy of a system of charges** (work to assemble).
7. **Motion of a charge in a uniform field** (projectile analogue: $a = qE/m$).

Shortcuts

SHORTCUT Symmetry kills vectors

Equal charges at the vertices of a regular polygon give $E = 0$ at the centre, but $V = nkq/r \neq 0$. Use symmetry to cancel field components before writing any vector sum. **Time saved:** 1–2 minutes on "square of charges" questions.

SHORTCUT Dielectric slab as series capacitors

A slab of thickness t in a gap d acts as an air capacitor $(d - t)$ in series with a dielectric capacitor (t, K) . Slabs placed side by side (each covering part of the area) act in **parallel**.

Question classification

Type	Difficulty	Target time
Coulomb / field superposition	E–M	1.5 min
Gauss flux (cube/face)	E	45 s
Dipole torque/energy	E	1 min
Capacitor networks	M	2 min
Dielectric with/without battery	M	1.5 min
System energy / work to assemble	M	2 min
Non-uniform charge density + Gauss	H	3 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Adding fields as scalars. Resolve into components first.
- Sheet field: $\sigma/2\epsilon_0$ for an isolated sheet but σ/ϵ_0 near a conductor's surface.
- Dielectric questions: always check whether the **battery stays connected**.
- Potential energy of a system: count **each pair once**, $\frac{n(n-1)}{2}$ pairs.
- Forgetting the sign of charges in potential sums.

Practice questions

P11 · Practice

1 Easy 1 min Basic

P11-01

Two point charges repel with force F . If each charge is doubled and the distance between them is also doubled, the force becomes:

- (A) F (B) $2F$ (C) $F/2$ (D) $4F$

2 Easy 0.75 min Speed

P11-02

A charge q sits at the centre of a cube. The electric flux through **one face** of the cube is:

- (A) q/ε_0 (B) $q/6\varepsilon_0$ (C) $q/8\varepsilon_0$ (D) $6q/\varepsilon_0$

3 Medium 1.5 min Concept

P11-03

A dipole of moment p is aligned with a uniform field E . The work needed to rotate it by 180° is:

- (A) 0 (B) pE (C) $2pE$ (D) $-2pE$

4 Medium 2 min JEE

P11-04

Three capacitors of $2\ \mu\text{F}$, $3\ \mu\text{F}$ and $6\ \mu\text{F}$ are in series across a 12 V battery. The charge on the $3\ \mu\text{F}$ capacitor and the energy it stores are:

- (A) $12\ \mu\text{C}$, $24\ \mu\text{J}$ (B) $12\ \mu\text{C}$, $48\ \mu\text{J}$ (C) $24\ \mu\text{C}$, $24\ \mu\text{J}$ (D) $6\ \mu\text{C}$, $6\ \mu\text{J}$

5 Medium 1.5 min Concept

P11-05

A charged parallel-plate capacitor is **disconnected** from its battery, and a dielectric slab ($K = 4$) is then inserted to fill the gap. The stored energy:

- (A) becomes 4 times (B) becomes $1/4$ (C) stays the same (D) becomes 16 times

6 Medium 1.5 min Concept

P11-06

A uniformly charged **non-conducting** sphere of radius R carries total charge Q . The field at $r = R/2$ is:

- (A) 0 (B) $\frac{kQ}{2R^2}$ (C) $\frac{4kQ}{R^2}$ (D) $\frac{kQ}{4R^2}$

7 Medium 2 min NV

P11-07

Three charges of $+1\ \mu\text{C}$ each are placed at the corners of an equilateral triangle of side 0.3 m. Find the electrostatic potential energy of the system (in mJ). ($k = 9 \times 10^9\ \text{SI}$.)

Your answer (numerical value):

8 Easy 1 min NV

P11-08

A $2\ \mu\text{F}$ capacitor charged to 100 V is connected in parallel to an uncharged $3\ \mu\text{F}$ capacitor. Find the common potential (in V).

Your answer (numerical value):

9 Easy 1 min Basic

P11-09

Charges $+4q$ and $+q$ are 30 cm apart. The point where the net field is zero lies on the line joining them, at a distance from $+4q$ of:

- (A) 10 cm (B) 15 cm (C) 20 cm (D) 24 cm

P11 · Chapter Test

1 Easy 0.5 min Speed

P11-T1

Inside a charged hollow conducting sphere, the electric field is:

- (A) maximum at the centre (B) zero
(C) kQ/R^2 (D) proportional to r

2 Easy 0.75 min Basic

P11-T2

If the plate separation of a parallel-plate capacitor is halved and its plate area is doubled, its capacitance becomes:

- (A) the same (B) 2 times (C) 4 times (D) 1/4 times

3 Medium 1 min Concept

P11-T3

At a point on the equatorial line of a short dipole:

- (A) $V = 0$ and $\vec{E}$ is parallel to $\vec{p}$ (B) $V = 0$ and $\vec{E}$ is antiparallel to $\vec{p}$
(C) $V \neq 0$ and $\vec{E} = 0$ (D) V and E are both zero

4 Easy 1 min NV

P11-T4

A $10 \mu\text{F}$ capacitor is charged to 20 V. Find the stored energy (in mJ).

Your answer (numerical value):

5 Medium 1.5 min JEE

P11-T5

A parallel-plate capacitor with gap d has capacitance C_0 . A dielectric slab of $K = 2$ and thickness $d/2$ is inserted parallel to the plates. The new capacitance is:

- (A) $\frac{4}{3}C_0$ (B) $\frac{3}{2}C_0$ (C) $2C_0$ (D) $\frac{5}{4}C_0$

Answers & Solutions

P11 · Practice

1 (A) 2 (B) 3 (C) 4 (A) 5 (B) 6 (B) 7 90 8 40 9 (C)

P11 · Chapter Test

1 (B) 2 (C) 3 (B) 4 2 5 (A)

P11 · PRACTICE

1 Answer: (A) Coulomb force change Easy 1 min Basic

P11-01

SOLUTION $F \propto q_1 q_2 / r^2 = \frac{(2)(2)}{2^2} = 1$, so the force is unchanged.

SHORTCUT Numerator $\times 4$, denominator $\times 4$.

2 Answer: (B) Flux through a cube face Easy 0.75 min Speed

P11-02

SOLUTION The total flux q/ϵ_0 divides equally among 6 faces by symmetry.

COMMON TRAP Choosing the corner-charge value $q/8\epsilon_0$.

3 Answer: (C) Work to rotate a dipole **Medium** ⌚ 1.5 min **Concept**

P11-03

SOLUTION $W = pE(\cos 0^\circ - \cos 180^\circ) = 2pE$.

SHORTCUT From stable ($U = -pE$) to unstable ($U = +pE$).

COMMON TRAP Using the torque $pE \sin \theta$, which is zero at both ends.

4 Answer: (A) Capacitor network **Medium** ⌚ 2 min **JEE**

P11-04

SOLUTION $\frac{1}{C} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1 \Rightarrow C = 1 \mu\text{F}$. $Q = 12 \mu\text{C}$ on each. $U_3 = \frac{Q^2}{2C_3} = \frac{144}{6} = 24 \mu\text{J}$.

SHORTCUT In series: same Q ; $V_3 = Q/C_3 = 4 \text{ V}$, so $U = \frac{1}{2}QV = 24 \mu\text{J}$.

COMMON TRAP Using $\frac{1}{2}CV^2$ with the full 12 V.

5 Answer: (B) Dielectric with battery disconnected **Medium** ⌚ 1.5 min **Concept**

P11-05

SOLUTION Q is fixed and $C \rightarrow 4C$, so $U = \frac{Q^2}{2C}$ falls to $1/4$.

SHORTCUT Isolated: $U \propto 1/C$. Connected: $U \propto C$.

COMMON TRAP Assuming the battery stays connected ($\times 4$).

6 Answer: (B) Field of a charged solid sphere **Medium** ⌚ 1.5 min **Concept**

P11-06

SOLUTION Inside: $E = \frac{kQr}{R^3} = \frac{kQ}{2R^2}$ at $r = R/2$.

SHORTCUT Inside $E \propto r$: at half the radius, half the surface field.

COMMON TRAP Treating it as a conductor ($E = 0$).

7 Answer: 90 Potential energy of a charge system **Medium** ⌚ 2 min **NV**

P11-07

SOLUTION Each pair: $U = \frac{kq^2}{r} = \frac{9 \times 10^9 \times 10^{-12}}{0.3} = 0.03 \text{ J}$. Three pairs: $0.09 \text{ J} = 90 \text{ mJ}$.

SHORTCUT Count the 3 pairs once each.

COMMON TRAP Counting 6 ordered pairs.

8 Answer: 40 Common potential after sharing **Easy** ⌚ 1 min **NV**

P11-08

SOLUTION $V = \frac{C_1 V_1}{C_1 + C_2} = \frac{200}{5} = 40 \text{ V}$.

SHORTCUT Charge is conserved; energy is not.

COMMON TRAP Conserving energy instead of charge.

9 Answer: (C) Null point of two charges **Easy** ⌚ 1 min **Basic**

P11-09

SOLUTION $\frac{4q}{x^2} = \frac{q}{(30-x)^2} \Rightarrow 2(30-x) = x \Rightarrow x = 20 \text{ cm}$.

SHORTCUT For like charges, the null point lies between them, closer to the smaller charge.

COMMON TRAP Measuring from the smaller charge.

P11 · CHAPTER TEST

1 Answer: (B) Field inside a conductor **Easy** ⌚ 0.5 min **Speed**

P11-T1

SOLUTION In electrostatic equilibrium, the field inside a conductor (and inside its cavity, if there's no charge in it) is zero.

2 Answer: (C) Parallel-plate capacitance **Easy** ⌚ 0.75 min **Basic**

P11-T2

SOLUTION $C \propto A/d = 2/(1/2) = 4$.

3 Answer: (B) Dipole on its equatorial line **Medium** ⌚ 1 min **Concept**

P11-T3

SOLUTION On the equatorial line the potentials of $+q$ and $-q$ cancel, but their fields add to a resultant pointing opposite to $\vec{p}$.

SHORTCUT Zero potential doesn't mean zero field.

COMMON TRAP Choosing (D).

4 Answer: 2 Energy of a capacitor **Easy** ⌚ 1 min **NV**

P11-T4

SOLUTION $U = \frac{1}{2}CV^2 = \frac{1}{2} \times 10^{-5} \times 400 = 2 \times 10^{-3} \text{ J} = 2 \text{ mJ}$.

COMMON TRAP Forgetting the $\frac{1}{2}$ (4 mJ).

5 Answer: (A) Partially filled capacitor **Medium** ⌚ 1.5 min JEE

SOLUTION $C = \frac{\epsilon_0 A}{d - t + t/K} = \frac{\epsilon_0 A}{d/2 + d/4} = \frac{4}{3} \frac{\epsilon_0 A}{d} = \frac{4}{3} C_0.$

SHORTCUT Effective gap: $d - t(1 - 1/K).$

COMMON TRAP Treating the halves as parallel gives $\frac{3}{2} C_0.$

 REVISION Revision tracker · Electrostatics

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 12

Current Electricity

12

2–3 per shift

TYPICAL QUESTIONS

Easy–Medium

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 3

NCERT

~12 hours

STUDY TIME

Concept summary

- **Current** is the rate of flow of charge. In metals, free electrons **drift** slowly ($\sim 10^{-4}$ m/s) against the field, while the field itself sets up almost instantly.
- **Ohm's law** $V = IR$ holds for ohmic conductors at constant temperature. Diodes and electrolytes are non-ohmic.
- **Resistance** depends on geometry and material: $R = \rho L/A$. Resistivity rises with temperature for metals and falls for semiconductors.
- **Kirchhoff's laws** (junction = charge conservation, loop = energy conservation) solve every circuit. The Wheatstone bridge and metre bridge are special cases.

⚠ WARNING Syllabus note

The 2024–2026 syllabus text for this unit ends at "**Kirchhoff's laws and their applications. Wheatstone bridge. Metre Bridge.**" The **potentiometer** and **colour code** are not listed. Keep them to a quick read only [NTA-SYL].

Formulas

Σ FORMULA Microscopic view

$$I = \frac{dq}{dt} = neAv_d \quad v_d = \frac{eE\tau}{m} \quad \mu = \frac{v_d}{E} = \frac{e\tau}{m} \quad J = \frac{I}{A} = \sigma E \quad \rho = \frac{m}{ne^2\tau}$$

Σ FORMULA Resistance

$$R = \frac{\rho L}{A} \quad R_T = R_0(1 + \alpha \Delta T) \quad \text{Series: } R = \sum R_i \quad \text{Parallel: } \frac{1}{R} = \sum \frac{1}{R_i}$$

- **Stretching a wire** (volume constant) to n times its length: $R' = n^2 R$. If its radius becomes r/n : $R' = n^4 R$.
- n equal resistors: $R_{\text{series}}/R_{\text{parallel}} = n^2$.

Σ FORMULA Cells

$$V_{\text{terminal}} = \varepsilon - Ir \text{ (discharging)} \quad V = \varepsilon + Ir \text{ (charging)} \quad I = \frac{\varepsilon}{R + r}$$

- n identical cells in series: $n\varepsilon, nr$. In parallel: $\varepsilon, r/n$. Mixed (m rows of n cells): $I = \frac{n\varepsilon}{R + nr/m}$, maximum when $R = nr/m$.
- Two different cells in parallel: $\varepsilon_{eq} = \frac{\varepsilon_1/r_1 + \varepsilon_2/r_2}{1/r_1 + 1/r_2}$, $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$.
- **Maximum power transfer:** $R = r$ gives $P_{\text{max}} = \frac{\varepsilon^2}{4r}$ (efficiency 50%).

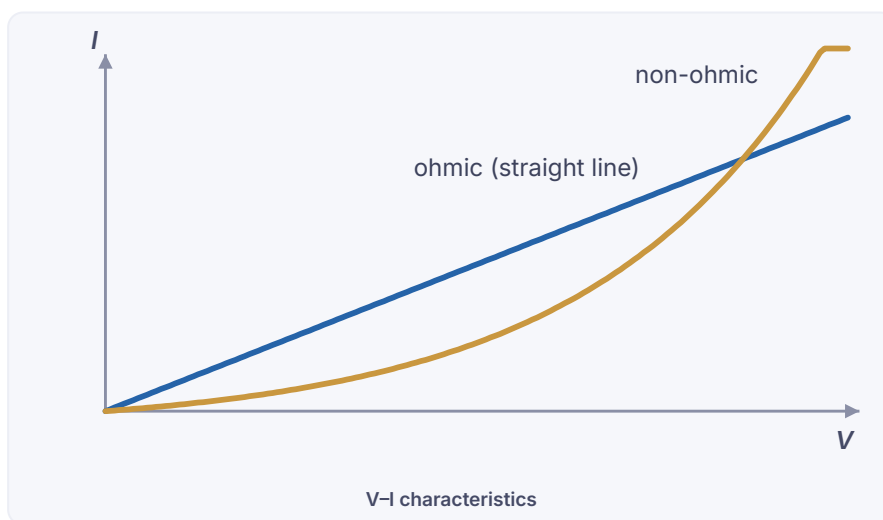
Σ FORMULA Power & bridges

$$P = VI = I^2 R = \frac{V^2}{R} \quad \text{rated bulb: } R = \frac{V_{\text{rated}}^2}{P_{\text{rated}}}$$

Wheatstone bridge balance: $\frac{P}{Q} = \frac{R}{S}$ (no current through the galvanometer; remove that arm). **Metre bridge:**

$$\frac{R}{S} = \frac{l}{100-l} \quad (l \text{ in cm from the end next to } R). \text{ Most sensitive near the middle.}$$

- Bulbs in **series**: the lower-wattage bulb (higher R) glows brighter. In **parallel**: the higher-wattage bulb glows brighter.
- Heater at a lower voltage: $P' = P(V'/V)^2$.



Kirchhoff's laws in practice

1. Mark currents in each branch with arbitrary directions.
2. **Junction rule:** $\sum I_{\text{in}} = \sum I_{\text{out}}$.
3. **Loop rule:** going around a loop, $\sum \Delta V = 0$. Across a resistor in the direction of current: $-IR$. Across a cell from $-$ to $+$: $+\epsilon$.
4. Solve. A negative current just means the real direction is opposite to the one you assumed.

⚡ SHORTCUT Nodal (potential) method

Set one node at 0 V. Write the other node potentials as unknowns, and use $\sum \frac{V - V_j}{R_j} = 0$ at each node. This usually needs fewer equations than loops. **Time saved:** 1–2 minutes on two-loop circuits.

⚡ SHORTCUT Symmetry & balanced bridges

If a network has a symmetry plane, points on it are at equal potential: join them or remove the resistors between them. Before solving, check $P/Q = R/S$.

Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **Equivalent resistance** of networks: bridges, symmetric cubes/triangles, infinite ladders.
2. **Cells**: terminal voltage, internal resistance, combinations, maximum power.
3. **Kirchhoff two-loop circuits**: current in a branch.
4. **Drift velocity / current density / mobility** numericals.
5. **Stretched wire** and temperature dependence of resistance.
6. **Metre-bridge balance** and interchange of resistances (part of the experimental skills).
7. **Rated bulbs** in series/parallel: brightness and power.

Question classification

Type	Difficulty	Target time
R from geometry / stretching	E	45 s
Series-parallel networks	E	1 min
Cells, terminal voltage, max power	E	1 min
Kirchhoff two loops	M	2 min
Bridge (balanced/unbalanced)	M	1.5 min
Infinite ladder / cube networks	M-H	2.5 min

Common mistakes

MISTAKE ALERT Mistake alerts

- Forgetting the internal resistance when finding the current.
- Using $P = I^2 R$ for bulbs in parallel when V is the common quantity. Use V^2/R .
- In the metre bridge, measuring l from the wrong end.
- Assuming drift speed equals the speed at which the signal travels.
- Sign errors in the loop rule when you cross a cell from + to -.

Practice questions

P12 · Practice

1 Medium ⌚ 1.5 min Calc P12-01

A copper wire of cross-section 1 mm^2 carries 1.6 A . With $n = 8 \times 10^{28} \text{ m}^{-3}$, the drift speed of the electrons is:

- (A) $1.25 \times 10^{-4} \text{ m s}^{-1}$ (B) $1.25 \times 10^{-3} \text{ m s}^{-1}$ (C) $2.5 \times 10^{-5} \text{ m s}^{-1}$ (D) $1.25 \times 10^{-2} \text{ m s}^{-1}$

2 Easy ⌚ 0.5 min Speed P12-02

A wire is stretched uniformly to twice its original length. Its resistance becomes:

- (A) 2 times (B) 4 times (C) $1/2$ (D) 8 times

3 Medium ⌚ 1.5 min NV P12-03

In a bridge network, $P = 1 \Omega$, $Q = 2 \Omega$ (in series on one side) and $R = 2 \Omega$, $S = 4 \Omega$ (in series on the other side). A 5Ω resistor connects the midpoints. Find the equivalent resistance (in Ω) between the two outer ends.

Your answer (numerical value):

4 Easy 1 min Basic

P12-04

A cell of emf 2 V and internal resistance $0.5\ \Omega$ drives current through a $3.5\ \Omega$ resistor. The terminal voltage is:

- (A) 2.0 V (B) 1.75 V (C) 1.5 V (D) 0.25 V

5 Easy 1 min NV

P12-05

A battery of emf 12 V and internal resistance $2\ \Omega$ is connected to a variable load. Find the maximum power (in W) the load can draw.

Your answer (numerical value):

6 Medium 2 min JEE

P12-06

Cells of 6 V and 4 V, each with internal resistance $1\ \Omega$, are connected in parallel (positive to positive) across a $2\ \Omega$ resistor. The current through the resistor is:

- (A) 1 A (B) 2 A (C) 2.5 A (D) 5 A

7 Easy 1 min Basic

P12-07

In a metre bridge, the balance point is at 40 cm from the end where a $20\ \Omega$ resistor is connected. The other resistance is:

- (A) $13.3\ \Omega$ (B) $30\ \Omega$ (C) $20\ \Omega$ (D) $50\ \Omega$

8 Medium 1 min Basic

P12-08

A wire's resistance is $10\ \Omega$ at 20°C and $12\ \Omega$ at 120°C . Taking the 20°C value as the reference, its temperature coefficient of resistance is:

- (A) $2 \times 10^{-3}\ ^\circ\text{C}^{-1}$ (B) $2 \times 10^{-2}\ ^\circ\text{C}^{-1}$ (C) $1.67 \times 10^{-3}\ ^\circ\text{C}^{-1}$ (D) $2 \times 10^{-4}\ ^\circ\text{C}^{-1}$

9 Easy 1 min Concept

P12-09

A 100 W and a 60 W bulb (both rated at 220 V) are connected in series across 220 V. Then:

- (A) the 100 W bulb glows brighter (B) the 60 W bulb glows brighter
(C) both glow equally (D) neither glows

P12 · Chapter Test

1 Easy 0.5 min Speed

P12-T1

Four identical resistors are connected first in series, then in parallel. The ratio $R_{\text{series}}/R_{\text{parallel}}$ is:

- (A) 4 (B) 8 (C) 16 (D) $1/16$

2 Easy 0.75 min Basic

P12-T2

A heater rated 1000 W at 220 V is operated at 110 V. Its power is:

- (A) 500 W (B) 250 W (C) 1000 W (D) 125 W

3 Easy 1 min NV

P12-T3

A current $I = (2 + 3t)$ A flows in a wire. Find the charge (in C) that passes in the first 2 s.

Your answer (numerical value):

4 Easy 1 min Basic

P12-T4

Five cells, each of emf 1.5 V and internal resistance $0.2\ \Omega$, are connected in series across a $4\ \Omega$ resistor. The current is:

- (A) 1.5 A (B) 1.875 A (C) 0.375 A (D) 7.5 A

5 Easy 0.5 min Concept

P12-T5

Which of the following is a non-ohmic device?

- (A) a nichrome wire at constant temperature (B) a copper wire at constant temperature
(C) a semiconductor diode (D) a manganin resistor

Answers & Solutions

P12 · Practice

1 (A) 2 (B) 3 2 4 (B) 5 18 6 (B) 7 (B) 8 (A) 9 (B)

P12 · Chapter Test

1 (C) 2 (B) 3 10 4 (A) 5 (C)

P12 · PRACTICE

1 Answer: (A) Drift velocity Medium 1.5 min Calc

P12-01

SOLUTION $v_d = \frac{I}{neA} = \frac{1.6}{8 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-6}} = \frac{1}{8 \times 10^3} = 1.25 \times 10^{-4} \text{ m s}^{-1}$.

SHORTCUT 1.6 cancels with e 's 1.6.

COMMON TRAP The $\text{mm}^2 \rightarrow \text{m}^2$ conversion.

2 Answer: (B) Stretching a wire Easy 0.5 min Speed

P12-02

SOLUTION Volume constant: A halves while L doubles, so $R \propto L/A$ rises $\times 4$.

SHORTCUT $R \propto L^2$ at constant volume.

COMMON TRAP Assuming A stays constant ($\times 2$).

3 Answer: 2 Balanced Wheatstone network Medium 1.5 min NV

P12-03

SOLUTION $P/Q = R/S = 1/2$: balanced, so the $5\ \Omega$ carries no current. $R_{eq} = (1+2) \parallel (2+4) = 3 \parallel 6 = 2\ \Omega$.

SHORTCUT Check the balance first.

COMMON TRAP Trying to include the middle resistor.

4 Answer: (B) Terminal voltage Easy 1 min Basic

P12-04

SOLUTION $I = 2/4 = 0.5\ \text{A}$; $V = \varepsilon - Ir = 2 - 0.25 = 1.75\ \text{V}$ (the same as IR).

SHORTCUT $V = \varepsilon \frac{R}{R+r}$.

COMMON TRAP Reporting the emf.

5 Answer: 18 Maximum power transfer Easy 1 min NV

P12-05

SOLUTION At $R = r$: $P_{max} = \frac{\varepsilon^2}{4r} = \frac{144}{8} = 18\ \text{W}$.

COMMON TRAP Using $\varepsilon^2/r = 72\ \text{W}$.

6 Answer: (B) Cells in parallel Medium 2 min JEE

P12-06

SOLUTION $\varepsilon_{eq} = \frac{6/1 + 4/1}{1+1} = 5\ \text{V}$; $r_{eq} = 0.5\ \Omega$. $I = \frac{5}{2+0.5} = 2\ \text{A}$.

SHORTCUT Nodal method: $\frac{V-6}{1} + \frac{V-4}{1} + \frac{V}{2} = 0 \Rightarrow V = 4\ \text{V}$, so $I = 2\ \text{A}$.

COMMON TRAP Adding the emfs (10 V).

7 Answer: (B) Metre bridge balance **Easy** ⌚ 1 min **Basic**

P12-07

SOLUTION $\frac{20}{S} = \frac{40}{60} \Rightarrow S = 30 \Omega.$

COMMON TRAP Inverting the ratio.

8 Answer: (A) Temperature coefficient **Medium** ⌚ 1 min **Basic**

P12-08

SOLUTION $\alpha = \frac{\Delta R}{R_0 \Delta T} = \frac{2}{10 \times 100} = 2 \times 10^{-3} ^\circ\text{C}^{-1}.$

COMMON TRAP Dividing by 12 gives (C).

9 Answer: (B) Rated bulbs in series **Easy** ⌚ 1 min **Concept**

P12-09

SOLUTION $R = V^2/P$, so the 60 W bulb has the larger R . In series (same I), $P = I^2 R$ is larger for it.

SHORTCUT Series: the lower rating glows brighter.

COMMON TRAP Thinking "100 W is always brighter".

P12 · CHAPTER TEST

1 Answer: (C) Series vs parallel ratio **Easy** ⌚ 0.5 min **Speed**

P12-T1

SOLUTION $4R \div (R/4) = 16.$

SHORTCUT n^2 .

2 Answer: (B) Heater at lower voltage **Easy** ⌚ 0.75 min **Basic**

P12-T2

SOLUTION $P \propto V^2$ (fixed R): $1000 \times \frac{1}{4} = 250 \text{ W}.$

COMMON TRAP Using $P \propto V$.

3 Answer: 10 Charge from variable current **Easy** ⌚ 1 min **NV**

P12-T3

SOLUTION $q = \int_0^2 (2 + 3t) dt = 4 + 6 = 10 \text{ C}.$

SHORTCUT Area under the I - t graph (a trapezium: $\frac{2+8}{2} \times 2$).

COMMON TRAP Using $I(2) \times 2 = 16 \text{ C}.$

4 Answer: (A) Cells in series **Easy** ⌚ 1 min **Basic**

P12-T4

SOLUTION $\varepsilon = 7.5 \text{ V}, r = 1 \Omega; I = 7.5/(4 + 1) = 1.5 \text{ A}.$

COMMON TRAP Ignoring the internal resistance gives 1.875 A.

5 Answer: (C) Non-ohmic devices **Easy** ⌚ 0.5 min **Concept**

P12-T5

SOLUTION A diode's I - V curve is non-linear and depends on the direction of the voltage.

REVISION Revision tracker · Current Electricity

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 13

Magnetic Effects of Current & Magnetism

13

2–3 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 4, 5

NCERT

~14 hours

STUDY TIME

Concept summary

- **Moving charges create magnetic fields** (Biot–Savart, Ampère). **Magnetic fields exert forces on moving charges** (Lorentz) and on currents.
- The magnetic force $q\vec{v} \times \vec{B}$ is always $\perp$ to the velocity. It **does no work** and changes only direction, which is why charges move on circles or helices.
- A **current loop is a magnetic dipole**, $\vec{m} = NI\vec{A}$. It feels a torque $\vec{m} \times \vec{B}$ in a uniform field. That is the working principle of the moving-coil galvanometer.
- **Magnetic materials** are dia-, para- or ferromagnetic, depending on their atomic moments and how those moments respond to a field and to temperature.

⚠ WARNING Syllabus note

Earth's magnetism (elements, dip, declination) and the tangent galvanometer are **not** in the 2024–2026 syllabus text [NTA-SYL].

Formulas

Σ FORMULA Fields of standard currents

Source	B
Finite straight wire (angles α, β at the ends, distance d)	$\frac{\mu_0 I}{4\pi d} (\sin \alpha + \sin \beta)$
Infinite straight wire	$\frac{\mu_0 I}{2\pi d}$
Semi-infinite wire, at a point level with its end	$\frac{\mu_0 I}{4\pi d}$
Centre of a circular loop (N turns)	$\frac{\mu_0 NI}{2R}$
Arc subtending angle θ at the centre	$\frac{\mu_0 I \theta}{4\pi R}$
Axis of a loop, distance x	$\frac{\mu_0 NI R^2}{2(R^2 + x^2)^{3/2}}$
Long solenoid (n turns per metre)	$\mu_0 nI$ inside; $\frac{1}{2}\mu_0 nI$ at an end
Toroid (N turns, radius r)	$\frac{\mu_0 NI}{2\pi r}$
Thick wire (radius R), inside at r	$\frac{\mu_0 I r}{2\pi R^2}$

$\mu_0 = 4\pi \times 10^{-7}$ T m/A. **Ampère's law:** $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$. Points lying on the line of a straight wire get **zero** field from it.

Σ FORMULA Forces

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \quad r = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mK}}{qB} = \frac{1}{B} \sqrt{\frac{2mV}{q}}$$

$$T = \frac{2\pi m}{qB} \text{ (independent of speed)} \quad \text{pitch of helix} = v \cos \theta \cdot T \quad \text{velocity selector: } v = \frac{E}{B}$$

$$\vec{F} = I\vec{L} \times \vec{B} \quad \frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} \text{ (parallel currents attract, antiparallel repel)}$$

A closed current loop in a **uniform** field feels zero net force.

Σ FORMULA Torque, dipoles, galvanometer

$$\vec{m} = NI\vec{A} \quad \vec{\tau} = \vec{m} \times \vec{B} \quad U = -\vec{m} \cdot \vec{B}$$

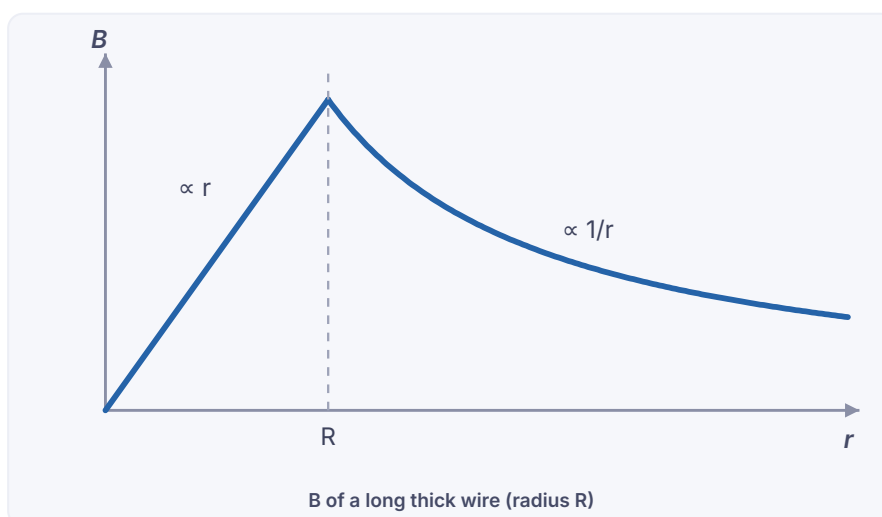
Moving-coil galvanometer: $NIAB = k\phi$. Current sensitivity $\frac{\phi}{I} = \frac{NAB}{k}$; voltage sensitivity $\frac{\phi}{V} = \frac{NAB}{kR_G}$.

Ammeter: shunt $S = \frac{I_g G}{I - I_g}$ in parallel. **Voltmeter:** series $R = \frac{V}{I_g} - G$. **Orbiting electron:** $\mu = \frac{evr}{2} = \frac{e}{2m_e} L$. Bohr magneton $\mu_B = \frac{eh}{4\pi m_e} \approx 9.27 \times 10^{-24} \text{ J T}^{-1}$. **Bar magnet (short dipole):** axial $B = \frac{\mu_0}{4\pi} \frac{2m}{r^3}$, equatorial $B = \frac{\mu_0}{4\pi} \frac{m}{r^3}$.

Σ FORMULA Magnetic materials

Type	χ	μ_r	Behaviour	Temperature	Examples
Diamagnetic	small, negative	< 1	weakly repelled, moves to weaker field	independent	Bi, Cu, water, superconductors ($\chi = -1$)
Paramagnetic	small, positive	> 1	weakly attracted	$\chi = C/T$ (Curie law)	Al, Na, O ₂
Ferromagnetic	large, positive	$\gg 1$	strongly attracted; domains	above the Curie temperature it becomes paramagnetic	Fe, Co, Ni

$B = \mu_0(H + M)$, $M = \chi H$, $\mu_r = 1 + \chi$. **Soft iron** (low coercivity, narrow hysteresis loop) suits electromagnets and transformer cores. **Steel** (high retentivity and coercivity) suits permanent magnets.



Frequently tested concepts & question patterns



PYQ PATTERN Recurring structures

1. **Field at the centre of composite shapes:** arcs + straight segments, loops of several radii, semicircles with radial leads.
2. **Charged particle in B:** radius comparisons (proton, deuteron, α), period, helix pitch, crossed E and B .
3. **Force between parallel wires;** force on a wire loop near a long wire.
4. **Galvanometer conversion** to ammeter/voltmeter.
5. **Torque on a coil** and its potential energy.
6. **Magnetic materials:** identifying types from χ or behaviour; Curie law.
7. **Solenoid/toroid fields** with Ampère's law.

Shortcuts



SHORTCUT Radius ratios for particles

Same **KE**: $r \propto \sqrt{m/q}$. Same **accelerating voltage**: $r \propto \sqrt{m/q}$. Same **momentum**: $r \propto 1/q$. Proton : deuteron : α with the same KE: $1 : \sqrt{2} : 1$.



SHORTCUT Composite-loop fields

Break the shape into arcs (use $\mu_0 I \theta / 4\pi R$) and straight pieces (radial pieces give **0**). Add with signs from the right-hand rule. That's a full answer in under a minute.

Question classification

Type	Difficulty	Target time
Standard-field formula	E	45 s
Composite loop field	M	1.5 min
Particle radius / period	E–M	1 min
Galvanometer conversion	E	1 min
Force between wires / on loops	M	1.5 min
Magnetic materials facts	E	30 s
Non-uniform B or combined E – B motion	H	3 min (Round 2)

Common mistakes



MISTAKE ALERT Mistake alerts

- Taking the magnetic force as doing work. It never changes the speed.
- For the shunt, using I instead of $(I - I_g)$ in the denominator.
- Diamagnetic susceptibility depends on temperature? **No**, it's independent.
- For a loop's field at its centre, forgetting the number of turns N .
- Using $B = \mu_0 I / 2\pi d$ for a finite wire.

Practice questions

P13 · Practice

1 Easy 0.75 min Calc P13-01

The magnetic field at the centre of a circular loop of radius 10 cm carrying 5 A is:

- (A) $\pi \times 10^{-5}$ T (B) $2\pi \times 10^{-5}$ T (C) 10^{-5} T (D) $\pi \times 10^{-6}$ T

2 Medium 1 min Concept P13-02

A proton and an α -particle with **equal kinetic energies** enter the same uniform magnetic field perpendicularly. The ratio of their radii $r_p : r_\alpha$ is:

- (A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) $1 : \sqrt{2}$

3 Easy 1 min Basic P13-03

Two long parallel wires 10 cm apart carry 10 A and 20 A in the same direction. The force per metre between them is:

- (A) 4×10^{-4} N/m, attractive (B) 4×10^{-4} N/m, repulsive
(C) 2×10^{-4} N/m, attractive (D) 8×10^{-4} N/m, attractive

4 Easy 1 min Basic P13-04

A galvanometer of resistance 50Ω gives full-scale deflection at 1 mA. The shunt needed to convert it into an ammeter of range 1 A is about:

- (A) 0.05Ω in parallel (B) 0.05Ω in series
(C) 50Ω in parallel (D) 0.5Ω in parallel

5 Easy 1 min NV P13-05

The same galvanometer (50Ω , 1 mA full scale) is to be converted into a voltmeter of range 10 V. Find the series resistance needed (in Ω).

Your answer (numerical value):

6 Easy 1 min Calc P13-06

A 0.5 m long solenoid has 500 turns and carries 2 A. The field inside is about:

- (A) 2.5×10^{-3} T (B) 1.26×10^{-3} T (C) 5.0×10^{-3} T (D) 2.5×10^{-4} T

7 Medium 1.5 min JEE P13-07

A wire runs radially inward to a point on a circle, follows a semicircular arc of radius R to the diametrically opposite point, then runs radially outward, carrying current I throughout. The field at the centre is:

- (A) $\frac{\mu_0 I}{2R}$ (B) $\frac{\mu_0 I}{4R}$ (C) $\frac{\mu_0 I}{4\pi R}$ (D) 0

8 Easy 0.5 min Concept P13-08

The time period of a charged particle moving in a circle in a uniform magnetic field:

- (A) increases with its speed (B) decreases with its speed
(C) is independent of its speed (D) is proportional to the radius

9 Easy 0.5 min Speed

P13-09

The susceptibility of a paramagnetic substance varies with absolute temperature T as:

- (A) T (B) $1/T$
(C) independent of T (D) T^2

10 Easy 1 min NV

P13-10

A 50-turn coil of area 0.02 m^2 carries 2 A in a uniform 1 T field, with the field **parallel to the plane** of the coil. Find the torque (in N m).

Your answer (numerical value):

P13 · Chapter Test

1 Easy 0.5 min Speed

P13-T1

A charge moves parallel to a uniform magnetic field. The magnetic force on it is:

- (A) qvB (B) zero
(C) $qvB/2$ (D) along the velocity

2 Medium 1 min Concept

P13-T2

A long straight wire of radius R carries current I uniformly distributed over its cross-section. The field at $r = R/2$ is:

- (A) $\frac{\mu_0 I}{4\pi R}$ (B) $\frac{\mu_0 I}{2\pi R}$ (C) $\frac{\mu_0 I}{\pi R}$ (D) 0

3 Medium 1 min Concept

P13-T3

A charge q (mass m) enters a uniform field B with speed v at 60° to the field. The pitch of its helical path is:

- (A) $\frac{\pi mv}{qB}$ (B) $\frac{2\pi mv}{qB}$ (C) $\frac{\sqrt{3}\pi mv}{qB}$ (D) $\frac{\pi mv}{2qB}$

4 Easy 0.5 min NV

P13-T4

A 100-turn coil of area 0.01 m^2 carries 5 A. Find its magnetic moment (in A m^2).

Your answer (numerical value):

5 Easy 0.5 min Speed

P13-T5

Above its Curie temperature, a ferromagnetic material becomes:

- (A) diamagnetic (B) paramagnetic
(C) a stronger ferromagnet (D) superconducting

Answers & Solutions

P13 · Practice

1 (A) 2 (A) 3 (A) 4 (A) 5 9950 6 (A) 7 (B) 8 (C) 9 (B) 10 2

P13 · Chapter Test

1 (B) 2 (A) 3 (A) 4 5 5 (B)

P13 · PRACTICE

1 Answer: (A) Field at the centre of a loop **Easy** ⌚ 0.75 min **Calc**

P13-01

SOLUTION $B = \frac{\mu_0 I}{2R} = \frac{4\pi \times 10^{-7} \times 5}{0.2} = \pi \times 10^{-5} \text{ T.}$

COMMON TRAP Using the diameter in place of R .

2 Answer: (A) Radii of proton and α particle **Medium** ⌚ 1 min **Concept**

P13-02

SOLUTION $r = \frac{\sqrt{2mK}}{qB} \propto \frac{\sqrt{m}}{q}$: proton $\sqrt{1}/1 = 1$, α $\sqrt{4}/2 = 1$.

COMMON TRAP Using the same-speed rule ($r \propto m/q$), which gives 1 : 2.

3 Answer: (A) Force between parallel wires **Easy** ⌚ 1 min **Basic**

P13-03

SOLUTION $\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} = \frac{2 \times 10^{-7} \times 200}{0.1} = 4 \times 10^{-4} \text{ N/m.}$ Like currents attract.

SHORTCUT $\frac{\mu_0}{2\pi} = 2 \times 10^{-7}$.

COMMON TRAP Saying "same direction repels", which is the opposite of electrostatics intuition.

4 Answer: (A) Galvanometer $\rightarrow$ ammeter **Easy** ⌚ 1 min **Basic**

P13-04

SOLUTION $S = \frac{I_g G}{I - I_g} = \frac{0.001 \times 50}{0.999} \approx 0.05 \Omega$, in parallel.

COMMON TRAP Putting the shunt in series.

5 Answer: 9950 Galvanometer $\rightarrow$ voltmeter **Easy** ⌚ 1 min **NV**

P13-05

SOLUTION $R = \frac{V}{I_g} - G = 10000 - 50 = 9950 \Omega$.

COMMON TRAP Forgetting to subtract G .

6 Answer: (A) Solenoid field **Easy** ⌚ 1 min **Calc**

P13-06

SOLUTION $n = 1000 \text{ m}^{-1}$; $B = \mu_0 n I = 4\pi \times 10^{-7} \times 1000 \times 2 = 8\pi \times 10^{-4} \approx 2.5 \times 10^{-3} \text{ T.}$

COMMON TRAP Using N in place of n .

7 Answer: (B) Semicircle with radial leads **Medium** ⌚ 1.5 min **JEE**

P13-07

SOLUTION The radial segments point through the centre, so they contribute nothing. The semicircle gives $\frac{\mu_0 I \pi}{4\pi R} = \frac{\mu_0 I}{4R}$.

SHORTCUT Radial leads contribute zero.

COMMON TRAP Adding contributions from the leads.

8 Answer: (C) Cyclotron frequency **Easy** ⌚ 0.5 min **Concept**

P13-08

SOLUTION $T = 2\pi m/(qB)$. A faster particle moves on a larger circle in the same time. This is the basis of the cyclotron.

9 Answer: (B) Paramagnetic susceptibility **Easy** ⌚ 0.5 min **Speed**

P13-09

SOLUTION Curie's law: $\chi = C/T$. Thermal agitation disturbs the alignment of the atomic moments.

COMMON TRAP Confusing it with diamagnetism (independent of T).

10 Answer: 2 Torque on a coil **Easy** ⌚ 1 min **NV**

P13-10

SOLUTION $\tau = NIAB \sin 90^\circ = 50 \times 2 \times 0.02 \times 1 = 2 \text{ N m.}$ The field lies in the plane, so it is perpendicular to $\vec{m}$.

SHORTCUT The torque is maximum when the field is parallel to the plane.

COMMON TRAP Using $\sin 0^\circ$ by confusing "plane" with "normal".

P13 · CHAPTER TEST

1 Answer: (B) Charge moving along B **Easy** ⌚ 0.5 min **Speed**

P13-T1

SOLUTION $\vec{v} \times \vec{B} = 0$ when they are parallel.

2 Answer: (A) Field inside a thick wire **Medium** ⌚ 1 min **Concept**

P13-T2

SOLUTION $B = \frac{\mu_0 I r}{2\pi R^2} = \frac{\mu_0 I}{4\pi R}$ at $r = R/2$.

SHORTCUT Half of the surface value $\mu_0 I / 2\pi R$.

COMMON TRAP Using the outside formula with $r = R/2$.

3 Answer: (A) Helix pitch **Medium** ⌚ 1 min **Concept**

P13-T3

SOLUTION Pitch = $v \cos 60^\circ \times \frac{2\pi m}{qB} = \frac{\pi mv}{qB}$.

SHORTCUT Parallel component $\times$ period.

COMMON TRAP Using $v \sin 60^\circ$.

4 Answer: 5 Magnetic moment of a coil **Easy** ⌚ 0.5 min **NV**

P13-T4

SOLUTION $m = NIA = 100 \times 5 \times 0.01 = 5 \text{ A m}^2$.

5 Answer: (B) Curie temperature **Easy** ⌚ 0.5 min **Speed**

P13-T5

SOLUTION Thermal agitation destroys the domain alignment, leaving paramagnetic behaviour.

 REVISION Revision tracker · Magnetism

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Electromagnetic Induction & Alternating Current

14

1–2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

High

PRIORITY

Class 12 · Ch 6, 7

NCERT

~12 hours

STUDY TIME

Concept summary

- **Faraday's law:** a changing magnetic flux induces an emf, $\varepsilon = -N \frac{d\Phi}{dt}$. The flux can change through B , the area, or the orientation.
- **Lenz's law:** the induced current opposes the *change* that produced it. It is a consequence of energy conservation.
- **Inductance** is the "inertia" of current: $L = N\Phi/I$. It stores energy $\frac{1}{2}LI^2$.
- **AC circuits:** resistors keep V and I in phase. In an inductor the current *lags* by 90° ; in a capacitor it *leads* by 90° . Impedance combines them like vectors (phasors).
- Only resistors dissipate average power. Pure L and C carry **wattless** current.

Formulas: Induction

Σ FORMULA Faraday–Lenz & motional emf

$$\Phi = \vec{B} \cdot \vec{A} = BA \cos \theta \quad \varepsilon = -N \frac{d\Phi}{dt} \quad \text{induced charge: } q = \frac{N \Delta \Phi}{R}$$

$$\text{rod moving } \perp B : \varepsilon = Blv \quad \text{rod rotating about one end: } \varepsilon = \frac{1}{2}B\omega l^2 \quad \text{coil rotating: } \varepsilon = NBA\omega \sin \omega t$$

The force needed to pull a rod at constant speed in a closed circuit of resistance R : $F = \frac{B^2 l^2 v}{R}$, with power $\frac{B^2 l^2 v^2}{R}$ (all dissipated as heat).

Σ FORMULA Inductance

$$L = \frac{N\Phi}{I} \quad \varepsilon = -L \frac{dI}{dt} \quad \text{solenoid: } L = \mu_0 n^2 Al = \frac{\mu_0 N^2 A}{l} \quad U = \frac{1}{2}LI^2$$

$$M = \mu_0 n_1 n_2 Al \quad M = k\sqrt{L_1 L_2} \quad (k \leq 1) \quad \text{series (no coupling): } L_1 + L_2 \quad \text{parallel: } \frac{L_1 L_2}{L_1 + L_2}$$

LR circuit: growth $I = I_0(1 - e^{-t/\tau})$, decay $I = I_0 e^{-t/\tau}$, with $\tau = L/R$. After one τ the current reaches 63% of its final value. Energy density of a magnetic field: $u = \frac{B^2}{2\mu_0}$.

Formulas: Alternating current

Σ FORMULA AC basics

$$V = V_0 \sin \omega t \quad V_{rms} = \frac{V_0}{\sqrt{2}} \quad V_{avg, half cycle} = \frac{2V_0}{\pi} \quad V_{avg, full cycle} = 0$$

Household mains in India: $V_{rms} = 220\text{--}230$ V, $f = 50$ Hz. AC meters read **rms** values.

Σ FORMULA Series LCR circuit

$$X_L = \omega L \quad X_C = \frac{1}{\omega C} \quad Z = \sqrt{R^2 + (X_L - X_C)^2} \quad \tan \phi = \frac{X_L - X_C}{R}$$

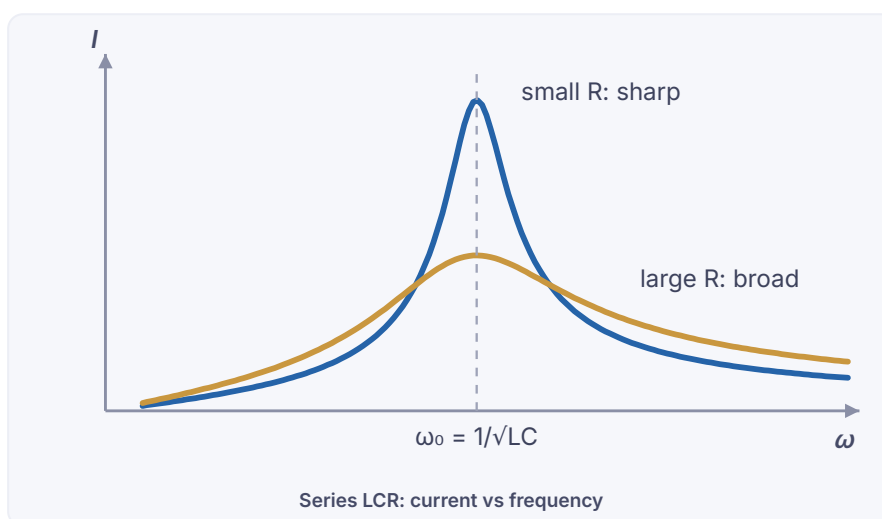
$$\text{Resonance: } \omega_0 = \frac{1}{\sqrt{LC}}, \quad Z = R, \quad I = I_{\max} \quad Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}} \quad \text{bandwidth } \Delta\omega = \frac{\omega_0}{Q} = \frac{R}{L}$$

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos \phi = I_{\text{rms}}^2 R \quad \cos \phi = \frac{R}{Z} \quad \text{wattless component} = I_{\text{rms}} \sin \phi$$

Element	Phase of I relative to V	Average power
R	in phase	$V_{\text{rms}} I_{\text{rms}}$
L	lags by 90°	0
C	leads by 90°	0

Σ FORMULA LC oscillations, generator, transformer

LC circuit: $\omega = \frac{1}{\sqrt{LC}}$; energy swaps between $\frac{q^2}{2C}$ and $\frac{1}{2}LI^2$ (analogue: $q \leftrightarrow x, L \leftrightarrow m, 1/C \leftrightarrow k$). **AC generator:** peak emf $\varepsilon_0 = NBA\omega$. **Ideal transformer:** $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$. Step-up raises V and lowers I . Losses: copper ($I^2 R$), eddy currents (reduced by laminating the core), hysteresis (reduced with soft iron), and flux leakage.

**Frequently tested concepts & question patterns****▮ PYQ PATTERN Recurring structures**

1. **Motional emf:** rods on rails, rotating rods and discs, force/power needed.
2. **Flux-change emf and induced charge** in coils ($q = N\Delta\Phi/R$, independent of time).
3. **LCR series:** impedance, phase, power factor, average power, resonance frequency, Q factor.
4. **rms/average values** from a given $V(t)$.
5. **Transformer** turns/voltage/current ratios and efficiency.
6. **LR growth/decay** and energy stored in inductors.
7. **LC oscillation** frequency and energy exchange.

Shortcuts**⚡ SHORTCUT Phasor triangle**

Draw R horizontally and $(X_L - X_C)$ vertically. The hypotenuse is Z , the angle is ϕ , and the power factor is the cosine. Use 3-4-5 triangles: numbers like $R = 30, X = 40$ are chosen deliberately. **Time saved:** about 1 minute.

⚡ **SHORTCUT** Resonance in rad/s vs Hz

$\omega_0 = 1/\sqrt{LC}$ is in **rad/s**. $f_0 = \omega_0/2\pi$. Options often include both. Check the unit asked for.

Question classification

Type	Difficulty	Target time
Motional emf / rotating rod	E	1 min
Induced charge / emf from $\Delta\Phi$	E	1 min
LCR impedance/power	M	2 min
Resonance / Q factor	E-M	1 min
Transformer	E	45 s
LR growth / energy	M	1.5 min
Rod on rails with mass (terminal velocity)	H	3 min (Round 2)

Common mistakes

⚠ **MISTAKE ALERT** Mistake alerts

- Using V_0 instead of V_{rms} in the power formula.
- Taking the full-cycle average of a sinusoid as $2V_0/\pi$. It's 0 over a full cycle.
- Missing the $\frac{1}{2}$ in the rotating-rod emf $\frac{1}{2}B\omega l^2$.
- Mixing up "lead" and "lag": **ELI the ICE man**. In L, E (voltage) leads I; in C, I leads E.
- Forgetting N (the number of turns) in Faraday's law.

Practice questions

🔗 **P14 · Practice**

1 Easy ⌚ 0.5 min Speed P14-01

A 0.5 m rod moves at 4 m s^{-1} perpendicular to a 0.2 T field (and perpendicular to its own length). The emf across its ends is:

(A) 0.1 V (B) 0.4 V (C) 0.8 V (D) 4 V

2 Easy ⌚ 1 min NV P14-02

A 100-turn coil of area 0.1 m^2 lies perpendicular to a field that falls uniformly from 0.5 T to zero in 0.1 s. Find the average induced emf (in V).

Your answer (numerical value):

3 Easy ⌚ 1 min Basic P14-03

The flux through each turn of a 100-turn coil changes by 0.05 Wb. The coil's total resistance is 10Ω . The charge that flows is:

(A) 0.005 C (B) 0.05 C (C) 0.5 C (D) 5 C

4 Easy ⌚ 0.5 min NV P14-04

Find the energy (in J) stored in a 2 H inductor carrying 3 A.

Your answer (numerical value):

5 Medium 2 min JEE

P14-05

A series LCR circuit has $R = 30\ \Omega$, $X_L = 80\ \Omega$, $X_C = 40\ \Omega$ and is connected to a 100 V (rms) source. The average power consumed is:

- (A) 200 W (B) 120 W (C) 160 W (D) 333 W

6 Medium 1 min Tricky

P14-06

In a series LCR circuit, $L = 0.1\text{ H}$ and $C = 10\ \mu\text{F}$. The resonant **angular** frequency is:

- (A) 1000 rad/s (B) 159 rad/s (C) 100 rad/s (D) 10 000 rad/s

7 Easy 0.5 min Speed

P14-07

For $V = 220\sqrt{2}\sin(100\pi t)$ volts, the rms voltage and frequency are:

- (A) 220 V, 50 Hz (B) $220\sqrt{2}$ V, 100 Hz (C) 220 V, 100 Hz (D) 311 V, 50 Hz

8 Easy 1 min Basic

P14-08

An ideal transformer has 200 primary and 1000 secondary turns. The primary is at 220 V, and the secondary delivers 0.2 A. The primary current is:

- (A) 0.04 A (B) 1 A (C) 0.2 A (D) 5 A

9 Easy 0.5 min Speed

P14-09

A 4 H inductor is in series with a $2\ \Omega$ resistor. The time constant of the circuit is:

- (A) 8 s (B) 2 s (C) 0.5 s (D) 6 s

10 Easy 1 min Basic

P14-10

A 1 m metal rod rotates at 20 rad/s about one end, in a plane perpendicular to a uniform 0.5 T field. The emf between its ends is:

- (A) 10 V (B) 5 V (C) 20 V (D) 2.5 V

P14 · Chapter Test

1 Easy 0.5 min Concept

P14-T1

Lenz's law is a consequence of the conservation of:

- (A) charge (B) momentum (C) energy (D) mass

2 Easy 0.5 min Speed

P14-T2

In a purely inductive AC circuit, the current:

- (A) leads the voltage by $\pi/2$ (B) lags the voltage by $\pi/2$
(C) is in phase with the voltage (D) lags by π

3 Easy 0.5 min Speed

P14-T3

If the frequency of an AC source is doubled, the capacitive reactance:

- (A) doubles (B) halves (C) stays the same (D) becomes 4 times

4 Easy 1 min NV

P14-T4

A 100-turn coil of area 0.01 m^2 rotates at 60 rad/s in a 0.5 T field. Find the peak emf (in V).

Your answer (numerical value):

5 Easy 0.5 min Speed

P14-T5

At resonance in a series LCR circuit, the power factor is:

- (A) 0 (B) 0.5 (C) 1 (D) $1/\sqrt{2}$

Answers & Solutions

P14 · Practice

1 (B) 2 50 3 (C) 4 9 5 (B) 6 (A) 7 (A) 8 (B) 9 (B) 10 (B)

P14 · Chapter Test

1 (C) 2 (B) 3 (B) 4 30 5 (C)

P14 · PRACTICE

1 Answer: (B) Motional emf Easy 0.5 min Speed

P14-01

SOLUTION $\varepsilon = Blv = 0.2 \times 0.5 \times 4 = 0.4 \text{ V}$.

2 Answer: 50 Faraday's law for a coil Easy 1 min NV

P14-02

SOLUTION $\varepsilon = N \frac{\Delta(BA)}{\Delta t} = 100 \times \frac{0.5 \times 0.1}{0.1} = 50 \text{ V}$.

COMMON TRAP Omitting N (0.5 V).

3 Answer: (C) Induced charge Easy 1 min Basic

P14-03

SOLUTION $q = \frac{N\Delta\Phi}{R} = \frac{100 \times 0.05}{10} = 0.5 \text{ C}$, regardless of how quickly the change happens.

SHORTCUT Charge depends on the change in flux, not its rate.

COMMON TRAP Thinking you need the time interval.

4 Answer: 9 Energy in an inductor Easy 0.5 min NV

P14-04

SOLUTION $U = \frac{1}{2}LI^2 = \frac{1}{2} \times 2 \times 9 = 9 \text{ J}$.

5 Answer: (B) LCR average power Medium 2 min JEE

P14-05

SOLUTION $Z = \sqrt{30^2 + 40^2} = 50 \Omega$; $I_{rms} = 2 \text{ A}$; $\cos \phi = 30/50 = 0.6$. $P = 100 \times 2 \times 0.6 = 120 \text{ W}$ (check: $I^2R = 4 \times 30$).

SHORTCUT $P = I_{rms}^2 R$ once you have I .

COMMON TRAP Using V^2/R (333 W).

6 Answer: (A) Resonance: rad/s vs Hz Medium 1 min Tricky

P14-06

SOLUTION $\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-6}}} = 1000 \text{ rad/s}$. (That's $f_0 \approx 159 \text{ Hz}$. The trap is option (B), with the wrong unit.)

COMMON TRAP Choosing 159 rad/s.

7 Answer: (A) Reading an AC expression Easy 0.5 min Speed

P14-07

SOLUTION $V_{rms} = V_0/\sqrt{2} = 220 \text{ V}$; $\omega = 100\pi \Rightarrow f = 50 \text{ Hz}$.

COMMON TRAP Reading $f = 100$ from 100π .

8 Answer: (B) Ideal transformer Easy 1 min Basic

P14-08

SOLUTION $\frac{I_p}{I_s} = \frac{N_s}{N_p} = 5 \Rightarrow I_p = 1 \text{ A}$. ($V_s = 1100 \text{ V}$; power $220 = 220 \text{ W}$.)

SHORTCUT Power in = power out.

COMMON TRAP Inverting the ratio (0.04 A).

9 Answer: (B) *LR time constant* **Easy** ⌚ 0.5 min **Speed**

P14-09

SOLUTION $\tau = L/R = 2 \text{ s}$.

COMMON TRAP R/L .

10 Answer: (B) *Rotating rod* **Easy** ⌚ 1 min **Basic**

P14-10

SOLUTION $\varepsilon = \frac{1}{2}B\omega l^2 = \frac{1}{2} \times 0.5 \times 20 \times 1 = 5 \text{ V}$.

COMMON TRAP Omitting the $\frac{1}{2}$.

P14 · CHAPTER TEST

1 Answer: (C) *Basis of Lenz's law* **Easy** ⌚ 0.5 min **Concept**

P14-T1

SOLUTION If the induced current aided the change, it would create energy from nothing.

2 Answer: (B) *Phase in a pure inductor* **Easy** ⌚ 0.5 min **Speed**

P14-T2

SOLUTION The inductor opposes changes in current, so the current lags behind the voltage by 90° .

SHORTCUT ELI.

3 Answer: (B) *Reactance vs frequency* **Easy** ⌚ 0.5 min **Speed**

P14-T3

SOLUTION $X_C = 1/(\omega C) \propto 1/f$.

COMMON TRAP Applying the inductive behaviour instead.

4 Answer: 30 *Generator peak emf* **Easy** ⌚ 1 min **NV**

P14-T4

SOLUTION $\varepsilon_0 = NBA\omega = 100 \times 0.5 \times 0.01 \times 60 = 30 \text{ V}$.

5 Answer: (C) *Power factor at resonance* **Easy** ⌚ 0.5 min **Speed**

P14-T5

SOLUTION $X_L = X_C$, so $Z = R$, $\phi = 0$ and $\cos \phi = 1$.

REVISION Revision tracker · EMI & AC

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 15

Electromagnetic Waves

15

~1 per shift
TYPICAL QUESTIONS

Easy
DIFFICULTY

Must-do (easy
marks)
PRIORITY

Class 12 · Ch 8
NCERT

~3 hours
STUDY TIME

Concept summary

- Maxwell added the **displacement current** $I_d = \epsilon_0 d\Phi_E/dt$ to Ampère's law. A changing E creates B , just as a changing B creates E . Together they give self-sustaining **electromagnetic waves**.
- EM waves are **transverse**: $\vec{E} \perp \vec{B} \perp$ direction of travel ($\vec{E} \times \vec{B}$ points along the propagation). They need no medium, and in vacuum they travel at $c = 1/\sqrt{\mu_0\epsilon_0}$.
- They carry energy and momentum, so they exert radiation pressure.
- The **spectrum** runs from radio to gamma rays. Only the wavelength or frequency differs.

Formulas

Σ FORMULA EM wave relations

$$c = \frac{1}{\sqrt{\mu_0\epsilon_0}} = 3 \times 10^8 \text{ m s}^{-1} \quad \frac{E_0}{B_0} = c \quad v_{\text{medium}} = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{\mu_r\epsilon_r}}$$

$$u = \frac{1}{2}\epsilon_0 E^2 + \frac{B^2}{2\mu_0} \text{ (equal shares)} \quad \langle u \rangle = \frac{1}{2}\epsilon_0 E_0^2 \quad I = \langle u \rangle c = \frac{1}{2}\epsilon_0 c E_0^2$$

$$\text{momentum } p = \frac{U}{c} \quad \text{radiation pressure: } \frac{I}{c} \text{ (absorbed), } \frac{2I}{c} \text{ (reflected)} \quad I_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

Inside the gap of a charging capacitor, the displacement current equals the conduction current in the wires.

The electromagnetic spectrum (NCERT ranges)

Type	Wavelength range	Produced by	Main uses
Radio	> 0.1 m	oscillating charges in antennas	radio/TV communication
Microwaves	0.1 m – 1 mm	klystron, magnetron, Gunn diode	radar, microwave ovens, satellite links
Infrared	1 mm – 700 nm	hot bodies, molecular vibrations	remote controls, night vision, greenhouse effect, physiotherapy
Visible	700 nm – 400 nm	electron transitions in atoms	vision, photography
Ultraviolet	400 nm – 1 nm	very hot bodies (Sun), discharge lamps	sterilisation, water purification, LASIK; mostly absorbed by ozone
X-rays	1 nm – 10^{-3} nm	X-ray tubes (fast electrons hitting metal)	medical imaging, crystallography
Gamma rays	< 10^{-3} nm	radioactive nuclei	cancer therapy, nuclear physics

⚡ SHORTCUT Memory order

"Raman's Mother Invited Very Unusual eXtra Guests": increasing frequency, decreasing wavelength. Energy per photon increases along the same order.

PYQ PATTERN Recurring structures

1. $E_0 \leftrightarrow B_0$ conversion, direction of $\vec{B}$ from a given $\vec{E}$ and propagation direction.
2. Intensity / energy density / radiation pressure numericals.
3. **Spectrum ordering** and matching uses (UV–sterilisation, IR–remote, microwave–radar).
4. Displacement current in a charging capacitor.
5. Speed in a medium from ϵ_r, μ_r .

MISTAKE ALERT Mistake alerts

- Energy is shared **equally** between the electric and magnetic fields, even though $B_0 = E_0/c$ looks small.
- Radiation pressure doubles for a perfectly reflecting surface.
- Direction: $\vec{E} \times \vec{B}$ gives the propagation direction. If E is along $\hat{x}$ and the wave travels along $+\hat{z}$, then B is along $+\hat{y}$.

Practice questions

P15 · Practice

1 Easy 0.5 min Speed P15-01

An EM wave in vacuum has electric field amplitude 30 V/m. Its magnetic field amplitude is:

- (A) 10^{-7} T (B) 9×10^9 T (C) 10^{-8} T (D) 3×10^{-7} T

2 Easy 0.5 min Speed P15-02

Which of the following has the shortest wavelength?

- (A) X-rays (B) microwaves (C) γ -rays (D) ultraviolet

3 Easy 0.5 min NV P15-03

A capacitor is charged by a steady 2 A current. Find the displacement current (in A) between its plates.

Your answer (numerical value):

4 Medium 1 min Calc P15-04

An EM wave has $E_0 = 100$ V/m. Its intensity is about ($\epsilon_0 = 8.85 \times 10^{-12}$ SI):

- (A) 13.3 W/m² (B) 26.6 W/m² (C) 1.33 W/m² (D) 133 W/m²

5 Medium 1 min Concept P15-05

An EM wave travels along $+z$ with $\vec{E}$ along $+x$. At that instant, $\vec{B}$ points along:

- (A) $+y$ (B) $-y$ (C) $+z$ (D) $-x$

6 Easy 0.75 min Basic P15-06

An EM wave has a wavelength of 3 cm. Its frequency and type are:

- (A) 10 GHz, microwave (B) 1 GHz, radio
(C) 100 GHz, infrared (D) 10 MHz, radio

P15 · Chapter Test

1 Easy 0.5 min Concept

P15-T1

In an EM wave, the average energy density associated with the magnetic field is:

- (A) greater than that of the electric field (B) smaller than that of the electric field
(C) equal to that of the electric field (D) zero

2 Easy 0.5 min Speed

P15-T2

The EM radiation used to sterilise surgical instruments is:

- (A) infrared (B) ultraviolet (C) microwaves (D) radio waves

3 Easy 0.5 min Basic

P15-T3

The speed of EM waves in a non-magnetic medium with $\epsilon_r = 4$ is:

- (A) 3×10^8 m/s (B) 1.5×10^8 m/s (C) 0.75×10^8 m/s (D) 6×10^8 m/s

4 Easy 0.75 min NV

P15-T4

Radiation of intensity 900 W/m^2 falls normally on a perfectly absorbing surface. The pressure is $x \times 10^{-6} \text{ Pa}$. Find x .

Your answer (numerical value):

Answers & Solutions

P15 · Practice

1 (A) 2 (C) 3 2 4 (A) 5 (A) 6 (A)

P15 · Chapter Test

1 (C) 2 (B) 3 (B) 4 3

P15 · PRACTICE

1 Answer: (A) *E-B amplitude relation* Easy 0.5 min Speed

P15-01

SOLUTION $B_0 = E_0/c = 30/(3 \times 10^8) = 10^{-7} \text{ T}$.

COMMON TRAP Multiplying by c .

2 Answer: (C) *Spectrum order* Easy 0.5 min Speed

P15-02

SOLUTION Gamma rays lie at the high-frequency end of the spectrum.

COMMON TRAP Choosing X-rays.

3 Answer: 2 *Displacement current* Easy 0.5 min NV

P15-03

SOLUTION Continuity of current: $I_d = I_c = 2 \text{ A}$.

COMMON TRAP Answering zero because "no charge crosses the gap".

4 Answer: (A) *Intensity from field amplitude* Medium 1 min Calc

P15-04

SOLUTION $I = \frac{1}{2} \epsilon_0 c E_0^2 = 0.5 \times 8.85 \times 10^{-12} \times 3 \times 10^8 \times 10^4 \approx 13.3 \text{ W m}^{-2}$.

COMMON TRAP Forgetting the $\frac{1}{2}$ (26.6).

5 Answer: (A) *Field directions* Medium 1 min Concept

P15-05

SOLUTION $\vec{E} \times \vec{B}$ must point along $+\hat{z}$: $\hat{x} \times \hat{y} = \hat{z}$, so $\vec{B}$ is along $+\hat{y}$.

COMMON TRAP Using $\vec{B} \times \vec{E}$.

6 Answer: (A) Identify the band **Easy** ⌚ 0.75 min Basic

P15-06

SOLUTION $f = c/\lambda = 3 \times 10^8 / 0.03 = 10^{10}$ Hz; 3 cm lies in the microwave band (1 mm – 0.1 m).

P15 · CHAPTER TEST

1 Answer: (C) Energy sharing **Easy** ⌚ 0.5 min Concept

P15-T1

SOLUTION $\frac{1}{2}\epsilon_0 E^2 = \frac{B^2}{2\mu_0}$ since $E = cB$ and $c^2 = 1/\mu_0\epsilon_0$.

2 Answer: (B) Uses of UV **Easy** ⌚ 0.5 min Speed

P15-T2

SOLUTION UV kills bacteria by damaging their DNA.

3 Answer: (B) Speed in a medium **Easy** ⌚ 0.5 min Basic

P15-T3

SOLUTION $v = c/\sqrt{\epsilon_r} = c/2$.

COMMON TRAP Dividing by 4.

4 Answer: 3 Radiation pressure **Easy** ⌚ 0.75 min NV

P15-T4

SOLUTION $P = I/c = 900/(3 \times 10^8) = 3 \times 10^{-6}$ Pa.

COMMON TRAP Using $2I/c$ (for a reflecting surface).



REVISION Revision tracker · EM Waves

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 16

Optics: Ray & Wave

16

2–3 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 9, 10

NCERT

~16 hours

STUDY TIME

Concept summary

- **Ray optics** treats light as rays. It covers reflection (mirrors), refraction (Snell's law, lenses, prisms), total internal reflection and optical instruments.
- **Wave optics** treats light as a wave. It covers Huygens' principle, interference (YDSE), diffraction (single slit) and polarisation (Malus, Brewster).
- **Sign convention (Cartesian)**: distances are measured from the pole or optical centre. Positive means in the direction of the incident light, negative means against it. Heights are positive upward.

Formulas: Ray optics

Σ FORMULA Mirrors & refraction

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad f = \frac{R}{2} \quad m = -\frac{v}{u}$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad n = \frac{c}{v} \quad \text{apparent depth} = \frac{\text{real depth}}{n} \quad \text{shift} = t \left(1 - \frac{1}{n} \right)$$

Total internal reflection (dense $\rightarrow$ rare, $i > C$): $\sin C = \frac{n_{\text{rare}}}{n_{\text{dense}}}$. Applications: optical fibres, totally reflecting prisms, sparkle of diamond, mirage. Plane mirror rotated by $\theta \rightarrow$ the reflected ray turns by 2θ . Two mirrors at angle θ form $\frac{360^\circ}{\theta} - 1$ images (when $360/\theta$ is even).

Σ FORMULA Spherical surfaces & lenses

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \quad \frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad m = \frac{v}{u}$$

- Lens in a medium: replace $(n - 1)$ with $\left(\frac{n_{\text{lens}}}{n_{\text{medium}}} - 1 \right)$. If $n_{\text{lens}} < n_{\text{medium}}$, a convex lens diverges.
- **Power** $P = \frac{1}{f \text{ (m)}}$ dioptre. **In contact**: $P = P_1 + P_2$. **Separated by d** : $P = P_1 + P_2 - dP_1P_2$.
- **Silvered lens** acts as a mirror with $P_{\text{eq}} = 2P_{\text{lens}} + P_{\text{mirror}}$. For a plano-convex lens (curved radius R): silvered on the **plane** face, $f = \frac{R}{2(n - 1)}$; silvered on the **curved** face, $f = \frac{R}{2n}$.
- **Displacement method**: $f = \frac{D^2 - x^2}{4D}$ (D = object–screen distance, x = distance between the two lens positions). Needs $D \geq 4f$.

Σ FORMULA Prism

$$A = r_1 + r_2 \quad \delta = i + e - A \quad n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}} \quad \text{thin prism: } \delta = (n - 1)A$$

At minimum deviation: $i = e$, $r_1 = r_2 = A/2$, and the ray passes symmetrically. The δ – i graph is U-shaped (see below). Dispersion: $\delta_v > \delta_r$ because $n_v > n_r$.



Σ FORMULA Optical instruments ($D = 25 \text{ cm}$)

Instrument	Image at infinity (normal adjustment)	Image at D
Simple microscope	$M = \frac{D}{f}$	$M = 1 + \frac{D}{f}$
Compound microscope	$M \approx \frac{L}{f_o} \cdot \frac{D}{f_e}$	$M = m_o \left(1 + \frac{D}{f_e}\right)$
Astronomical telescope	$M = \frac{f_o}{f_e}$, tube length $f_o + f_e$	$M = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D}\right)$

Telescope: large-aperture objective (more light, better resolution), short- f eyepiece. **Reflecting telescopes** (Newtonian, Cassegrain) avoid chromatic aberration, and mirrors are easier to support than large lenses.

Formulas: Wave optics

Σ FORMULA Interference (YDSE)

$$\Delta x = d \sin \theta \approx \frac{yd}{D} \quad \beta = \frac{\lambda D}{d} \quad \text{bright: } \Delta x = n\lambda \quad \text{dark: } \Delta x = (2n-1)\frac{\lambda}{2}$$

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi \quad \text{equal slits: } I = 4I_0 \cos^2 \frac{\phi}{2} \quad \frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2$$

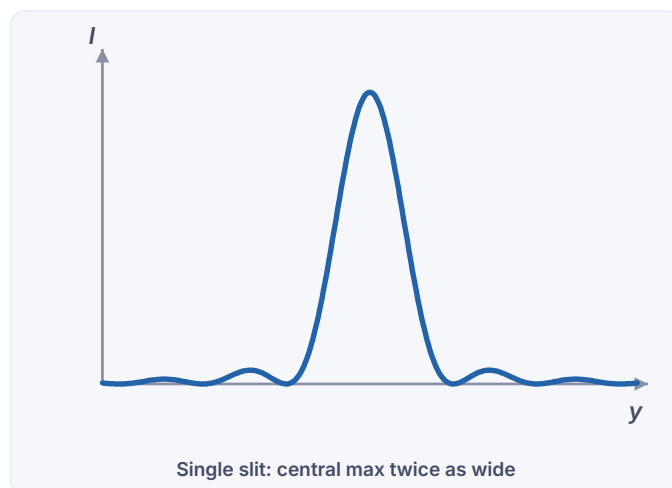
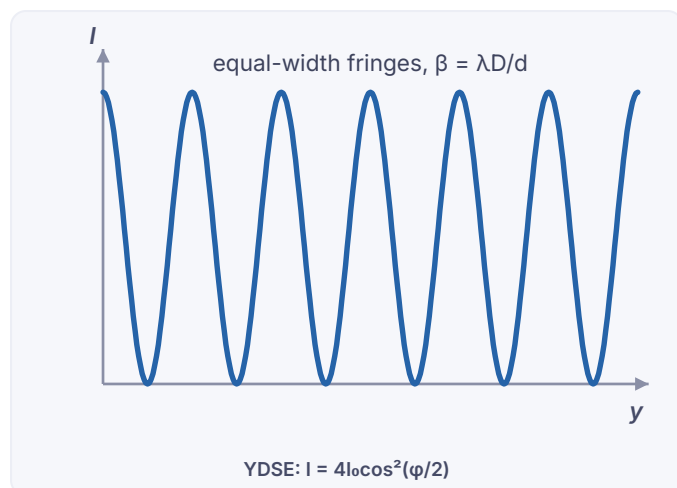
- Apparatus immersed in a medium of index n : $\beta' = \beta/n$.
- A thin film (index μ , thickness t) over one slit shifts the pattern **towards that slit** by $\frac{(\mu-1)tD}{d}$. The number of fringes shifted is $\frac{(\mu-1)t}{\lambda}$.
- Sustained interference needs **coherent** sources (constant phase difference).

Σ FORMULA Diffraction (single slit, width a)

Minima: $a \sin \theta = n\lambda$ ($n = \pm 1, \pm 2, \dots$). Angular width of the central maximum = $\frac{2\lambda}{a}$; linear width = $\frac{2\lambda D}{a}$.
Secondary maxima get rapidly weaker. Narrower slit $\rightarrow$ wider central maximum.

Σ FORMULA Polarisation

Malus's law: $I = I_0 \cos^2 \theta$. Unpolarised light through one polaroid $\rightarrow I_0/2$. **Brewster's law:** $\tan \theta_B = n_i$; at θ_B the reflected light is completely polarised, and the reflected and refracted rays are **perpendicular**. Polarisation proves light is **transverse**.



Frequently tested concepts & question patterns

PYQ PATTERN Recurring structures

1. **Mirror/lens formula** with sign conventions: nature, position and size of the image.
2. **Lens maker's formula**, including a lens in a liquid, and combinations (contact/separated).
3. **TIR/critical angle**, apparent depth, glass slab shift.
4. **Prism minimum deviation**; thin-prism deviation.
5. **Microscope/telescope magnification** and tube length.
6. **YDSE**: fringe width changes (medium, D , d , λ), intensity at a point, thin-film shift, I_{max}/I_{min} .
7. **Single-slit width** of the central maximum; **Malus/Brewster** numericals.

Shortcuts

SHORTCUT Image-nature table (concave mirror / convex lens)

Object position	Image
beyond $2f$ (C)	between f and $2f$, real, inverted, diminished
at $2f$	at $2f$, real, inverted, same size
between f and $2f$	beyond $2f$, real, inverted, magnified
inside f	virtual, erect, magnified

Convex mirrors and concave lenses: **always virtual, erect, diminished**. This settles many MCQs without any calculation.

SHORTCUT Fringe-width checks

$\beta \propto \lambda D/d$. Any change $\rightarrow$ multiply the old fringe width by the ratios. For a medium: divide by n . Changing the slit width doesn't change β , only the intensity and visibility.

Question classification

Type	Difficulty	Target time
Mirror/lens formula	E	1 min
Lens maker / lens in liquid	M	1.5 min
Prism min deviation	E	1 min
TIR / apparent depth	E	1 min
Instruments	E	1 min

Type	Difficulty	Target time
YDSE fringe/intensity	E-M	1.5 min
Thin film shift / multiple wavelengths coinciding	M	2 min
Combined systems (lens + mirror, refraction at two surfaces)	H	3 min (Round 2)

Common mistakes

MISTAKE ALERT Mistake alerts

- Sign convention: for a concave mirror $f < 0$. Object distance u is **negative** for real objects.
- Lens formula has a **minus** ($1/v - 1/u$); mirror formula has a **plus**.
- Critical angle is defined only for dense $\rightarrow$ rare.
- The intensity after the first polaroid is $I_0/2$, **before** applying $\cos^2 \theta$ at the second.
- In a medium, the wavelength changes but the frequency doesn't.

Practice questions

P16 · Practice

1 Medium 1.5 min Concept

P16-01

An object is 10 cm in front of a concave mirror of focal length 15 cm. The image is:

- (A) 30 cm behind the mirror, virtual, erect, $3\times$ magnified
 (B) 30 cm in front, real, inverted, $3\times$ magnified
 (C) 6 cm behind, virtual, erect, diminished
 (D) at infinity

2 Easy 0.5 min Speed

P16-02

The critical angle for glass of refractive index $\sqrt{2}$ (against air) is:

- (A) 30° (B) 45° (C) 60° (D) 90°

3 Medium 1.5 min JEE

P16-03

A biconvex glass lens ($n = 1.5$) with both radii 20 cm has focal length 20 cm in air. Immersed in water ($n = 4/3$), its focal length becomes:

- (A) 20 cm (B) 40 cm (C) 60 cm (D) 80 cm

4 Easy 1 min NV

P16-04

A convex lens of focal length 20 cm is placed in contact with a concave lens of focal length 40 cm. Find the focal length (in cm) of the combination.

Your answer (numerical value):

5 Medium 1 min NV

P16-05

A prism with $A = 60^\circ$ is made of glass with $n = \sqrt{2}$. Find its angle of minimum deviation (in degrees).

Your answer (numerical value):

6 Easy 0.75 min Basic

P16-06

A coin lies at the bottom of a tank filled with water ($n = 4/3$) to a depth of 12 cm. Viewed from above, the coin appears raised by:

- (A) 9 cm (B) 3 cm (C) 4 cm (D) 16 cm

7 Easy 0.75 min Calc

P16-07

In YDSE, $d = 1$ mm, $D = 1$ m and $\lambda = 600$ nm. The fringe width is:

- (A) 0.6 mm (B) 6 mm (C) 0.06 mm (D) 1.2 mm

8 Medium 1 min NV

P16-08

In YDSE the intensities from the two slits are in the ratio 9 : 1. Find I_{max}/I_{min} .

Your answer (numerical value):

9 Easy 1 min Basic

P16-09

Light of 500 nm falls on a slit 0.2 mm wide, and the screen is 1 m away. The width of the central bright band is:

- (A) 2.5 mm (B) 5 mm (C) 10 mm (D) 1.25 mm

10 Medium 1 min Concept

P16-10

Unpolarised light of intensity I_0 passes through two polaroids whose axes are at 60° . The emerging intensity is:

- (A) $I_0/2$ (B) $I_0/4$ (C) $I_0/8$ (D) $3I_0/8$

11 Easy 0.5 min Speed

P16-11

The Brewster angle for glass of refractive index $\sqrt{3}$ is:

- (A) 30° (B) 45° (C) 60° (D) 90°

12 Easy 0.75 min Basic

P16-12

An astronomical telescope has $f_o = 100$ cm and $f_e = 5$ cm. In normal adjustment, its magnifying power and tube length are:

- (A) 20, 105 cm (B) 20, 95 cm (C) 500, 105 cm (D) 0.05, 105 cm

P16 · Chapter Test

1 Easy 0.5 min Speed

P16-T1

A plane mirror is rotated through θ while the incident ray stays fixed. The reflected ray turns through:

- (A) θ (B) 2θ (C) $\theta/2$ (D) 0

2 Easy 0.5 min Speed

P16-T2

The speed of light in glass of refractive index 1.5 is:

- (A) 4.5×10^8 m/s (B) 2×10^8 m/s (C) 3×10^8 m/s (D) 1.5×10^8 m/s

3 Easy 0.5 min Speed

P16-T3

A lens has focal length -25 cm. Its power is:

- (A) +4 D (B) -4 D (C) -0.04 D (D) -25 D

4 Easy 0.75 min Basic

P16-T4

A YDSE apparatus is immersed in water ($n = 4/3$). The fringe width becomes:

- (A) $4/3$ of the original (B) $3/4$ of the original
(C) unchanged (D) $9/16$ of the original

5 Easy 0.5 min NV

P16-T5

A simple microscope has focal length 5 cm. Find its magnifying power when the image forms at the near point ($D = 25$ cm).

Your answer (numerical value):

6 Medium 1.5 min JEE

P16-T6

A thin film of refractive index 1.5 placed over one slit of a YDSE shifts the central fringe by 5 fringe widths ($\lambda = 600$ nm). The film's thickness is:

- (A) $3\ \mu\text{m}$ (B) $6\ \mu\text{m}$ (C) $2\ \mu\text{m}$ (D) $12\ \mu\text{m}$

Answers & Solutions

P16 · Practice

- 1 (A) 2 (B) 3 (D) 4 40 5 30 6 (B) 7 (A) 8 4 9 (B) 10 (C)
11 (C) 12 (A)

P16 · Chapter Test

- 1 (B) 2 (B) 3 (B) 4 (B) 5 6 6 (B)

P16 · PRACTICE

1 Answer: (A) Concave mirror, object inside f Medium 1.5 min Concept

P16-01

SOLUTION $u = -10$, $f = -15$: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = -\frac{1}{15} + \frac{1}{10} = \frac{1}{30}$, so $v = +30$ cm (behind the mirror). $m = -v/u = 3$: virtual, erect.

SHORTCUT Object inside f of a concave mirror $\rightarrow$ virtual, erect, magnified. Only (A) fits.

COMMON TRAP Using $+15$ for f .

2 Answer: (B) Critical angle Easy 0.5 min Speed

P16-02

SOLUTION $\sin C = 1/\sqrt{2} \Rightarrow C = 45^\circ$.

3 Answer: (D) Lens in water Medium 1.5 min JEE

P16-03

SOLUTION In water: $\frac{1}{f'} = \left(\frac{1.5}{4/3} - 1\right) \frac{2}{20} = 0.125 \times 0.1 \Rightarrow f' = 80$ cm.

SHORTCUT $\frac{f'}{f} = \frac{n-1}{n/n_m-1} = \frac{0.5}{0.125} = 4$.

COMMON TRAP Assuming f is unchanged.

4 Answer: 40 Lenses in contact Easy 1 min NV

P16-04

SOLUTION $P = \frac{100}{20} - \frac{100}{40} = 5 - 2.5 = 2.5$ D, so $f = 40$ cm (converging).

COMMON TRAP Adding focal lengths.

5 Answer: 30 Minimum deviation Medium 1 min NV

P16-05

SOLUTION $\sqrt{2} = \frac{\sin \frac{60^\circ + \delta_m}{2}}{\sin 30^\circ} \Rightarrow \sin \frac{60^\circ + \delta_m}{2} = \frac{\sqrt{2}}{2} \Rightarrow \frac{60^\circ + \delta_m}{2} = 45^\circ \Rightarrow \delta_m = 30^\circ$.

COMMON TRAP Using the thin-prism formula: $(n-1)A \approx 24.9^\circ$.

6 Answer: (B) Apparent depth **Easy** ⌚ 0.75 min Basic

P16-06

SOLUTION Apparent depth = $12/(4/3) = 9$ cm, so the rise is $12 - 9 = 3$ cm.**SHORTCUT** Shift = $t(1 - 1/n)$.**COMMON TRAP** Answering the apparent depth (9 cm).**7 Answer: (A)** YDSE fringe width **Easy** ⌚ 0.75 min Calc

P16-07

SOLUTION $\beta = \frac{\lambda D}{d} = \frac{6 \times 10^{-7} \times 1}{10^{-3}} = 6 \times 10^{-4} \text{ m} = 0.6 \text{ mm}.$ **8 Answer: 4** Maximum to minimum intensity **Medium** ⌚ 1 min NV

P16-08

SOLUTION Amplitude ratio 3 : 1. $\frac{I_{max}}{I_{min}} = \left(\frac{3+1}{3-1}\right)^2 = 4.$ **SHORTCUT** Work with amplitudes ($\sqrt{I}$).**COMMON TRAP** Using intensities directly: $(9+1)/(9-1).$ **9 Answer: (B)** Single-slit central maximum **Easy** ⌚ 1 min Basic

P16-09

SOLUTION Width = $\frac{2\lambda D}{a} = \frac{2 \times 5 \times 10^{-7}}{2 \times 10^{-4}} = 5 \times 10^{-3} \text{ m}.$ **COMMON TRAP** Forgetting the 2 (half-width).**10 Answer: (C)** Two polaroids **Medium** ⌚ 1 min Concept

P16-10

SOLUTION After the first: $I_0/2$. After the second: $\frac{I_0}{2} \cos^2 60^\circ = I_0/8.$ **COMMON TRAP** Skipping the first halving ($I_0/4$).**11 Answer: (C)** Brewster's angle **Easy** ⌚ 0.5 min Speed

P16-11

SOLUTION $\tan \theta_B = \sqrt{3} \Rightarrow \theta_B = 60^\circ.$ **COMMON TRAP** Using sin (critical-angle thinking).**12 Answer: (A)** Astronomical telescope **Easy** ⌚ 0.75 min Basic

P16-12

SOLUTION $M = f_o/f_e = 20$; $L = f_o + f_e = 105 \text{ cm}.$ **P16 · CHAPTER TEST****1 Answer: (B)** Rotating a plane mirror **Easy** ⌚ 0.5 min Speed

P16-T1

SOLUTION Both the angle of incidence and the angle of reflection change by θ .**2 Answer: (B)** Speed of light in glass **Easy** ⌚ 0.5 min Speed

P16-T2

SOLUTION $v = c/n = 2 \times 10^8 \text{ m/s}.$ **3 Answer: (B)** Power of a lens **Easy** ⌚ 0.5 min Speed

P16-T3

SOLUTION $P = 1/f(\text{m}) = 1/(-0.25) = -4 \text{ D (diverging)}.$ **COMMON TRAP** Using cm (-0.04).**4 Answer: (B)** YDSE in water **Easy** ⌚ 0.75 min Basic

P16-T4

SOLUTION $\lambda' = \lambda/n$, so $\beta' = \beta/n = \frac{3}{4}\beta.$ **5 Answer: 6** Simple microscope **Easy** ⌚ 0.5 min NV

P16-T5

SOLUTION $M = 1 + D/f = 1 + 5 = 6.$ **COMMON TRAP** Using $D/f = 5$ (the normal-adjustment case).**6 Answer: (B)** Film over one slit **Medium** ⌚ 1.5 min JEE

P16-T6

SOLUTION Extra path $(\mu - 1)t = 5\lambda \Rightarrow t = \frac{5 \times 600 \times 10^{-9}}{0.5} = 6 \times 10^{-6} \text{ m}.$ **COMMON TRAP** Using μt instead of $(\mu - 1)t$ gives $2 \mu\text{m}.$ **REVISION** Revision tracker · Optics☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 17

Dual Nature of Matter & Radiation

17

~1 per shift
TYPICAL QUESTIONS

Easy
DIFFICULTY

Must-do (easy
marks)
PRIORITY

Class 12 · Ch 11
NCERT

~4 hours
STUDY TIME

Concept summary

- The **photoelectric effect** (observed by Hertz and Lenard, explained by Einstein) shows that light is absorbed in quanta: **photons** of energy $h\nu$.
- One photon ejects at most one electron. The electron's **maximum KE** depends on frequency, not intensity. Intensity controls the **number** of electrons, and so the photocurrent.
- Matter has wave nature** (de Broglie): $\lambda = h/p$. This was confirmed by electron diffraction in the Davisson–Germer experiment.

Formulas

Σ FORMULA Photons

$$E = h\nu = \frac{hc}{\lambda} \quad hc \approx 1240 \text{ eV nm} \quad p = \frac{h}{\lambda} = \frac{E}{c} \quad \text{photon flux } n = \frac{P}{h\nu} = \frac{P\lambda}{hc}$$

$h = 6.63 \times 10^{-34} \text{ J s}; 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}.$

Σ FORMULA Photoelectric effect

$$K_{max} = h\nu - \phi = eV_0 \quad \nu_0 = \frac{\phi}{h} \quad \lambda_0 = \frac{hc}{\phi} \quad V_0 = \frac{h}{e}\nu - \frac{\phi}{e}$$

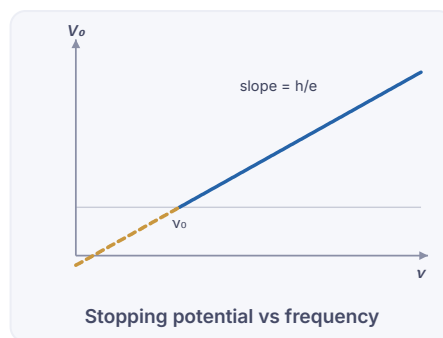
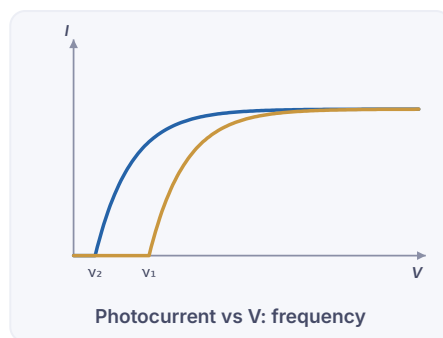
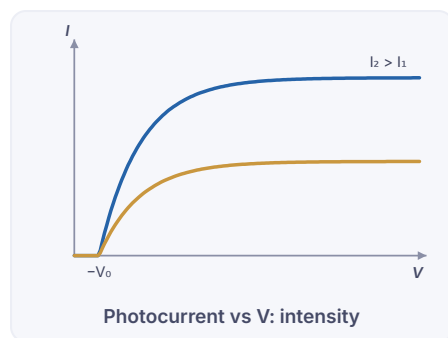
Change	Saturation current	Stopping potential V_0	K_{max}
Intensity $\uparrow$ (same ν)	$\uparrow$ (proportional)	unchanged	unchanged
Frequency $\uparrow$ (same intensity)	about the same (fewer photons per joule)	$\uparrow$	$\uparrow$
$\nu < \nu_0$	zero, whatever the intensity	—	—

- Emission is **instantaneous** (no time lag), even for very weak light.
- The V_0 vs ν graph is a straight line with slope h/e (**the same for every metal**) and ν -intercept ν_0 .

Σ FORMULA de Broglie waves

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}}$$

- Electron through V volts: $\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}.$
- Thermal particle at temperature T : $\lambda = \frac{h}{\sqrt{3mk_B T}}.$
- Davisson–Germer: 54 V electrons diffracted by a nickel crystal, $\lambda \approx 0.167 \text{ nm}$, matching de Broglie's prediction.



Frequently tested concepts & question patterns



PYQ PATTERN Recurring structures

1. K_{max} , V_0 , threshold wavelength for given λ and ϕ (use 1240 eV nm).
2. **Graph questions:** I - V for different intensities/frequencies; V_0 - ν slope and intercept.
3. **Two-wavelength problems:** V_1 , V_2 for λ_1 , $\lambda_2 \rightarrow$ find ϕ or h .
4. **de Broglie wavelength ratios** for particles accelerated through the same potential, or with the same KE/momentum.
5. **Photon flux** from a source of given power.

Shortcuts



SHORTCUT 1240 rule

$E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$. 620 nm $\rightarrow$ 2 eV, 400 nm $\rightarrow$ 3.1 eV, 310 nm $\rightarrow$ 4 eV, 248 nm $\rightarrow$ 5 eV. **Time saved:** avoids $6.63 \times 10^{-34} \times 3 \times 10^8$ arithmetic. **Common mistake:** using Å (divide by 10 first).



SHORTCUT de Broglie ratios

Same V : $\lambda \propto 1/\sqrt{mq}$. Same K : $\lambda \propto 1/\sqrt{m}$. Same p : equal λ .

Question classification

Type	Difficulty	Target time
Photon energy / flux	E	45 s
Einstein equation numericals	E	1 min
Graph interpretation	E	45 s
Two-wavelength elimination	M	1.5 min
de Broglie ratios	E	1 min

Common mistakes



MISTAKE ALERT Mistake alerts

- Thinking higher intensity gives faster electrons. It gives **more** electrons.
- Using λ in Å with 1240 (which needs nm).
- For de Broglie of a charged particle through V , forgetting the charge q (α -particle: $q = 2e$).
- Stopping potential in volts = K_{max} in eV, numerically. Don't multiply by e again.

Practice questions

P17 · Practice

1 Easy 0.5 min Speed

P17-01

The energy of a photon of wavelength 620 nm is about:

- (A) 1 eV (B) 2 eV (C) 3.1 eV (D) 0.5 eV

2 Easy 1 min Basic

P17-02

Light of 400 nm falls on a metal with work function 2.3 eV. The stopping potential is:

- (A) 0.8 V (B) 3.1 V (C) 2.3 V (D) 5.4 V

3 Easy 0.75 min NV

P17-03

A metal has work function 4.13 eV. Find its threshold wavelength (in nm). Take $hc = 1240$ eV nm and give the nearest integer.

Your answer (numerical value):

4 Easy 0.5 min Concept

P17-04

The intensity of light on a photo-cell is doubled at the same frequency. Then:

- (A) the stopping potential doubles
(B) the saturation current doubles and the stopping potential stays the same
(C) both double
(D) neither changes

5 Easy 0.75 min Basic

P17-05

An electron is accelerated from rest through 100 V. Its de Broglie wavelength is about:

- (A) 1.23 nm (B) 0.123 nm (C) 12.3 nm (D) 0.0123 nm

6 Medium 1 min Concept

P17-06

A proton and an α -particle are accelerated from rest through the same potential difference. The ratio $\lambda_p : \lambda_\alpha$ is:

- (A) $2\sqrt{2} : 1$ (B) $1 : 2\sqrt{2}$ (C) $2 : 1$ (D) $1 : 1$

7 Easy 0.5 min Concept

P17-07

The slope of the stopping potential vs frequency graph for a photoelectric metal is:

- (A) h (B) h/e (C) e/h (D) ϕ/e

8 Medium 1 min Calc

P17-08

A 10 W source emits light of wavelength 662 nm. The number of photons it emits per second is about ($h = 6.62 \times 10^{-34}$ J s):

- (A) 3.3×10^{19} (B) 3.3×10^{18} (C) 1.0×10^{20} (D) 6.6×10^{19}

P17 · Chapter Test

1 Easy 0.5 min Concept

P17-T1

In the photoelectric effect, the time lag between incidence of light and emission of electrons is:

- (A) about 1 s (B) negligible ($< 10^{-9}$ s)
(C) proportional to intensity (D) inversely proportional to frequency

2 Easy 0.5 min Speed

P17-T2

A photon of energy E has momentum:

- (A) E/c (B) Ec (C) E/c^2 (D) $\sqrt{2mE}$

3 Easy 0.5 min NV

P17-T3

An electron's de Broglie wavelength after acceleration through V volts is 0.1227 nm. Find V .

Your answer (numerical value):

4 Medium 1 min Concept

P17-T4

An electron and a photon have the same wavelength, 0.1 nm. Then:

- (A) they have equal energies
(B) the photon has more energy than the electron's kinetic energy
(C) the electron's KE is greater
(D) they have different momenta

Answers & Solutions

P17 · Practice

1 (B) 2 (A) 3 300 4 (B) 5 (B) 6 (A) 7 (B) 8 (A)

P17 · Chapter Test

1 (B) 2 (A) 3 100 4 (B)

P17 · PRACTICE

1 Answer: (B) Photon energy Easy 0.5 min Speed

P17-01

SOLUTION $E = 1240/620 = 2$ eV.

2 Answer: (A) Stopping potential Easy 1 min Basic

P17-02

SOLUTION $h\nu = 1240/400 = 3.1$ eV; $K_{max} = 0.8$ eV, so $V_0 = 0.8$ V.

COMMON TRAP Adding the work function.

3 Answer: 300 Threshold wavelength Easy 0.75 min NV

P17-03

SOLUTION $\lambda_0 = hc/\phi = 1240/4.13 \approx 300$ nm.

4 Answer: (B) Effect of intensity Easy 0.5 min Concept

P17-04

SOLUTION Twice as many photons eject twice as many electrons, but each electron's maximum energy is unchanged.

COMMON TRAP Choosing (A).

5 Answer: (B) Electron de Broglie wavelength Easy 0.75 min Basic

P17-05

SOLUTION $\lambda = 1.227/\sqrt{100}$ nm = 0.1227 nm ≈ 1.23 Å.

COMMON TRAP Forgetting the square root.

6 Answer: (A) de Broglie ratio: proton vs α **Medium** ⌚ 1 min **Concept**

P17-06

SOLUTION $\lambda \propto 1/\sqrt{mq}$: $\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}} = \sqrt{4 \times 2} = 2\sqrt{2}$.

COMMON TRAP Ignoring the charge ($\sqrt{4} = 2$).

7 Answer: (B) Slope of V_0 - v graph **Easy** ⌚ 0.5 min **Concept**

P17-07

SOLUTION $V_0 = \frac{h}{e}\nu - \frac{\phi}{e}$. The slope is h/e , the same for all metals.

8 Answer: (A) Photon flux **Medium** ⌚ 1 min **Calc**

P17-08

SOLUTION $n = \frac{P\lambda}{hc} = \frac{10 \times 662 \times 10^{-9}}{6.62 \times 10^{-34} \times 3 \times 10^8} = \frac{6.62 \times 10^{-6}}{1.986 \times 10^{-25}} \approx 3.3 \times 10^{19} \text{ s}^{-1}$.

SHORTCUT Photon energy $\approx 1240/662 \approx 1.87 \text{ eV} \approx 3 \times 10^{-19} \text{ J}$; $10/(3 \times 10^{-19}) \approx 3.3 \times 10^{19}$.

COMMON TRAP Powers-of-ten slips.

P17 · CHAPTER TEST

1 Answer: (B) Time lag **Easy** ⌚ 0.5 min **Concept**

P17-T1

SOLUTION Each photon transfers its energy to one electron at once.

2 Answer: (A) Photon momentum **Easy** ⌚ 0.5 min **Speed**

P17-T2

SOLUTION $p = h/\lambda = h\nu/c = E/c$.

COMMON TRAP Using the massive-particle formula (D).

3 Answer: 100 Accelerating voltage from wavelength **Easy** ⌚ 0.5 min **NV**

P17-T3

SOLUTION $0.1227 = 1.227/\sqrt{V} \Rightarrow \sqrt{V} = 10 \Rightarrow V = 100 \text{ V}$.

4 Answer: (B) Electron vs photon, same λ **Medium** ⌚ 1 min **Concept**

P17-T4

SOLUTION The momenta are equal (h/λ). Photon: $E = pc \approx 12.4 \text{ keV}$. Electron: $K = p^2/2m \approx 150 \text{ eV}$. The photon's energy is much larger.

SHORTCUT Same $\lambda \Rightarrow$ same p ; $pc \gg p^2/2m$ when $v \ll c$.

COMMON TRAP Choosing (D).

REVISION Revision tracker · Dual Nature

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 18

Atoms & Nuclei

18

1-2 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

Must-do (easy marks)

PRIORITY

Class 12 · Ch 12, 13

NCERT

~6 hours

STUDY TIME

Concept summary

- **Rutherford's α -scattering:** most α -particles pass straight through gold foil, and a few bounce back. So an atom is mostly empty space, with a tiny, dense, positive nucleus.
- **Bohr's model** (for H-like atoms) quantises angular momentum, $mvr = n\hbar/2\pi$. It gives discrete orbits and energies, and explains the hydrogen spectrum.
- A **nucleus** of Z protons and N neutrons weighs **less** than its parts. The **mass defect** is the binding energy, Δmc^2 .
- **Binding energy per nucleon** peaks near $A \approx 56$. Heavy nuclei release energy by **fission** and light ones by **fusion**.

WARNING Syllabus note

The 2024–2026 syllabus text for Unit 18 lists α -scattering, Rutherford and Bohr models, the hydrogen spectrum, nuclear composition and size, mass–energy relation, mass defect, BE/A, fission and fusion. **Radioactivity (decay law, half-life, $\alpha/\beta/\gamma$ properties) is not listed** [NTA-SYL]. Give it a short read only.

Formulas: Atoms

FORMULA Bohr model (hydrogen-like ion, atomic number Z)

$$r_n = 0.529 \frac{n^2}{Z} \text{ \AA} \quad v_n = 2.18 \times 10^6 \frac{Z}{n} \text{ m s}^{-1} \quad E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$$

$$K = -E \quad U = 2E \quad mvr = \frac{nh}{2\pi} \quad 2\pi r_n = n\lambda_{dB}$$

Scaling: $r \propto n^2/Z$, $v \propto Z/n$, $E \propto Z^2/n^2$, time period $\propto n^3/Z^2$. **Rutherford closest approach** of an α -particle with kinetic energy K : $d = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{K}$.

FORMULA Hydrogen spectrum

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad R = 1.097 \times 10^7 \text{ m}^{-1} \quad \Delta E = h\nu$$

Series	n_1	Region
Lyman	1	ultraviolet
Balmer	2	visible
Paschen	3	infrared
Brackett	4	infrared
Pfund	5	infrared

- Lines from level n down to ground: $\frac{n(n-1)}{2}$.
- Ionisation energy of H: 13.6 eV. Excitation energies: 10.2 eV ($1 \rightarrow 2$), 12.09 eV ($1 \rightarrow 3$).
- Lyman series limit ≈ 91 nm; Balmer limit ≈ 365 nm; first Balmer line (H- α) ≈ 656 nm.
- **Bohr's limitations:** works only for single-electron atoms; can't explain line intensities or fine structure.

Formulas: Nuclei

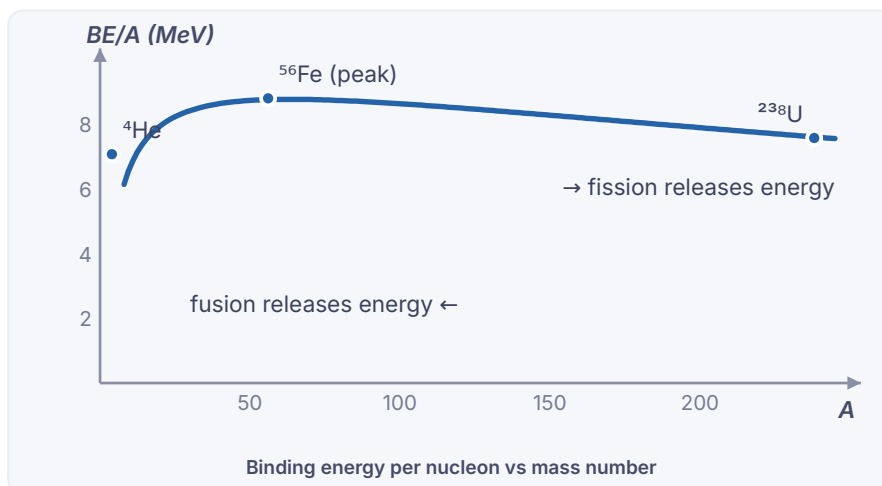
Σ FORMULA Size, mass defect, binding energy

$$R = R_0 A^{1/3}, R_0 \approx 1.2 \text{ fm} \quad \rho_{\text{nucleus}} \approx 2.3 \times 10^{17} \text{ kg m}^{-3} \text{ (independent of } A)$$

$$\Delta m = [Zm_p + (A - Z)m_n] - M_{\text{nucleus}} \quad BE = \Delta m c^2 \quad 1 \text{ u} = 931.5 \text{ MeV}/c^2$$

$$Q\text{-value} = \left(\sum m_{\text{reactants}} - \sum m_{\text{products}} \right) c^2 \quad Q > 0 \Rightarrow \text{energy released}$$

- **Nuclear force:** strongest force, short range (a few fm), charge independent, saturates.
- **Fission:** $^{235}\text{U} + n \rightarrow$ two medium nuclei + 2–3 neutrons + ~200 MeV. **Fusion:** light nuclei combine ($4 \text{ } ^1\text{H} \rightarrow \text{}^4\text{He}$ releases ~26.7 MeV in stars). Fusion needs very high temperatures to overcome Coulomb repulsion.



Frequently tested concepts & question patterns

▮ PYQ PATTERN Recurring structures

1. **Bohr scaling** for H, He^+ , Li^{2+} : radius, speed, energy, ratios between orbits.
2. **Spectral wavelengths:** ratio of first lines of two series, series limits, number of lines.
3. **Energy-level diagrams:** which transition emits the longest/shortest wavelength.
4. **Binding energy/nucleon** from given masses; Q-values of reactions.
5. **Nuclear radius/density** scaling with A .
6. **Closest approach** in α -scattering.

Shortcuts

⚡ SHORTCUT Wavelength ratios without R

$$\lambda \propto \frac{1}{\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)}. \text{ For the first lines: Lyman } \propto 4/3, \text{ Balmer } \propto 36/5, \text{ Paschen } \propto 144/7. \text{ Take ratios directly.}$$

⚡ SHORTCUT BE/A curve reading

Anything that moves nucleons **towards the peak** at $A \approx 56$ releases energy. Energy released = $\sum (BE)_{\text{products}} - \sum (BE)_{\text{reactants}}$.

Question classification

Type	Difficulty	Target time
Bohr formulas	E	45 s
Spectral lines / ratios	E	1 min

Type	Difficulty	Target time
BE per nucleon	E	1 min
Q-value	E-M	1.5 min
Closest approach	M	1.5 min

Common mistakes

MISTAKE ALERT Mistake alerts

- Using Z instead of Z^2 in energy scaling.
- The **longest** wavelength in a series is its **first** line (smallest ΔE).
- Mass defect uses the **nucleus** mass. If atomic masses are given, electron masses cancel only when you use hydrogen-atom masses for the protons.
- Converting u to MeV with 931 **before** multiplying by the mass defect in u.

Practice questions

P18 · Practice

1 Easy 0.75 min Basic P18-01

The radius of the first Bohr orbit of hydrogen is $0.53 \text{ \AA}$. The radius of the $n = 3$ orbit of Li^{2+} is:

- (A) $0.53 \text{ \AA}$ (B) $1.59 \text{ \AA}$ (C) $4.77 \text{ \AA}$ (D) $0.18 \text{ \AA}$

2 Easy 0.5 min Speed P18-02

The energy of the electron in the $n = 2$ state of He^+ is:

- (A) -3.4 eV (B) -13.6 eV (C) -54.4 eV (D) -6.8 eV

3 Medium 1 min Concept P18-03

The ratio of the wavelength of the first line of the Balmer series to that of the first line of the Lyman series in hydrogen is:

- (A) $27 : 5$ (B) $5 : 27$ (C) $4 : 1$ (D) $9 : 4$

4 Easy 0.5 min NV P18-04

Electrons in a sample of hydrogen are excited to $n = 5$. Find the maximum number of different spectral lines emitted as they return to the ground state.

Your answer (numerical value):

5 Easy 0.5 min Speed P18-05

The ratio of the nuclear radii of nuclei with $A = 216$ and $A = 27$ is:

- (A) 8 (B) 2 (C) 4 (D) $2^{1/3}$

6 Easy 1 min Calc P18-06

The mass defect of ${}^4_2\text{He}$ is 0.0304 u . Its binding energy per nucleon is about:

- (A) 28.3 MeV (B) 7.1 MeV (C) 14.2 MeV (D) 3.5 MeV

7

Medium

⌚ 1.5 min

Calc

P18-07

A 5 MeV α -particle is aimed head-on at a gold nucleus ($Z = 79$). Its distance of closest approach is about:

- (A) 4.5×10^{-14} m (B) 4.5×10^{-15} m (C) 2.3×10^{-14} m (D) 9.1×10^{-14} m

8

Easy

⌚ 0.5 min

NV

P18-08

Find the ionisation energy (in eV, nearest integer) of Li^{2+} in its ground state.

Your answer (numerical value):

9

Easy

⌚ 0.75 min

Concept

P18-09

Fission of a heavy nucleus releases energy because:

- (A) the products have a smaller binding energy per nucleon
 (B) the products have a larger binding energy per nucleon
 (C) the number of nucleons decreases
 (D) neutrons are destroyed



P18 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

P18-T1

The Balmer series of hydrogen lies in the:

- (A) ultraviolet (B) visible (C) infrared (D) X-ray region

2

Easy

⌚ 0.5 min

Concept

P18-T2

Nuclear density:

- (A) increases with A (B) decreases with A
 (C) is nearly the same for all nuclei (D) is maximum for $A \approx 56$

3

Easy

⌚ 0.5 min

Speed

P18-T3

The energy equivalent of 1 atomic mass unit is about:

- (A) 9.31 MeV (B) 931.5 MeV (C) 1.6 MeV (D) 0.511 MeV

4

Easy

⌚ 0.75 min

NV

P18-T4

Using $R = 1.097 \times 10^7 \text{ m}^{-1}$, find the series-limit wavelength of the Lyman series (in nm, nearest integer).

Your answer (numerical value):

5

Easy

⌚ 0.5 min

Concept

P18-T5

In the Bohr model, as n increases:

- (A) KE increases and total energy decreases
 (B) KE decreases and total energy increases (becomes less negative)
 (C) both increase
 (D) both decrease

Answers & Solutions

P18 · Practice

1 (B) 2 (B) 3 (A) 4 10 5 (B) 6 (B) 7 (A) 8 122 9 (B)

P18 · Chapter Test

1 (B) 2 (C) 3 (B) 4 91 5 (B)

P18 · PRACTICE

1 **Answer: (B)** Bohr radius scaling **Easy** ⌚ 0.75 min **Basic**

P18-01

SOLUTION $r = 0.53 \times \frac{n^2}{Z} = 0.53 \times \frac{9}{3} = 1.59 \text{ Å}.$

COMMON TRAP Using n instead of n^2 .

2 **Answer: (B)** Energy in He^+ **Easy** ⌚ 0.5 min **Speed**

P18-02

SOLUTION $E = -13.6 \times \frac{Z^2}{n^2} = -13.6 \times \frac{4}{4} = -13.6 \text{ eV}.$

COMMON TRAP Using Z instead of Z^2 (-6.8).

3 **Answer: (A)** Ratio of first lines **Medium** ⌚ 1 min **Concept**

P18-03

SOLUTION $\frac{\lambda_B}{\lambda_L} = \frac{1 - \frac{1}{4}}{\frac{1}{4} - \frac{1}{9}} = \frac{3/4}{5/36} = \frac{27}{5}.$

COMMON TRAP Inverting the ratio.

4 **Answer: 10** Number of spectral lines **Easy** ⌚ 0.5 min **NV**

P18-04

SOLUTION $\frac{n(n-1)}{2} = \frac{5 \times 4}{2} = 10.$

COMMON TRAP Counting only direct jumps to $n = 1$ (4 lines).

5 **Answer: (B)** Nuclear radius scaling **Easy** ⌚ 0.5 min **Speed**

P18-05

SOLUTION $R \propto A^{1/3}$: $(216/27)^{1/3} = 8^{1/3} = 2.$

6 **Answer: (B)** Binding energy per nucleon **Easy** ⌚ 1 min **Calc**

P18-06

SOLUTION $BE = 0.0304 \times 931.5 \approx 28.3 \text{ MeV}.$ Per nucleon: $28.3/4 \approx 7.1 \text{ MeV}.$

COMMON TRAP Stopping at the total BE.

7 **Answer: (A)** Distance of closest approach **Medium** ⌚ 1.5 min **Calc**

P18-07

SOLUTION $d = \frac{k(2e)(79e)}{K} = \frac{9 \times 10^9 \times 158 \times (1.6 \times 10^{-19})^2}{5 \times 1.6 \times 10^{-13}} \approx 4.55 \times 10^{-14} \text{ m}.$

SHORTCUT In MeV·fm units: $ke^2 = 1.44 \text{ MeV fm}$, so $d = 1.44 \times 158/5 \approx 45.5 \text{ fm}.$

COMMON TRAP Forgetting the α -particle's charge $2e$ gives (C).

8 **Answer: 122** Ionisation energy of Li^{2+} **Easy** ⌚ 0.5 min **NV**

P18-08

SOLUTION $13.6 \times Z^2 = 13.6 \times 9 = 122.4 \text{ eV}.$

9 **Answer: (B)** Why fission releases energy **Easy** ⌚ 0.75 min **Concept**

P18-09

SOLUTION Middle-mass nuclei are more tightly bound (higher BE/A). The increase in total binding energy is released.

P18 · CHAPTER TEST

1 **Answer: (B)** Balmer region **Easy** ⌚ 0.5 min **Speed**

P18-T1

SOLUTION Transitions to $n = 2$ emit 656 nm to 365 nm, mostly visible.

2 **Answer: (C)** Nuclear density **Easy** ⌚ 0.5 min **Concept**

P18-T2

SOLUTION $R^3 \propto A$ and mass $\propto A$, so the density is constant.

COMMON TRAP Confusing it with BE/A (D).

3 **Answer: (B)** Energy equivalent of 1 u **Easy** ⌚ 0.5 min **Speed**

P18-T3

SOLUTION $1 \text{ u} \times c^2 \approx 931.5 \text{ MeV}.$ (0.511 MeV is the electron's rest energy.)

4 Answer: 91 Lyman series limit **Easy** ⌚ 0.75 min **NV**

P18-T4

SOLUTION $\lambda = 1/R = 9.116 \times 10^{-8} \text{ m} \approx 91 \text{ nm}$.

5 Answer: (B) Trends with n **Easy** ⌚ 0.5 min **Concept**

P18-T5

SOLUTION $K \propto 1/n^2$ decreases; $E = -K$ rises towards 0.

 REVISION Revision tracker · Atoms & Nuclei

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 19

Electronic Devices

19

1–2 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

Must-do (easy marks)

PRIORITY

Class 12 · Ch 14

NCERT

~5 hours

STUDY TIME

Concept summary

- In **semiconductors** (Si, Ge) a small band gap lets thermal energy create electron–hole pairs. **Doping** controls conductivity: pentavalent donors give **n-type**, trivalent acceptors give **p-type**.
- A **p–n junction** forms a depletion layer with a barrier potential. It conducts easily in **forward bias** and hardly at all in **reverse bias**, which makes it a rectifier.
- Special diodes**: Zener (voltage regulator in reverse breakdown), LED (light from recombination, forward bias), photodiode (light-controlled reverse current), solar cell (generates emf with no bias).
- Logic gates** process binary signals. NAND and NOR are **universal** gates.

**WARNING** Syllabus note

Transistors (BJT action, amplifiers, oscillators) are **not** in the 2024–2026 syllabus text. Unit 19 covers semiconductors, diode I–V characteristics, rectifiers, LED, photodiode, solar cell, Zener diode and logic gates (OR, AND, NOT, NAND, NOR) [NTA-SYL].

Formulas & facts

**FORMULA** Semiconductors

Type	Dopant (group)	Majority carriers	Examples
Intrinsic	none	$n_e = n_h = n_i$	pure Si, Ge
n-type	pentavalent (15)	electrons	P, As, Sb in Si
p-type	trivalent (13)	holes	B, Al, Ga, In in Si

Mass-action law: $n_e n_h = n_i^2$. A doped semiconductor is still electrically **neutral**. Band gaps: conductor ≈ 0 (bands overlap), Si ≈ 1.1 eV, Ge ≈ 0.7 eV, insulator > 3 eV. At 0 K an intrinsic semiconductor behaves like an insulator. Conductivity **rises** with temperature (resistance falls).

**FORMULA** p–n junction & diodes

Barrier potential: Si ≈ 0.7 V, Ge ≈ 0.3 V. **Forward bias** narrows the depletion layer. **Reverse bias** widens it, leaving a tiny reverse saturation current (minority carriers) until breakdown.

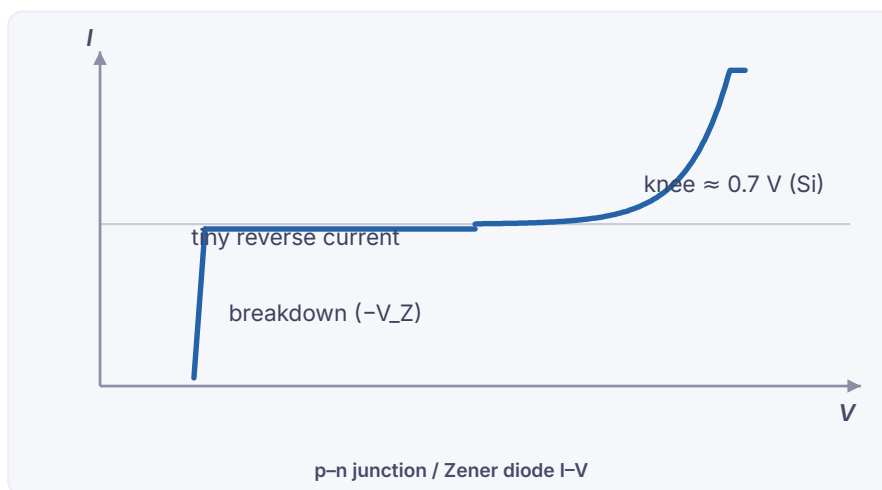
Device	Bias	Principle / use
Rectifier diode	forward conducts	half-wave: output ripple at f ; full-wave: at $2f$
Zener diode	reverse (breakdown)	constant V_Z across the load $\rightarrow$ voltage regulator
LED	forward	recombination emits photons, $\lambda = \frac{hc}{E_g}$
Photodiode	reverse	light raises the reverse current (light detector)
Solar cell	none (self-generating)	I–V curve in the 4th quadrant; V_{oc} , I_{sc}

Zener regulator: $I_R = \frac{V_{in} - V_Z}{R_s}$, $I_L = \frac{V_Z}{R_L}$, $I_Z = I_R - I_L$. **Diode in a circuit:** ideal = switch; real Si = 0.7 V drop when conducting.

Σ FORMULA Logic gates

A	B	AND	OR	NAND	NOR	XOR
0	0	0	0	1	1	0
0	1	0	1	1	0	1
1	0	0	1	1	0	1
1	1	1	1	0	0	0

NOT: $\bar{A}$. **De Morgan:** $\overline{A+B} = \bar{A} \cdot \bar{B}$ and $\overline{A \cdot B} = \bar{A} + \bar{B}$. With NAND gates: NOT = 1 gate (inputs tied), AND = 2 gates, OR = 3 gates. With NOR gates: NOT = 1, OR = 2, AND = 3.



Frequently tested concepts & question patterns

▮ PYQ PATTERN Recurring structures

1. **Gate combinations:** identify the output or the equivalent gate of a 2–3 gate circuit (very frequent).
2. **Zener regulator currents.**
3. **Diode circuits:** which diode conducts; current with ideal or 0.7 V diodes.
4. **Rectifier** output frequency and waveform.
5. **Carrier concentration** with $n_e n_h = n_i^2$.
6. **LED colour ↔ band gap;** photodiode and solar-cell bias facts.

Shortcuts

⚡ SHORTCUT Boolean algebra over truth tables

For gate puzzles, write the Boolean expression and simplify with De Morgan. It's usually faster than a 4-row table and less error-prone. Example: NOR followed by NOT is $\overline{\overline{A+B}} = A+B$ (OR).

Question classification

Type	Difficulty	Target time
Gate identification	E	45 s
Zener currents	E	1 min
Diode circuit current	E–M	1.5 min
Doping / carrier concentration	E	45 s
Special diode facts	E	30 s

Common mistakes



MISTAKE ALERT Mistake alerts

- The Zener is used in **reverse** bias, the LED in **forward** bias, the photodiode in **reverse** bias.
- n-type doesn't mean negatively charged. The material stays neutral.
- Full-wave rectifier ripple frequency is **2f** (100 Hz for 50 Hz mains).
- Forgetting the 0.7 V drop when the question says "silicon diode".

Practice questions

P19 · Practice

1

Easy

0.5 min

Speed

P19-01

Silicon doped with arsenic is:

- (A) p-type, with holes as majority carriers
 (B) n-type, with electrons as majority carriers
 (C) intrinsic
 (D) an insulator

2

Medium

1 min

Calc

P19-02

An intrinsic semiconductor has $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$. After doping, the electron concentration is $4.5 \times 10^{22} \text{ m}^{-3}$. The hole concentration is:

- (A) $5 \times 10^9 \text{ m}^{-3}$ (B) $3 \times 10^6 \text{ m}^{-3}$ (C) $1.5 \times 10^{16} \text{ m}^{-3}$ (D) $4.5 \times 10^{22} \text{ m}^{-3}$

3

Easy

0.5 min

Speed

P19-03

The ripple frequency at the output of a full-wave rectifier fed from 50 Hz mains is:

- (A) 25 Hz (B) 50 Hz (C) 100 Hz (D) 200 Hz

4

Medium

1.5 min

NV

P19-04

A Zener diode with $V_Z = 5 \text{ V}$ is connected, through a 500Ω series resistor, to a 15 V supply. The load across the Zener is $1 \text{ k}\Omega$. Find the Zener current (in mA).

Your answer (numerical value):

5

Easy

0.75 min

Basic

P19-05

An LED has a band gap of 2.0 eV. The wavelength of the light it emits is about:

- (A) 400 nm (B) 500 nm (C) 620 nm (D) 900 nm

6

Easy

0.5 min

Concept

P19-06

A NAND gate with both inputs joined together acts as:

- (A) AND (B) OR (C) NOT (D) NOR

7

Easy

0.75 min

Concept

P19-07

The output of a NOR gate is fed into a NOT gate. The combination is equivalent to:

- (A) AND (B) OR (C) NAND (D) XOR

8 Easy 1 min Basic

P19-08

A silicon diode (0.7 V drop when conducting) is forward biased in series with a $1\text{ k}\Omega$ resistor across a 5 V battery. The current is:

- (A) 5.0 mA (B) 4.3 mA (C) 5.7 mA (D) 0.7 mA

9 Easy 0.5 min Concept

P19-09

A photodiode is normally operated in:

- (A) forward bias (B) reverse bias (C) zero bias only (D) breakdown

P19 · Chapter Test

1 Easy 0.5 min Speed

P19-T1

The correct order of band gaps is:

- (A) conductor > semiconductor > insulator (B) insulator > semiconductor > conductor
(C) semiconductor > insulator > conductor (D) all equal

2 Easy 0.5 min Speed

P19-T2

In reverse bias, the depletion layer of a p-n junction:

- (A) narrows (B) widens (C) disappears (D) is unchanged

3 Easy 0.5 min Concept

P19-T3

A device whose working I-V characteristic lies in the fourth quadrant is a:

- (A) Zener diode (B) LED (C) solar cell (D) rectifier diode

4 Easy 0.5 min NV

P19-T4

What is the minimum number of NAND gates needed to realise an OR gate?

Your answer (numerical value):

5 Easy 0.5 min Speed

P19-T5

At 0 K, a pure semiconductor behaves as:

- (A) a perfect conductor (B) an insulator
(C) a superconductor (D) an n-type semiconductor

Answers & Solutions

P19 · Practice

1 (B) 2 (A) 3 (C) 4 15 5 (C) 6 (C) 7 (B) 8 (B) 9 (B)

P19 · Chapter Test

1 (B) 2 (B) 3 (C) 4 3 5 (B)

P19 · PRACTICE

1 Answer: (B) Type of doping Easy 0.5 min Speed

P19-01

SOLUTION As is pentavalent (group 15), so it donates electrons.

2 Answer: (A) Mass-action law **Medium** ⌚ 1 min **Calc**

P19-02

SOLUTION $n_h = \frac{n_i^2}{n_e} = \frac{2.25 \times 10^{32}}{4.5 \times 10^{22}} = 5 \times 10^9 \text{ m}^{-3}.$

COMMON TRAP Assuming $n_h = n_i$.

3 Answer: (C) Rectifier ripple **Easy** ⌚ 0.5 min **Speed**

P19-03

SOLUTION Both half-cycles are rectified, so the output repeats at $2f$.

COMMON TRAP Using the half-wave value.

4 Answer: 15 Zener regulator **Medium** ⌚ 1.5 min **NV**

P19-04

SOLUTION $I_R = (15 - 5)/500 = 20 \text{ mA}$; $I_L = 5/1000 = 5 \text{ mA}$; $I_Z = 15 \text{ mA}$.

COMMON TRAP Forgetting to subtract the load current.

5 Answer: (C) LED wavelength from band gap **Easy** ⌚ 0.75 min **Basic**

P19-05

SOLUTION $\lambda = 1240/2.0 = 620 \text{ nm}$ (red–orange).

6 Answer: (C) NAND as NOT **Easy** ⌚ 0.5 min **Concept**

P19-06

SOLUTION $\overline{A \cdot A} = \bar{A}.$

7 Answer: (B) Gate combination **Easy** ⌚ 0.75 min **Concept**

P19-07

SOLUTION $\overline{A + B} = A \cdot B.$

8 Answer: (B) Silicon diode current **Easy** ⌚ 1 min **Basic**

P19-08

SOLUTION $I = (5 - 0.7)/1000 = 4.3 \text{ mA}.$

COMMON TRAP Ignoring the diode drop.

9 Answer: (B) Photodiode bias **Easy** ⌚ 0.5 min **Concept**

P19-09

SOLUTION In reverse bias the small current is very sensitive to light-generated carriers, so the change is easy to measure.

COMMON TRAP Confusing it with the LED.

P19 · CHAPTER TEST

1 Answer: (B) Band gaps **Easy** ⌚ 0.5 min **Speed**

P19-T1

SOLUTION Insulator ($> 3 \text{ eV}$) > semiconductor ($\sim 1 \text{ eV}$) > conductor (overlapping bands, ~ 0).

2 Answer: (B) Depletion width **Easy** ⌚ 0.5 min **Speed**

P19-T2

SOLUTION The applied field adds to the barrier field, pulling carriers away from the junction.

3 Answer: (C) Fourth-quadrant device **Easy** ⌚ 0.5 min **Concept**

P19-T3

SOLUTION A solar cell delivers power: current flows opposite to the voltage it develops across itself.

4 Answer: 3 Universal gates **Easy** ⌚ 0.5 min **NV**

P19-T4

SOLUTION $A + B = \overline{\bar{A} \cdot \bar{B}}$: two NANDs as inverters for $\bar{A}$ and $\bar{B}$, then one NAND.

5 Answer: (B) Semiconductor at 0 K **Easy** ⌚ 0.5 min **Speed**

P19-T5

SOLUTION No thermal energy is available to excite electrons across the gap.



REVISION Revision tracker · Electronic Devices

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART II · PHYSICS · UNIT 20

Experimental Skills

20

0–2 direct per shift
TYPICAL QUESTIONS

Easy
DIFFICULTY

High (predictable)
PRIORITY

Class 11 & 12
Physics Lab
Manuals
NCERT

~6 hours
STUDY TIME

Concept summary

The official syllabus lists **18 experiments** [NTA-SYL]. Questions test three things: **(1)** reading an instrument correctly (least count, zero error), **(2)** the formula and graph behind an experiment, and **(3)** which measurement contributes the most error. Learn each experiment as a one-line card: *aim* → *formula* → *graph* → *main error source*.

Measuring instruments

Σ FORMULA Vernier callipers & screw gauge

Vernier: $LC = 1 \text{ MSD} - 1 \text{ VSD}$. If $N \text{ VSD} = (N - 1) \text{ MSD}$, then $LC = \frac{\text{MSD}}{N}$. Reading = MSR + (coinciding VSD) × LC. Zero error: vernier zero to the **right** of the main zero is positive; to the **left** is negative. **Corrected = observed – zero error**. Parts: outer jaws (external diameter), inner jaws (internal diameter), depth strip (depth).

Screw gauge: $LC = \frac{\text{pitch}}{\text{no. of circular divisions}}$. Reading = MSR + CSR × LC. **Backlash error** comes from play in the screw. Avoid it by always turning in one direction.

The 18 syllabus experiments on one page

#	Experiment	Key formula / relation	Graph	Main error source / tip
1	Vernier callipers: internal/external diameter, depth	LC, zero error	—	zero-error sign
2	Screw gauge: thickness/diameter of sheet/wire	LC = pitch/divisions	—	backlash, zero error
3	Simple pendulum: dissipation of energy	$A^2 \propto e^{-bt/m}$; $T = 2\pi\sqrt{L/g}$	A^2 vs t : exponential decay; L vs T^2 : straight line, slope $g/4\pi^2$	amplitude reading, air drag
4	Metre scale: mass by principle of moments	$m_1 l_1 = m_2 l_2$	—	knife-edge position
5	Young's modulus of a wire (Searle's)	$Y = \frac{MgL}{\pi r^2 \Delta L}$	load vs extension: line through origin	radius (r^2) and ΔL (smallest quantities)
6	Surface tension by capillary rise; effect of detergent	$T = \frac{r h \rho g}{2 \cos \theta}$	—	radius of bore; detergent lowers h
7	Viscosity by terminal velocity	$\eta = \frac{2r^2(\rho - \sigma)g}{9v_t}$	—	timing only after terminal speed is reached; wall effects
8	Speed of sound, resonance tube	$v = 2f(L_2 - L_1)$; $e = 0.3d$	—	end correction cancels in $L_2 - L_1$
9	Specific heat of a solid/liquid (mixtures)	$m_s c_s (T_s - T) = (m_w c_w + W)(T - T_w)$	—	heat loss to the surroundings
10	Resistivity using a metre bridge	$\frac{R}{S} = \frac{l}{100 - l}$, $\rho = \frac{\pi r^2 R}{L}$	—	end errors (interchange R and S and average); balance near 50 cm

#	Experiment	Key formula / relation	Graph	Main error source / tip
1 1	Resistance by Ohm's law	$R = V/I$	$V-I$ straight line, slope R	heating of the wire
1 2	Galvanometer resistance & figure of merit (half deflection)	$G = \frac{RS}{R - S}; k = \frac{E}{(R + G)\theta}$	—	use $R \gg G$
1 3	Focal length by parallax: convex mirror, concave mirror, convex lens	$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ (mirror), $\frac{1}{v} - \frac{1}{u}$ (lens)	$u-v$ hyperbola; $1/u$ vs $1/v$ line with intercepts $1/f$	parallax removal; for a lens, $u = v = 2f$ point
1 4	$i-\delta$ curve for a prism	$n = \frac{\sin \frac{A+\delta_m}{2}}{\sin \frac{A}{2}}$	U-shaped, minimum at δ_m	pin alignment
1 5	Refractive index of a slab with a travelling microscope	$n = \frac{\text{real}}{\text{apparent}} = \frac{R_3 - R_1}{R_3 - R_2}$	—	focusing
1 6	p-n junction diode, forward & reverse characteristics	knee ≈ 0.7 V (Si)	exponential forward curve; tiny reverse current	use mA in forward, μ A in reverse
1 7	Zener diode characteristics, breakdown voltage	V_Z at the sharp rise in reverse current	vertical reverse breakdown	series resistor to limit current
1 8	Identify diode, LED, resistor, capacitor (multimeter)	—	—	see the table below

☆ IMPORTANT Component identification (Experiment 18)

Component	Multimeter behaviour
Resistor	same resistance in both directions
Diode	low resistance one way, very high the other way
LED	conducts one way and glows
Capacitor	momentary deflection (charging), then very high resistance

▮ PYQ PATTERN Recurring structures

1. Vernier/screw-gauge readings with zero error (often combined with the error in a derived quantity).
2. "Which measurement has the largest percentage error?" (usually the one raised to a power or the smallest reading).
3. Metre-bridge null points, interchange; galvanometer half-deflection.
4. Resonance tube: v and end correction from two resonance lengths.
5. Pendulum graphs ($L-T^2$ slope; A^2-t decay).
6. Travelling microscope readings $\rightarrow$ refractive index.

⚠ MISTAKE ALERT Mistake alerts

- Screw gauge: "the reference line is below/above zero" statements are orientation-dependent. Convert them into a **signed reading with the jaws closed** first.
- Metre bridge: l is measured from the end connected to the **unknown** in its gap, or whichever the question says. Read carefully.
- End correction of the resonance tube is $0.3 \times \text{diameter}$ ($0.6 \times \text{radius}$).

Practice questions

P20 · Practice

1 **Easy** ⌚ 0.5 min **Speed** P20-01

In a Vernier calliper, 20 VSD coincide with 19 MSD, and 1 MSD = 1 mm. The least count is:

- (A) 0.1 mm (B) 0.05 mm (C) 0.02 mm (D) 0.5 mm

2 **Medium** ⌚ 1.5 min **JEE** P20-02

A screw gauge has pitch 1 mm and 100 circular divisions. With the studs closed it reads -0.03 mm. Measuring a wire, the main-scale reading is 4 mm and the circular-scale reading is 52. The corrected diameter is:

- (A) 4.49 mm (B) 4.52 mm (C) 4.55 mm (D) 4.82 mm

3 **Medium** ⌚ 1.5 min **Calc** P20-03

In the half-deflection method, the galvanometer shows deflection θ with $5000\ \Omega$ in series. Connecting a $50\ \Omega$ shunt across the galvanometer reduces the deflection to $\theta/2$. The galvanometer resistance is about:

- (A) $50\ \Omega$ (B) $50.5\ \Omega$ (C) $49.5\ \Omega$ (D) $100\ \Omega$

4 **Easy** ⌚ 0.5 min **Concept** P20-04

In the simple-pendulum energy-dissipation experiment, the graph of (amplitude)² against time is:

- (A) a straight line with positive slope (B) an exponentially decaying curve
(C) a parabola (D) constant

5 **Easy** ⌚ 1 min **Calc** P20-05

Water rises 3 cm in a capillary of radius 0.5 mm (contact angle 0° , $\rho = 1000$, $g = 10$). The surface tension is:

- (A) 0.075 N/m (B) 0.15 N/m (C) 0.0375 N/m (D) 0.75 N/m

6 **Medium** ⌚ 1 min **Basic** P20-06

Travelling-microscope readings: mark on paper (no slab) 10.20 mm, mark seen through the slab 13.20 mm, top surface of the slab 19.20 mm. The refractive index of the slab is:

- (A) 1.33 (B) 1.50 (C) 1.60 (D) 2.00

7 **Medium** ⌚ 1.5 min **Calc** P20-07

A 0.1 kg metal piece at 100°C is dropped into 0.2 kg of water at 20°C (negligible calorimeter heat capacity), and the final temperature is 24°C . With $c_w = 4200\ \text{J kg}^{-1}\text{K}^{-1}$, the metal's specific heat is about:

- (A) 221 J/kg K (B) 442 J/kg K (C) 884 J/kg K (D) 336 J/kg K

8 **Easy** ⌚ 0.5 min **Speed** P20-08

The end correction of a resonance tube of internal diameter 5 cm is about:

- (A) 0.6 cm (B) 1.5 cm (C) 3.0 cm (D) 5.0 cm

9 **Easy** ⌚ 0.5 min **Concept** P20-09

Tested with a multimeter, a component shows a momentary deflection that then falls to (almost) zero current. The component is a:

- (A) resistor (B) diode (C) capacitor (D) LED

10 Easy 0.75 min NV

P20-10

In the $u-v$ experiment with a convex lens, the object and image distances are equal at 40 cm. Find the focal length (in cm).

Your answer (numerical value):

P20 · Chapter Test

1 Easy 0.75 min Concept

P20-T1

In Searle's experiment for Young's modulus, the largest contribution to the percentage error usually comes from measuring:

- (A) the length L of the wire (B) the load M
(C) the radius r of the wire (D) g

2 Easy 0.5 min Speed

P20-T2

A screw gauge has pitch 0.5 mm and 50 circular divisions. Its least count is:

- (A) 0.1 mm (B) 0.01 mm (C) 0.001 mm (D) 0.05 mm

3 Easy 0.5 min Speed

P20-T3

In the viscosity experiment, the radius of the ball is doubled (same material). Its terminal velocity becomes:

- (A) 2 times (B) 4 times (C) 8 times (D) half

4 Easy 0.5 min NV

P20-T4

Find the least count (in μm) of a screw gauge with pitch 1 mm and 200 circular divisions.

Your answer (numerical value):

Answers & Solutions

P20 · Practice

1 (B) 2 (C) 3 (B) 4 (B) 5 (A) 6 (B) 7 (B) 8 (B) 9 (C) 10 20

P20 · Chapter Test

1 (C) 2 (B) 3 (B) 4 5

P20 · PRACTICE

1 Answer: (B) Vernier least count Easy 0.5 min Speed

P20-01

SOLUTION $LC = MSD/N = 1/20 = 0.05 \text{ mm}$.

2 Answer: (C) Screw gauge, negative zero error Medium 1.5 min JEE

P20-02

SOLUTION $LC = 0.01 \text{ mm}$; observed = $4 + 0.52 = 4.52 \text{ mm}$; corrected = $4.52 - (-0.03) = 4.55 \text{ mm}$.

SHORTCUT Subtracting a negative error means adding.

COMMON TRAP Subtracting 0.03 gives 4.49.

3 Answer: (B) Half-deflection method Medium 1.5 min Calc

P20-03

SOLUTION $G = \frac{RS}{R+S} = \frac{5000 \times 50}{4950} \approx 50.5 \Omega$.

SHORTCUT When $R \gg G$, $G \approx S$.

4 Answer: (B) Pendulum energy dissipation graph Easy 0.5 min Concept

P20-04

SOLUTION Damping makes the energy ($\propto A^2$) decay exponentially: $A^2 \propto e^{-bt/m}$.

5 Answer: (A) Surface tension from capillary rise **Easy** ⌚ 1 min **Calc**

P20-05

SOLUTION $T = \frac{r h \rho g}{2} = \frac{5 \times 10^{-4} \times 0.03 \times 1000 \times 10}{2} = 0.075 \text{ N/m.}$

COMMON TRAP Forgetting the 2.

6 Answer: (B) Travelling microscope **Medium** ⌚ 1 min **Basic**

P20-06

SOLUTION Real thickness = $19.20 - 10.20 = 9.00 \text{ mm}$. Apparent = $19.20 - 13.20 = 6.00 \text{ mm}$. $n = 9/6 = 1.5$.

COMMON TRAP Using $13.20 - 10.20$ as the apparent thickness (that's the shift).

7 Answer: (B) Method of mixtures **Medium** ⌚ 1.5 min **Calc**

P20-07

SOLUTION $0.1c(76) = 0.2(4200)(4) = 3360 \Rightarrow c = 3360/7.6 \approx 442 \text{ J/kg K.}$

COMMON TRAP Using $100 - 20$ instead of $100 - 24$.

8 Answer: (B) End correction **Easy** ⌚ 0.5 min **Speed**

P20-08

SOLUTION $e = 0.3d = 1.5 \text{ cm.}$

COMMON TRAP Using $0.6d$.

9 Answer: (C) Identifying a capacitor **Easy** ⌚ 0.5 min **Concept**

P20-09

SOLUTION The capacitor charges briefly, then blocks DC.

10 Answer: 20 $u = v$ point of a convex lens **Easy** ⌚ 0.75 min **NV**

P20-10

SOLUTION $|u| = |v| = 2f \Rightarrow f = 20 \text{ cm.}$

P20 · CHAPTER TEST

1 Answer: (C) Largest error in Searle's experiment **Easy** ⌚ 0.75 min **Concept**

P20-T1

SOLUTION r is small and appears squared in $Y = MgL/(\pi r^2 \Delta L)$, so its percentage error is doubled. (The extension ΔL is also critical, which is why a micrometer is used.)

SHORTCUT Small quantity $\times$ high power $\Rightarrow$ largest error.

COMMON TRAP Choosing L (large, easy to measure precisely).

2 Answer: (B) Screw gauge least count **Easy** ⌚ 0.5 min **Speed**

P20-T2

SOLUTION $0.5/50 = 0.01 \text{ mm.}$

3 Answer: (B) Terminal velocity scaling **Easy** ⌚ 0.5 min **Speed**

P20-T3

SOLUTION $v_t \propto r^2$.

4 Answer: 5 Least count in micrometres **Easy** ⌚ 0.5 min **NV**

P20-T4

SOLUTION $1 \text{ mm}/200 = 0.005 \text{ mm} = 5 \mu\text{m.}$

REVISION Revision tracker · Experimental Skills

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

Chemistry



Chemistry strategy and the NCERT method, then Physical, Inorganic and Organic modules with reaction maps, reagent tables and NCERT fact sheets.

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09	Classification of Elements & Periodicity	206
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11	d- and f-Block Elements	217
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13	Purification & Characterisation of Organic Compounds	230
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PART III · CHEMISTRY

Chemistry: The Complete Strategy

Why Chemistry decides 99 percentile

Chemistry is usually the **fastest and most accurate** section for well-prepared students. In 2026 analyses it was repeatedly described as the section where NCERT-strong students gained their edge, with several questions taken almost directly from NCERT text and lab manuals [||| THIRD-PARTY ANALYSIS](#) [T-ANALYSIS26]. A realistic 99+ plan treats Chemistry as the **mark bank**: 80–90/100 in about 45 minutes, leaving more time for Mathematics.

VERIFIED FACT Syllabus structure (official)

The official Chemistry syllabus has **20 units**: 8 Physical (Units 1–8), 4 Inorganic (Units 9–12) and 8 Organic (Units 13–20, including Unit 20, Principles Related to Practical Chemistry). The 2024 revision removed States of Matter, Surface Chemistry, Hydrogen, s-Block, Metallurgy, Environmental Chemistry, Polymers and Chemistry in Everyday Life [NTA-SYL], [T-DELETED].

Unit-wise weight and priority

"Typical questions per shift" is compiled from published 2024–2026 analyses [||| THIRD-PARTY ANALYSIS](#) [T-WT-VED], [T-WT-CD]. Priority is this book's recommendation [📌 RECOMMENDATION](#).

#	Unit	Branch	Typical Q / shift	Priority	Hours
1	Some Basic Concepts	Physical	~1	High	5
2	Atomic Structure	Physical	1–2	Must-do	7
3	Chemical Bonding	Physical/Inorganic	1–2	Must-do	12
4	Chemical Thermodynamics	Physical	1–2	Must-do	9
5	Solutions	Physical	~1	High	6
6	Equilibrium (chemical + ionic)	Physical	1–2	Must-do	12
7	Redox & Electrochemistry	Physical	1–2	Must-do	9
8	Chemical Kinetics	Physical	1–2	Must-do	7
9	Periodicity	Inorganic	~1	Must-do	5
10	p-Block (Groups 13–18)	Inorganic	1–2	Must-do	12
11	d- & f-Block	Inorganic	1–2	Must-do	8
12	Coordination Compounds	Inorganic	2–3	Must-do	10
13	Purification & Characterisation	Organic	0–1	High	4
14	Basic Principles (GOC)	Organic	1–2	Must-do	12
15	Hydrocarbons	Organic	~1	High	8
16	Halogen compounds	Organic	~1	High	6
17	Oxygen compounds	Organic	2–3	Must-do	14
18	Nitrogen compounds	Organic	~1	High	6
19	Biomolecules	Organic	~1	Must-do (fast)	4
20	Practical Chemistry	Organic/Inorganic	~1	High	5

PYQ PATTERN Published 2026 counts (all Paper-1 shifts, both sessions)

Chapter counts for JEE Main 2026 reported by Collegedunia from the ExamSIDE PYQ archive

 **THIRD-PARTY ANALYSIS** [T-WT-CD]: **Coordination Compounds 36** questions (7.58%) · **Aldehydes, Ketones & Carboxylic Acids 35** (7.37%) · **GOC 27** (5.68%) · **Chemical Kinetics 27** (5.68%) · **Thermodynamics 22** (4.63%) · **Electrochemistry 21** (4.42%) · **Salt analysis 5** (1.05%). On about 19 shifts, that's roughly **2 questions per shift** each for Coordination and Carbonyls.

How to maximise marks in Chemistry

Strong student (≥ 80)

- Target 23–24 attempts at 95% accuracy in ≤ 40 minutes.
- Remaining losses usually come from NCERT exceptions and unusual organic conversions. Keep an **exceptions notebook** and read it weekly.
- Solve 2024–2026 PYQs **line by line against NCERT**, and highlight the NCERT sentence each one came from.

Average student (55–80)

- Physical Chemistry formulas + Coordination + Biomolecules + Practical Chemistry = the quickest 20 marks to add.
- Organic: learn the reaction maps (not isolated reactions) for Units 15–18.
- Inorganic: one NCERT reading **every 10 days** for Units 9–12 (spaced repetition).

Weak student (< 55)

- Start with **NCERT reading**, not question banks. Chemistry is where reading converts to marks most directly.
- Order: Biomolecules → Practical Chemistry → Coordination → Periodicity → Atomic Structure → Mole concept → Kinetics → Solutions → Electrochemistry → GOC → carbonyl compounds.
- Attempt only questions you recognise within 20 seconds. Chemistry questions reward recall, and hesitation usually signals a gap.

PART III · CHEMISTRY

The NCERT Strategy

Why NCERT, and how exactly

VERIFIED FACT Alignment

The 2024 syllabus revision aligned JEE Main with the **rationalised NCERT** textbooks [NTA-SYL]. Inorganic Chemistry, Biomolecules and Practical Chemistry questions are frequently answerable from NCERT text, tables and in-text examples  **THIRD-PARTY ANALYSIS** [T-ANALYSIS26].

The 4-read method RECOMMENDATION

- Read 1 · Understand** (with the module): read the whole chapter, including boxes, tables and in-text questions.
- Read 2 · Mark** (Day 7): highlight every fact that could be an MCQ: exceptions, colours, shapes, orders, named reactions, uses.
- Read 3 · Extract** (Day 14): copy the highlights into this book's NCERT checklists or your own fact sheet.
- Read 4+ · Recall** (every 10–14 days): cover the page and recall the table, then check. Only the fact sheet is re-read after this point.

NCERT chapter checklist (map to JEE units)

JEE unit	NCERT source (rationalised editions)	Read 1	Read 2	Read 3	Recall
1 Basic Concepts	Class 11 · Ch 1 Some Basic Concepts of Chemistry	☐	☐	☐	☐
2 Atomic Structure	Class 11 · Ch 2 Structure of Atom	☐	☐	☐	☐
3 Bonding	Class 11 · Ch 4 Chemical Bonding & Molecular Structure	☐	☐	☐	☐
4 Thermodynamics	Class 11 · Ch 5 Thermodynamics	☐	☐	☐	☐
5 Solutions	Class 12 · Ch 1 Solutions	☐	☐	☐	☐
6 Equilibrium	Class 11 · Ch 6 Equilibrium	☐	☐	☐	☐
7 Redox & Electrochemistry	Class 11 · Ch 7 Redox Reactions; Class 12 · Ch 2 Electrochemistry	☐	☐	☐	☐
8 Kinetics	Class 12 · Ch 3 Chemical Kinetics	☐	☐	☐	☐
9 Periodicity	Class 11 · Ch 3 Classification of Elements	☐	☐	☐	☐
10 p-Block (Groups 13–18)	Not in the rationalised books. Use the p-Block chapters of the pre-2023 NCERT editions (Class 11 Groups 13–14; Class 12 Groups 15–18) for trends	☐	☐	☐	☐
11 d- & f-Block	Class 12 · Ch 4 The d- and f-Block Elements	☐	☐	☐	☐
12 Coordination	Class 12 · Ch 5 Coordination Compounds	☐	☐	☐	☐
13 Purification & Characterisation	Class 11 · Ch 8 (sections on purification, qualitative & quantitative analysis)	☐	☐	☐	☐
14 GOC	Class 11 · Ch 8 Organic Chemistry: Some Basic Principles & Techniques	☐	☐	☐	☐
15 Hydrocarbons	Class 11 · Ch 9 Hydrocarbons	☐	☐	☐	☐
16 Halogen compounds	Class 12 · Ch 6 Haloalkanes & Haloarenes	☐	☐	☐	☐

JEE unit	NCERT source (rationalised editions)	Read 1	Read 2	Read 3	Recall
17 Oxygen compounds	Class 12 · Ch 7 Alcohols, Phenols & Ethers; Ch 8 Aldehydes, Ketones & Carboxylic Acids	☐	☐	☐	☐
18 Nitrogen compounds	Class 12 · Ch 9 Amines	☐	☐	☐	☐
19 Biomolecules	Class 12 · Ch 10 Biomolecules	☐	☐	☐	☐
20 Practical Chemistry	NCERT Laboratory Manual Chemistry, Class 11 & 12 (salt analysis, titrations, preparations, functional-group tests)	☐	☐	☐	☐

⚠ WARNING p-Block needs extra care

The syllabus lists **Groups 13 to 18: general trends and the unique behaviour of the first element**. The rationalised NCERT books no longer contain these chapters. Use the p-Block module in this book (page [211](#)) together with the older NCERT chapters, and **stay at the level of trends and anomalies**. Detailed compound preparations were not the focus of the revised syllabus text.

Important factual areas (what to extract from NCERT)

Tables & trends to memorise

- Periodic trends: atomic/ionic radii, IE, EGE, EN, with every exception
- Shapes & hybridisation (VSEPR table)
- MO diagrams for B_2 to Ne_2 : bond order, magnetism
- d-block: electronic configurations (Cr, Cu anomalies), oxidation states, colours, magnetic moments
- Lanthanoid contraction and its consequences
- Spectrochemical series; crystal field splitting (Δ_o vs pairing)
- Acid/base strengths: carboxylic acids, phenols, amines (gas vs water)
- Vitamins and deficiency diseases; bases in DNA vs RNA

Reactions, exceptions & lines

- Preparation of $K_2Cr_2O_7$ and $KMnO_4$ (steps and equations)
- All named reactions in Class 12 organic chapters (list on page [252](#))
- Tests: Lucas, iodoform, Tollens', Fehling's, Hinsberg, carbylamine, $FeCl_3$ (phenol)
- Reimer-Tiemann, Kolbe, Cannizzaro, aldol, Gabriel phthalimide, Hofmann bromamide, Sandmeyer, diazo coupling
- Glucose: reactions showing its structure (HI, NH_2OH , Br_2 water, HNO_3 , acetylation)
- Anomalous behaviour of the first element of each p-block group
- Uses of coordination compounds (EDTA, cisplatin, Wilkinson's catalyst, estimation of hardness)
- Every "Why?" statement in the in-text questions

Chemistry revision method

- Physical:** formula sheet + 10 numerical questions per chapter per week. Formula recall on a blank sheet every 5 days.
- Inorganic:** fact sheets read on a **spaced schedule** (Day 1, 3, 7, 14, 30). 15-minute "flash review" every night in the final 45 days.
- Organic:** redraw each **reaction map** from memory once a week; solve 20 conversion questions per week.
- Mistake Book:** every Chemistry error tagged *fact* / *concept* / *exception* / *calculation*. Exceptions become flash cards.

PART III · CHEMISTRY · UNIT 01

Some Basic Concepts in Chemistry

01

~1 per shift (+
used in every
numerical)

TYPICAL QUESTIONS

Easy

DIFFICULTY

High

PRIORITY

Class 11 · Ch 1

NCERT

~5 hours

STUDY TIME

Important concepts

- **Laws of chemical combination:** conservation of mass; definite proportions (a compound always has the same composition); multiple proportions (masses of B combining with a fixed mass of A are in small whole-number ratios, as in CO/CO₂); Gay-Lussac's law of gaseous volumes; Avogadro's law (equal volumes of gases at the same T , P contain equal numbers of molecules).
- **Dalton's atomic theory:** matter is made of indivisible atoms. Atoms of an element are identical, and compounds form in fixed whole-number ratios.
- **Atomic mass unit:** $1 \text{ u} = \frac{1}{12}$ of the mass of one ^{12}C atom $= 1.66 \times 10^{-24} \text{ g}$.
- **Mole:** 6.022×10^{23} entities (N_A). Molar mass (g/mol) is numerically equal to the atomic or molecular mass in u.

Formula sheet

Σ FORMULA Mole relations

$$n = \frac{m}{M} = \frac{N}{N_A} = \frac{V_{\text{gas}}}{V_m} \quad V_m = 22.4 \text{ L (273 K, 1 atm)} \quad \text{or} \quad 22.7 \text{ L (273.15 K, 1 bar, NCERT STP)}$$

$$\text{Mass \% of element} = \frac{(\text{atoms}) \times (\text{atomic mass})}{\text{molar mass}} \times 100 \quad \text{Molecular formula} = n \times \text{empirical formula}, \quad n = \frac{M}{M_{\text{empirical}}}$$

Σ FORMULA Concentration terms

Term	Definition	Temperature dependent?
Mass % (w/w)	$\frac{m_{\text{solute}}}{m_{\text{solution}}} \times 100$	No
Molarity M	mol solute / L solution	Yes (volume changes)
Molality m	mol solute / kg solvent	No
Mole fraction x	$n_i / \sum n$	No
ppm	$\frac{m_{\text{solute}}}{m_{\text{solution}}} \times 10^6$	No

Conversions: $m = \frac{1000M}{1000d - MM_B}$ (d in g/mL, M_B = molar mass of solute). Dilution: $M_1V_1 = M_2V_2$. Mixing: $M_{\text{mix}} = \frac{M_1V_1 + M_2V_2}{V_1 + V_2}$. Mole fraction of solute in a 1 molal aqueous solution: $\frac{1}{1 + 55.5} \approx 0.018$.

Numerical-solving method: stoichiometry in 4 steps

1. Write and **balance** the equation.
2. Convert every given quantity to **moles**.
3. Find the **limiting reagent**: divide moles by the stoichiometric coefficient; the smallest value is limiting.
4. Use mole ratios from the limiting reagent, then convert to the unit asked.

⚡ SHORTCUT Empirical formula in 30 seconds

Divide each mass % by its atomic mass, then divide by the smallest result. If you get x.5, double everything; x.33 or x.67, triple; x.25, quadruple. Example: C 40, H 6.67, O 53.3 $\rightarrow$ 3.33 : 6.67 : 3.33 $\rightarrow$ 1 : 2 : 1, so CH_2O .

Common traps**⚠ MISTAKE ALERT Mistake alerts**

- Molarity uses the volume of **solution**; molality uses the mass of **solvent** (not solution).
- Counting atoms vs molecules: 1 mol of NH_3 contains 4 mol of atoms.
- Using 22.4 L for gases **not** at STP. Use $PV = nRT$ instead.
- Forgetting to check the limiting reagent when both reactant amounts are given.
- Mass % of an element: multiply by the **number of atoms** of that element in the formula.

Question patterns**📊 PYQ PATTERN Recurring structures**

1. Moles/atoms/molecules counting; mass of one molecule.
2. Empirical and molecular formula from % composition or combustion data.
3. Limiting reagent and yield.
4. Concentration interconversion (molarity $\leftrightarrow$ molality $\leftrightarrow$ mole fraction) with density.
5. Volume of gases in reactions (usually as a Section-B numerical).

Practice questions**✍ C01 · Practice**

1 Easy ⌚ 1 min Basic C01-01

The total number of atoms in 4.25 g of NH_3 is about:

- (A) 1.5×10^{23} (B) 6.0×10^{23} (C) 2.4×10^{24} (D) 3.0×10^{23}

2 Easy ⌚ 1 min Basic C01-02

A compound contains C 40.0%, H 6.67% and O 53.3% by mass. Its empirical formula is:

- (A) CHO (B) CH_2O (C) $\text{C}_2\text{H}_4\text{O}$ (D) CH_2O_2

3 Medium ⌚ 1.5 min NV C01-03

3 g of H_2 reacts with 32 g of O_2 to form water. Find the mass of water formed (in g).

Your answer (numerical value):

4 Easy ⌚ 0.5 min Speed C01-04

5.85 g of NaCl is dissolved in water to make 500 mL of solution. The molarity is:

- (A) 0.1 M (B) 0.2 M (C) 0.01 M (D) 2 M

5 Medium ⌚ 1.5 min Calc C01-05

A 2 M NaOH solution has density 1.1 g/mL. Its molality is about:

- (A) 1.82 m (B) 1.96 m (C) 2.00 m (D) 2.20 m

6 Easy 1 min Basic

C01-06

The mole fraction of solute in a 1 molal aqueous solution is about:

- (A) 0.018 (B) 0.1 (C) 0.5 (D) 0.001

7 Easy 0.5 min Concept

C01-07

In CO and CO₂, the masses of oxygen that combine with a fixed mass of carbon are in the ratio:

- (A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 3 : 8

8 Easy 0.5 min Speed

C01-08

A compound with empirical formula CH₂O has molar mass 180 g/mol. Its molecular formula is:

- (A) C₃H₆O₃ (B) C₆H₁₂O₆ (C) C₁₂H₂₂O₁₁ (D) C₂H₄O₂

9 Easy 1 min Basic

C01-09

The volume of oxygen at STP (take 22.4 L/mol) needed to burn 1 mol of methane completely is:

- (A) 22.4 L (B) 44.8 L (C) 67.2 L (D) 11.2 L

C01 · Chapter Test

1 Easy 0.5 min Speed

C01-T1

The mass of one ¹²C atom is about:

- (A) 12 g (B) 1.99×10^{-23} g (C) 1.66×10^{-24} g (D) 6.02×10^{-23} g

2 Easy 0.5 min Speed

C01-T2

The percentage of nitrogen in urea, NH₂CONH₂, is about:

- (A) 23.3% (B) 46.7% (C) 28.0% (D) 60.0%

3 Easy 0.5 min NV

C01-T3

Find the number of moles of CO₂ formed by the complete combustion of 2 mol of propane (C₃H₈).

Your answer (numerical value):

4 Easy 0.5 min Speed

C01-T4

100 mL of 0.5 M HCl is diluted to 250 mL. The new molarity is:

- (A) 0.25 M (B) 0.2 M (C) 1.25 M (D) 0.5 M

5 Easy 0.75 min Concept

C01-T5

Which of the following contains the largest number of atoms?

- (A) 1 g H₂ (B) 1 g He (C) 1 g O₂ (D) 1 g N₂

Answers & Solutions

C01 · Practice

1 (B) 2 (B) 3 27 4 (B) 5 (B) 6 (A) 7 (B) 8 (B) 9 (B)

C01 · Chapter Test

1 (B) 2 (B) 3 6 4 (B) 5 (A)

C01 · PRACTICE

1 Answer: (B) Counting atoms **Easy** ⌚ 1 min **Basic** C01-01

SOLUTION $n = 4.25/17 = 0.25$ mol of molecules; atoms = $0.25 \times 4 \times N_A = N_A \approx 6.0 \times 10^{23}$.

SHORTCUT $0.25 \text{ mol} \times 4 \text{ atoms} = 1 \text{ mol of atoms}$.

COMMON TRAP Counting molecules only (1.5×10^{23}).

2 Answer: (B) Empirical formula **Easy** ⌚ 1 min **Basic** C01-02

SOLUTION Moles: C $40/12 = 3.33$, H $6.67/1 = 6.67$, O $53.3/16 = 3.33$. Ratio 1 : 2 : 1.

3 Answer: 27 Limiting reagent **Medium** ⌚ 1.5 min **NV** C01-03

SOLUTION $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. H_2 : 1.5 mol ($\div 2 = 0.75$); O_2 : 1 mol ($\div 1 = 1$). H_2 is limiting, so it gives 1.5 mol $\text{H}_2\text{O} = 27 \text{ g}$.

SHORTCUT Divide moles by coefficients; the smaller value is limiting.

COMMON TRAP Using O_2 as limiting gives 36 g.

4 Answer: (B) Molarity **Easy** ⌚ 0.5 min **Speed** C01-04

SOLUTION $n = 5.85/58.5 = 0.1$ mol; $M = 0.1/0.5 = 0.2 \text{ M}$.

COMMON TRAP Dividing by 500 mL as if it were 1 L.

5 Answer: (B) Molarity $\rightarrow$ molality **Medium** ⌚ 1.5 min **Calc** C01-05

SOLUTION Per litre: solution mass 1100 g, solute $2 \times 40 = 80 \text{ g}$, solvent 1020 g. $m = 2/1.020 \approx 1.96 \text{ m}$.

SHORTCUT $m = \frac{1000M}{1000d - MM_B} = \frac{2000}{1100 - 80}$.

COMMON TRAP Using the solution mass (1.1 kg) gives 1.82.

6 Answer: (A) Mole fraction from molality **Easy** ⌚ 1 min **Basic** C01-06

SOLUTION 1 mol solute per 1000 g water (55.5 mol): $x = 1/56.5 \approx 0.0177$.

7 Answer: (B) Law of multiple proportions **Easy** ⌚ 0.5 min **Concept** C01-07

SOLUTION 12 g C combines with 16 g O (CO) and 32 g O (CO_2), a ratio of 1 : 2.

COMMON TRAP Using mass percentages (3 : 8 is the C : O mass ratio in CO_2).

8 Answer: (B) Molecular formula **Easy** ⌚ 0.5 min **Speed** C01-08

SOLUTION $n = 180/30 = 6$.

9 Answer: (B) Gas volume in combustion **Easy** ⌚ 1 min **Basic** C01-09

SOLUTION $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$: 2 mol $\text{O}_2 = 44.8 \text{ L}$.

COMMON TRAP Not balancing the equation.

C01 · CHAPTER TEST

1 Answer: (B) Mass of one atom **Easy** ⌚ 0.5 min **Speed** C01-T1

SOLUTION $12/(6.022 \times 10^{23}) \approx 1.99 \times 10^{-23} \text{ g}$.

COMMON TRAP Choosing (C), which is the value of 1 u.

2 Answer: (B) Percentage composition **Easy** ⌚ 0.5 min **Speed** C01-T2

SOLUTION $28/60 \times 100 \approx 46.7\%$.

COMMON TRAP Counting only one N (23.3%).

3 Answer: 6 Combustion stoichiometry **Easy** ⌚ 0.5 min **NV** C01-T3

SOLUTION $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$: $2 \times 3 = 6 \text{ mol}$.

SHORTCUT Carbon balance: moles of CO_2 = moles of C atoms.

4 Answer: (B) Dilution **Easy** ⌚ 0.5 min **Speed** C01-T4

SOLUTION $M_2 = 0.5 \times 100/250 = 0.2 \text{ M}$.

5 Answer: (A) *Maximum number of atoms* **Easy** **0.75 min** **Concept**

C01-T5

SOLUTION Moles of atoms: H_2 : $0.5 \times 2 = 1$; He : 0.25 ; O_2 : $2/32 = 0.0625$; N_2 : $2/28 \approx 0.071$.

SHORTCUT Lowest mass per atom wins.

 **REVISION** Revision tracker · Basic Concepts

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III • CHEMISTRY • UNIT 02

Atomic Structure

02

1-2 per shift

TYPICAL QUESTIONS

Easy-Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 • Ch 2

NCERT

~7 hours

STUDY TIME

Important concepts

- **Electromagnetic radiation:** $c = \nu\lambda$, wavenumber $\bar{\nu} = 1/\lambda$. Planck's quantum: $E = h\nu$. The photoelectric effect supports the particle nature of light.
- **Bohr model** (one-electron species): quantised orbits. It explains the hydrogen spectrum but fails for multi-electron atoms, the Zeeman/Stark effects and the uncertainty principle.
- **Dual nature:** $\lambda = h/mv$ (de Broglie). **Heisenberg:** $\Delta x \cdot \Delta p \geq h/4\pi$, so exact orbits are impossible. Hence orbitals.
- **Quantum-mechanical model:** ψ is a one-electron wave function, and ψ^2 is the probability density. Orbitals are described by four quantum numbers.
- **Filling rules:** Aufbau ($n + l$ rule), Pauli exclusion, Hund's rule of maximum multiplicity. Half-filled and fully filled subshells give extra stability.

Formula sheet

Σ FORMULA Spectra & Bohr model

$$E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J} = -13.6 \frac{Z^2}{n^2} \text{ eV} \quad r_n = 52.9 \frac{n^2}{Z} \text{ pm} \quad v_n = 2.18 \times 10^6 \frac{Z}{n} \text{ m s}^{-1}$$

$$\bar{\nu} = \frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad R_H = 109\,677 \text{ cm}^{-1} \quad \text{lines} = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

Series: Lyman ($n_1 = 1$, UV), Balmer (2, visible), Paschen (3), Brackett (4), Pfund (5) (IR).

Σ FORMULA Quantum-mechanical results

$$\lambda = \frac{h}{p} \quad \Delta x \Delta p \geq \frac{h}{4\pi} \quad \text{orbital angular momentum} = \sqrt{l(l+1)} \frac{h}{2\pi} \quad \mu_{spin} = \sqrt{n(n+2)} \text{ BM}$$

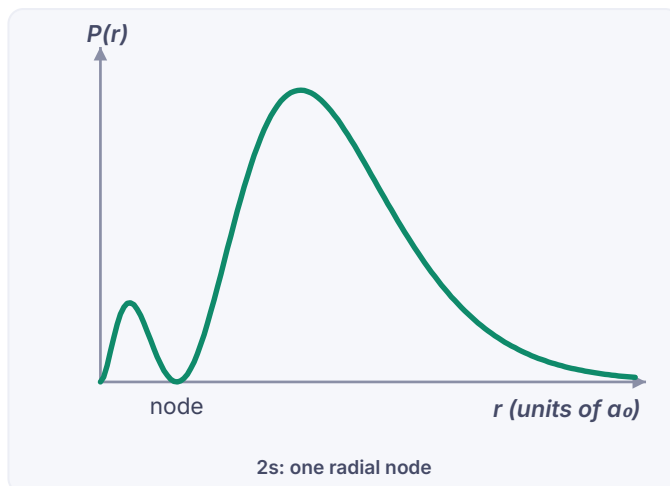
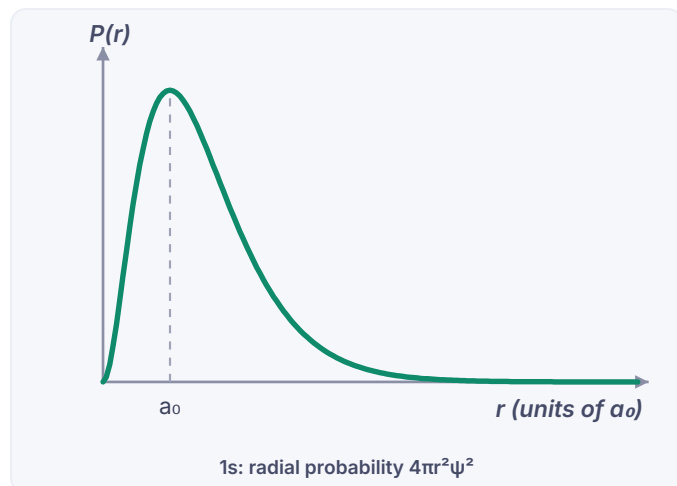
Radial nodes = $n - l - 1$; angular nodes = l ; total nodes = $n - 1$. Maximum electrons: shell $2n^2$; subshell $2(2l + 1)$; orbital 2. Orbitals in a shell = n^2 .

Σ FORMULA Quantum numbers

Symbol	Name	Values	Determines
n	principal	1, 2, 3, ...	size, energy (for H)
l	azimuthal	0 ... ($n - 1$)	shape (s, p, d, f)
m_l	magnetic	$-l \dots +l$	orientation
m_s	spin	$\pm \frac{1}{2}$	spin

Electronic configuration: rules and exceptions

- **Order ($n + l$ rule):** 1s 2s 2p 3s 3p **4s 3d** 4p 5s 4d 5p 6s 4f 5d 6p ... For equal $n + l$, the lower n fills first.
- **Anomalies to memorise:** Cr $[\text{Ar}]3d^5 4s^1$, Cu $[\text{Ar}]3d^{10} 4s^1$. Similar: Mo $4d^5 5s^1$, Ag $4d^{10} 5s^1$, Au $5d^{10} 6s^1$, Pd $4d^{10} 5s^0$.
- **Ions of d-block elements lose 4s electrons first:** Fe^{2+} is $[\text{Ar}]3d^6$ and Fe^{3+} is $[\text{Ar}]3d^5$.



☆ IMPORTANT ψ^2 plots (NCERT)

For 1s, ψ^2 is maximum at the nucleus and decays. For 2s, ψ^2 has **one radial node**. The *radial probability* $4\pi r^2 \psi^2$ for 1s peaks at $a_0 = 52.9$ pm. The d_{z^2} orbital has two lobes and a ring (torus). The other four d orbitals have four lobes each.

Common traps

⚠ MISTAKE ALERT Mistake alerts

- l can't equal n : "3f" and "2d" don't exist.
- Using the **atomic** configuration for ions: remove 4s electrons before 3d.
- Unpaired electrons in Cr: **6** ($3d^5 4s^1$), not 4.
- The Heisenberg product uses $\hbar/4\pi$ (NCERT). Some books use $\hbar/2$, which is the same value.
- Wavenumber in cm^{-1} vs m^{-1} : $R_H = 1.097 \times 10^7 \text{ m}^{-1}$.

Question patterns

▮ PYQ PATTERN Recurring structures

1. Valid/invalid quantum-number sets; number of electrons with given n, l, m_l, m_s .
2. Nodes of orbitals (radial/angular/total).
3. Bohr energies, radii, wavelengths for H, He^+ , Li^{2+} ; number of spectral lines.
4. de Broglie / Heisenberg numericals.
5. Configurations and magnetic moments of ions (feeds into d-block and coordination questions).

Practice questions

🔗 C02 · Practice

1

Easy

⌚ 0.5 min

Speed

C02-01

Which set of quantum numbers is **not** allowed?

- (A) $n = 2, l = 1, m_l = 0, m_s = +\frac{1}{2}$ (B) $n = 3, l = 3, m_l = 1, m_s = -\frac{1}{2}$
 (C) $n = 4, l = 2, m_l = -2, m_s = +\frac{1}{2}$ (D) $n = 1, l = 0, m_l = 0, m_s = -\frac{1}{2}$

2 Easy 0.5 min Speed

C02-02

The maximum number of electrons that can have $n = 3$ and $l = 2$ is:

- (A) 2 (B) 6 (C) 10 (D) 18

3 Medium 1 min Calc

C02-03

The wavelength emitted when the electron in a hydrogen atom falls from $n = 3$ to $n = 2$ is about:

- (A) 486 nm (B) 656 nm (C) 122 nm (D) 1875 nm

4 Medium 1.5 min Calc

C02-04

An electron's position is known to within $1 \text{ \AA}$. The minimum uncertainty in its velocity is about ($m_e = 9.1 \times 10^{-31} \text{ kg}$, $h = 6.626 \times 10^{-34} \text{ J s}$):

- (A) $5.8 \times 10^5 \text{ m/s}$ (B) $5.8 \times 10^3 \text{ m/s}$ (C) $1.2 \times 10^6 \text{ m/s}$ (D) $5.8 \times 10^7 \text{ m/s}$

5 Easy 0.5 min NV

C02-05

Find the total number of nodes in a 4d orbital.

Your answer (numerical value): **6** Easy 0.5 min Speed

C02-06

The ground-state electronic configuration of copper ($Z = 29$) is:

- (A) $[\text{Ar}]3d^9 4s^2$ (B) $[\text{Ar}]3d^{10} 4s^1$ (C) $[\text{Ar}]3d^{10} 4s^2$ (D) $[\text{Ar}]3d^8 4s^2 4p^1$

7 Easy 0.75 min Basic

C02-07

The orbital angular momentum of an electron in a d orbital is:

- (A) $\sqrt{2} \frac{h}{2\pi}$ (B) $\sqrt{6} \frac{h}{2\pi}$ (C) $2 \frac{h}{2\pi}$ (D) $\sqrt{12} \frac{h}{2\pi}$

8 Easy 0.75 min Basic

C02-08

The spin-only magnetic moment of Fe^{2+} is about:

- (A) 2.83 BM (B) 3.87 BM (C) 4.90 BM (D) 5.92 BM

9 Easy 0.5 min NV

C02-09

How many unpaired electrons are there in a gaseous Mn^{2+} ion ($Z = 25$)?Your answer (numerical value): **C02 · Chapter Test****1** Easy 0.5 min Speed

C02-T1

Which orbital is filled first?

- (A) 3d (B) 4s
(C) 4p (D) they fill together

2 Easy 0.5 min Speed

C02-T2

The energy of an electron in the second Bohr orbit of hydrogen is:

- (A) -13.6 eV (B) -3.4 eV (C) -1.51 eV (D) -6.8 eV

3 Easy 0.5 min NV

C02-T3

Find the maximum number of electrons that can be accommodated in the shell with $n = 4$.Your answer (numerical value): **4** Easy 0.5 min Concept

C02-T4

Which orbital is spherically symmetric (non-directional)?

- (A) p_x (B) d_{z^2} (C) s (D) d_{xy}

5 Easy 0.5 min Concept

C02-T5

According to Hund's rule, the three 2p electrons of nitrogen are arranged as:

- (A) two paired, one unpaired (B) all three unpaired, with parallel spins
(C) all three in one orbital (D) all three unpaired, with alternating spins

Answers & Solutions

C02 · Practice

1 (B) 2 (C) 3 (B) 4 (A) 5 3 6 (B) 7 (B) 8 (C) 9 5

C02 · Chapter Test

1 (B) 2 (B) 3 32 4 (C) 5 (B)

C02 · PRACTICE

1 Answer: (B) Allowed quantum numbers Easy 0.5 min Speed

C02-01

SOLUTION l must be $\leq n - 1$. With $n = 3$, $l = 3$ is impossible.**2** Answer: (C) Capacity of a subshell Easy 0.5 min Speed

C02-02

SOLUTION $l = 2$ is a d subshell: $2(2l + 1) = 10$.

COMMON TRAP Answering 18 (the whole shell).

3 Answer: (B) Wavelength of H- α Medium 1 min Calc

C02-03

SOLUTION $\Delta E = 13.6 \left(\frac{1}{4} - \frac{1}{9} \right) = 13.6 \times \frac{5}{36} \approx 1.89 \text{ eV}$; $\lambda = 1240/1.89 \approx 656 \text{ nm}$.

SHORTCUT The first Balmer line is red, at 656 nm.

COMMON TRAP Using $n_1 = 1$ (Lyman).**4** Answer: (A) Uncertainty principle Medium 1.5 min Calc

C02-04

SOLUTION $\Delta v = \frac{h}{4\pi m \Delta x} = \frac{6.626 \times 10^{-34}}{4\pi \times 9.1 \times 10^{-31} \times 10^{-10}} \approx 5.8 \times 10^5 \text{ m/s}$.SHORTCUT $4\pi \times 9.1 \approx 114$.COMMON TRAP Using $h/2\pi$ gives about 1.2×10^6 .**5** Answer: 3 Total nodes Easy 0.5 min NV

C02-05

SOLUTION Total = $n - 1 = 3$ (radial $4 - 2 - 1 = 1$, angular 2).

COMMON TRAP Counting only radial nodes (1).

6 Answer: (B) Anomalous configuration Easy 0.5 min Speed

C02-06

SOLUTION A completely filled 3d subshell gives extra stability.

COMMON TRAP Applying Aufbau blindly (A).

7 Answer: (B) Orbital angular momentum Easy 0.75 min Basic

C02-07

SOLUTION $\sqrt{l(l+1)} \hbar$ with $l = 2$: $\sqrt{6} \hbar$.COMMON TRAP Using Bohr's $n\hbar/2\pi$.

8 Answer: (C) Spin-only magnetic moment **Easy** ⌚ 0.75 min **Basic**

C02-08

SOLUTION Fe^{2+} : $3d^6$ with 4 unpaired electrons (free ion). $\sqrt{4 \times 6} = \sqrt{24} \approx 4.90$ BM.

SHORTCUT Memorise $\sqrt{n(n+2)}$: 1.73, 2.83, 3.87, 4.90, 5.92 for $n = 1-5$.

COMMON TRAP Removing 3d before 4s electrons.

9 Answer: 5 Unpaired electrons in an ion **Easy** ⌚ 0.5 min **NV**

C02-09

SOLUTION Mn is $[\text{Ar}]3d^54s^2$, so Mn^{2+} is $3d^5$: 5 unpaired electrons.

C02 · CHAPTER TEST

1 Answer: (B) $(n + l)$ rule **Easy** ⌚ 0.5 min **Speed**

C02-T1

SOLUTION $n + l$: 4s = 4, 3d = 5, 4p = 5. The lowest value fills first.

2 Answer: (B) Energy of the second orbit **Easy** ⌚ 0.5 min **Speed**

C02-T2

SOLUTION $-13.6/4 = -3.4$ eV.

3 Answer: 32 Shell capacity **Easy** ⌚ 0.5 min **NV**

C02-T3

SOLUTION $2n^2 = 32$.

4 Answer: (C) Spherical orbital **Easy** ⌚ 0.5 min **Concept**

C02-T4

SOLUTION s orbitals have $l = 0$, so there are no angular nodes and no directional dependence.

5 Answer: (B) Hund's rule **Easy** ⌚ 0.5 min **Concept**

C02-T5

SOLUTION Degenerate orbitals are singly occupied, with parallel spins, before any pairing.

REVISION Revision tracker · Atomic Structure

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 03

Chemical Bonding & Molecular Structure

03

1–2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 4

NCERT

~12 hours

STUDY TIME

Important concepts

- **Kossel–Lewis approach:** atoms attain noble-gas configurations by transferring electrons (ionic bonds) or sharing them (covalent bonds). **Octet exceptions:** incomplete (BeCl_2 , BF_3), expanded (PCl_5 , SF_6), odd-electron (NO , NO_2).
- **Formal charge** = $V - N_{\text{alone}} - \frac{1}{2}N_{\text{bonding}}$ picks the best Lewis structure.
- **Ionic bonding:** favoured by low IE, high electron-gain enthalpy and high **lattice enthalpy**. Lattice enthalpy rises with ionic charge and falls with ionic size.
- **Fajans' rules:** covalent character of an ionic bond rises with a **small, highly charged cation**, a **large anion**, and a cation with a pseudo-noble-gas (18-electron) configuration.
- **VSEPR:** electron pairs repel. Order of repulsion: $\text{lp-lp} > \text{lp-bp} > \text{bp-bp}$.
- **Valence bond theory & hybridisation** explain geometry. **MO theory** explains bond order, magnetism and species like O_2 (paramagnetic).
- **Hydrogen bonding** (with F, O, N) raises boiling points. Intramolecular H-bonding lowers them compared with the isomer that has intermolecular H-bonding.

Formula sheet

Σ FORMULA Useful counts

$$\text{Steric number (hybridisation)} = \frac{1}{2}(V + M - C + A) \quad \text{Bond order} = \frac{1}{2}(N_b - N_a) \quad \mu = q \times d$$

V = valence electrons of the central atom, M = monovalent atoms attached, C = cation charge, A = anion charge. $1 \text{ D} = 3.336 \times 10^{-30} \text{ C m}$. Percentage ionic character $\approx \frac{\mu_{\text{observed}}}{\mu_{\text{ionic}}} \times 100$.

VSEPR shape table

Steric no.	Bond pairs + lone pairs	Shape	Examples
2 (sp)	2 + 0	linear	BeCl_2 , CO_2
3 (sp ²)	3 + 0	trigonal planar	BF_3 , NO_3^-
3	2 + 1	bent	SO_2 , O_3
4 (sp ³)	4 + 0	tetrahedral	CH_4 , NH_4^+
4	3 + 1	trigonal pyramidal	NH_3 , PCl_3
4	2 + 2	bent	H_2O
5 (sp ³ d)	5 + 0	trigonal bipyramidal	PCl_5
5	4 + 1	see-saw	SF_4
5	3 + 2	T-shaped	ClF_3
5	2 + 3	linear	XeF_2 , I_3^-
6 (sp ³ d ²)	6 + 0	octahedral	SF_6
6	5 + 1	square pyramidal	BrF_5 , XeOF_4
6	4 + 2	square planar	XeF_4 , ICl_4^-

Steric no.	Bond pairs + lone pairs	Shape	Examples
7 (sp ³ d ³)	7 + 0	pentagonal bipyramidal	IF ₇
7	6 + 1	distorted octahedral	XeF ₆

In sp³d the **lone pairs go equatorial**. In sp³d², two lone pairs sit **trans** (opposite).

☆ IMPORTANT Bond angles & dipole moments to remember

- CH₄ 109.5° > NH₃ 107° > H₂O 104.5°. NH₃ (107°) > NF₃ (~102°). PH₃ ~93.5°, H₂S ~92° (little hybridisation).
- μ: NH₃ (1.47 D) ≫ NF₃ (0.23 D). In NH₃ the lone-pair moment adds to the bond moments; in NF₃ it opposes them.
- μ = 0: CO₂, BF₃, CCl₄, PCl₅, SF₆, XeF₄, *trans*-dichloroethene, *p*-dichlorobenzene.
- CH₃Cl > CH₂Cl₂ > CHCl₃ > CCl₄ (= 0).

Molecular orbital theory

Σ FORMULA MO energy order

Up to N₂ (14 electrons or fewer): $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^*) < \sigma^* 2p_z$ O₂ and **beyond:** $\sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

Species	Bond order	Magnetic
H ₂	1	dia
He ₂	0 (doesn't exist)	—
B ₂	1	para (2 unpaired π)
C ₂	2	dia
N ₂	3	dia
N ₂ ⁺ , N ₂ ⁻	2.5	para
O ₂ ⁺	2.5	para
O ₂	2	para (2 unpaired π*)
O ₂ ⁻ (superoxide)	1.5	para
O ₂ ²⁻ (peroxide)	1	dia
F ₂	1	dia

Higher bond order → shorter, stronger bond. Isoelectronic species (e.g. N₂, CO, CN⁻, NO⁺) have the same bond order (3).

Memory tricks

⚡ SHORTCUT Bond order by total electron count (2nd-period diatomics)

Total electrons	10	11	12	13	14	15	16	17	18	19	20
Bond order	1	1.5	2	2.5	3	2.5	2	1.5	1	0.5	0

The bond order rises by 0.5 per electron up to 14 (N₂, CO), then falls by 0.5 per electron. Examples: B₂ (10) = 1, O₂ (16) = 2, O₂⁻ (17) = 1.5, F₂ (18) = 1. **Time saved:** no MO diagram needed.

⚡ SHORTCUT Is it paramagnetic?

Odd total electrons → always paramagnetic. Even electrons → paramagnetic only for B₂ and O₂ in the common set.

Common traps

MISTAKE ALERT Mistake alerts

- Lone pairs occupy positions but are **not** included in the "shape" name: NH_3 is pyramidal, not tetrahedral.
- Using the N_2 -type MO order for O_2 gives the wrong magnetism for B_2/C_2 if you mix them up.
- The most covalent halide of a metal is its **iodide**. The most ionic is its **fluoride**.
- Intramolecular H-bonding (o-nitrophenol) → **lower** boiling point, steam-volatile. Intermolecular (p-nitrophenol) → higher boiling point.
- Resonance structures are not real. The actual structure is the hybrid, with equal bond lengths (e.g. all C–O bonds in CO_3^{2-} are equal).

Question patterns

PYQ PATTERN Recurring structures

- Shape/hybridisation of a list of species; "which pair has the same shape?"
- Bond order/magnetism of O_2 , N_2 family ions; ordering bond lengths.
- Dipole-moment ordering and zero-dipole identification.
- Fajans' rules / lattice-enthalpy comparisons.
- Number of lone pairs, σ and π bonds in a molecule.
- Hydrogen bonding and its effect on properties.
- Formal charge on atoms in O_3 , CO , NO_3^- .

Practice questions

C03 · Practice

1 Easy 0.5 min Speed

C03-01

The shape of XeF_4 is:

- (A) tetrahedral (B) square planar (C) see-saw (D) octahedral

2 Easy 0.75 min Basic

C03-02

The bond order and magnetic nature of O_2^- are:

- (A) 2, paramagnetic (B) 1.5, paramagnetic
(C) 1.5, diamagnetic (D) 2.5, paramagnetic

3 Easy 0.75 min Basic

C03-03

The hybridisation and shape of SF_4 are:

- (A) sp^3 , tetrahedral (B) sp^3d , see-saw
(C) sp^3d^2 , square planar (D) dsp^2 , square planar

4 Easy 0.75 min Concept

C03-04

Which compound has the highest lattice enthalpy?

- (A) NaCl (B) MgO (C) KCl (D) CaO

5 Easy 0.75 min Concept

C03-05

Which compound has the most covalent character?

- (A) LiF (B) LiI (C) KI (D) NaCl

6 Medium 1 min Concept

C03-06

NH_3 has a much larger dipole moment than NF_3 because:

- (A) N-F bonds are non-polar
(B) in NH_3 the lone-pair moment reinforces the bond moments, while in NF_3 it opposes them
(C) NF_3 is planar
(D) F is less electronegative than H

7 Easy 0.5 min NV

C03-07

Find the number of lone pairs on the Xe atom in XeF_2 .

Your answer (numerical value):

8 Easy 0.5 min Speed

C03-08

The correct order of bond angles is:

- (A) $\text{H}_2\text{O} < \text{NH}_3 < \text{CH}_4$ (B) $\text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$ (C) $\text{NH}_3 < \text{H}_2\text{O} < \text{CH}_4$ (D) all equal

9 Easy 0.5 min Speed

C03-09

Which molecule is paramagnetic?

- (A) N_2 (B) F_2 (C) O_2 (D) C_2

10 Medium 1 min Concept

C03-10

o-Nitrophenol is steam-volatile but p-nitrophenol is not, because:

- (A) o-nitrophenol has intermolecular H-bonding (B) o-nitrophenol has intramolecular H-bonding
(C) p-nitrophenol is more acidic (D) p-nitrophenol has lower molar mass

11 Medium 1 min Concept

C03-11

In the Lewis structure of O_3 , the formal charge on the central oxygen is:

- (A) 0 (B) +1 (C) -1 (D) +2

C03 · Chapter Test

1 Easy 0.5 min Speed

C03-T1

The hybridisation of carbon in CO_2 is:

- (A) sp (B) sp^2 (C) sp^3 (D) sp^3d

2 Easy 0.75 min NV

C03-T2

Find the number of σ bonds in ethene (C_2H_4).

Your answer (numerical value):

3 Easy 0.5 min Speed

C03-T3

Which species is isoelectronic with N_2 and has bond order 3?

- (A) O_2^+ (B) CO (C) NO (D) C_2^{2+}

4 Easy 0.5 min Speed

C03-T4

Which molecule has zero dipole moment?

- (A) $CHCl_3$ (B) CH_2Cl_2 (C) CCl_4 (D) CH_3Cl

5 Easy 0.5 min Speed

C03-T5

The shape of ClF_3 is:

- (A) trigonal planar (B) pyramidal (C) T-shaped (D) see-saw

Answers & Solutions

C03 · Practice

- 1 (B) 2 (B) 3 (B) 4 (B) 5 (B) 6 (B) 7 3 8 (A) 9 (C) 10 (B)
11 (B)

C03 · Chapter Test

- 1 (A) 2 5 3 (B) 4 (C) 5 (C)

C03 · PRACTICE

1 Answer: (B) Shape from VSEPR Easy 0.5 min Speed

C03-01

SOLUTION Xe has 8 valence electrons: 4 bond pairs + 2 lone pairs, steric number 6 (sp^3d^2). The lone pairs sit trans, leaving a square planar shape.

COMMON TRAP Naming the electron geometry (octahedral).

2 Answer: (B) Bond order of superoxide Easy 0.75 min Basic

C03-02

SOLUTION O_2^- has 17 electrons: one more π^* electron than O_2 . Bond order = $\frac{1}{2}(10 - 7) = 1.5$, with one unpaired electron.

SHORTCUT Odd electron count $\rightarrow$ paramagnetic.

3 Answer: (B) Hybridisation of SF_4 Easy 0.75 min Basic

C03-03

SOLUTION S: 6 valence electrons; 4 bonds + 1 lone pair, steric number 5, so sp^3d . The lone pair sits equatorial, giving a see-saw.

SHORTCUT $H = \frac{1}{2}(6 + 4) = 5$.

4 Answer: (B) Lattice enthalpy Easy 0.75 min Concept

C03-04

SOLUTION Lattice enthalpy $\propto \frac{q^+q^-}{r^+ + r^-}$. MgO and CaO have $2+/2-$ ions; Mg^{2+} is smaller than Ca^{2+} .

SHORTCUT Charge first, then size.

5 Answer: (B) Fajans' rules Easy 0.75 min Concept

C03-05

SOLUTION The small Li^+ polarises the large, easily polarisable I^- strongly.

SHORTCUT Small cation + large anion.

6 Answer: (B) Dipole of NH_3 vs NF_3 Medium 1 min Concept

C03-06

SOLUTION In NH_3 the N-H bond moments point towards N, the same direction as the lone pair's moment. In NF_3 they point towards F, away from the lone pair.

7 Answer: 3 Lone pairs on Xe Easy 0.5 min NV

C03-07

SOLUTION 8 valence electrons – 2 used in bonds = 6 electrons, i.e. 3 lone pairs (linear shape).

8 Answer: (A) Bond-angle order **Easy** ⌚ 0.5 min Speed

C03-08

SOLUTION More lone pairs → more compression: $104.5^\circ < 107^\circ < 109.5^\circ$.**9 Answer: (C)** Paramagnetic molecule **Easy** ⌚ 0.5 min Speed

C03-09

SOLUTION O_2 has two unpaired electrons in degenerate π^* orbitals.**10 Answer: (B)** Intramolecular hydrogen bonding **Medium** ⌚ 1 min Concept

C03-10

SOLUTION Intramolecular H-bonding in the ortho isomer prevents association between molecules, so it is more volatile. The para isomer associates through intermolecular H-bonds.**11 Answer: (B)** Formal charge in ozone **Medium** ⌚ 1 min Concept

C03-11

SOLUTION Central O: $V = 6$, one lone pair (2 electrons), 3 bonds (6 bonding electrons). $FC = 6 - 2 - 3 = +1$.**C03 · CHAPTER TEST****1 Answer: (A)** Hybridisation of C in CO_2 **Easy** ⌚ 0.5 min Speed

C03-T1

SOLUTION Two σ bonds, no lone pairs: linear, sp.**2 Answer: 5** Counting σ bonds **Easy** ⌚ 0.75 min NV

C03-T2

SOLUTION $4\text{ C-H } \sigma + 1\text{ C-C } \sigma = 5$ (plus 1π).**COMMON TRAP** Counting the double bond as two σ bonds.**3 Answer: (B)** Isoelectronic with N_2 **Easy** ⌚ 0.5 min Speed

C03-T3

SOLUTION CO has 14 electrons, the same as N_2 , so its bond order is 3.**4 Answer: (C)** Zero dipole moment **Easy** ⌚ 0.5 min Speed

C03-T4

SOLUTION A symmetric tetrahedron with four identical substituents.**5 Answer: (C)** Shape of ClF_3 **Easy** ⌚ 0.5 min Speed

C03-T5

SOLUTION 3 bond pairs + 2 lone pairs (sp^3d). Both lone pairs are equatorial, leaving a T shape.**REVISION** Revision tracker · Chemical Bonding☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 04

Chemical Thermodynamics

04

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 5

NCERT

~9 hours

STUDY TIME

Important concepts

- **System & surroundings:** open, closed and isolated systems. **Extensive** properties depend on amount (V, H, S, G, U). **Intensive** properties don't (T, P , density, molar quantities).
- **State functions:** U, H, S, G, P, V, T . Heat and work are path functions.
- **First law:** $\Delta U = q + w$. **Chemistry (IUPAC) sign convention:** w is work done **on** the system; $w = -P_{ext}\Delta V$.
- **Enthalpy** $H = U + PV$; $\Delta H = \Delta U + \Delta n_g RT$ for reactions involving gases.
- **Hess's law:** ΔH is path-independent, so thermochemical equations can be added and subtracted.
- **Second law:** a spontaneous process increases the entropy of the universe. At constant T, P : $\Delta G < 0$ means spontaneous.

⚠ WARNING Sign convention differs from Physics

In Chemistry (NCERT Class 11), $\Delta U = q + w$ with w = work **done on** the system. In Physics, $\Delta U = Q - W$ with W = work **done by** the system. The physics is the same; only the sign of w differs. Stay consistent within each subject.

Formula sheet

Σ FORMULA Work, heat, enthalpy

$$w_{irrev} = -P_{ext} \Delta V \quad w_{rev, isothermal} = -2.303 nRT \log \frac{V_2}{V_1} \quad q_V = \Delta U \quad q_P = \Delta H$$

$$\Delta H = \Delta U + \Delta n_g RT \quad C_p - C_v = R \quad q = nC\Delta T$$

Free expansion (into vacuum): $w = 0$. Isothermal ideal gas: $\Delta U = \Delta H = 0$.

Σ FORMULA Thermochemistry

$$\Delta_r H^\circ = \sum \Delta_f H^\circ_{products} - \sum \Delta_f H^\circ_{reactants} \quad \Delta_r H = \sum BE_{reactants} - \sum BE_{products}$$

$\Delta_f H^\circ$ of an element in its standard state = 0 (e.g. $O_2(g)$, $C(\text{graphite})$, $Br_2(l)$). Enthalpies to know by definition: formation, combustion (always negative), atomisation, bond dissociation, sublimation ($= \Delta_{fus}H + \Delta_{vap}H$), phase transition, hydration, ionisation, solution. Strong acid + strong base neutralisation: $\Delta H \approx -57.1$ kJ per mol of water. It is less exothermic for a weak acid or base, because some energy goes into ionisation.

Σ FORMULA Entropy & Gibbs energy

$$\Delta S = \frac{q_{rev}}{T} \quad \Delta S_{fusion/vap} = \frac{\Delta H_{trans}}{T_{trans}} \quad \Delta S_{total} = \Delta S_{sys} + \Delta S_{surr} > 0 \text{ (spontaneous)}$$

$$\Delta G = \Delta H - T\Delta S \quad \Delta G^\circ = -RT \ln K = -2.303RT \log K \quad \Delta G = \Delta G^\circ + RT \ln Q$$

Isothermal reversible expansion of an ideal gas: $\Delta S = nR \ln \frac{V_2}{V_1}$.

Spontaneity table

ΔH	ΔS	$\Delta G = \Delta H - T\Delta S$	Spontaneous?
–	+	always –	at all temperatures

ΔH	ΔS	$\Delta G = \Delta H - T\Delta S$	Spontaneous?
+	-	always +	never
-	-	- at low T	below $T = \Delta H/\Delta S$
+	+	- at high T	above $T = \Delta H/\Delta S$

⚡ SHORTCUT Δn_g in 5 seconds

Δn_g = (moles of gaseous products) – (moles of gaseous reactants). **Ignore liquids and solids.** Example:
 $C(s) + O_2(g) \longrightarrow CO_2(g)$ has $\Delta n_g = 0$, so $\Delta H = \Delta U$.

Common traps

⚠ MISTAKE ALERT Mistake alerts

- Mixing J and kJ in $\Delta G = \Delta H - T\Delta S$ (ΔS is usually given in J/K).
- Using log vs ln incorrectly in $\Delta G^\circ = -2.303RT \log K$.
- Bond-energy method: **reactants minus products** (the reverse of the formation-enthalpy method).
- $\Delta_f H^\circ$ of O_3 , diamond or $H_2O(g)$ is **not** zero.
- The entropy of a pure perfect crystal at 0 K is zero (third law). Entropy increases: solid < liquid < gas; dissolving a solid usually increases S .

Question patterns

📊 PYQ PATTERN Recurring structures

- ΔH vs ΔU with Δn_g (frequent Section-B numerical).
- Hess's law manipulation; enthalpy of formation or combustion from given data.
- Bond-enthalpy calculations.
- Spontaneity temperature; sign analysis.
- $\Delta G^\circ \leftrightarrow K$ conversions.
- Work in isothermal reversible vs irreversible expansion.
- Intensive/extensive and state/path classification.

Practice questions

📝 C04 · Practice

1 Medium ⌚ 1.5 min NV

C04-01

For $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$ at 300 K, find $\Delta H - \Delta U$ in kJ/mol. Take $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$ and give the answer to the nearest integer (with sign).

Your answer (numerical value):

2 Medium ⌚ 1.5 min JEE

C04-02

Given: $C(s) + O_2(g) \longrightarrow CO_2(g)$, $\Delta H = -393.5 \text{ kJ}$; $CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$, $\Delta H = -283.0 \text{ kJ}$. The enthalpy of formation of $CO(g)$ is:

- (A) -110.5 kJ/mol (B) -676.5 kJ/mol (C) $+110.5 \text{ kJ/mol}$ (D) -221.0 kJ/mol

3 Medium 1.5 min Calc

C04-03

Using bond enthalpies (kJ/mol) H-H 436, Cl-Cl 243, H-Cl 431, $\Delta_r H$ for $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ is:

- (A) -183 kJ (B) +183 kJ (C) -248 kJ (D) -431 kJ

4 Medium 1 min Concept

C04-04

For a reaction, $\Delta H = +30 \text{ kJ/mol}$ and $\Delta S = +100 \text{ J K}^{-1} \text{ mol}^{-1}$. It is spontaneous:

- (A) above 300 K (B) below 300 K
(C) at all temperatures (D) at no temperature

5 Medium 1 min Calc

C04-05

For a reaction at 300 K with $K = 10$, ΔG° is about ($R = 8.314$):

- (A) -5.74 kJ/mol (B) +5.74 kJ/mol (C) -2.49 kJ/mol (D) -57.4 kJ/mol

6 Easy 0.75 min Basic

C04-06

A gas expands against a constant external pressure of 2 bar from 1 L to 6 L. The work done on the gas is (1 L bar = 100 J):

- (A) +1000 J (B) -1000 J (C) -1200 J (D) -500 J

7 Easy 0.5 min Speed

C04-07

Which is an intensive property?

- (A) enthalpy (B) volume
(C) molar heat capacity (D) internal energy

8 Easy 1 min NV

C04-08

The enthalpy of vaporisation of a liquid is 30 kJ/mol at its boiling point of 300 K. Find $\Delta_{vap} S$ (in $\text{J K}^{-1} \text{ mol}^{-1}$).

Your answer (numerical value):

9 Easy 0.75 min Concept

C04-09

For which species is $\Delta_f H^\circ = 0$?

- (A) $\text{O}_3(\text{g})$ (B) C(diamond) (C) $\text{Br}_2(\text{l})$ (D) $\text{H}_2\text{O}(\text{l})$

C04 · Chapter Test

1 Easy 0.5 min Speed

C04-T1

At constant T and P , a process is spontaneous when:

- (A) $\Delta G > 0$ (B) $\Delta G < 0$ (C) $\Delta H > 0$ (D) $\Delta S < 0$

2 Easy 0.5 min Concept

C04-T2

The enthalpy of neutralisation of HCl by NaOH is -57.1 kJ/mol. For CH_3COOH with NaOH it is:

- (A) exactly -57.1 kJ/mol (B) less negative than -57.1 kJ/mol
(C) more negative than -57.1 kJ/mol (D) zero

3 Easy 0.5 min Speed

C04-T3

For the free expansion of an ideal gas into a vacuum:

- (A) $w = 0, q = 0, \Delta U = 0$ (B) $w < 0, q > 0$
 (C) $\Delta U > 0$ (D) $\Delta S = 0$

4 Medium 1 min NV

C04-T4

2 mol of an ideal gas expands isothermally and reversibly at 300 K from 1 L to 10 L. Find the magnitude of the work done (in kJ, nearest integer). ($R = 8.314$)

Your answer (numerical value):

5 Easy 0.5 min Speed

C04-T5

Which process involves a decrease in entropy?

- (A) melting of ice (B) evaporation of water
 (C) freezing of water (D) dissolving NaCl in water

Answers & Solutions

C04 · Practice

1 -5 2 (A) 3 (A) 4 (A) 5 (A) 6 (B) 7 (C) 8 100 9 (C)

C04 · Chapter Test

1 (B) 2 (B) 3 (A) 4 11 5 (C)

C04 · PRACTICE

1 Answer: -5 $\Delta H - \Delta U$ relation Medium 1.5 min NV

C04-01

SOLUTION $\Delta n_g = 2 - 4 = -2$. $\Delta H - \Delta U = \Delta n_g RT = -2 \times 8.3 \times 300 = -4980 \text{ J} \approx -5 \text{ kJ/mol}$.

COMMON TRAP Counting Δn_g as +2.

2 Answer: (A) Hess's law Medium 1.5 min JEE

C04-02

SOLUTION Subtract the second equation from the first: $\text{C} + \frac{1}{2} \text{O}_2 \longrightarrow \text{CO}$, $\Delta H = -393.5 + 283.0 = -110.5 \text{ kJ/mol}$.

COMMON TRAP Adding the two equations.

3 Answer: (A) Bond-enthalpy calculation Medium 1.5 min Calc

C04-03

SOLUTION $\Delta H = (436 + 243) - 2(431) = 679 - 862 = -183 \text{ kJ}$.

SHORTCUT Bonds broken – bonds formed.

COMMON TRAP Doing products – reactants (+183).

4 Answer: (A) Spontaneity temperature Medium 1 min Concept

C04-04

SOLUTION Both positive, so entropy-driven: spontaneous when $T > \Delta H / \Delta S = 30\,000 / 100 = 300 \text{ K}$.

COMMON TRAP Unit mismatch (0.3 K).

5 Answer: (A) ΔG° from K Medium 1 min Calc

C04-05

SOLUTION $\Delta G^\circ = -2.303 \times 8.314 \times 300 \times \log 10 \approx -5744 \text{ J/mol}$.

SHORTCUT $2.303RT \approx 5.74 \text{ kJ}$ at 300 K: memorise it.

COMMON TRAP Using $\ln 10 = 1$.

6 Answer: (B) Irreversible work Easy 0.75 min Basic

C04-06

SOLUTION $w = -P_{\text{ext}} \Delta V = -2 \times 5 = -10 \text{ L bar} = -1000 \text{ J}$.

SHORTCUT Expansion $\rightarrow w < 0$ in the chemistry convention.

COMMON TRAP Positive sign.

7 Answer: (C) *Intensive properties* **Easy** ⌚ 0.5 min **Speed**

C04-07

SOLUTION Molar (per-mole) quantities are intensive.

SHORTCUT Any extensive quantity ÷ amount → intensive.

8 Answer: 100 *Entropy of vaporisation* **Easy** ⌚ 1 min **NV**

C04-08

SOLUTION $\Delta S = \Delta H/T = 30\,000/300 = 100 \text{ J K}^{-1} \text{ mol}^{-1}$.

9 Answer: (C) *Standard enthalpy of formation* **Easy** ⌚ 0.75 min **Concept**

C04-09

SOLUTION Bromine's standard state is liquid, so $\text{Br}_2(\text{l})$ is an element in its standard state.

COMMON TRAP Diamond (graphite is carbon's standard state).

C04 · CHAPTER TEST

1 Answer: (B) *Condition for spontaneity* **Easy** ⌚ 0.5 min **Speed**

C04-T1

SOLUTION At constant T and P , the criterion for a spontaneous process is $\Delta G = \Delta H - T\Delta S < 0$. Neither ΔH nor ΔS_{sys} alone decides it.

2 Answer: (B) *Heat of neutralisation* **Easy** ⌚ 0.5 min **Concept**

C04-T2

SOLUTION Some heat is used to ionise the weak acid.

3 Answer: (A) *Free expansion* **Easy** ⌚ 0.5 min **Speed**

C04-T3

SOLUTION No external pressure → no work. Adiabatic/ideal → $\Delta U = 0$. (But $\Delta S > 0$, since the process is spontaneous.)

COMMON TRAP Choosing (D).

4 Answer: 11 *Isothermal reversible work* **Medium** ⌚ 1 min **NV**

C04-T4

SOLUTION $|w| = 2.303nRT \log(V_2/V_1) = 2.303 \times 2 \times 8.314 \times 300 \times 1 \approx 11\,488 \text{ J} \approx 11 \text{ kJ}$.

5 Answer: (C) *Entropy increase* **Easy** ⌚ 0.5 min **Speed**

C04-T5

SOLUTION Liquid → solid is more ordered.

REVISION Revision tracker · Chemical Thermodynamics

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 05

Solutions

05

~1 per shift

TYPICAL QUESTIONS

Easy-Medium

DIFFICULTY

High

PRIORITY

Class 12 · Ch 1

NCERT

~6 hours

STUDY TIME

Important concepts

- **Concentration terms:** molarity, molality, mole fraction, mass %, volume %, ppm. For definitions and conversions see Unit 1 (page 163).
- **Henry's law:** the solubility of a gas is proportional to its partial pressure: $p = K_H x$. A **larger K_H means lower solubility**. Solubility of gases falls as temperature rises.
- **Raoult's law:** for volatile components, $p_i = p_i^\circ x_i$. For a non-volatile solute, the solution's vapour pressure is $p = p_1^\circ x_1$.
- **Ideal solutions** obey Raoult's law exactly: $\Delta_{mix}H = 0$, $\Delta_{mix}V = 0$ (benzene–toluene, n-hexane–n-heptane).
- **Colligative properties** depend on the **number** of solute particles, not their nature: RLVP, ΔT_b , ΔT_f , π .
- **van't Hoff factor i** corrects for dissociation ($i > 1$) or association ($i < 1$).

Formula sheet

Σ FORMULA Colligative properties

$$\frac{p^\circ - p}{p^\circ} = x_B \quad \Delta T_b = iK_b m \quad \Delta T_f = iK_f m \quad \pi = iCRT$$

$$M_B = \frac{K_b w_B \times 1000}{\Delta T_b w_A} \text{ (same form for } K_f) \quad M_B = \frac{w_B RT}{\pi V}$$

Water: $K_b = 0.52 \text{ K kg mol}^{-1}$, $K_f = 1.86 \text{ K kg mol}^{-1}$. $R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$ (or $0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$).

Σ FORMULA van't Hoff factor

$$i = \frac{\text{observed colligative property}}{\text{calculated (normal) value}} = \frac{\text{normal molar mass}}{\text{observed molar mass}}$$

Dissociation into n ions with degree α : $i = 1 + (n - 1)\alpha$. Association of n molecules into one: $i = 1 - \alpha \left(1 - \frac{1}{n}\right)$.

Complete dissociation: NaCl 2, CaCl_2 3, K_2SO_4 3, $\text{K}_4[\text{Fe}(\text{CN})_6]$ 5, $\text{Al}_2(\text{SO}_4)_3$ 5. Benzoic/acetic acid fully dimerised in benzene: $i = 0.5$.

Σ FORMULA Vapour pressure of binary liquid mixtures

$$P_{\text{total}} = p_A^\circ x_A + p_B^\circ x_B \quad y_A = \frac{p_A^\circ x_A}{P_{\text{total}}} \text{ (vapour-phase mole fraction)}$$

The vapour is always richer in the **more volatile** component.

Deviations from Raoult's law

	Positive deviation	Negative deviation
A–B interactions	weaker than A–A, B–B	stronger than A–A, B–B
$\Delta_{mix}H$, $\Delta_{mix}V$	> 0 , > 0	< 0 , < 0
Vapour pressure	higher than ideal	lower than ideal
Azeotrope	minimum-boiling (e.g. ethanol–water, ~95% ethanol)	maximum-boiling (e.g. HNO_3 –water, ~68% acid)

	Positive deviation	Negative deviation
Examples	ethanol + acetone, CS_2 + acetone, ethanol + water	chloroform + acetone (H-bond), HNO_3 + water, phenol + aniline



Numerical-solving method

1. Identify the colligative property and write its formula **with i** .
2. Molality needs moles of solute $\div$ **kg of solvent**. Convert grams to kg.
3. For electrolytes, decide on complete dissociation vs a given α .
4. For osmotic pressure use C in mol/L and $R = 0.0821$ L atm (answer in atm).

SHORTCUT Comparing colligative effects

For equal molal concentrations, compare $i \times m$ only. Largest $im \rightarrow$ largest ΔT_b , ΔT_f , π and lowest vapour pressure/freezing point.

Common traps

MISTAKE ALERT Mistake alerts

- Forgetting i for electrolytes.
- **Higher $K_H \rightarrow$ lower solubility.** It's easy to reverse this.
- Using the mass of the **solution** in molality.
- Osmotic pressure units: with $R = 0.0821$ the answer is in atm. With $R = 8.314$ you need SI units throughout.
- Freezing point depression is **positive ΔT_f** ; the new freezing point is $T_f^\circ - \Delta T_f$.

Question patterns

PYQ PATTERN Recurring structures

1. Molar mass from ΔT_b , ΔT_f , π (common Section-B numerical).
2. i and degree of dissociation/association.
3. Ordering boiling/freezing points of several solutions.
4. Raoult's law with two volatile liquids; vapour composition.
5. Positive/negative deviation identification; azeotropes.
6. Henry's law applications (scuba diving, soft drinks).

Practice questions

C05 · Practice

1 Easy 0.75 min Basic C05-01

Pure water has a vapour pressure of 100 mm Hg at some temperature. A non-volatile solute is added so that its mole fraction is 0.02. The solution's vapour pressure is:

- (A) 102 mm (B) 98 mm (C) 2 mm (D) 80 mm

2 Easy 1 min Basic C05-02

The freezing-point depression of a 0.1 molal aqueous NaCl solution, assuming complete dissociation ($K_f = 1.86$), is:

- (A) 0.186 K (B) 0.372 K (C) 0.558 K (D) 0.093 K

3 Easy 1 min Calc C05-03

The osmotic pressure of 0.1 M glucose at 300 K is about ($R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$):

- (A) 2.46 atm (B) 24.6 atm (C) 0.246 atm (D) 4.92 atm

4 Medium 1.5 min NV C05-04

1.8 g of a non-volatile non-electrolyte dissolved in 100 g of water raises its boiling point by 0.052 K ($K_b = 0.52$). Find the molar mass of the solute (in g/mol).

Your answer (numerical value):

5 Easy 0.5 min NV C05-05

Find the van't Hoff factor of $\text{K}_4[\text{Fe}(\text{CN})_6]$ in dilute solution, assuming complete dissociation.

Your answer (numerical value):

6 Easy 0.75 min Concept C05-06

Acetic acid in benzene is completely dimerised. Its van't Hoff factor is:

- (A) 2 (B) 1 (C) 0.5 (D) 0.25

7 Easy 0.5 min Concept C05-07

Which mixture shows a **positive** deviation from Raoult's law?

- (A) chloroform + acetone (B) ethanol + acetone
(C) nitric acid + water (D) benzene + toluene

8 Easy 0.5 min Concept C05-08

At the same temperature, gas X has a larger Henry's-law constant K_H than gas Y in water. Therefore:

- (A) X is more soluble than Y (B) X is less soluble than Y
(C) both are equally soluble (D) solubility doesn't depend on K_H

9 Easy 1 min Basic C05-09

Which NaCl concentration (complete dissociation) is isotonic with 0.1 M glucose?

- (A) 0.1 M (B) 0.05 M (C) 0.2 M (D) 0.025 M

C05 · Chapter Test

1 Easy 0.5 min Speed

C05-T1

Colligative properties depend on:

- (A) the nature of the solute (B) the number of solute particles
(C) the molar mass of the solvent only (D) the chemical reactivity of the solute

2 Easy 0.75 min Concept

C05-T2

Among 0.1 m aqueous solutions of glucose, NaCl, CaCl₂ and Al₂(SO₄)₃ (complete dissociation), the lowest freezing point belongs to:

- (A) glucose (B) NaCl (C) CaCl₂ (D) Al₂(SO₄)₃

3 Easy 0.75 min NV

C05-T3

The elevation of boiling point of 0.5 m aqueous urea ($K_b = 0.52$) is $x \times 0.01$ K. Find x .

Your answer (numerical value):

4 Easy 0.5 min Concept

C05-T4

Reverse osmosis requires:

- (A) a pressure greater than the osmotic pressure applied on the solution side
(B) heating the solution
(C) a pressure smaller than the osmotic pressure
(D) no semipermeable membrane

5 Easy 0.5 min Concept

C05-T5

A solution showing large positive deviation from Raoult's law forms:

- (A) a maximum-boiling azeotrope (B) a minimum-boiling azeotrope
(C) no azeotrope (D) an ideal solution

Answers & Solutions

C05 · Practice

1 (B) 2 (B) 3 (A) 4 180 5 5 6 (C) 7 (B) 8 (B) 9 (B)

C05 · Chapter Test

1 (B) 2 (D) 3 26 4 (A) 5 (B)

C05 · PRACTICE

1 Answer: (B) Relative lowering of vapour pressure Easy 0.75 min Basic

C05-01

SOLUTION $p = p^\circ x_{\text{solvent}} = 100 \times 0.98 = 98$ mm Hg.

COMMON TRAP Answering the lowering (2 mm).

2 Answer: (B) Freezing-point depression Easy 1 min Basic

C05-02

SOLUTION $\Delta T_f = iK_f m = 2 \times 1.86 \times 0.1 = 0.372$ K.

COMMON TRAP Forgetting $i = 2$.

3 Answer: (A) Osmotic pressure Easy 1 min Calc

C05-03

SOLUTION $\pi = CRT = 0.1 \times 0.0821 \times 300 \approx 2.46$ atm.

4 Answer: 180 Molar mass from elevation of boiling point **Medium** ⌚ 1.5 min **NV** C05-04

SOLUTION $m = \Delta T_b / K_b = 0.1 \text{ mol/kg}$, so moles of solute $= 0.1 \times 0.1 = 0.01$. $M = 1.8 / 0.01 = 180 \text{ g/mol}$.

SHORTCUT $M = \frac{K_b w_B \times 1000}{\Delta T_b w_A} = \frac{0.52 \times 1.8 \times 1000}{0.052 \times 100}$.

COMMON TRAP Using 100 g as 1 kg.

5 Answer: 5 van't Hoff factor of a complex salt **Easy** ⌚ 0.5 min **NV** C05-05

SOLUTION $\text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow 4\text{K}^+ + [\text{Fe}(\text{CN})_6]^{4-}$: 5 ions. The complex ion doesn't break up.

COMMON TRAP Counting the CN^- ligands (11).

6 Answer: (C) Association in benzene **Easy** ⌚ 0.75 min **Concept** C05-06

SOLUTION Two molecules form one particle, so $i = 1/2$. The observed molar mass is double.

7 Answer: (B) Positive deviation example **Easy** ⌚ 0.5 min **Concept** C05-07

SOLUTION Acetone breaks up ethanol's hydrogen bonding and forms weaker A–B interactions, so the vapour pressure is higher than ideal.

SHORTCUT Chloroform–acetone forms new H-bonds, which gives a negative deviation.

8 Answer: (B) Henry's constant **Easy** ⌚ 0.5 min **Concept** C05-08

SOLUTION $x = p/K_H$: at a given p , a larger K_H means a smaller mole fraction dissolved.

9 Answer: (B) Isotonic solutions **Easy** ⌚ 1 min **Basic** C05-09

SOLUTION Isotonic means equal iC : $1 \times 0.1 = 2 \times C \Rightarrow C = 0.05 \text{ M}$.

C05 · CHAPTER TEST

1 Answer: (B) What colligative properties depend on **Easy** ⌚ 0.5 min **Speed** C05-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (D) Lowest freezing point **Easy** ⌚ 0.75 min **Concept** C05-T2

SOLUTION $i = 1, 2, 3, 5$. The largest i gives the largest depression.

3 Answer: 26 Boiling-point elevation **Easy** ⌚ 0.75 min **NV** C05-T3

SOLUTION $\Delta T_b = 0.52 \times 0.5 = 0.26 \text{ K}$, so $x = 26$.

4 Answer: (A) Reverse osmosis **Easy** ⌚ 0.5 min **Concept** C05-T4

SOLUTION This forces the solvent out of the solution through the membrane, as in desalination.

5 Answer: (B) Azeotrope type **Easy** ⌚ 0.5 min **Concept** C05-T5

SOLUTION Higher vapour pressure $\rightarrow$ lower boiling point at the azeotropic composition.

REVISION Revision tracker · Solutions

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 06

Equilibrium: Chemical & Ionic

06

1-2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 6

NCERT

~12 hours

STUDY TIME

Important concepts

- **Dynamic equilibrium:** forward and backward rates are equal, and concentrations stay constant. It can be reached from either side.
- **Law of mass action:** for $aA + bB \rightleftharpoons cC + dD$, $K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$. Pure solids and liquids are omitted.
- **Reaction quotient Q** has the same form at any moment. $Q < K \rightarrow$ forward; $Q > K \rightarrow$ backward; $Q = K \rightarrow$ at equilibrium.
- **Le Chatelier's principle:** a system at equilibrium shifts to partly counteract any imposed change. Only **temperature** changes the value of K .
- **Ionic equilibria:** acids/bases (Arrhenius, Brønsted–Lowry, Lewis), K_w , pH, weak electrolytes, common-ion effect, salt hydrolysis, buffers, solubility product.

Formula sheet: Chemical equilibrium

Σ FORMULA K relations

$$K_p = K_c(RT)^{\Delta n_g} \quad \Delta G^\circ = -RT \ln K \quad \log \frac{K_2}{K_1} = \frac{\Delta H^\circ}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Operation on the equation	New K
reverse	$1/K$
multiply by n	K^n
add two equations	$K_1 K_2$

Degree of dissociation from vapour density ($A \rightarrow nB$): $\alpha = \frac{D-d}{(n-1)d}$, where D = initial and d = equilibrium vapour density.

Σ FORMULA Le Chatelier summary

Change	Effect
add reactant / remove product	shifts forward (K unchanged)
increase pressure (decrease V)	shifts towards fewer gas moles
increase temperature	shifts in the endothermic direction; K rises for endothermic reactions
catalyst	no shift; equilibrium reached faster
inert gas at constant volume	no effect
inert gas at constant pressure	shifts towards more gas moles

Formula sheet: Ionic equilibrium

Σ FORMULA pH and weak electrolytes

$$K_w = [\text{H}^+][\text{OH}^-] = 10^{-14} \text{ (298 K)} \quad \text{pH} = -\log[\text{H}^+] \quad \text{pH} + \text{pOH} = 14$$

$$\text{Weak acid: } \alpha = \sqrt{\frac{K_a}{C}}, [\text{H}^+] = \sqrt{K_a C}, \text{pH} = \frac{1}{2}(\text{p}K_a - \log C) \quad K_a K_b = K_w \text{ (conjugate pair)}$$

Ostwald's dilution law: α increases on dilution. K_w increases with temperature (neutral pH < 7 above 25 °C).

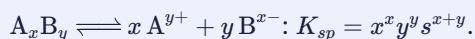
Very dilute strong acid (< 10^{-6} M): include water's H^+ . 10^{-8} M HCl has pH $\approx$ 6.98, not 8.

Σ FORMULA Buffers & hydrolysis

$$\text{Acidic buffer: } \text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]} \quad \text{Basic buffer: } \text{pOH} = \text{p}K_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

Salt of	Nature	pH
strong acid + strong base (NaCl)	neutral	7
weak acid + strong base (CH_3COONa)	basic	$7 + \frac{1}{2}\text{p}K_a + \frac{1}{2}\log C$
strong acid + weak base (NH_4Cl)	acidic	$7 - \frac{1}{2}\text{p}K_b - \frac{1}{2}\log C$
weak acid + weak base ($\text{CH}_3\text{COONH}_4$)	depends	$7 + \frac{1}{2}\text{p}K_a - \frac{1}{2}\text{p}K_b$
Maximum buffer capacity at $[\text{salt}] = [\text{acid}]$, i.e. $\text{pH} = \text{p}K_a$.		

Σ FORMULA Solubility product



Type	K_{sp}	Example
AB	s^2	AgCl, BaSO_4
AB_2 / A_2B	$4s^3$	CaF_2 , Ag_2CrO_4
AB_3	$27s^4$	$\text{Fe}(\text{OH})_3$
A_2B_3	$108s^5$	Bi_2S_3

Precipitation occurs when the ionic product > K_{sp} . **Common ion** lowers solubility: AgCl in c M NaCl has $s \approx K_{sp}/c$.

Numerical-solving method (ICE table)

1. Write Initial, Change ($-x$, $+x$ with stoichiometric coefficients) and Equilibrium rows.
2. Substitute into K . If K is small relative to C , approximate $C - x \approx C$.
3. Check: if $x/C > 5\%$, solve the quadratic.

⚡ SHORTCUT pH of a buffer in your head

If $[\text{salt}] = [\text{acid}]$, $\text{pH} = \text{p}K_a$. Every 10× increase in $[\text{salt}]/[\text{acid}]$ adds 1 pH unit. Acetic acid $\text{p}K_a = 4.74$; NH_3 $\text{p}K_b = 4.74$ (NH_4^+ $\text{p}K_a = 9.26$).

Common traps

MISTAKE ALERT Mistake alerts

- Including solids/liquids in K expressions.
- Δn_g for K_p/K_c : gaseous moles **products – reactants**.
- Catalysts don't change K or the equilibrium composition.
- Diprotic bases: $\text{Ba}(\text{OH})_2$ gives $[\text{OH}^-] = 2C$.
- K_{sp} of Ag_2CrO_4 is $4s^3$ (not s^2): $[\text{Ag}^+] = 2s$.
- Applying the Henderson equation to a solution with no conjugate pair.

Question patterns

PYQ PATTERN Recurring structures

1. K_p – K_c relation; K for combined/reversed reactions.
2. Le Chatelier predictions (pressure, temperature, inert gas).
3. pH of weak acids/bases, very dilute strong acids, mixtures of acid and base (after neutralisation).
4. Buffer pH (Henderson); buffer made by partial neutralisation.
5. $K_{sp} \leftrightarrow$ solubility; common-ion effect; precipitation condition.
6. Salt hydrolysis pH.
7. Degree of dissociation from vapour density or pressure data.

Practice questions

C06 · Practice

1 Easy 0.75 min Basic C06-01

For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$:

- (A) $K_p = K_c$ (B) $K_p = K_c(RT)^2$ (C) $K_p = K_c(RT)^{-2}$ (D) $K_p = K_c(RT)^{-1}$

2 Easy 0.5 min Concept C06-02

At some instant $Q < K$ for a reaction. The reaction will:

- (A) proceed backward (B) proceed forward
(C) be at equilibrium (D) stop

3 Easy 0.5 min Speed C06-03

For $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, increasing the total pressure (by compression) at constant temperature:

- (A) shifts the equilibrium towards NO_2 (B) shifts it towards N_2O_4
(C) has no effect (D) changes K_p

4 Medium 1 min Calc C06-04

The pH of 0.1 M acetic acid ($K_a = 1.8 \times 10^{-5}$) is about:

- (A) 2.87 (B) 1.00 (C) 4.74 (D) 3.87

5 Medium 1 min Basic C06-05

A buffer contains 0.2 M CH_3COONa and 0.02 M CH_3COOH ($\text{p}K_a = 4.74$). Its pH is:

- (A) 3.74 (B) 4.74 (C) 5.74 (D) 6.74

6

Medium

⌚ 1 min

Calc

C06-06

The solubility of CaF_2 is 2×10^{-4} mol/L. Its K_{sp} is:

- (A) 4×10^{-8} (B) 8×10^{-12} (C) 3.2×10^{-11} (D) 1.6×10^{-11}

7

Medium

⌚ 1 min

Concept

C06-07

The solubility of AgCl ($K_{sp} = 1.8 \times 10^{-10}$) in 0.1 M NaCl is about:

- (A) 1.34×10^{-5} M (B) 1.8×10^{-9} M (C) 1.8×10^{-11} M (D) 0.1 M

8

Hard

⌚ 2 min

NV

C06-08

Find the pH of 10^{-8} M HCl at 25°C , to the nearest integer.

Your answer (numerical value):

9

Medium

⌚ 1.5 min

JEE

C06-09

The pH of 0.1 M CH_3COONa ($\text{p}K_a$ of acetic acid = 4.74) is about:

- (A) 8.87 (B) 9.37 (C) 7.00 (D) 5.13

10

Easy

⌚ 0.5 min

Speed

C06-10

The conjugate base of HSO_4^- is:

- (A) H_2SO_4 (B) SO_4^{2-} (C) HSO_3^- (D) SO_3^{2-}

11

Easy

⌚ 0.75 min

Basic

C06-11

For $\text{A} \rightleftharpoons \text{B}$, $K = 4$. For $2\text{B} \rightleftharpoons 2\text{A}$, K is:

- (A) $1/16$ (B) 16 (C) $1/8$ (D) -8



C06 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

C06-T1

A catalyst added to a reaction at equilibrium:

- (A) increases K
 (B) shifts the equilibrium forward
 (C) doesn't change the equilibrium composition, but equilibrium is reached faster
 (D) decreases K

2

Easy

⌚ 0.5 min

Speed

C06-T2

Which is a Lewis acid?

- (A) NH_3 (B) BF_3 (C) H_2O (D) Cl^-

3

Easy

⌚ 1 min

NV

C06-T3

Find the pH of 0.001 M $\text{Ba}(\text{OH})_2$ (complete dissociation) to the nearest integer.

Your answer (numerical value):

4 Easy 0.5 min Speed

C06-T4

For Ag_2CrO_4 with molar solubility s , K_{sp} is:

- (A) s^2 (B) $2s^3$ (C) $4s^3$ (D) $27s^4$

5 Easy 0.5 min Concept

C06-T5

On diluting a weak acid, its degree of dissociation:

- (A) decreases (B) increases (C) stays constant (D) becomes zero

Answers & Solutions

C06 · Practice

- 1 (C) 2 (B) 3 (B) 4 (A) 5 (C) 6 (C) 7 (B) 8 7 9 (A) 10 (B)
11 (A)

C06 · Chapter Test

- 1 (C) 2 (B) 3 11 4 (C) 5 (B)

C06 · PRACTICE

1 Answer: (C) K_p and K_c Easy 0.75 min Basic

C06-01

SOLUTION $\Delta n_g = 2 - 4 = -2$.COMMON TRAP Reversing the sign of Δn_g .

2 Answer: (B) Reaction quotient Easy 0.5 min Concept

C06-02

SOLUTION Too few products relative to equilibrium, so the net reaction goes forward until $Q = K$.

3 Answer: (B) Effect of pressure Easy 0.5 min Speed

C06-03

SOLUTION The shift is towards fewer gas moles: 1 mol N_2O_4 vs 2 mol NO_2 . K_p is unchanged.

4 Answer: (A) pH of a weak acid Medium 1 min Calc

C06-04

SOLUTION $[\text{H}^+] = \sqrt{1.8 \times 10^{-6}} \approx 1.34 \times 10^{-3}$; pH ≈ 2.87 .SHORTCUT pH = $\frac{1}{2}(4.74 + 1) = 2.87$.

COMMON TRAP Treating it as a strong acid (pH 1).

5 Answer: (C) Buffer pH Medium 1 min Basic

C06-05

SOLUTION pH = $4.74 + \log(0.2/0.02) = 4.74 + 1 = 5.74$.

COMMON TRAP Inverting the ratio.

6 Answer: (C) K_{sp} from solubility Medium 1 min Calc

C06-06

SOLUTION $K_{sp} = (s)(2s)^2 = 4s^3 = 4 \times 8 \times 10^{-12} = 3.2 \times 10^{-11}$.COMMON TRAP Using s^2 .

7 Answer: (B) Common-ion effect Medium 1 min Concept

C06-07

SOLUTION $s(0.1 + s) = 1.8 \times 10^{-10} \Rightarrow s \approx 1.8 \times 10^{-9} \text{ M}$.SHORTCUT $s \approx K_{sp}/[\text{common ion}]$.

COMMON TRAP Answering the solubility in pure water (A).

8 Answer: 7 pH of very dilute HCl Hard 2 min NV

C06-08

SOLUTION $[\text{H}^+] = 10^{-8} + x$ with $x(10^{-8} + x) = 10^{-14}$. So $x \approx 9.5 \times 10^{-8}$ and $[\text{H}^+] \approx 1.05 \times 10^{-7}$, giving pH $\approx 6.98 \rightarrow 7$.

SHORTCUT An acid solution can't have pH > 7. At this dilution, water's ions dominate.

COMMON TRAP Answering 8.

9 Answer: (A) *Hydrolysis of sodium acetate* **Medium** ⌚ 1.5 min JEE

C06-09

SOLUTION $\text{pH} = 7 + \frac{1}{2}(4.74) + \frac{1}{2}\log 0.1 = 7 + 2.37 - 0.5 = 8.87$.

SHORTCUT Salt of a weak acid and strong base is basic.

COMMON TRAP Sign of the $\log C$ term.

10 Answer: (B) *Conjugate base* **Easy** ⌚ 0.5 min Speed

C06-10

SOLUTION Remove one H^+ .

COMMON TRAP Giving the conjugate acid.

11 Answer: (A) *K for reversed and doubled reaction* **Easy** ⌚ 0.75 min Basic

C06-11

SOLUTION Reverse: $1/4$. Double: $(1/4)^2 = 1/16$.

C06 · CHAPTER TEST

1 Answer: (C) *Role of a catalyst* **Easy** ⌚ 0.5 min Speed

C06-T1

SOLUTION It speeds up both directions equally.

2 Answer: (B) *Lewis acid* **Easy** ⌚ 0.5 min Speed

C06-T2

SOLUTION B has an incomplete octet and accepts an electron pair.

3 Answer: 11 *pH of a diacidic base* **Easy** ⌚ 1 min NV

C06-T3

SOLUTION $[\text{OH}^-] = 0.002 \text{ M}$, so $\text{pOH} = 2.70$ and $\text{pH} = 11.30 \approx 11$.

COMMON TRAP Forgetting the factor 2 (still pH 11, but the exact value is 11.0 rather than 11.3).

4 Answer: (C) *Ksp expression* **Easy** ⌚ 0.5 min Speed

C06-T4

SOLUTION $[\text{Ag}^+] = 2s$, $[\text{CrO}_4^{2-}] = s$.

5 Answer: (B) *Ostwald's dilution law* **Easy** ⌚ 0.5 min Concept

C06-T5

SOLUTION $\alpha = \sqrt{K_a/C}$ rises as C falls.

REVISION Revision tracker · Equilibrium

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 07

Redox Reactions & Electrochemistry

07

1–2 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 7;

Class 12 · Ch 2

NCERT

~9 hours

STUDY TIME

Important concepts

- **Oxidation** = loss of electrons, or increase in oxidation number. **Reduction** = gain of electrons, or decrease in oxidation number. The oxidising agent is itself reduced.
- **Types of redox reactions:** combination, decomposition, displacement, and **disproportionation** (the same element is both oxidised and reduced).
- **Galvanic cell:** a spontaneous redox reaction produces electricity. **Anode = oxidation (–), cathode = reduction (+)**. In electrolytic cells the signs reverse, but the anode is still where oxidation happens.
- **Nernst equation** relates cell emf to concentrations. $\Delta G = -nFE$ links electrochemistry and thermodynamics.
- **Conductance:** conductivity κ falls on dilution, while molar conductivity Λ_m rises. **Kohlrausch's law** gives Λ_m° of weak electrolytes.

Formula sheet

Σ FORMULA Oxidation-number rules (in order of priority)

F is always –1. O is –2, except in peroxides (–1), superoxides (–½), OF₂ (+2). H is +1 (–1 in metal hydrides). Alkali metals +1, alkaline-earth metals +2. The sum equals the charge on the species. Watch for **peroxide linkages**: S is +6 in both H₂SO₅ and H₂S₂O₈. Cr is +6 in CrO₅ (butterfly structure).

Σ FORMULA Titration & equivalents

n-factor = electrons exchanged per formula unit: KMnO₄ 5 (acidic), 3 (neutral), 1 (strongly basic); K₂Cr₂O₇ 6 (acidic); oxalic acid 2 (redox) or 2 (acid–base); Fe²⁺ → Fe³⁺ 1. At the equivalence point: $n_1M_1V_1 = n_2M_2V_2$.

Σ FORMULA Cells

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ \quad E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.0591}{n} \log Q \quad (298 \text{ K})$$

$$\Delta G = -nFE \quad \Delta G^\circ = -nFE^\circ = -2.303RT \log K \quad \log K = \frac{nE^\circ}{0.0591}$$

$F = 96\,500 \text{ C mol}^{-1}$. Cell notation: anode | anode solution || cathode solution | cathode. Concentration cell $M | M^{n+}(C_1) || M^{n+}(C_2) | M$; $E = \frac{0.0591}{n} \log \frac{C_2}{C_1}$ (spontaneous when $C_2 > C_1$). Hydrogen electrode: $E = -0.0591 \text{ pH}$ (at 1 bar H₂).

Σ FORMULA Electrolysis (Faraday's laws)

$$m = ZIt = \frac{M}{nF}It \quad \frac{m_1}{m_2} = \frac{E_1}{E_2} \text{ (same charge; } E = M/n \text{)}$$

Electrolyte (inert electrodes)	Cathode	Anode
molten NaCl	Na	Cl ₂
aqueous NaCl (concentrated)	H ₂	Cl ₂ (overpotential of O ₂)
aqueous CuSO ₄	Cu	O ₂
aqueous CuSO ₄ , Cu electrodes	Cu	Cu dissolves (purification)
dilute H ₂ SO ₄	H ₂	O ₂

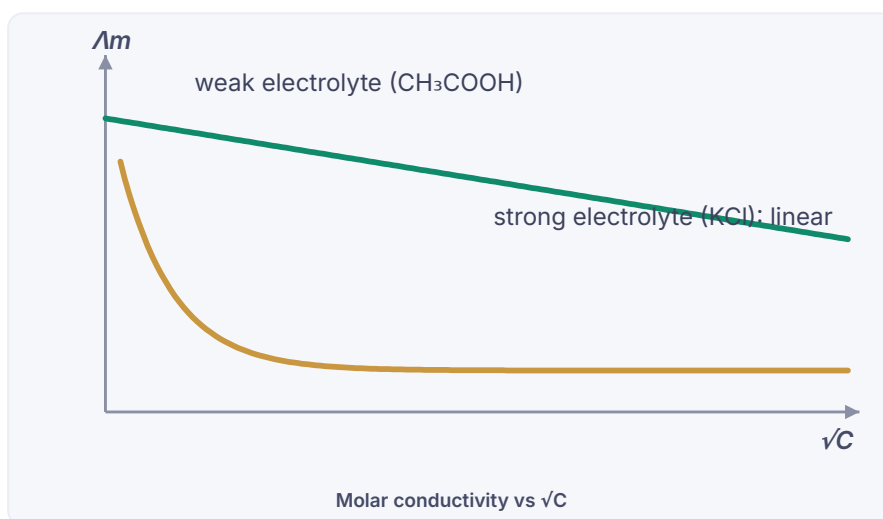
Σ FORMULA Conductance

$$G = \frac{1}{R} \quad \kappa = G \cdot \frac{l}{A} = \frac{G^*}{R} \quad \Lambda_m = \frac{\kappa \times 1000}{M} \text{ (S cm}^2\text{ mol}^{-1}, \kappa \text{ in S cm}^{-1}\text{)}$$

Strong electrolyte: $\Lambda_m = \Lambda_m^\circ - A\sqrt{C}$ (linear in $\sqrt{C}$). **Weak electrolyte:** steep rise at low C ; use Kohlrausch.

Kohlrausch: $\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$; e.g. $\Lambda^\circ(\text{CH}_3\text{COOH}) = \Lambda^\circ(\text{CH}_3\text{COONa}) + \Lambda^\circ(\text{HCl}) - \Lambda^\circ(\text{NaCl})$. **Weak**

electrolyte: $\alpha = \Lambda_m / \Lambda_m^\circ$; $K_a = \frac{C\alpha^2}{1 - \alpha}$.


Batteries & fuel cells (NCERT reactions)

Cell	Anode	Cathode	Notes
Dry (Leclanché)	$\text{Zn} \longrightarrow \text{Zn}^{2+} + 2\text{e}^-$	$\text{MnO}_2 + \text{NH}_4^+ + \text{e}^- \longrightarrow \text{MnO}(\text{OH}) + \text{NH}_3$	~1.5 V; primary
Mercury	$\text{Zn}(\text{Hg}) + 2\text{OH}^- \longrightarrow \text{ZnO} + \text{H}_2\text{O} + 2\text{e}^-$	$\text{HgO} + \text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{Hg} + 2\text{OH}^-$	1.35 V, constant (no ions change in overall reaction)
Lead storage	$\text{Pb} + \text{SO}_4^{2-} \longrightarrow \text{PbSO}_4 + 2\text{e}^-$	$\text{PbO}_2 + \text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^- \longrightarrow \text{PbSO}_4 + 2\text{H}_2\text{O}$	secondary; H ₂ SO ₄ is consumed on discharge (density falls)
H ₂ -O ₂ fuel cell	$2\text{H}_2 + 4\text{OH}^- \longrightarrow 4\text{H}_2\text{O} + 4\text{e}^-$	$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \longrightarrow 4\text{OH}^-$	~70% efficient; used in Apollo missions; product is water

Common traps

MISTAKE ALERT Mistake alerts

- E° is **intensive**. Don't multiply it by coefficients when balancing (but ΔG° does scale).
- Nernst Q uses products over reactants; solids and pure liquids are omitted.
- For KMnO_4 , the n -factor depends on the medium (5/3/1).
- Molar conductivity units: with κ in S cm^{-1} , use $\times 1000/M$ to get $\text{S cm}^2 \text{ mol}^{-1}$.
- The reducing agent is the one with the **more negative** reduction potential.

Question patterns

PYQ PATTERN Recurring structures

1. Oxidation numbers (including peroxide linkages); balancing and n -factor; titration stoichiometry.
2. E°_{cell} , spontaneity, Nernst calculations (a very common NV type).
3. ΔG° and K from E° .
4. Faraday's laws: mass deposited, time, charge; comparative deposition.
5. Conductivity $\leftrightarrow$ molar conductivity; Kohlrausch; degree of dissociation.
6. Battery and fuel-cell reactions (NCERT text).

Practice questions

C07 · Practice

1 Medium 1 min Concept

C07-01

The oxidation number of sulphur in $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulphuric acid) is:

- (A) +7 (B) +6 (C) +8 (D) +5

2 Easy 0.75 min Concept

C07-02

Which reaction is a disproportionation?

- (A) $\text{Zn} + \text{CuSO}_4 \longrightarrow \text{ZnSO}_4 + \text{Cu}$ (B) $\text{Cl}_2 + 2\text{OH}^- \longrightarrow \text{Cl}^- + \text{ClO}^- + \text{H}_2\text{O}$
 (C) $2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$ (D) $\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2$

3 Easy 0.5 min Speed

C07-03

$E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$ and $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$. The standard emf of the Daniell cell is:

- (A) 0.42 V (B) 1.10 V (C) -1.10 V (D) 0.34 V

4 Medium 1.5 min Calc

C07-04

For $\text{Zn} | \text{Zn}^{2+}(0.01 \text{ M}) || \text{Cu}^{2+}(1 \text{ M}) | \text{Cu}$ at 298 K ($E^\circ = 1.10 \text{ V}$), the cell emf is about:

- (A) 1.041 V (B) 1.100 V (C) 1.159 V (D) 1.218 V

5 Easy 1 min Calc

C07-05

For a cell with $n = 2$ and $E^\circ = 1.10 \text{ V}$, ΔG° is about:

- (A) -212 kJ/mol (B) -106 kJ/mol (C) +212 kJ/mol (D) -21.2 kJ/mol

6

Medium

1 min

Calc

C07-06

The mass of copper deposited by a current of 2 A passed for 965 s through CuSO_4 solution is ($\text{Cu} = 63.5$):

- (A) 0.635 g (B) 1.27 g (C) 0.3175 g (D) 6.35 g

7

Easy

1 min

NV

C07-07

The conductivity of 0.1 M KCl is $1.29 \times 10^{-2} \text{ S cm}^{-1}$. Find its molar conductivity (in $\text{S cm}^2 \text{ mol}^{-1}$).

Your answer (numerical value):

8

Medium

1 min

Basic

C07-08

Λ_m° ($\text{S cm}^2 \text{ mol}^{-1}$): CH_3COONa 91, HCl 426, NaCl 126. Λ_m° of CH_3COOH is:

- (A) 391 (B) 643 (C) 461 (D) 209

9

Easy

0.75 min

Basic

C07-09

Acetic acid at some concentration has $\Lambda_m = 39.1 \text{ S cm}^2 \text{ mol}^{-1}$, and $\Lambda_m^\circ = 391$. Its degree of dissociation is:

- (A) 0.01 (B) 0.1 (C) 0.5 (D) 10

10

Easy

1 min

NV

C07-10

For a cell reaction with $n = 2$ and $E^\circ = 0.0591 \text{ V}$ at 298 K, find the equilibrium constant.

Your answer (numerical value):

11

Easy

0.5 min

Concept

C07-11

Electrolysis of concentrated aqueous NaCl with inert electrodes gives:

- (A) Na at the cathode, Cl_2 at the anode (B) H_2 at the cathode, Cl_2 at the anode
(C) H_2 at the cathode, O_2 at the anode (D) Na at the cathode, O_2 at the anode

12

Easy

0.75 min

Concept

C07-12

When a lead storage battery discharges:

- (A) H_2SO_4 is produced
(B) PbSO_4 forms at both electrodes and the acid's density falls
(C) Pb is deposited on the cathode
(D) PbO_2 forms at the anode



C07 · Chapter Test

1

Easy

0.5 min

Speed

C07-T1

Which is the strongest reducing agent?

- (A) Li (B) Na (C) Zn (D) Cu

2

Easy

0.5 min

Speed

C07-T2

The n-factor of KMnO_4 in acidic medium is:

- (A) 1 (B) 3 (C) 5 (D) 7

3 Easy 0.5 min NV

C07-T3

How many moles of electrons are needed to reduce 1 mol of $\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+} ?Your answer (numerical value): **4** Easy 0.5 min Concept

C07-T4

For a strong electrolyte, a plot of Λ_m against $\sqrt{C}$ is:

- (A) a straight line with positive slope (B) a straight line with negative slope
(C) a curve rising steeply at low C (D) horizontal

5 Easy 0.5 min Speed

C07-T5

The product of an $\text{H}_2\text{-O}_2$ fuel cell is:

- (A) H_2O_2 (B) H_2O (C) O_3 (D) OH^- only

Answers & Solutions

C07 · Practice

- 1 (B) 2 (B) 3 (B) 4 (C) 5 (A) 6 (A) 7 129 8 (A) 9 (B) 10 100
11 (B) 12 (B)

C07 · Chapter Test

- 1 (A) 2 (C) 3 6 4 (B) 5 (B)

C07 · PRACTICE

1 Answer: (B) Peroxide linkage Medium 1 min Concept

C07-01

SOLUTION The molecule has one O-O (peroxide) bond. Six O are -2 and two are -1: $2(+1) + 2x + 6(-2) + 2(-1) = 0 \Rightarrow x = +6$.

SHORTCUT S can't exceed +6 (group 16).

COMMON TRAP Assuming all O are -2 gives +7.

2 Answer: (B) Disproportionation Easy 0.75 min Concept

C07-02

SOLUTION Cl goes from 0 to -1 (reduced) and to +1 (oxidised).

3 Answer: (B) Standard cell emf Easy 0.5 min Speed

C07-03

SOLUTION $E^\circ = 0.34 - (-0.76) = 1.10 \text{ V}$.

4 Answer: (C) Nernst equation Medium 1.5 min Calc

C07-04

SOLUTION $Q = [\text{Zn}^{2+}]/[\text{Cu}^{2+}] = 0.01$. $E = 1.10 - \frac{0.0591}{2} \log 0.01 = 1.10 + 0.059 = 1.159 \text{ V}$.

SHORTCUT A smaller Q (fewer products) raises E.

COMMON TRAP Inverting Q gives 1.041 V.

5 Answer: (A) ΔG° from E° Easy 1 min Calc

C07-05

SOLUTION $\Delta G^\circ = -2 \times 96\,500 \times 1.10 \approx -212\,300 \text{ J/mol}$.

6 Answer: (A) Faraday's first law Medium 1 min Calc

C07-06

SOLUTION $Q = 1930 \text{ C} = 0.02 \text{ F}$. Cu^{2+} needs 2 F per mol: $0.01 \text{ mol} = 0.635 \text{ g}$.

COMMON TRAP Using $n = 1$ (1.27 g).

7 Answer: 129 Molar conductivity Easy 1 min NV

C07-07

SOLUTION $\Lambda_m = \frac{\kappa \times 1000}{M} = \frac{1.29 \times 10^{-2} \times 1000}{0.1} = 129$.

8 Answer: (A) Kohlrausch's law **Medium** ⌚ 1 min Basic

C07-08

SOLUTION $91 + 426 - 126 = 391$.

SHORTCUT Add the two salts that contain the ions you want; subtract the one containing the unwanted ions.

9 Answer: (B) Degree of dissociation **Easy** ⌚ 0.75 min Basic

C07-09

SOLUTION $\alpha = 39.1/391 = 0.1$.

10 Answer: 100 Equilibrium constant from E° **Easy** ⌚ 1 min NV

C07-10

SOLUTION $\log K = \frac{nE^\circ}{0.0591} = 2 \Rightarrow K = 100$.

11 Answer: (B) Electrolysis of brine **Easy** ⌚ 0.5 min Concept

C07-11

SOLUTION Water is reduced more easily than Na^+ . Cl^- is oxidised in preference to water because of oxygen's overpotential.

COMMON TRAP Using the molten-NaCl result (A).

12 Answer: (B) Lead storage battery **Easy** ⌚ 0.75 min Concept

C07-12

SOLUTION Both electrodes turn to PbSO_4 , and H_2SO_4 is consumed, producing water.

C07 · CHAPTER TEST

1 Answer: (A) Strongest reducing agent **Easy** ⌚ 0.5 min Speed

C07-T1

SOLUTION Li has the most negative E° (-3.05 V), because of its high hydration enthalpy.

COMMON TRAP Choosing Cs/Na by atomic-size reasoning alone.

2 Answer: (C) n -factor of permanganate **Easy** ⌚ 0.5 min Speed

C07-T2

SOLUTION $\text{MnO}_4^- (+7) \rightarrow \text{Mn}^{2+} (+2)$.

3 Answer: 6 Electrons for dichromate reduction **Easy** ⌚ 0.5 min NV

C07-T3

SOLUTION Each Cr goes from +6 to +3 (3 electrons); $\times 2 = 6$.

4 Answer: (B) Λ_m vs $\sqrt{C}$ **Easy** ⌚ 0.5 min Concept

C07-T4

SOLUTION Debye-Hückel-Onsager: $\Lambda_m = \Lambda_m^\circ - A\sqrt{C}$.

COMMON TRAP Choosing (C), which describes a weak electrolyte.

5 Answer: (B) Fuel cell **Easy** ⌚ 0.5 min Speed

C07-T5

SOLUTION Overall: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$.

REVISION Revision tracker · Redox & Electrochemistry

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 08

Chemical Kinetics

08

1-2 per shift

TYPICAL QUESTIONS

Easy-Medium

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 3

NCERT

~7 hours

STUDY TIME

Important concepts

- **Rate** of $aA + bB \rightarrow cC$: $-\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt}$.
- **Rate law** $r = k[A]^x[B]^y$ is **experimental**. The order $x + y$ can be zero or fractional.
- **Molecularity** is the number of species colliding in an **elementary** step (1, 2 or 3, never 0 or fractional). For complex reactions, the slowest step determines the rate law.
- **Temperature** raises the rate sharply, roughly doubling it per 10 °C rise for many reactions. Arrhenius: $k = Ae^{-E_a/RT}$.
- **Catalysts** lower E_a by providing an alternative path. They don't change ΔH , ΔG or K .
- **Collision theory**: rate = $PZ_{AB}e^{-E_a/RT}$. Only collisions with enough energy **and** the proper orientation (steric factor P) react.

Formula sheet

Σ FORMULA Integrated rate laws

Order	Integrated law	Half-life	Linear plot	Units of k
0	$[A] = [A]_0 - kt$	$\frac{[A]_0}{2k}$	$[A]$ vs t	$\text{mol L}^{-1} \text{s}^{-1}$
1	$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$	$\frac{0.693}{k}$	$\ln[A]$ vs t (slope $-k$)	s^{-1}
n (general)	—	$\propto [A]_0^{1-n}$	—	$(\text{mol L}^{-1})^{1-n} \text{s}^{-1}$

First order: time for 99.9% completion $\approx 10 t_{1/2}$; 99% $\approx 6.64 t_{1/2}$; 75% $= 2t_{1/2}$; 87.5% $= 3t_{1/2}$. **Gas phase**

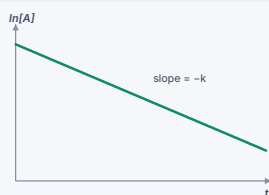
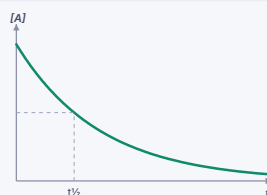
$A(g) \rightarrow B(g) + C(g)$ (first order, total pressure p_t): $k = \frac{2.303}{t} \log \frac{p_0}{2p_0 - p_t}$. **Pseudo-first-order**: acid hydrolysis of an ester in excess water; inversion of cane sugar.

Σ FORMULA Temperature dependence

$$k = Ae^{-E_a/RT} \quad \ln k = \ln A - \frac{E_a}{RT} \quad \log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Plot of $\ln k$ vs $1/T$: slope = $-E_a/R$. Fraction of molecules with energy $\geq E_a$ is $e^{-E_a/RT}$. Energy profile:

$E_a(\text{backward}) = E_a(\text{forward}) - \Delta H$. For an exothermic reaction, the backward barrier is larger.

Zero order: $[A]$ vs t First order: $\ln[A]$ vs t First order: $[A]$ vs t Arrhenius: $\ln k$ vs $1/T$

Numerical-solving method

1. From data tables, find the order by comparing two runs where only one concentration changes: $\frac{r_2}{r_1} = \left(\frac{C_2}{C_1} \right)^x$.

- For first-order time questions, convert "x% completed" into the fraction remaining, then use half-lives if the number is a power of $\frac{1}{2}$.
- For Arrhenius, use $2.303R = 19.15 \text{ J K}^{-1} \text{ mol}^{-1}$ and keep $1/T$ in K^{-1} .

⚡ SHORTCUT The 10-degree rule

If the rate doubles for every 10°C rise, a rise of ΔT multiplies the rate by $2^{\Delta T/10}$. Also memorise: a rate that doubles between 300 K and 310 K corresponds to $E_a \approx 53.6 \text{ kJ/mol}$.

Common traps

⚠ MISTAKE ALERT Mistake alerts

- Order $\neq$ stoichiometric coefficient (except in elementary steps).
- The half-life of a zero-order reaction **decreases** as $[A]_0$ decreases ($\propto [A]_0$). For first order it is independent of $[A]_0$.
- Mixing up log and ln: $t_{1/2} = 0.693/k$ uses ln 2.
- A catalyst changes k (and E_a), but **not** K_{eq} .
- Rate-of-appearance questions: divide by the stoichiometric coefficients.

Question patterns

📊 PYQ PATTERN Recurring structures

- Order from initial-rate tables (Section-B favourite).
- First-order time/half-life calculations; % completion.
- Arrhenius: E_a from two temperatures; effect of a catalyst on E_a and rate.
- Units of $k \leftrightarrow$ order.
- Graph identification (which plot is linear for which order).
- Relations between rates of disappearance/appearance.
- Kinetic study of the iodide– H_2O_2 reaction (practical chemistry).

Practice questions

🔗 C08 · Practice

1

Easy

⌚ 0.5 min

Speed

C08-01

The units of the rate constant of a second-order reaction are:

- (A) s^{-1} (B) $\text{mol L}^{-1} \text{s}^{-1}$ (C) $\text{L mol}^{-1} \text{s}^{-1}$ (D) $\text{L}^2 \text{mol}^{-2} \text{s}^{-1}$

2

Medium

⌚ 1 min

Calc

C08-02

A first-order reaction has $k = 0.0693 \text{ min}^{-1}$. The time for 99% completion is about:

- (A) 10 min (B) 46 min (C) 66.5 min (D) 100 min

3

Easy

⌚ 1 min

NV

C08-03

A zero-order reaction has $[A]_0 = 0.4 \text{ M}$ and $k = 0.01 \text{ M min}^{-1}$. Find its half-life (in minutes).

Your answer (numerical value):

4

Medium

⌚ 1.5 min

Calc

C08-04

The rate constant of a reaction doubles when the temperature rises from 300 K to 310 K. Its activation energy is about ($R = 8.314$, $\log 2 = 0.301$):

- (A) 26.8 kJ/mol (B) 53.6 kJ/mol (C) 107 kJ/mol (D) 5.36 kJ/mol

5

Easy

⌚ 0.5 min

NV

C08-05

When the concentration of a reactant is doubled, the rate becomes 8 times larger. Find the order with respect to that reactant.

Your answer (numerical value):

6

Medium

⌚ 1 min

Concept

C08-06

For $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$, the rate of formation of O_2 is $1 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$. The rate of formation of NO_2 is:

- (A) 1×10^{-3} (B) 2×10^{-3} (C) 4×10^{-3} (D) 0.25×10^{-3}

7

Easy

⌚ 0.5 min

Concept

C08-07

A catalyst increases the rate of a reaction by:

- (A) increasing ΔH (B) lowering the activation energy
(C) increasing the equilibrium constant (D) increasing the number of collisions only

8

Easy

⌚ 0.5 min

Concept

C08-08

The acid-catalysed hydrolysis of ethyl acetate in a large excess of water is:

- (A) zero order
(B) pseudo-first order
(C) truly second order in the observed rate law
(D) third order

9

Easy

⌚ 0.5 min

Concept

C08-09

A plot of $\ln k$ against $1/T$ is a straight line with slope:

- (A) E_a/R (B) $-E_a/R$ (C) $-E_a/2.303R$ (D) $\ln A$

10

Easy

⌚ 0.5 min

Concept

C08-10

Which statement is correct?

- (A) Molecularity can be fractional. (B) Order can be zero or fractional.
(C) Order is always equal to molecularity. (D) Molecularity is determined experimentally.



C08 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

C08-T1

The half-life of a first-order reaction:

- (A) is proportional to the initial concentration
(B) is independent of the initial concentration
(C) is inversely proportional to the initial concentration
(D) depends on the square of the initial concentration

2 Easy 0.5 min NV

C08-T2

A first-order reaction is 75% complete in 30 min. Find its half-life (in minutes).

Your answer (numerical value):

3 Easy 0.5 min Speed

C08-T3

The units of k for a zero-order reaction are:

- (A) s^{-1} (B) $\text{mol L}^{-1} \text{s}^{-1}$ (C) $\text{L mol}^{-1} \text{s}^{-1}$ (D) dimensionless

4 Easy 0.75 min Basic

C08-T4

If a reaction's rate doubles for every 10°C rise, raising the temperature from 20°C to 50°C multiplies the rate by:

- (A) 3 (B) 6 (C) 8 (D) 9

5 Medium 0.75 min Concept

C08-T5

For an exothermic reaction, the activation energy of the backward reaction is:

- (A) smaller than that of the forward reaction (B) larger than that of the forward reaction
(C) equal to that of the forward reaction (D) zero

Answers & Solutions

C08 · Practice

1 (C) 2 (C) 3 20 4 (B) 5 3 6 (C) 7 (B) 8 (B) 9 (B) 10 (B)

C08 · Chapter Test

1 (B) 2 15 3 (B) 4 (C) 5 (B)

C08 · PRACTICE

1 Answer: (C) Units of k Easy 0.5 min Speed

C08-01

SOLUTION $(\text{mol L}^{-1})^{1-n} \text{s}^{-1}$ with $n = 2$.

2 Answer: (C) 99% completion time Medium 1 min Calc

C08-02

SOLUTION $t = \frac{2.303}{0.0693} \log 100 = 33.23 \times 2 \approx 66.5 \text{ min.}$

SHORTCUT $t_{99\%} \approx 6.64 t_{1/2} = 6.64 \times 10$.

3 Answer: 20 Zero-order half-life Easy 1 min NV

C08-03

SOLUTION $t_{1/2} = [A]_0 / 2k = 0.4 / 0.02 = 20 \text{ min.}$

COMMON TRAP Using $0.693/k$.

4 Answer: (B) Arrhenius two-temperature Medium 1.5 min Calc

C08-04

SOLUTION $0.301 = \frac{E_a}{2.303 \times 8.314} \left(\frac{10}{300 \times 310} \right) \Rightarrow E_a = \frac{0.301 \times 19.15 \times 93000}{10} \approx 53600 \text{ J/mol.}$

SHORTCUT Standard result: doubling between 300 K and 310 K means $E_a \approx 53.6 \text{ kJ/mol.}$

5 Answer: 3 Order from rate change Easy 0.5 min NV

C08-05

SOLUTION $2^x = 8 \Rightarrow x = 3$.

6 Answer: (C) Relating rates Medium 1 min Concept

C08-06

SOLUTION $\frac{1}{4} \frac{d[\text{NO}_2]}{dt} = \frac{d[\text{O}_2]}{dt} \Rightarrow \frac{d[\text{NO}_2]}{dt} = 4 \times 10^{-3}$.

COMMON TRAP Dividing instead of multiplying.

7 Answer: (B) *Effect of catalyst* **Easy** ⌚ 0.5 min **Concept**

C08-07

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

8 Answer: (B) *Pseudo-first-order reaction* **Easy** ⌚ 0.5 min **Concept**

C08-08

SOLUTION Water's concentration is effectively constant, so the rate law reduces to $r = k'[\text{ester}]$.

9 Answer: (B) *Arrhenius plot* **Easy** ⌚ 0.5 min **Concept**

C08-09

SOLUTION $\ln k = \ln A - \frac{E_a}{R} \cdot \frac{1}{T}$. (For $\log k$ the slope is $-E_a/2.303R$.)

COMMON TRAP Mixing up $\ln$ and $\log$ plots.

10 Answer: (B) *Molecularity vs order* **Easy** ⌚ 0.5 min **Concept**

C08-10

SOLUTION Order is experimental and can take any value. Molecularity is a theoretical positive integer for an elementary step.

C08 · CHAPTER TEST

1 Answer: (B) *First-order half-life* **Easy** ⌚ 0.5 min **Speed**

C08-T1

SOLUTION $t_{1/2} = 0.693/k$.

2 Answer: 15 *Half-life from 75% completion* **Easy** ⌚ 0.5 min **NV**

C08-T2

SOLUTION 75% complete means two half-lives have passed, so $t_{1/2} = 15$ min.

3 Answer: (B) *Units of zero-order k* **Easy** ⌚ 0.5 min **Speed**

C08-T3

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

4 Answer: (C) *Temperature-coefficient rule* **Easy** ⌚ 0.75 min **Basic**

C08-T4

SOLUTION $2^{30/10} = 8$.

COMMON TRAP Multiplying 2×3 .

5 Answer: (B) *Barrier heights* **Medium** ⌚ 0.75 min **Concept**

C08-T5

SOLUTION $E_a(b) = E_a(f) - \Delta H$, and $\Delta H < 0$.

REVISION Revision tracker · Chemical Kinetics

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 09

Classification of Elements & Periodicity

09

~1 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

Must-do

PRIORITY

Class 11 · Ch 3

NCERT

~5 hours

STUDY TIME

Important concepts

- **Modern periodic law (Moseley):** properties are periodic functions of **atomic number**. The long form has 18 groups and 7 periods, divided into s, p, d and f blocks by the subshell being filled.
- **Elements with $Z > 100$** get IUPAC temporary names from digit roots: 0 nil, 1 un, 2 bi, 3 tri, 4 quad, 5 pent, 6 hex, 7 sept, 8 oct, 9 enn, plus "-ium". So 119 = ununennium (Uue), 120 = unbilium (Ubn).
- **Periodic trends** come from effective nuclear charge and shell number: radius, ionisation enthalpy, electron gain enthalpy, electronegativity, valence, metallic character, nature of oxides.

Trends at a glance

Property	Across a period (→)	Down a group (↓)	Key exceptions
Atomic radius	decreases	increases	noble gases (van der Waals radii) look large; Ga < Al (d-contraction)
Ionisation enthalpy	increases	decreases	Be > B, N > O (also Mg > Al, P > S); group 13 order B > Tl > Ga > Al > In
Electron gain enthalpy (more negative)	increases	decreases	Cl more negative than F ; S more negative than O; noble gases, Be, Mg, N ≈ positive
Electronegativity	increases	decreases	F highest (4.0 Pauling)
Metallic character	decreases	increases	—
Oxide nature	basic → amphoteric → acidic	more basic	amphoteric: BeO, Al ₂ O ₃ , ZnO, Ga ₂ O ₃ , SnO ₂ , PbO ₂ ...

☆ IMPORTANT Ionic radii rules

- Cation < parent atom < anion.
- **Isoelectronic species:** radius falls as Z rises. $N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$ (all 10 electrons).
- Down a group, ions of the same charge get larger.

☆ IMPORTANT Ionisation enthalpy essentials

- $IE_1 < IE_2 < IE_3 < \dots$ A **huge jump** means you've reached a noble-gas core, which tells you the group (e.g. a big jump after IE_2 means group 2).
- Half-filled (N: $2p^3$) and fully filled (Be: $2s^2$) subshells are extra stable, which causes the period-2 exceptions.
- Group 13 anomaly: Ga's poor 3d shielding and Tl's poor 4f/5d shielding raise their IE values.

Σ FORMULA Oxides across period 3

Na₂O (strongly basic) · MgO (basic) · Al₂O₃ (amphoteric) · SiO₂ (weakly acidic) · P₄O₁₀ (acidic) · SO₃ (strongly acidic) · Cl₂O₇ (most acidic). For the same element, a higher oxidation state gives a more acidic oxide (Mn₂O₇ acidic, MnO basic).

Memory tricks

SHORTCUT Exceptions sentence

"Before Boron, Nitrogen beats Oxygen, and Clever chlorine grabs electrons better than fluorine." That covers the three most-asked exceptions: IE ($Be > B$, $N > O$) and $\Delta_{eg}H$ ($Cl > F$).

Common traps

MISTAKE ALERT Mistake alerts

- Electron gain enthalpy is **negative** when energy is released. "Highest electron affinity" means most negative $\Delta_{eg}H$: Cl, not F.
- The smallest atom of period 2 is F, but **He is the smallest atom overall**.
- Radii of noble gases can't be compared with covalent radii.
- IE_2 of Na is exceptionally high (it would break the neon core).

Question patterns

PYQ PATTERN Recurring structures

- Ordering IE / EGE / radius / EN for 3–4 elements (with the exceptions above).
- Isoelectronic ion radii.
- Identifying group/period/block from Z or configuration.
- IUPAC names for $Z > 100$.
- Acidic/basic/amphoteric oxides.
- Successive IE data $\rightarrow$ group identification.

Practice questions

C09 · Practice

1

Easy

 0.5 min

Speed

C09-01

The IUPAC temporary name of the element with $Z = 119$ is:

- (A) ununennium (B) unennium (C) ununoctium (D) unbiennium

2

Easy

 0.5 min

Speed

C09-02

The correct order of ionic radii is:

- (A) $N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+}$ (B) $Mg^{2+} > Na^+ > F^- > O^{2-} > N^{3-}$
 (C) $Na^+ > Mg^{2+} > F^- > O^{2-} > N^{3-}$ (D) all equal (isoelectronic)

3

Easy

 0.5 min

Speed

C09-03

Which element has the most negative electron gain enthalpy?

- (A) F (B) Cl (C) O (D) Br

4

Medium

 1 min

Concept

C09-04

The correct order of first ionisation enthalpies is:

- (A) $Na < Mg < Al < Si$ (B) $Na < Al < Mg < Si$
 (C) $Al < Na < Mg < Si$ (D) $Si < Al < Mg < Na$

5 Easy 0.5 min Speed

C09-05

Which oxide is amphoteric?

- (A) Na_2O (B) Al_2O_3 (C) SO_3 (D) P_4O_{10}

6 Medium 1 min Concept

C09-06

The atomic radius of Ga (135 pm) is slightly smaller than that of Al (143 pm) because:

- (A) Ga has more shells
 (B) the 3d electrons in Ga shield the nucleus poorly, raising the effective nuclear charge
 (C) Ga is a metalloid
 (D) Al has a completely filled d subshell

7 Easy 0.5 min Speed

C09-07

The most electronegative element is:

- (A) O (B) Cl (C) F (D) N

8 Easy 0.75 min NV

C09-08

Element $Z = 34$ belongs to which group of the periodic table?Your answer (numerical value):

9 Medium 1 min Concept

C09-09

Which element has the highest second ionisation enthalpy?

- (A) Na (B) Mg (C) Al (D) Si

10 Easy 0.75 min Basic

C09-10

The correct order of acidic strength is:

- (A) $\text{Cl}_2\text{O}_7 > \text{SO}_3 > \text{P}_4\text{O}_{10} > \text{SiO}_2$ (B) $\text{SiO}_2 > \text{P}_4\text{O}_{10} > \text{SO}_3 > \text{Cl}_2\text{O}_7$
 (C) $\text{SO}_3 > \text{Cl}_2\text{O}_7 > \text{SiO}_2 > \text{P}_4\text{O}_{10}$ (D) all equally acidic

C09 · Chapter Test

1 Easy 0.5 min Speed

C09-T1

An element with configuration $[\text{Ar}]3d^{10}4s^24p^1$ belongs to the:

- (A) s-block (B) p-block (C) d-block (D) f-block

2 Easy 0.5 min Speed

C09-T2

The correct order of size is:

- (A) $\text{Mg}^{2+} < \text{Na}^+ < \text{F}^- < \text{O}^{2-}$ (B) $\text{O}^{2-} < \text{F}^- < \text{Na}^+ < \text{Mg}^{2+}$
 (C) $\text{Na}^+ < \text{Mg}^{2+} < \text{O}^{2-} < \text{F}^-$ (D) $\text{F}^- < \text{O}^{2-} < \text{Mg}^{2+} < \text{Na}^+$

3 Easy 0.5 min Speed

C09-T3

The electron gain enthalpies of the noble gases are:

- (A) large and negative (B) positive
 (C) zero exactly (D) the most negative in their period

4 Easy 0.5 min NV

C09-T4

How many elements are there in the 4th period of the periodic table?

Your answer (numerical value):

5 Easy 0.5 min Speed

C09-T5

Which element is the most metallic?

- (A) Li (B) Na (C) K (D) Cs

Answers & Solutions

C09 · Practice

1 (A) 2 (A) 3 (B) 4 (B) 5 (B) 6 (B) 7 (C) 8 16 9 (A) 10 (A)

C09 · Chapter Test

1 (B) 2 (A) 3 (B) 4 18 5 (D)

C09 · PRACTICE

1 Answer: (A) IUPAC name for $Z = 119$ Easy 0.5 min Speed

C09-01

SOLUTION 1-1-9 → un-un-enn + ium, which is ununennium (Uue).

2 Answer: (A) Isoelectronic radii Easy 0.5 min Speed

C09-02

SOLUTION With the same 10 electrons, a larger nuclear charge pulls them in more tightly.

COMMON TRAP Choosing (D).

3 Answer: (B) Most negative electron gain enthalpy Easy 0.5 min Speed

C09-03

SOLUTION F's small 2p shell has high electron–electron repulsion, which makes its $\Delta_{eg}H$ less negative than Cl's.

COMMON TRAP Choosing F.

4 Answer: (B) First ionisation enthalpy order Medium 1 min Concept

C09-04

SOLUTION Mg ($3s^2$, filled) > Al ($3s^23p^1$), where the 3p electron is easier to remove.

COMMON TRAP Ignoring the Mg > Al exception.

5 Answer: (B) Amphoteric oxide Easy 0.5 min Speed

C09-05

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

6 Answer: (B) Gallium vs aluminium Medium 1 min Concept

C09-06

SOLUTION The ten 3d electrons added between Al and Ga are poor shielders, so the valence electrons feel a higher Z_{eff} .

7 Answer: (C) Highest electronegativity Easy 0.5 min Speed

C09-07

SOLUTION F: 4.0 on the Pauling scale.

8 Answer: 16 Group from atomic number Easy 0.75 min NV

C09-08

SOLUTION $[\text{Ar}]3d^{10}4s^24p^4$: 6 valence (s + p) electrons in the p-block, so group $10 + 6 = 16$ (Se).

SHORTCUT p-block group number = $10 + (\text{valence s} + \text{p electrons})$.

COMMON TRAP Answering 6 (old group notation VI).

9 Answer: (A) Second ionisation enthalpy Medium 1 min Concept

C09-09

SOLUTION Na^+ has a neon configuration, so removing a second electron breaks a noble-gas core.

COMMON TRAP Choosing Si by the general trend.

10 Answer: (A) Acidic strength of oxides Easy 0.75 min Basic

C09-10

SOLUTION Across period 3, non-metallic character and the oxidation state of the central atom increase, so the oxides become more acidic.

C09 · CHAPTER TEST

1 **Answer: (B)** Block from configuration **Easy** ⌚ 0.5 min **Speed**

C09-T1

SOLUTION The last electron enters a p subshell (it is Ga).

2 **Answer: (A)** Size order **Easy** ⌚ 0.5 min **Speed**

C09-T2

SOLUTION Isoelectronic (10 electrons): smaller Z gives a larger ion.

3 **Answer: (B)** Noble-gas EGE **Easy** ⌚ 0.5 min **Speed**

C09-T3

SOLUTION The added electron must enter the next shell, which requires energy.

4 **Answer: 18** Period length **Easy** ⌚ 0.5 min **NV**

C09-T4

SOLUTION $4s(2) + 3d(10) + 4p(6) = 18$.

5 **Answer: (D)** Most metallic element **Easy** ⌚ 0.5 min **Speed**

C09-T5

SOLUTION Metallic character increases down a group, because the IE falls.

 REVISION Revision tracker · Periodicity

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III • CHEMISTRY • UNIT 10

p-Block Elements (Groups 13–18)

10

1–2 per shift

TYPICAL QUESTIONS

Easy–Medium
(fact-based)

DIFFICULTY

Must-do

PRIORITY

Pre-2023
editions: Class 11
p-Block (Gr 13–
14); Class 12 p-
Block (Gr 15–18)

NCERT

~12 hours

STUDY TIME

 VERIFIED FACT What the syllabus asks

Unit 10: "Group-13 to Group-18 elements: electronic configuration and general trends in physical and chemical properties of elements across the periods and down the groups; unique behaviour of the first element in each group" [NTA-SYL]. This module stays at that level: **trends, anomalies and a few landmark compounds** that questions use to test the trends.

Cross-group principles

- **Inert pair effect:** down Groups 13–15, the ns^2 pair becomes reluctant to take part in bonding, so the **lower** oxidation state becomes more stable: $Tl^+ > Tl^{3+}$, $Pb^{2+} > Pb^{4+}$, $Bi^{3+} > Bi^{5+}$. So Pb^{4+} and Bi^{5+} are strong oxidising agents, while Sn^{2+} is a reducing agent.
- **First element anomalies** (B, C, N, O, F): small size, high electronegativity, **no d orbitals** (maximum covalency 4), and the ability to form $p\pi-p\pi$ multiple bonds.
- **Hydrides:** thermal stability **decreases** down a group. Reducing character **increases**. H-bonding makes NH_3 , H_2O and HF boil anomalously high.

Group-by-group trend cards**Group 13 • B Al Ga In Tl • ns^2np^1**

- Oxidation states +3 and +1 (+1 more stable for Tl).
- B is a non-metal (a metalloid); the rest are metals.
- Radius: **Ga < Al** (poor 3d shielding). IE: $B > Tl > Ga > Al > In$.
- B is **electron-deficient**: BX_3 are Lewis acids. Acidity $BF_3 < BCl_3 < BBr_3 < BI_3$ because $p\pi-p\pi$ back-bonding is strongest in BF_3 .
- **Diborane** B_2H_6 : 4 terminal B–H bonds + **2 three-centre two-electron** (banana) B–H–B bridges.
- **Boric acid** $B(OH)_3$ is a weak **monobasic Lewis acid**: $B(OH)_3 + H_2O \longrightarrow [B(OH)_4]^- + H^+$.
- $AlCl_3$ dimerises to Al_2Cl_6 . Al_2O_3 is amphoteric.
- B anomaly: max covalency 4 (e.g. $[BF_4]^-$), vs $[AlF_6]^{3-}$ for Al.

Group 14 • C Si Ge Sn Pb • ns^2np^2

- Oxidation states +4 and +2 (+2 more stable for Pb). $SnCl_2$ is reducing; PbO_2 is oxidising.
- **Catenation:** $C \gg Si > Ge \approx Sn$ (C–C bond strength). Pb shows virtually none.
- C anomaly: multiple bonds (CO_2 is molecular while SiO_2 is a network), max covalency 4, allotropes.
- **Allotropes of C:** diamond (sp^3 , hard, insulator), graphite (sp^2 , layered, conducts), fullerene C_{60} (sp^2 , 20 six- and 12 five-membered rings).
- CO is neutral and toxic (binds haemoglobin); CO_2 is acidic. SiO_2 is a 3D network.
- CCl_4 is not hydrolysed (no d orbitals on C), while $SiCl_4$ hydrolyses readily.
- Silicones $(R_2SiO)_n$ are water-repellent. Silicates are built from SiO_4^{4-} tetrahedra.

Group 15 · N P As Sb Bi · ns^2np^3

- Oxidation states –3 to +5. +5 stability falls down the group (Bi^{5+} is a strong oxidant).
- Hydrides** EH_3 : basicity $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3$; thermal stability falls; reducing power rises. Bond angle NH_3 $107.8^\circ > \text{PH}_3$ $93.6^\circ > \dots > \text{BiH}_3 \sim 90^\circ$.
- Boiling points: $\text{BiH}_3 > \text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$ (NH_3 is raised by H-bonding).
- N anomaly: $\text{N} \equiv \text{N}$ triple bond (very inert N_2); no NCl_5 (no d orbitals), whereas PCl_5 exists.
- Oxides of N: N_2O , NO neutral; N_2O_3 , NO_2 , N_2O_5 acidic. NO_2 is brown and paramagnetic, and dimerises to N_2O_4 .
- Oxoacids of P: H_3PO_2 (monobasic, 2 P–H, strong reducer), H_3PO_3 (dibasic, 1 P–H, reducer), H_3PO_4 (tribasic). **Basicity = number of P–OH groups.**
- White P (P_4 , reactive, glows) vs red P (polymeric, stable). PCl_5 is trigonal bipyramidal; in the solid it is $[\text{PCl}_4]^+[\text{PCl}_6]^-$.

Group 16 · O S Se Te Po · ns^2np^4

- O shows –2 (–1 in peroxides, +2 in OF_2). S, Se, Te: +2, +4, +6.
- Hydrides** H_2E : acidity $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$; thermal stability falls; boiling point $\text{H}_2\text{O} \gg \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$.
- O anomaly: H-bonding, O_2 paramagnetic, max covalency 4. SF_6 exists but " OF_6 " doesn't.
- Ozone** O_3 : bent, diamagnetic, strong oxidant; formed by silent electric discharge.
- Sulphur allotropes**: rhombic (α , stable below 369 K) and monoclinic (β); S_8 puckered crown ring.
- SF_6 is exceptionally inert (steric protection). H_2SO_4 is dibasic, dehydrating and oxidising.

Group 17 · F Cl Br I At · ns^2np^5

- F shows only –1. The others show –1, +1, +3, +5, +7.
- Oxidising power: $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$.
- Bond dissociation enthalpy**: $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$. F–F is anomalously weak (lone-pair repulsion).
- Acidity of HX (aq)**: $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$. HF has the highest boiling point (H-bonding).
- Colours: F_2 pale yellow, Cl_2 greenish yellow, Br_2 red-brown, I_2 violet (vapour).
- Oxoacid strength: $\text{HOCl} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$. Oxidising power runs the other way.
- Interhalogens**: XX' (linear), XX'_3 (T-shaped, e.g. ClF_3), XX'_5 (square pyramidal, BrF_5), XX'_7 (pentagonal bipyramidal, IF_7). They are more reactive than the halogens (except F_2).
- F anomaly: most electronegative element, no d orbitals, only one oxoacid (HOF).

Group 18 · He Ne Ar Kr Xe Rn · ns^2np^6

- Very high IE (decreasing down the group); positive electron gain enthalpy; monatomic gases.
- First noble-gas compound**: $\text{Xe}^+[\text{PtF}_6]^-$ (Bartlett, 1962). The idea came from $\text{O}_2^+[\text{PtF}_6]^-$, because the IE of $\text{O}_2 \approx \text{IE of Xe}$.
- Xenon fluorides: XeF_2 linear, XeF_4 square planar, XeF_6 distorted octahedral. Oxides: XeO_3 pyramidal, XeOF_4 square pyramidal.
- Hydrolysis of XeF_6 gives XeO_3 (complete).
- Uses: He (balloons, diving gas mixtures, low-temperature work), Ar (inert atmosphere, bulbs), Ne (signs).
- He has the lowest boiling point of any substance.

Unique behaviour of the first element: summary

Group	First element	Key anomalies
13	B	non-metal; covalent compounds; max covalency 4; electron-deficient hydrides (diborane)
14	C	catenation; $p\pi-p\pi$ bonds; allotropes; max covalency 4
15	N	diatomic gas with a triple bond; no pentahalides; H-bonding in NH_3
16	O	diatomic paramagnetic gas; H-bonding; mainly –2 state
17	F	only –1; weak F–F bond; HF weak acid with H-bonds; strongest oxidant

Common traps

MISTAKE ALERT Mistake alerts

- "Stronger Lewis acid = more electronegative halogen" is **wrong** for BX_3 . Back-bonding reverses the order.
- Basicity of oxoacids of P = number of **P–OH**, not the total number of H atoms.
- Bond energy of F_2 is **lower** than Cl_2 and Br_2 .
- Acidity of hydrides increases down Groups 16 and 17 (bond strength decreases), even though electronegativity decreases.

Question patterns

PYQ PATTERN Recurring structures

1. Ordering properties (acidity, basicity, bond energy, boiling point, stability) of hydrides/halides/oxoacids down a group.
2. Inert-pair-effect consequences (oxidising/reducing behaviour).
3. Shapes and bonding of landmark compounds (B_2H_6 , PCl_5 , xenon fluorides, interhalogens).
4. Basicity of phosphorus oxoacids.
5. Anomalous behaviour of the first element.

Practice questions

C10 · Practice

1 Easy 0.5 min Concept C10-01

Which ion is the most stable?

- (A) Tl^{3+} (B) Tl^+ (C) Al^+ (D) B^+

2 Medium 1 min Concept C10-02

The correct order of Lewis acid strength is:

- (A) $BF_3 > BCl_3 > BBr_3$ (B) $BBr_3 > BCl_3 > BF_3$
 (C) $BCl_3 > BF_3 > BBr_3$ (D) all equal

3 Easy 0.5 min Speed C10-03

The most basic hydride is:

- (A) NH_3 (B) PH_3 (C) AsH_3 (D) SbH_3

4 Medium 1 min Concept C10-04

The correct order of bond dissociation enthalpy is:

- (A) $F_2 > Cl_2 > Br_2 > I_2$ (B) $Cl_2 > Br_2 > F_2 > I_2$
 (C) $I_2 > Br_2 > Cl_2 > F_2$ (D) $Cl_2 > F_2 > Br_2 > I_2$

5 Easy 0.5 min Speed C10-05

Which is the strongest acid in water?

- (A) HF (B) HCl (C) HBr (D) HI

6

Easy

⌚ 0.5 min

NV

C10-06

Find the basicity of phosphorous acid, H_3PO_3 .Your answer (numerical value):

7

Easy

⌚ 0.5 min

Speed

C10-07

The shape of XeO_3 is:

- (A) trigonal planar (B) pyramidal (C) T-shaped (D) tetrahedral

8

Easy

⌚ 0.75 min

Basic

C10-08

The correct order of boiling points is:

- (A) $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$ (B) $\text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S} > \text{H}_2\text{O}$
 (C) $\text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te} > \text{H}_2\text{O}$ (D) $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$

9

Easy

⌚ 0.5 min

Concept

C10-09

Nitrogen can't form NCl_5 while phosphorus forms PCl_5 , because:

- (A) N is less electronegative
 (B) N has no vacant d orbitals in its valence shell
 (C) N is larger
 (D) Cl_2 doesn't react with N

10

Easy

⌚ 0.75 min

NV

C10-10

How many three-centre two-electron ($3c-2e$) bonds are there in B_2H_6 ?Your answer (numerical value):

11

Easy

⌚ 0.5 min

Speed

C10-11

The strongest acid is:

- (A) HOCl (B) HClO_2 (C) HClO_3 (D) HClO_4



C10 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

C10-T1

The first noble-gas compound prepared (Bartlett, 1962) was:

- (A) XeF_2 (B) $\text{Xe}^+[\text{PtF}_6]^-$ (C) XeO_3 (D) KrF_2

2

Easy

⌚ 0.5 min

Speed

C10-T2

Which element shows the greatest tendency for catenation?

- (A) Si (B) C (C) Ge (D) Sn

3

Easy

⌚ 0.5 min

Speed

C10-T3

The maximum covalency of nitrogen is:

- (A) 3 (B) 4 (C) 5 (D) 6

4 Easy 0.5 min NV

C10-T4

How many P–OH groups are there in hypophosphorous acid, H_3PO_2 ?

Your answer (numerical value):

5 Easy 0.5 min Speed

C10-T5

Iodine vapour is:

- (A) yellow (B) green (C) violet (D) red-brown

Answers & Solutions

C10 · Practice

- 1 (B) 2 (B) 3 (A) 4 (B) 5 (D) 6 2 7 (B) 8 (A) 9 (B) 10 2
11 (D)

C10 · Chapter Test

- 1 (B) 2 (B) 3 (B) 4 1 5 (C)

C10 · PRACTICE

1 **Answer: (B)** *Inert pair effect* Easy 0.5 min Concept C10-01

SOLUTION The inert pair effect makes +1 the more stable state for thallium.

2 **Answer: (B)** *Lewis acidity of boron halides* Medium 1 min Concept C10-02

SOLUTION $p\pi$ – $p\pi$ back-donation from F to B (similar-sized 2p orbitals) is strongest in BF_3 , which reduces boron's electron deficiency the most.

COMMON TRAP Using electronegativity alone.

3 **Answer: (A)** *Basicity of group 15 hydrides* Easy 0.5 min Speed C10-03

SOLUTION N's lone pair is in a small, compact orbital with high electron density, so it is the most available.

4 **Answer: (B)** *Halogen bond energies* Medium 1 min Concept C10-04

SOLUTION F–F is weakened by repulsion between the lone pairs on the two small F atoms.

COMMON TRAP Assuming a simple trend.

5 **Answer: (D)** *Strongest hydrohalic acid* Easy 0.5 min Speed C10-05

SOLUTION H–I is the weakest H–X bond.

6 **Answer: 2** *Basicity of phosphorous acid* Easy 0.5 min NV C10-06

SOLUTION Structure HP(O)(OH)_2 : two ionisable P–OH groups. The P–H hydrogen isn't acidic.

COMMON TRAP Answering 3.

7 **Answer: (B)** *Shape of XeO_3* Easy 0.5 min Speed C10-07

SOLUTION 3 bond pairs + 1 lone pair on Xe.

8 **Answer: (A)** *Boiling points of group 16 hydrides* Easy 0.75 min Basic C10-08

SOLUTION H_2O is raised by H-bonding. For the rest, van der Waals forces increase with molar mass.

9 **Answer: (B)** *Pentahalides* Easy 0.5 min Concept C10-09

SOLUTION N's maximum covalency is 4.

10 **Answer: 2** *Bonds in diborane* Easy 0.75 min NV C10-10

SOLUTION Two B–H–B bridges, each a 3c–2e bond.

COMMON TRAP Counting the 4 terminal bonds.

- 11 Answer: (D)** Strongest oxoacid of chlorine **Easy** ⌚ 0.5 min Speed C10-11
SOLUTION More O atoms spread the negative charge of the conjugate base better.

C10 · CHAPTER TEST

- 1 Answer: (B)** First noble-gas compound **Easy** ⌚ 0.5 min Speed C10-T1
SOLUTION Direct recall: see the concept summary and formula box of this chapter.

- 2 Answer: (B)** Catenation **Easy** ⌚ 0.5 min Speed C10-T2
SOLUTION The C–C bond is strong, and carbon is small.

- 3 Answer: (B)** Maximum covalency of N **Easy** ⌚ 0.5 min Speed C10-T3
SOLUTION Only 4 valence orbitals (one 2s + three 2p), e.g. in NH_4^+ .

- 4 Answer: 1** Hypophosphorous acid **Easy** ⌚ 0.5 min NV C10-T4
SOLUTION $\text{H}_2\text{P}(\text{O})(\text{OH})$: one P–OH and two P–H, so it is monobasic.

- 5 Answer: (C)** Colour of iodine vapour **Easy** ⌚ 0.5 min Speed C10-T5
SOLUTION Direct recall: see the concept summary and formula box of this chapter.

 REVISION Revision tracker · p-Block

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 11

d- and f-Block Elements

11

1-2 per shift

TYPICAL QUESTIONS

Easy-Medium
(NCERT-heavy)

DIFFICULTY

Must-do

PRIORITY

Class 12 · Ch 4

NCERT

~8 hours

STUDY TIME

Important concepts

- **Transition elements:** partly filled d subshell in the atom or in a common ion. General configuration $(n-1)d^{1-10}ns^{1-2}$. **Zn, Cd, Hg** (d^{10} in both the atom and the common ion) are not regarded as transition elements.
- **Characteristic properties:** metallic, high melting points (maximum near the middle of each series; W is highest; Zn, Cd, Hg are low, and Hg is liquid), variable oxidation states, coloured ions, paramagnetism, catalytic activity, complex formation, interstitial compounds, alloys.
- **Lanthanoid contraction:** the steady decrease in size from La to Lu, caused by the poor shielding of 4f electrons.
- **Actinoids:** all radioactive; wider range of oxidation states because 5f, 6d and 7s have comparable energies.

Trends & NCERT facts (first transition series)

Σ FORMULA Oxidation states & stability

- Sc shows only +3; Zn only +2. **Mn shows the maximum (+7)** in the first series. Os/Ru reach +8 (OsO_4).
- Stable half-filled d^5 : Mn^{2+} and Fe^{3+} . Mn^{3+} (d^4) is a strong oxidant; Cr^{2+} (d^4) is a strong reductant.
- $E^\circ(\text{M}^{2+}/\text{M})$: **Cu is the only first-series metal with a positive value** (+0.34 V), so it doesn't liberate H_2 from dilute acids.
- Selected $E^\circ(\text{M}^{3+}/\text{M}^{2+})$: Co +1.97, Mn +1.57, Fe +0.77, Cr -0.41 V.
- Higher oxidation states are stabilised by F and O (e.g. MnO_4^- , CrO_4^{2-} , VF_5).

Σ FORMULA Colour & magnetism

$\mu_{\text{spin-only}} = \sqrt{n(n+2)}$ BM. Colour arises from **d-d transitions**, so d^0 and d^{10} ions are colourless (Sc^{3+} , Ti^{4+} , Zn^{2+} , Cu^+).

Aqueous ion	Colour	Aqueous ion	Colour
Ti^{3+} (d^1)	purple	Fe^{2+} (d^6)	pale green
V^{3+} (d^2)	green	Fe^{3+} (d^5)	yellow
Cr^{3+} (d^3)	violet	Co^{2+} (d^7)	pink
Mn^{2+} (d^5)	pale pink	Ni^{2+} (d^8)	green
Cu^{2+} (d^9)	blue	Zn^{2+} (d^{10})	colourless

The intense colours of MnO_4^- (purple) and $\text{Cr}_2\text{O}_7^{2-}$ (orange) come from **charge transfer**, not d-d transitions. Both have d^0 metal centres.

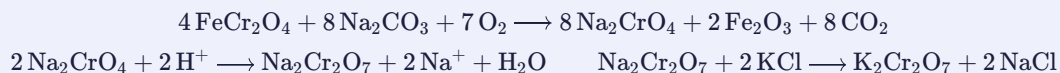
☆ IMPORTANT Catalysts, interstitials, alloys (NCERT)

- Catalysts: V_2O_5 (contact process, $\text{SO}_2 \rightarrow \text{SO}_3$), Fe (Haber process), Ni (hydrogenation), $\text{TiCl}_4 + \text{Al}(\text{CH}_3)_3$ (Ziegler-Natta), MnO_2 (KClO_3 decomposition). They work through variable oxidation states and adsorption on surfaces.
- Interstitial compounds (TiC , Mn_4N , Fe_3H , TiH_2): hard, very high melting points, conducting, chemically inert.
- Alloys form easily because the atomic radii are similar: brass (Cu-Zn), bronze (Cu-Sn), stainless steel.

Potassium dichromate and potassium permanganate (NCERT)

Σ FORMULA $K_2Cr_2O_7$

Preparation from chromite:



pH-dependent equilibrium: $2CrO_4^{2-}$ (yellow) + $2H^+ \rightleftharpoons Cr_2O_7^{2-}$ (orange) + H_2O . **Oxidising action (acid):** $Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$ ($E^\circ = 1.33$ V). It oxidises $I^- \rightarrow I_2$, $Fe^{2+} \rightarrow Fe^{3+}$, $H_2S \rightarrow S$, $Sn^{2+} \rightarrow Sn^{4+}$. Structure: two CrO_4 tetrahedra sharing one O (Cr–O–Cr angle $\sim 126^\circ$).

Σ FORMULA $KMnO_4$

Preparation: $2MnO_2 + 4KOH + O_2 \longrightarrow 2K_2MnO_4$ (green) + $2H_2O$; then $3MnO_4^{2-} + 4H^+ \longrightarrow 2MnO_4^- + MnO_2 + 2H_2O$ (disproportionation). Commercially, manganate is oxidised electrolytically. **Oxidising action:**

Medium	Half-reaction	n-factor
acidic	$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$ (E° 1.52 V)	5
neutral / faintly alkaline	$MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$	3

Acidic: $I^- \rightarrow I_2$; $Fe^{2+} \rightarrow Fe^{3+}$; $C_2O_4^{2-} \rightarrow CO_2$ (at 333 K); $S^{2-} \rightarrow S$; $SO_3^{2-} \rightarrow SO_4^{2-}$; $NO_2^- \rightarrow NO_3^-$. Neutral/alkaline: $I^- \rightarrow IO_3^-$; thiosulphate $\rightarrow$ sulphate; $Mn^{2+} \rightarrow MnO_2$. $KMnO_4$ is dark purple and diamagnetic (d^0), with weak temperature-dependent paramagnetism. Titrations with it use **dilute** H_2SO_4 , not HCl (which it would oxidise to Cl_2).

Inner transition elements

Lanthanoids (Ce–Lu)

- Configuration $[Xe]4f^{1-14}5d^{0-1}6s^2$. The common oxidation state is **+3**.
- Other stable states track f^0 , f^7 , f^{14} : Ce^{4+} (f^0 , oxidant), Eu^{2+} (f^7 , reductant), Yb^{2+} (f^{14}), Tb^{4+} (f^7).
- Lanthanoid contraction consequences:** Zr $\approx$ Hf in size (they occur together and are hard to separate); lanthanoids are hard to separate from each other; basicity of $Ln(OH)_3$ decreases from La to Lu; the 5d series has higher IE and density.
- Misch metal ($\sim 95\%$ lanthanoids + Fe) is used in lighter flints.

Actinoids (Th–Lr)

- Configuration $[Rn]5f^{1-14}6d^{0-1}7s^2$. All are radioactive.
- Wider range of oxidation states** (+3 to +7; e.g. Np, Pu up to +7), because 5f, 6d and 7s have comparable energies.
- Actinoid contraction is **greater** than lanthanoid contraction (5f shields even more poorly).
- Early actinoids resemble transition metals more than the lanthanoids do.

Common traps

⚠ MISTAKE ALERT Mistake alerts

- Configurations of ions: remove 4s electrons first ($Fe^{3+} = 3d^5$).
- Cr and Cu atomic anomalies: $3d^54s^1$ and $3d^{10}4s^1$.
- Colour of $KMnO_4/K_2Cr_2O_7$ is **charge transfer**, not d–d.
- $KMnO_4$ titrations: never acidify with HCl .
- Zn, Cd, Hg: d-block, but not transition elements.

Question patterns

PYQ PATTERN Recurring structures

1. Spin-only magnetic moment of ions; colour/colourless identification.
2. Oxidation-state trends; stability (Mn^{2+} vs Mn^{3+} , Cr^{2+} reducing); E° trends.
3. $\text{K}_2\text{Cr}_2\text{O}_7/\text{KMnO}_4$ preparation steps, structures, products in each medium, titration stoichiometry.
4. Lanthanoid contraction consequences; stable oxidation states of Ce, Eu, Yb.
5. Catalysts and interstitial compounds (NCERT lines).

Practice questions

C11 · Practice

1 Easy 0.5 min Speed C11-01

Which first-row transition element shows the highest oxidation state?

- (A) Cr (B) Mn (C) Fe (D) V

2 Easy 0.5 min Speed C11-02

Which aqueous ion is colourless?

- (A) Ti^{3+} (B) Cr^{3+} (C) Zn^{2+} (D) Cu^{2+}

3 Easy 0.5 min Speed C11-03

The spin-only magnetic moment of Cr^{3+} is about:

- (A) 1.73 BM (B) 2.83 BM (C) 3.87 BM (D) 4.90 BM

4 Medium 1.5 min NV C11-04

Find the volume (in mL) of 0.1 M $\text{K}_2\text{Cr}_2\text{O}_7$ (acidified) needed to oxidise 60 mL of 0.1 M Fe^{2+} completely.

Your answer (numerical value):

5 Easy 0.5 min Concept C11-05

Adding acid to a yellow solution of K_2CrO_4 :

- (A) turns it orange as $\text{Cr}_2\text{O}_7^{2-}$ forms (B) turns it green as Cr^{3+} forms
(C) causes no change (D) turns it colourless

6 Easy 0.5 min Speed C11-06

In neutral or faintly alkaline medium, MnO_4^- is reduced to:

- (A) Mn^{2+} (B) MnO_2 (C) MnO_4^{2-} (D) Mn

7 Easy 0.5 min Concept C11-07

A consequence of the lanthanoid contraction is that:

- (A) Zr and Hf have nearly the same atomic radius
(B) lanthanoid hydroxides become more basic from La to Lu
(C) lanthanoids are easy to separate
(D) the 5d elements have lower IE than the 4d elements

8 Easy 0.75 min Concept

C11-08

Copper doesn't liberate H_2 from dilute HCl because:

- (A) Cu is a transition metal
(B) $E^\circ(Cu^{2+}/Cu)$ is positive
(C) Cu forms a protective oxide layer
(D) HCl is a weak acid

9 Easy 0.5 min Concept

C11-09

Actinoids show a wider range of oxidation states than lanthanoids because:

- (A) they are radioactive
(B) the 5f, 6d and 7s orbitals are close in energy
(C) they are larger
(D) they have fewer electrons

10 Easy 0.5 min NV

C11-10

Find the number of unpaired electrons in a gaseous Co^{2+} ion ($Z = 27$).

Your answer (numerical value):

C11 · Chapter Test

1 Easy 0.5 min Speed

C11-T1

Zinc is not regarded as a transition element because:

- (A) it is too reactive
(B) it has a completely filled d subshell in the atom and in Zn^{2+}
(C) it has no d electrons
(D) it forms no complexes

2 Easy 0.5 min Speed

C11-T2

The catalyst used in the contact process for H_2SO_4 is:

- (A) Fe (B) Ni (C) V_2O_5 (D) Pt only

3 Easy 0.5 min Concept

C11-T3

The intense purple colour of $KMnO_4$ is due to:

- (A) d-d transitions in Mn^{7+}
(B) ligand-to-metal charge transfer
(C) unpaired electrons
(D) the K^+ ion

4 Easy 0.5 min NV

C11-T4

Find the oxidation state of Mn in K_2MnO_4 .

Your answer (numerical value):

5 Easy 0.5 min Speed

C11-T5

The most common oxidation state of the lanthanoids is:

- (A) +2 (B) +3 (C) +4 (D) +5

Answers & Solutions

C11 · Practice

1 (B) 2 (C) 3 (C) 4 10 5 (A) 6 (B) 7 (A) 8 (B) 9 (B) 10 3

C11 · Chapter Test

1 (B) 2 (C) 3 (B) 4 6 5 (B)

C11 · PRACTICE

1 **Answer: (B)** Highest oxidation state, 3d series **Easy** ⌚ 0.5 min **Speed** C11-01

SOLUTION Mn reaches +7 (in MnO_4^-), using all of its 3d and 4s electrons.

2 **Answer: (C)** Colourless ion **Easy** ⌚ 0.5 min **Speed** C11-02

SOLUTION Zn^{2+} is d^{10} , so no d-d transition is possible.

SHORTCUT d^0 or $d^{10} \rightarrow$ colourless.

3 **Answer: (C)** Magnetic moment of Cr^{3+} **Easy** ⌚ 0.5 min **Speed** C11-03

SOLUTION d^3 : 3 unpaired, so $\sqrt{15} \approx 3.87$ BM.

4 **Answer: 10** Dichromate titration **Medium** ⌚ 1.5 min **NV** C11-04

SOLUTION $\text{Cr}_2\text{O}_7^{2-} + 6\text{Fe}^{2+} + 14\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$. Moles of $\text{Fe}^{2+} = 0.006$, so dichromate = $0.001 \text{ mol} = 10 \text{ mL}$ of 0.1 M.

SHORTCUT Equivalents: $6 \times 0.1 \times V = 1 \times 0.1 \times 60$.

COMMON TRAP Using a 1 : 1 ratio (60 mL).

5 **Answer: (A)** Chromate-dichromate equilibrium **Easy** ⌚ 0.5 min **Concept** C11-05

SOLUTION $2\text{CrO}_4^{2-} + 2\text{H}^+ \rightleftharpoons \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$ shifts right. This is not a redox change: Cr stays +6.

COMMON TRAP Choosing (B), which would need a reducing agent.

6 **Answer: (B)** KMnO_4 in neutral medium **Easy** ⌚ 0.5 min **Speed** C11-06

SOLUTION $\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \rightarrow \text{MnO}_2 + 4\text{OH}^-$.

7 **Answer: (A)** Lanthanoid contraction **Easy** ⌚ 0.5 min **Concept** C11-07

SOLUTION The 4f contraction cancels the size increase expected from the extra shell.

COMMON TRAP Choosing (B): basicity actually **decreases**.

8 **Answer: (B)** Copper and dilute acid **Easy** ⌚ 0.75 min **Concept** C11-08

SOLUTION +0.34 V: Cu is a weaker reducing agent than H_2 .

9 **Answer: (B)** Actinoid oxidation states **Easy** ⌚ 0.5 min **Concept** C11-09

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

10 **Answer: 3** Unpaired electrons in Co^{2+} **Easy** ⌚ 0.5 min **NV** C11-10

SOLUTION Co^{2+} : $3d^7$, which is 3 unpaired electrons (free ion, Hund's rule).

COMMON TRAP Using the atomic configuration.

C11 · CHAPTER TEST

1 **Answer: (B)** Zinc's status **Easy** ⌚ 0.5 min **Speed** C11-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 **Answer: (C)** Contact-process catalyst **Easy** ⌚ 0.5 min **Speed** C11-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 **Answer: (B)** Origin of KMnO_4 colour **Easy** ⌚ 0.5 min **Concept** C11-T3

SOLUTION Mn(VII) is d^0 , so there are no d-d transitions. The colour comes from $\text{O} \rightarrow \text{Mn}$ charge transfer.

4 Answer: 6 Oxidation state in manganate **Easy** ⌚ 0.5 min **NV**

C11-T4

SOLUTION $2(+1) + x + 4(-2) = 0 \Rightarrow x = +6$.

5 Answer: (B) Common lanthanoid state **Easy** ⌚ 0.5 min **Speed**

C11-T5

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

 REVISION Revision tracker · d- & f-Block

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 12

Coordination Compounds

12

2–3 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

Must-do (top-weight chapter)

PRIORITY

Class 12 · Ch 5

NCERT

~10 hours

STUDY TIME

PYQ PATTERN Weight evidence

36 questions across the 2026 Paper-1 shifts ($\approx 7.6\%$ of Chemistry, the highest single chapter in that dataset)

 **THIRD-PARTY ANALYSIS** [T-WT-CD]. Coordination compounds are also the base for colour, magnetism and isomerism questions in the d-block unit.

Important concepts

- **Werner's theory:** metals have **primary valency** (ionisable, equal to the oxidation state, satisfied by anions) and **secondary valency** (non-ionisable, equal to the coordination number, directed in space).
- **Ligands:** unidentate (Cl^- , NH_3 , H_2O), bidentate (en, $\text{C}_2\text{O}_4^{2-}$), polydentate (EDTA^{4-} , hexadentate). **Ambidentate** ligands bind through two different atoms: $\text{NO}_2^-/\text{ONO}^-$, $\text{SCN}^-/\text{NCS}^-$, CN^-/NC^- . **Chelates** (ring-forming) are extra stable (the chelate effect).
- **Homoleptic** complexes have one kind of ligand ($[\text{Co}(\text{NH}_3)_6]^{3+}$); **heteroleptic** complexes have more than one ($[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$).
- **Bonding:** VBT gives hybridisation and geometry. **CFT** explains colour, magnetism and the spectrochemical series.

IUPAC nomenclature: the rules

1. Name the **cation first**, then the anion (as for any salt).
2. Inside the complex: **ligands in alphabetical order** (ignoring multiplying prefixes), then the metal.
3. Anionic ligands end in **-o**: chlorido, cyanido, hydroxido, oxido, oxalato, nitrito-N (NO_2), nitrito-O (ONO), thiocyanato-S/-N. Neutral: **aqua**, **ammine**, **carbonyl**, **nitrosyl**, en (ethane-1,2-diamine).
4. Prefixes di-, tri-, tetra-; for ligands whose names already contain a number, bis-, tris-, tetrakis-.
5. Oxidation state of the metal in Roman numerals. In an **anionic complex** the metal name ends in **-ate**: ferrate (Fe), cuprate (Cu), argentate (Ag), aurate (Au), plumbate (Pb), stannate (Sn), cobaltate, nickelate, zincate.

Formula	IUPAC name
$[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$	pentaamminechloridocobalt(III) chloride
$\text{K}_3[\text{Fe}(\text{CN})_6]$	potassium hexacyanidoferrate(III)
$\text{K}_4[\text{Fe}(\text{CN})_6]$	potassium hexacyanidoferrate(II)
$[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$	diamminedichloridoplatinum(II)
$\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$	potassium trioxalatoferrate(III)
$[\text{Co}(\text{en})_3]^{3+}$	tris(ethane-1,2-diamine)cobalt(III) ion
$[\text{Ni}(\text{CO})_4]$	tetracarbonylnickel(0)

Isomerism

Σ FORMULA Types

Structural: ionisation ($[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ / $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$), linkage ($-\text{NO}_2$ / $-\text{ONO}$), coordination ($[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ / $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$), solvate/hydrate ($[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ violet / $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ grey-green / $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ dark green). **Stereo:** geometrical (cis/trans, fac/mer) and optical (non-superimposable mirror images).

Complex type	Geometrical isomers	Optical?
square planar MA_2B_2	2 (cis, trans)	no
square planar MABCD	3	no
tetrahedral (any)	0	only if 4 different groups (rare)
octahedral MA_4B_2	2 (cis, trans)	no
octahedral MA_3B_3	2 (fac, mer)	no
octahedral $\text{M}(\text{AA})_2\text{B}_2$	2	cis form is chiral (1 pair)
octahedral $\text{M}(\text{AA})_3$	—	yes (Δ and Λ)
octahedral $\text{MA}_2\text{B}_2\text{C}_2$	5	one of them is chiral (6 stereoisomers in total)

Valence bond theory: standard examples

Complex	Metal d count	Hybridisation	Shape	Unpaired e^-	Magnetic
$[\text{Ni}(\text{CN})_4]^{2-}$	d^8	dsp^2	square planar	0	dia
$[\text{NiCl}_4]^{2-}$	d^8	sp^3	tetrahedral	2	para (2.83 BM)
$[\text{Ni}(\text{CO})_4]$	d^{10} (Ni^0)	sp^3	tetrahedral	0	dia
$[\text{Fe}(\text{CN})_6]^{3-}$	d^5	$d^2\text{sp}^3$ (inner)	octahedral	1	para (1.73 BM)
$[\text{FeF}_6]^{3-}$	d^5	sp^3d^2 (outer)	octahedral	5	para (5.92 BM)
$[\text{Co}(\text{NH}_3)_6]^{3+}$	d^6	$d^2\text{sp}^3$	octahedral	0	dia
$[\text{CoF}_6]^{3-}$	d^6	sp^3d^2	octahedral	4	para (4.90 BM)
$[\text{Fe}(\text{CN})_6]^{4-}$	d^6	$d^2\text{sp}^3$	octahedral	0	dia

Crystal field theory

Σ FORMULA Splitting & CFSE

Octahedral: t_{2g} (3 orbitals, $-0.4\Delta_o$ each electron) below e_g (2 orbitals, $+0.6\Delta_o$). **Tetrahedral:** reversed (e below t_2), $\Delta_t = \frac{4}{9}\Delta_o$, so tetrahedral complexes are almost always **high spin**.

$$\text{CFSE} = (-0.4 n_{t_{2g}} + 0.6 n_{e_g})\Delta_o \text{ (+ pairing energy terms)}$$

If $\Delta_o > P$ (pairing energy), electrons pair up in t_{2g} (low spin). If $\Delta_o < P$, they occupy e_g first (high spin).

Spectrochemical series (weak → strong field): $\text{I}^- < \text{Br}^- < \text{SCN}^- < \text{Cl}^- < \text{S}^{2-} < \text{F}^- < \text{OH}^- < \text{C}_2\text{O}_4^{2-} < \text{H}_2\text{O} < \text{NCS}^- < \text{edta}^{4-} < \text{NH}_3 < \text{en} < \text{CN}^- < \text{CO}$

☆ IMPORTANT Colour

The colour seen is **complementary** to the light absorbed in the d-d transition. $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ (t_{2g}^1) absorbs in the blue-green region (~498 nm) and looks violet. Anhydrous CuSO_4 is white; $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is blue (water ligands produce the splitting). Stronger-field ligands give a larger Δ_o , so absorption shifts to shorter wavelength.

☆ IMPORTANT Metal carbonyls

Bonding is **synergic**: CO donates a lone pair to the metal (σ), and the metal back-donates d electrons into CO's π^* orbital (π back-bonding). Result: the M–C bond gains double-bond character, while the C–O bond weakens and lengthens. Examples: $[\text{Ni}(\text{CO})_4]$ (tetrahedral), $[\text{Fe}(\text{CO})_5]$ (trigonal bipyramidal), $[\text{Cr}(\text{CO})_6]$ (octahedral). The metal is in oxidation state 0.

Importance of coordination compounds (NCERT list)

Area	Example
Qualitative analysis	$\text{Ni}^{2+} + \text{dimethylglyoxime} \rightarrow \text{red precipitate}$; $\text{Cu}^{2+} + \text{NH}_3 \rightarrow \text{deep blue } [\text{Cu}(\text{NH}_3)_4]^{2+}$
Hardness of water	EDTA titration of $\text{Ca}^{2+}/\text{Mg}^{2+}$
Metallurgy	extraction of Ag and Au as cyanido complexes, e.g. $[\text{Au}(\text{CN})_2]^-$; purification of Ni via $[\text{Ni}(\text{CO})_4]$
Biology	chlorophyll (Mg), haemoglobin (Fe), vitamin B ₁₂ (Co)
Medicine	cisplatin (anticancer); EDTA for lead poisoning
Catalysis	Wilkinson's catalyst $[(\text{Ph}_3\text{P})_3\text{RhCl}]$ for hydrogenation of alkenes
Photography	$[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$ formed when fixing with hypo

Common traps

⚠ MISTAKE ALERT Mistake alerts

- Oxidation state vs coordination number: in $[\text{Co}(\text{en})_3]^{3+}$ the CN is **6** (3 bidentate ligands), not 3.
- Only ions **outside** the square brackets precipitate with AgNO_3 .
- A strong-field ligand does **not** always mean inner-orbital hybridisation. Check the d count: $[\text{Ni}(\text{NH}_3)_6]^{2+}$ (d^8) is sp^3d^2 whatever the ligand.
- Tetrahedral complexes: essentially always high spin, and no geometrical isomers.
- Alphabetical order of ligands ignores prefixes: "tetraammine" is ordered under **a**.

Question patterns

📊 PYQ PATTERN Recurring structures

1. IUPAC names $\leftrightarrow$ formulae (including the -ate endings).
2. Isomer counts (geometrical/optical) for octahedral and square-planar complexes.
3. Hybridisation, geometry, magnetic moment (VBT) of standard complexes.
4. CFT: CFSE, high/low spin, colour, spectrochemical series ordering.
5. Werner-type questions (moles of AgCl precipitated, conductivity/number of ions).
6. Carbonyl bonding; applications (NCERT list).

Practice questions

📝 C12 · Practice

1 Easy ⌚ 0.75 min Basic

C12-01

The IUPAC name of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ is:

- ☐ (A) potassium trioxalatoferrate(III)
 ☐ (B) potassium trioxalatoiron(III)
 ☐ (C) tripotassium oxalatoferrate(II)
 ☐ (D) potassium ferrioxalate(III)

2 Easy 0.75 min NV

C12-02

How many moles of AgCl precipitate when excess AgNO_3 is added to a solution containing 1 mol of $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$?

Your answer (numerical value):

3 Easy 0.5 min Speed

C12-03

The number of geometrical isomers of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ is:

- (A) 1 (B) 2 (C) 3 (D) 4

4 Medium 1 min Concept

C12-04

Which complex is optically active?

- (A) $\text{trans}-[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (B) $\text{cis}-[\text{Co}(\text{en})_2\text{Cl}_2]^+$
(C) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (D) $\text{trans}-[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

5 Easy 0.75 min Basic

C12-05

$[\text{Ni}(\text{CN})_4]^{2-}$ is:

- (A) sp^3 , tetrahedral, paramagnetic (B) dsp^2 , square planar, diamagnetic
(C) sp^3 , tetrahedral, diamagnetic (D) dsp^2 , square planar, paramagnetic

6 Easy 0.75 min Basic

C12-06

The spin-only magnetic moment of $[\text{NiCl}_4]^{2-}$ is about:

- (A) 0 BM (B) 1.73 BM (C) 2.83 BM (D) 3.87 BM

7 Easy 0.5 min Speed

C12-07

Which ligand produces the largest crystal-field splitting?

- (A) H_2O (B) NH_3 (C) CO (D) F^-

8 Medium 1 min Calc

C12-08

Ignoring pairing energy, the CFSE of a low-spin octahedral d^6 complex is:

- (A) $-0.4\Delta_o$ (B) $-2.4\Delta_o$ (C) $-1.2\Delta_o$ (D) 0

9 Easy 0.5 min Concept

C12-09

Linkage isomerism is possible with the ligand:

- (A) NH_3 (B) NO_2^- (C) H_2O (D) en

10 Easy 0.5 min NV

C12-10

Find the denticity of the EDTA^{4-} ligand.

Your answer (numerical value):

11 Easy 0.75 min Concept

C12-11

In $[\text{Fe}(\text{CO})_5]$, the oxidation state and coordination number of Fe are:

- (A) 0 and 5 (B) +2 and 5 (C) 0 and 10 (D) +5 and 5

12 Medium 1 min Concept

C12-12

The violet colour of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is due to:

- (A) charge transfer from water
 (B) a d-d transition $t_{2g}^1 \rightarrow e_g^1$
 (C) an unpaired electron alone
 (D) the Ti^{3+} ion's high charge

C12 · Chapter Test

1 Easy 0.5 min Speed

C12-T1

Which ligand can form a chelate?

- (A) Cl^-
 (B) NH_3
 (C) ethane-1,2-diamine (en)
 (D) H_2O

2 Easy 0.75 min Concept

C12-T2

$[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ are:

- (A) linkage isomers
 (B) ionisation isomers
 (C) coordination isomers
 (D) geometrical isomers

3 Easy 0.5 min NV

C12-T3

Find the coordination number of Co in $[\text{Co}(\text{en})_3]^{3+}$.

Your answer (numerical value):

4 Easy 0.5 min Speed

C12-T4

Cisplatin, $\text{cis-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$, is used:

- (A) as a hydrogenation catalyst
 (B) in the treatment of cancer
 (C) to treat lead poisoning
 (D) in photography

5 Medium 0.75 min Concept

C12-T5

In metal carbonyls, π back-bonding from the metal to CO:

- (A) strengthens the C-O bond
 (B) weakens the C-O bond and strengthens the M-C bond
 (C) has no effect on bond strengths
 (D) breaks the M-C σ bond

Answers & Solutions

C12 · Practice

- 1 (A) 2 2 3 (B) 4 (B) 5 (B) 6 (C) 7 (C) 8 (B) 9 (B) 10 6
 11 (A) 12 (B)

C12 · Chapter Test

- 1 (C) 2 (B) 3 6 4 (B) 5 (B)

C12 · PRACTICE

1 Answer: (A) IUPAC name Easy 0.75 min Basic

C12-01

SOLUTION Anionic complex, so "ferrate". Fe is +3 because $3(+1) + x + 3(-2) = 0$.

COMMON TRAP Using "iron" in an anionic complex.

- 2 Answer: 2** *Werner-type precipitation* **Easy** ⌚ 0.75 min **NV** C12-02
SOLUTION Only the two chloride ions outside the coordination sphere are ionisable.
COMMON TRAP Counting 3.
- 3 Answer: (B)** *Geometrical isomers* **Easy** ⌚ 0.5 min **Speed** C12-03
SOLUTION Octahedral MA_4B_2 : cis and trans.
- 4 Answer: (B)** *Optical activity* **Medium** ⌚ 1 min **Concept** C12-04
SOLUTION The cis isomer has no plane of symmetry, so its mirror image is non-superimposable. The trans isomer has a mirror plane.
- 5 Answer: (B)** *VBT of $[Ni(CN)_4]^{2-}$* **Easy** ⌚ 0.75 min **Basic** C12-05
SOLUTION Ni^{2+} is d^8 . Strong-field CN^- pairs the electrons, freeing one 3d orbital for dsp^2 hybridisation.
COMMON TRAP Mixing it up with $[NiCl_4]^{2-}$.
- 6 Answer: (C)** *Magnetic moment of $[NiCl_4]^{2-}$* **Easy** ⌚ 0.75 min **Basic** C12-06
SOLUTION Tetrahedral, weak field: d^8 with 2 unpaired electrons, giving $\sqrt{8} \approx 2.83$ BM.
- 7 Answer: (C)** *Strongest field ligand* **Easy** ⌚ 0.5 min **Speed** C12-07
SOLUTION CO is at the strong end of the spectrochemical series (π -acceptor).
- 8 Answer: (B)** *CFSE of low-spin d^6* **Medium** ⌚ 1 min **Calc** C12-08
SOLUTION $t_{2g}^6 e_g^0$: $6 \times (-0.4) = -2.4\Delta_o$.
COMMON TRAP Using the high-spin value ($t_{2g}^4 e_g^2 = -0.4\Delta_o$).
- 9 Answer: (B)** *Linkage isomerism* **Easy** ⌚ 0.5 min **Concept** C12-09
SOLUTION NO_2^- is ambidentate: it can bind through N (nitrito-N) or through O (nitrito-O).
- 10 Answer: 6** *Denticity of EDTA* **Easy** ⌚ 0.5 min **NV** C12-10
SOLUTION 2 N donors + 4 carboxylate O donors.
- 11 Answer: (A)** *Metal carbonyls* **Easy** ⌚ 0.75 min **Concept** C12-11
SOLUTION CO is neutral, so Fe is 0. Five CO ligands give CN 5 (trigonal bipyramidal).
- 12 Answer: (B)** *Origin of colour* **Medium** ⌚ 1 min **Concept** C12-12
SOLUTION The single d electron absorbs light around 498 nm, and the complementary colour (violet) is seen.

C12 · CHAPTER TEST

- 1 Answer: (C)** *Chelating ligand* **Easy** ⌚ 0.5 min **Speed** C12-T1
SOLUTION A bidentate ligand forms a 5-membered ring with the metal.
- 2 Answer: (B)** *Ionisation isomers* **Easy** ⌚ 0.75 min **Concept** C12-T2
SOLUTION They swap an ion inside the coordination sphere with one outside it, so they give different ions in solution.
- 3 Answer: 6** *Coordination number with bidentates* **Easy** ⌚ 0.5 min **NV** C12-T3
SOLUTION 3 bidentate ligands $\times$ 2 donor atoms = 6.
COMMON TRAP Answering 3.
- 4 Answer: (B)** *Cisplatin* **Easy** ⌚ 0.5 min **Speed** C12-T4
SOLUTION Direct recall: see the concept summary and formula box of this chapter.
COMMON TRAP Confusing it with EDTA (lead poisoning) or Wilkinson's catalyst (hydrogenation).
- 5 Answer: (B)** *Back-bonding effect* **Medium** ⌚ 0.75 min **Concept** C12-T5
SOLUTION Electron density enters CO's π^* (antibonding) orbital, which lowers the C–O bond order.

**REVISION** Revision tracker · Coordination Compounds

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III • CHEMISTRY • UNIT 13

Purification & Characterisation of Organic Compounds

13

0–1 per shift

TYPICAL QUESTIONS

Easy

DIFFICULTY

High (fast marks)

PRIORITY

Class 11 • Ch 8
(purification,
qualitative &
quantitative
analysis)

NCERT

~4 hours

STUDY TIME

Purification methods

Method	Principle	Classic examples
Crystallisation	different solubilities at high/low temperature	sugar, benzoic acid; <i>fractional</i> crystallisation for mixtures
Sublimation	solid → vapour directly	camphor, naphthalene, anthracene, benzoic acid
Simple distillation	boiling points differ widely	chloroform (334 K) + aniline (457 K)
Fractional distillation	boiling points close	crude oil fractions
Distillation under reduced pressure	lowers the boiling point; avoids decomposition	glycerol from spent lye (soap industry)
Steam distillation	steam-volatile, water-immiscible compounds; the mixture boils when $p_{\text{water}} + p_{\text{organic}} = p_{\text{atm}}$ (below 373 K)	aniline , o-nitrophenol, essential oils
Differential extraction	distribution between two immiscible solvents	organic compound from water using ether (separating funnel)
Chromatography	differential adsorption/partition between stationary and mobile phases	column, TLC (adsorption); paper (partition)

Retardation factor: $R_f = \frac{\text{distance moved by compound}}{\text{distance moved by solvent front}}$ (always < 1). Colourless spots are made visible with UV light, iodine vapour or reagent sprays (ninhydrin for amino acids).

Qualitative analysis: Lassaigne's (sodium fusion) test

Σ FORMULA Reactions in the sodium extract



Element	Test	Observation
N	extract + FeSO_4 , boil, acidify (H_2SO_4)	Prussian blue , $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
S	+ lead acetate / acetic acid	black PbS
S	+ sodium nitroprusside	violet colour
N and S together	+ FeCl_3	blood red , $[\text{Fe}(\text{SCN})]^{2+}$
Halogens	boil with HNO_3 (removes $\text{CN}^-/\text{S}^{2-}$), then AgNO_3	AgCl white (soluble in NH_3), AgBr pale yellow (sparingly soluble), AgI yellow (insoluble)
P	oxidise with Na_2O_2 , then HNO_3 + ammonium molybdate	yellow ammonium phosphomolybdate

Quantitative analysis (basic principles)

Σ FORMULA Percentage formulas (m = mass of compound)

Element	Method	Formula
C	Liebig combustion $\rightarrow$ CO_2 (absorbed by KOH)	$\%C = \frac{12}{44} \times \frac{m_{\text{CO}_2}}{m} \times 100$
H	Liebig $\rightarrow$ H_2O (absorbed by anhydrous CaCl_2)	$\%H = \frac{2}{18} \times \frac{m_{\text{H}_2\text{O}}}{m} \times 100$
N	Dumas $\rightarrow$ N_2 volume at STP	$\%N = \frac{28}{22400} \times \frac{V_{\text{N}_2} (\text{mL, STP})}{m} \times 100$
N	Kjeldahl $\rightarrow$ NH_3 absorbed in acid	$\%N = \frac{1.4 \times N_{\text{acid}} \times V_{\text{acid used}} (\text{mL})}{m}$
Halogen X	Carius $\rightarrow$ AgX	$\%X = \frac{A_X}{M_{\text{AgX}}} \times \frac{m_{\text{AgX}}}{m} \times 100$
S	Carius $\rightarrow$ BaSO_4	$\%S = \frac{32}{233} \times \frac{m_{\text{BaSO}_4}}{m} \times 100$
P	$\rightarrow$ $\text{Mg}_2\text{P}_2\text{O}_7$	$\%P = \frac{62}{222} \times \frac{m_{\text{Mg}_2\text{P}_2\text{O}_7}}{m} \times 100$
O	by difference	$100 - (\text{sum of the others})$

Kjeldahl's method fails for nitro and azo compounds and for N in a ring (pyridine), because that N isn't converted to ammonium sulphate.

Common traps

⚠ MISTAKE ALERT Mistake alerts

- In the Kjeldahl formula, N_{acid} is the **normality**: $1 \text{ M H}_2\text{SO}_4 = 2 \text{ N}$.
- Blood-red colour (N and S together) arises only if there is **less** sodium than needed for separate NaCN and Na_2S .
- Boil with HNO_3 before the halogen test only when N or S is present.
- Steam distillation works for compounds that are **immiscible** with water and volatile in steam.

Question patterns

📊 PYQ PATTERN Recurring structures

- Matching method $\leftrightarrow$ mixture (glycerol, aniline, camphor, crude oil).
- Lassaigne colours and the species responsible.
- $\%N$ (Kjeldahl/Dumas), $\%X$ (Carius), $\%C/\%H$ numericals: frequent Section-B items.
- R_f calculation and interpretation.
- Limitations of Kjeldahl's method.

Practice questions

✍ C13 · Practice

1

Easy

⌚ 0.5 min

Speed

C13-01

Glycerol is recovered from spent lye in the soap industry by:

- (A) steam distillation (B) distillation under reduced pressure
(C) sublimation (D) crystallisation

2 Easy 0.5 min Speed

C13-02

Aniline is purified by:

- (A) steam distillation (B) sublimation
(C) chromatography only (D) simple crystallisation

3 Easy 0.5 min Speed

C13-03

The Prussian blue colour in Lassaigne's test for nitrogen is due to:

- (A) $\text{Na}_4[\text{Fe}(\text{CN})_6]$ (B) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (C) $\text{Fe}(\text{SCN})_3$ (D) $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$

4 Easy 0.5 min Concept

C13-04

A blood-red colour with FeCl_3 in the sodium extract shows the presence of:

- (A) N only (B) S only (C) both N and S (D) halogen

5 Medium 1.5 min NV

C13-05

In a Kjeldahl estimation, the ammonia from 0.5 g of an organic compound neutralises 12.5 mL of 0.5 M H_2SO_4 . Find the percentage of nitrogen.

Your answer (numerical value):

6 Medium 1.5 min NV

C13-06

0.1 g of an organic compound gives 0.287 g of AgCl in the Carius method. Find the percentage of chlorine (nearest integer).

Your answer (numerical value):

7 Easy 0.5 min Speed

C13-07

In a TLC run, a spot moves 3 cm while the solvent front moves 5 cm. The R_f value is:

- (A) 0.6 (B) 1.67 (C) 0.4 (D) 2

8 Easy 0.5 min Concept

C13-08

Kjeldahl's method can't be used to estimate nitrogen in:

- (A) urea (B) acetamide (C) nitrobenzene (D) aniline

9 Easy 1 min Basic

C13-09

0.2 g of an organic compound gives 0.44 g of CO_2 on combustion. The percentage of carbon is:

- (A) 30% (B) 60% (C) 44% (D) 12%



C13 · Chapter Test

1 Easy 0.5 min Speed

C13-T1

Which substance is purified by sublimation?

- (A) sugar (B) naphthalene (C) glycerol (D) aniline

2 Easy 0.5 min Speed

C13-T2

Silver bromide precipitate is:

- (A) white (B) pale yellow
(C) yellow, insoluble in ammonia (D) black

3 Medium 1.5 min NV

C13-T3

0.3 g of an organic compound gives 50 mL of N_2 at STP in the Dumas method. Find the percentage of nitrogen (nearest integer).

Your answer (numerical value):

4 Easy 0.5 min Concept

C13-T4

In steam distillation the mixture boils:

- (A) at the boiling point of the organic compound
(B) below 373 K, when the sum of the vapour pressures equals atmospheric pressure
(C) above 373 K
(D) only under reduced pressure

5 Easy 0.5 min Speed

C13-T5

Phosphorus in an organic compound is detected as a yellow precipitate with:

- (A) lead acetate
(B) ammonium molybdate (after oxidation to phosphate)
(C) silver nitrate
(D) sodium nitroprusside

Answers & Solutions

C13 · Practice

1 (B) 2 (A) 3 (B) 4 (C) 5 35 6 71 7 (A) 8 (C) 9 (B)

C13 · Chapter Test

1 (B) 2 (B) 3 21 4 (B) 5 (B)

C13 · PRACTICE

1 Answer: (B) Glycerol from spent lye Easy 0.5 min Speed

C13-01

SOLUTION Glycerol decomposes at its normal boiling point, so it is distilled at lower pressure (lower boiling point).

2 Answer: (A) Purifying aniline Easy 0.5 min Speed

C13-02

SOLUTION Aniline is steam-volatile and immiscible with water.

3 Answer: (B) Lassaigne test for N Easy 0.5 min Speed

C13-03

SOLUTION Ferric ferrocyanide.

4 Answer: (C) Blood-red colour Easy 0.5 min Concept

C13-04

SOLUTION $NaSCN$ forms, which gives $[Fe(SCN)]^{2+}$ with Fe^{3+} .

5 Answer: 35 Kjeldahl calculation Medium 1.5 min NV

C13-05

SOLUTION $0.5\text{ M H}_2\text{SO}_4 = 1\text{ N}$. $\%N = \frac{1.4 \times 1 \times 12.5}{0.5} = 35\%$.

COMMON TRAP Using molarity (0.5) as normality gives 17.5%.

6 Answer: 71 *Carius method* **Medium** ⌚ 1.5 min **NV**

C13-06

SOLUTION $\%Cl = \frac{35.5}{143.5} \times \frac{0.287}{0.1} \times 100 = 0.2474 \times 287 \approx 71\%$.

7 Answer: (A) *R_f value* **Easy** ⌚ 0.5 min **Speed**

C13-07

SOLUTION $R_f = 3/5$.

8 Answer: (C) *Kjeldahl limitation* **Easy** ⌚ 0.5 min **Concept**

C13-08

SOLUTION The N of nitro groups (and of azo compounds and pyridine) isn't converted into ammonium sulphate.

9 Answer: (B) *Percentage of carbon* **Easy** ⌚ 1 min **Basic**

C13-09

SOLUTION $\frac{12}{44} \times \frac{0.44}{0.2} \times 100 = 60\%$.

C13 · CHAPTER TEST

1 Answer: (B) *Sublimation* **Easy** ⌚ 0.5 min **Speed**

C13-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (B) *Silver halide colours* **Easy** ⌚ 0.5 min **Speed**

C13-T2

SOLUTION AgCl is white, AgBr pale yellow, AgI yellow.

3 Answer: 21 *Dumas method* **Medium** ⌚ 1.5 min **NV**

C13-T3

SOLUTION $\%N = \frac{28 \times 50}{22400 \times 0.3} \times 100 \approx 20.8 \approx 21\%$.

4 Answer: (B) *Steam distillation principle* **Easy** ⌚ 0.5 min **Concept**

C13-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (B) *Detection of phosphorus* **Easy** ⌚ 0.5 min **Speed**

C13-T5

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

REVISION Revision tracker · Purification & Characterisation

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 14

Basic Principles of Organic Chemistry (GOC)

14

1–2 per shift (+
underlies all
organic)

TYPICAL QUESTIONS

Medium
DIFFICULTY

Must-do
PRIORITY

Class 11 · Ch 8
NCERT

~12 hours
STUDY TIME

PYQ PATTERN Weight evidence

27 GOC questions across the 2026 Paper-1 shifts ($\approx 5.7\%$ of Chemistry)  THIRD-PARTY ANALYSIS [T-WT-CD].
GOC logic (stability, acidity, basicity, electronic effects) also decides most reaction-mechanism questions in Units 15–18.

Important concepts

- **Tetravalency & hybridisation:** sp^3 (25% s, tetrahedral, 109.5°), sp^2 (33% s, trigonal, 120°), sp (50% s, linear, 180°). More s-character makes carbon more electronegative, so sp C–H bonds are more acidic.
- **Classification & homologous series:** functional groups; successive members differ by $-\text{CH}_2-$.
- **Isomerism:** structural (chain, position, functional, metamerism, tautomerism) and stereoisomerism (geometrical, optical).
- **Bond fission:** homolytic $\rightarrow$ free radicals; heterolytic $\rightarrow$ carbocations/carbanions.
- **Electronic effects:** inductive (permanent, through σ bonds, falls off with distance), electromeric (temporary, on demand), resonance/mesomeric (π delocalisation), hyperconjugation (σ – π delocalisation).
- **Reaction types:** substitution, addition, elimination, rearrangement.

IUPAC nomenclature essentials

1. **Principal chain:** the longest chain containing the principal functional group (and as many multiple bonds as possible).
2. **Lowest locants:** principal group first, then multiple bonds, then substituents (lowest-set rule).
3. **Suffix priority** (highest first): carboxylic acid > sulphonic acid > anhydride > ester > acyl halide > amide > nitrile > aldehyde > ketone > alcohol > amine. Halo, nitro and alkoxy groups are always prefixes.
4. Substituent prefixes in **alphabetical** order (di-, tri- ignored).

Examples: $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{COOH}$ = 3-hydroxybutanoic acid; $\text{CH}_2=\text{CHCHO}$ = prop-2-enal; $\text{CH}_3\text{COCH}_2\text{CH}_2\text{Cl}$ = 4-chlorobutan-2-one.

Stability orders (the core of GOC)

FORMULA Reactive intermediates

Intermediate	Hybridisation / shape	Stability order	Stabilised by
Carbocation	sp^2 , planar	benzyl $\approx$ allyl (resonance) > 3° > 2° > 1° > methyl > vinyl, aryl	+I, hyperconjugation, +R
Free radical	sp^2 , planar (nearly)	benzyl $\approx$ allyl > 3° > 2° > 1° > methyl	hyperconjugation, resonance
Carbanion	sp^3 , pyramidal	methyl > 1° > 2° > 3° (reverse)	–I, –R (e.g. next to C=O, NO_2)

Triphenylmethyl cation/radical/anion are exceptionally stable (extensive resonance).

Σ FORMULA Electronic-effect rankings

-I: $-\text{NO}_2 > -\text{CN} > -\text{COOH} > -\text{F} > -\text{Cl} > -\text{Br} > -\text{I} > -\text{OH} > -\text{OR} > -\text{C}_6\text{H}_5$ **+I:** $-\text{C}(\text{CH}_3)_3 > -\text{CH}(\text{CH}_3)_2 > -\text{CH}_2\text{CH}_3 > -\text{CH}_3$ **+R (electron-donating by resonance):** $-\text{O}- > -\text{NH}_2 > -\text{OH} > -\text{OR} > -\text{NHCOR}$, halogens (weak +R, strong -I) **-R (electron-withdrawing):** $-\text{NO}_2, -\text{CN}, -\text{CHO}, -\text{COR}, -\text{COOH}, -\text{SO}_3\text{H}$
Hyperconjugation: the number of α -hydrogens decides it: $\text{CH}_3^+ 0 < \text{CH}_3\text{CH}_2^+ 3 < (\text{CH}_3)_2\text{CH}^+ 6 < (\text{CH}_3)_3\text{C}^+ 9$.

☆ IMPORTANT Electrophiles & nucleophiles

Electrophiles (electron-seeking): $\text{H}^+, \text{NO}_2^+, \text{Cl}^+$, carbocations, $\text{BF}_3, \text{AlCl}_3$, carbonyl carbon. **Nucleophiles** (nucleus-seeking): $\text{OH}^-, \text{RO}^-, \text{CN}^-, \text{Cl}^-, \text{NH}_3, \text{H}_2\text{O}, \text{ROH}, \text{RNH}_2$, carbanions.

Isomerism quick reference

Type	Example
Chain	butane / 2-methylpropane
Position	propan-1-ol / propan-2-ol
Functional	ethanol / methoxymethane; propanal / propanone
Metamerism	diethyl ether / methyl propyl ether
Tautomerism	keto-enol: propanone $\rightleftharpoons$ prop-1-en-2-ol (needs an α -H)
Geometrical	<i>cis/trans</i> -but-2-ene (needs different groups on each double-bond carbon)
Optical	butan-2-ol (one chiral C); lactic acid

Counting stereoisomers: up to 2^n for n chiral centres, fewer when **meso** forms exist (2,3-dibromobutane and tartaric acid have 3: a d/l pair + meso). Meso compounds have an internal mirror plane and are optically inactive. A racemic mixture (50 : 50) is optically inactive by external compensation.

Common traps**⚠ MISTAKE ALERT Mistake alerts**

- Halogens: -I dominates +R, so they deactivate rings, but they still direct ortho/para.
- Vinyl and aryl carbocations are **very unstable** (sp carbon, no resonance help).
- Carbanion stability order is the **reverse** of the carbocation order.
- Geometrical isomerism: but-1-ene has none (two H on C1).
- A molecule with chiral centres may still be achiral (meso).

Question patterns**📊 PYQ PATTERN Recurring structures**

1. Ranking carbocation/radical/carbanion stability.
2. Ranking acidity/basicity using I, R and hyperconjugation effects.
3. IUPAC names of polyfunctional compounds.
4. Counting isomers (structural, stereo); identifying chiral/meso compounds.
5. Classifying reagents (electrophile/nucleophile) and reaction types.
6. Hyperconjugation structures/ α -H counting; resonance structures and their relative stability.

Practice questions

C14 · Practice

1 Medium 1 min Concept C14-01

In allene, $\text{CH}_2=\text{C}=\text{CH}_2$, the hybridisations of C1, C2 and C3 are:

- (A) $\text{sp}^2, \text{sp}^2, \text{sp}^2$ (B) $\text{sp}^2, \text{sp}, \text{sp}^2$ (C) $\text{sp}, \text{sp}, \text{sp}$ (D) $\text{sp}^3, \text{sp}, \text{sp}^3$

2 Easy 0.75 min Basic C14-02

The IUPAC name of $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{COOH}$ is:

- (A) 3-hydroxybutanoic acid (B) 2-hydroxybutanoic acid
(C) 3-carboxybutan-2-ol (D) β -hydroxybutyric acid

3 Easy 0.75 min Concept C14-03

The most stable carbocation is:

- (A) CH_3CH_2^+ (B) $(\text{CH}_3)_2\text{CH}^+$ (C) $\text{C}_6\text{H}_5\text{CH}_2^+$ (D) $\text{CH}_2=\text{CH}^+$

4 Easy 0.5 min Speed C14-04

The correct order of $-\text{I}$ effect is:

- (A) $-\text{NO}_2 > -\text{CN} > -\text{F} > -\text{Cl}$ (B) $-\text{F} > -\text{NO}_2 > -\text{CN} > -\text{Cl}$
(C) $-\text{Cl} > -\text{F} > -\text{CN} > -\text{NO}_2$ (D) $-\text{CN} > -\text{NO}_2 > -\text{Cl} > -\text{F}$

5 Medium 1 min NV C14-05

Find the total number of stereoisomers of 2,3-dibromobutane.

Your answer (numerical value):

6 Easy 0.5 min NV C14-06

How many α -hydrogens are available for hyperconjugation in the tert-butyl cation, $(\text{CH}_3)_3\text{C}^+$?

Your answer (numerical value):

7 Easy 0.5 min Speed C14-07

Which compound shows geometrical isomerism?

- (A) but-1-ene (B) but-2-ene (C) 2-methylpropene (D) ethene

8 Easy 0.5 min Speed C14-08

Which species is an electrophile?

- (A) NH_3 (B) NO_2^+ (C) OH^- (D) CN^-

9 Easy 0.75 min Concept C14-09

Among these alkyl carbanions, the most stable is:

- (A) CH_3^- (B) CH_3CH_2^- (C) $(\text{CH}_3)_2\text{CH}^-$ (D) $(\text{CH}_3)_3\text{C}^-$

10 Easy 0.5 min Concept

C14-10

Which compound shows keto–enol tautomerism?

- (A) $(\text{CH}_3)_3\text{C}-\text{CHO}$ (B) CH_3COCH_3 (C) $\text{C}_6\text{H}_5\text{CHO}$ (D) HCHO

11 Easy 0.5 min Speed

C14-11

Which molecule is chiral?

- (A) propan-2-ol (B) butan-2-ol
(C) 2-methylpropan-2-ol (D) pentan-3-ol

C14 · Chapter Test

1 Easy 0.5 min Speed

C14-T1

Diethyl ether and methyl propyl ether are:

- (A) chain isomers (B) position isomers
(C) metamers (D) functional isomers

2 Easy 0.5 min Speed

C14-T2

Homolytic cleavage of a covalent bond produces:

- (A) carbocations (B) carbanions (C) free radicals (D) ion pairs

3 Easy 0.5 min NV

C14-T3

How many structural isomers does C_5H_{12} have?Your answer (numerical value):

4 Easy 0.5 min Speed

C14-T4

Which functional group gets the highest priority as the suffix in IUPAC naming?

- (A) $-\text{OH}$ (B) $-\text{CHO}$ (C) $-\text{COOH}$ (D) $>\text{C}=\text{O}$

5 Easy 0.5 min Speed

C14-T5

Which group shows a +R (electron-donating resonance) effect?

- (A) $-\text{NO}_2$ (B) $-\text{CHO}$ (C) $-\text{OH}$ (D) $-\text{CN}$

Answers & Solutions

C14 · Practice

- 1 (B) 2 (A) 3 (C) 4 (A) 5 3 6 9 7 (B) 8 (B) 9 (A) 10 (B)
11 (B)

C14 · Chapter Test

- 1 (C) 2 (C) 3 3 4 (C) 5 (C)

C14 · PRACTICE

1 Answer: (B) Hybridisation in allene Medium 1 min Concept

C14-01

SOLUTION The central carbon forms two π bonds (sp). The terminal carbons form one π bond each (sp^2).

2 Answer: (A) *IUPAC naming* **Easy** ⌚ 0.75 min **Basic** C14-02

SOLUTION -COOH is the principal group, so it is C1. The OH is on C3.

COMMON TRAP Choosing (D), which is the common name.

3 Answer: (C) *Carbocation stability* **Easy** ⌚ 0.75 min **Concept** C14-03

SOLUTION The benzyl cation is resonance-stabilised over the ring. Vinyl is the least stable.

4 Answer: (A) *-I effect order* **Easy** ⌚ 0.5 min **Speed** C14-04

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: 3 *Stereoisomers with a meso form* **Medium** ⌚ 1 min **NV** C14-05

SOLUTION Two chiral centres, but the molecule is symmetric: a pair of enantiomers + one meso form = 3.

COMMON TRAP Answering $2^2 = 4$.

6 Answer: 9 *α -Hydrogens* **Easy** ⌚ 0.5 min **NV** C14-06

SOLUTION Three methyl groups $\times$ 3 H.

7 Answer: (B) *Geometrical isomerism* **Easy** ⌚ 0.5 min **Speed** C14-07

SOLUTION Each double-bond carbon carries two different groups (H and CH_3).

8 Answer: (B) *Electrophile* **Easy** ⌚ 0.5 min **Speed** C14-08

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

9 Answer: (A) *Carbanion stability* **Easy** ⌚ 0.75 min **Concept** C14-09

SOLUTION Alkyl groups (+I) destabilise a negative charge.

SHORTCUT Carbanion order is the reverse of the carbocation order.

10 Answer: (B) *Keto-enol tautomerism* **Easy** ⌚ 0.5 min **Concept** C14-10

SOLUTION Tautomerism needs an α -hydrogen next to the carbonyl group. Acetone has six. The others have none.

11 Answer: (B) *Chirality* **Easy** ⌚ 0.5 min **Speed** C14-11

SOLUTION C2 of butan-2-ol carries H, OH, CH_3 and C_2H_5 : four different groups.

C14 · CHAPTER TEST

1 Answer: (C) *Metamers* **Easy** ⌚ 0.5 min **Speed** C14-T1

SOLUTION Same functional group, different alkyl groups on either side of it.

2 Answer: (C) *Homolytic fission* **Easy** ⌚ 0.5 min **Speed** C14-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 3 *Chain isomers of pentane* **Easy** ⌚ 0.5 min **NV** C14-T3

SOLUTION n-pentane, isopentane (2-methylbutane), neopentane (2,2-dimethylpropane).

4 Answer: (C) *Principal group priority* **Easy** ⌚ 0.5 min **Speed** C14-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (C) *+R group* **Easy** ⌚ 0.5 min **Speed** C14-T5

SOLUTION O's lone pair delocalises into the π system.

REVISION Revision tracker · GOC

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 15

Hydrocarbons

15

~1 per shift
TYPICAL QUESTIONSMedium
DIFFICULTYHigh
PRIORITYClass 11 · Ch 9
NCERT~8 hours
STUDY TIME

Alkanes

Σ FORMULA Preparation & reactions

Preparation: hydrogenation of alkenes/alkynes (Ni, Pt or Pd); alkyl halide + Zn/HCl; **Wurtz** $2\text{RX} + 2\text{Na} \xrightarrow{\text{dry ether}} \text{R-R} + 2\text{NaX}$ (poor for unsymmetrical alkanes: gives a mixture); **decarboxylation** $\text{RCOONa} + \text{NaOH} \xrightarrow{\text{CaO}, \Delta} \text{RH} + \text{Na}_2\text{CO}_3$ (one carbon fewer); **Kolbe electrolysis** $2\text{RCOONa} + 2\text{H}_2\text{O} \longrightarrow \text{R-R} + 2\text{CO}_2 + \text{H}_2 + 2\text{NaOH}$ (R-R at the anode). **Reactions:** free-radical **halogenation** (hv). Reactivity $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$; H abstraction $3^\circ > 2^\circ > 1^\circ$. Combustion; controlled oxidation; isomerisation (AlCl₃/HCl); **aromatisation** (n-hexane → benzene over Cr₂O₃, V₂O₅ or Mo₂O₃ at 773 K); pyrolysis (cracking).

☆ IMPORTANT Free-radical chlorination of methane: mechanism

Initiation: $\text{Cl}_2 \xrightarrow{h\nu} 2\text{Cl}\cdot$ **Propagation:** $\text{Cl}\cdot + \text{CH}_4 \longrightarrow \text{HCl} + \text{CH}_3\cdot$; $\text{CH}_3\cdot + \text{Cl}_2 \longrightarrow \text{CH}_3\text{Cl} + \text{Cl}\cdot$ **Termination:** radicals combine ($\text{Cl}\cdot + \text{Cl}\cdot$, $\text{CH}_3\cdot + \text{CH}_3\cdot \rightarrow$ ethane, which explains traces of ethane in the product).**Conformations of ethane:** staggered (most stable) and eclipsed (least stable), differing by about 12.5 kJ/mol (torsional strain). They are drawn as **Sawhorse** and **Newman** projections. Free rotation at room temperature means the conformers can't be isolated.

Alkenes

Σ FORMULA Preparation

- Alkynes: **Lindlar's catalyst** (Pd/CaCO₃ poisoned with S or quinoline) → **cis**-alkene. **Na in liquid NH₃** → **trans**-alkene.
- Dehydrohalogenation:** alkyl halide + alcoholic KOH → alkene (Saytzeff: more substituted alkene is major).
- Dehalogenation** of vicinal dihalides with Zn.
- Acidic dehydration** of alcohols: conc. H₂SO₄, 443 K ($3^\circ > 2^\circ > 1^\circ$ ease).

🔬 REACTION MAP Reaction map · propene CH₃CH=CH₂

H ₂ / Ni	→ propane
Br ₂ in CCl ₄ (anti addition; decolourises)	→ 1,2-dibromopropane
HBr (no peroxide) · Markovnikov	→ 2-bromopropane
HBr + peroxide · anti-Markovnikov (only HBr)	→ 1-bromopropane
H ₂ O / H ⁺ · Markovnikov	→ propan-2-ol
cold dil. alkaline KMnO ₄ (Baeyer's test)	→ propane-1,2-diol (purple → brown)
hot acidic KMnO ₄	→ ethanoic acid + CO ₂
O ₃ then Zn/H ₂ O (ozonolysis)	→ ethanal + methanal
polymerisation	→ polypropene

☆ IMPORTANT Markovnikov & peroxide effect: mechanisms

- **Markovnikov:** H^+ adds first to give the **more stable carbocation** (2° here), and then Br^- attacks it.
- **Peroxide (Kharasch) effect:** $\text{HBr} + \text{ROOR}$ gives a free-radical chain. $\text{Br}\cdot$ adds first to give the more stable **radical**, so Br ends up on the less substituted carbon. It doesn't occur with HCl or HI (their propagation steps are not both exothermic).

Alkynes

Σ FORMULA Key facts

Preparation: $\text{CaC}_2 + 2\text{H}_2\text{O} \longrightarrow \text{C}_2\text{H}_2 + \text{Ca(OH)}_2$; vicinal dihalide + alc. KOH , then NaNH_2 . **Acidity:** terminal alkynes are weakly acidic (sp C-H): $\text{HC}\equiv\text{CH} + \text{Na} \longrightarrow \text{HC}\equiv\text{CNa} + \frac{1}{2}\text{H}_2$. Acidity order: ethyne > ethene > ethane. **Additions:** H_2 , X_2 , HX (Markovnikov, twice $\rightarrow$ gem-dihalide). **Hydration:** $\text{HgSO}_4/\text{dil. H}_2\text{SO}_4$, 333 K. Ethyne $\rightarrow$ **ethanal**; propyne $\rightarrow$ **propanone** (via the enol). **Cyclic polymerisation:** $3\text{C}_2\text{H}_2 \rightarrow$ benzene (red-hot iron tube, 873 K).

Aromatic hydrocarbons

Σ FORMULA Benzene & aromaticity

All C-C bonds are equal (139 pm) because of resonance. **Hückel's rule:** planar, cyclic, fully conjugated, with $(4n + 2)$ π electrons $\rightarrow$ aromatic. Examples: benzene (6), naphthalene (10), cyclopentadienyl **anion** (6), tropylium **cation** (6), pyridine. The cyclopentadienyl cation (4) is antiaromatic; cyclooctatetraene is non-planar (tub-shaped), so non-aromatic. **Preparation:** cyclic polymerisation of ethyne; decarboxylation of sodium benzoate (soda lime); reduction of phenol with Zn dust.

Σ FORMULA Electrophilic aromatic substitution (EAS)

Mechanism: (1) generate E^+ $\rightarrow$ (2) attack forms the **arenium (σ) complex** (slow, rate-determining) $\rightarrow$ (3) loss of H^+ restores aromaticity.

Reaction	Reagents	Electrophile
Nitration	conc. HNO_3 + conc. H_2SO_4	NO_2^+
Halogenation	$\text{Cl}_2/\text{Br}_2 + \text{FeCl}_3/\text{AlCl}_3$ (dark, cold)	Cl^+ (Br^+)
Sulphonation	fuming H_2SO_4	SO_3
Friedel-Crafts alkylation	$\text{RX} + \text{anhydrous AlCl}_3$	R^+ (rearranges ; polyalkylation possible)
Friedel-Crafts acylation	$\text{RCOCl} + \text{anhydrous AlCl}_3$	$\text{RC}\equiv\text{O}^+$ (acylium; no rearrangement)

Addition to benzene: H_2/Ni (high T , P) $\rightarrow$ cyclohexane; $\text{Cl}_2 + \text{UV} \rightarrow$ benzene hexachloride (BHC, $\text{C}_6\text{H}_6\text{Cl}_6$).

Σ FORMULA Directive influence of substituents

Group type	Examples	Directs to	Ring activity
Strongly activating	$-\text{O}-$, $-\text{NH}_2$, $-\text{NHR}$, $-\text{OH}$, $-\text{OR}$	ortho/para	activating
Mildly activating	$-\text{NHCOCH}_3$, $-\text{CH}_3$, alkyl, $-\text{C}_6\text{H}_5$	ortho/para	activating
Halogens	$-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$	ortho/para	deactivating ($-\text{I} > +\text{R}$)
Meta directors	$-\text{NO}_2$, $-\text{CN}$, $-\text{CHO}$, $-\text{COR}$, $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{NR}_3^+$	meta	deactivating

Friedel-Crafts reactions **fail** on strongly deactivated rings (e.g. nitrobenzene) and on aniline ($-\text{NH}_2$ complexes with AlCl_3). Polynuclear hydrocarbons such as 3,4-benzpyrene are **carcinogenic** (NCERT).

Common traps

MISTAKE ALERT Mistake alerts

- The peroxide effect works **only with HBr**.
- Friedel–Crafts alkylation with n-propyl chloride gives mainly **isopropylbenzene** (the 1° carbocation rearranges to 2°).
- Lindlar gives cis; Na/liquid NH_3 gives trans.
- Hydration of ethyne gives an **aldehyde** (ethanal). Every other alkyne gives ketones.
- Wurtz with two different halides gives **three** alkanes.

Question patterns

PYQ PATTERN Recurring structures

1. Products of addition to unsymmetrical alkenes (Markovnikov/anti-Markovnikov), ozonolysis → identifying the parent alkene.
2. Stereochemistry of reductions (Lindlar vs Na/NH_3).
3. Wurtz/Kolbe/decarboxylation products.
4. EAS reagents, electrophiles, and directive effects in disubstituted benzenes.
5. Aromaticity identification.
6. Conformations (Newman/Sawhorse) and their relative energies.

Practice questions

C15 · Practice

1

Medium

 1 min

NV

C15-01

A mixture of CH_3Br and $\text{C}_2\text{H}_5\text{Br}$ is treated with sodium in dry ether. Find the number of different alkanes formed.

Your answer (numerical value):

2

Easy

 0.5 min

Speed

C15-02

Propene reacts with HBr in the presence of benzoyl peroxide to give mainly:

- (A) 2-bromopropane (B) 1-bromopropane
(C) 1,2-dibromopropane (D) propan-2-ol

3

Easy

 0.75 min

Basic

C15-03

Ozonolysis of but-2-ene (with $\text{Zn}/\text{H}_2\text{O}$ work-up) gives:

- (A) two molecules of ethanal (B) propanal + methanal
(C) propanone (D) ethanoic acid

4

Easy

 0.75 min

Basic

C15-04

Propyne with $\text{HgSO}_4/\text{dil. H}_2\text{SO}_4$ gives:

- (A) propanal (B) propanone (C) propan-1-ol (D) propanoic acid

5

Easy

 0.5 min

Speed

C15-05

The most acidic hydrocarbon is:

- (A) ethane (B) ethene (C) ethyne (D) methane

6 Medium 1 min Concept

C15-06

Which species is aromatic?

- (A) cyclopentadienyl cation (B) cyclopentadienyl anion
(C) cyclooctatetraene (D) cyclobutadiene

7 Easy 0.5 min Speed

C15-07

Which substituent directs an incoming electrophile to the meta position?

- (A) $-\text{OH}$ (B) $-\text{CH}_3$ (C) $-\text{NO}_2$ (D) $-\text{Cl}$

8 Easy 0.5 min Concept

C15-08

In ethane, the staggered conformation is more stable than the eclipsed one by about:

- (A) 1.25 kJ/mol (B) 12.5 kJ/mol
(C) 125 kJ/mol (D) 0 (they are equally stable)

9 Easy 0.75 min Basic

C15-09

Kolbe electrolysis of aqueous sodium propanoate gives at the anode:

- (A) ethane (B) propane (C) butane (D) hexane

10 Easy 0.5 min Speed

C15-10

In the chlorination of methane, the chain-initiation step is:

- (A) $\text{Cl}\cdot + \text{CH}_4 \longrightarrow \text{CH}_3\cdot + \text{HCl}$ (B) $\text{Cl}_2 \xrightarrow{h\nu} 2\text{Cl}\cdot$
(C) $\text{CH}_3\cdot + \text{Cl}_2 \longrightarrow \text{CH}_3\text{Cl} + \text{Cl}\cdot$ (D) $\text{CH}_3\cdot + \text{CH}_3\cdot \longrightarrow \text{C}_2\text{H}_6$

11 Medium 1 min Tricky

C15-11

Benzene reacts with 1-chloropropane and anhydrous AlCl_3 . The major product is:

- (A) n-propylbenzene (B) isopropylbenzene (cumene)
(C) propiophenone (D) no reaction

C15 · Chapter Test

1 Easy 0.5 min Speed

C15-T1

Baeyer's reagent, used to test for unsaturation, is:

- (A) bromine water (B) cold, dilute, alkaline KMnO_4
(C) acidic $\text{K}_2\text{Cr}_2\text{O}_7$ (D) ammoniacal AgNO_3

2 Easy 0.5 min Speed

C15-T2

Propene + HCl gives mainly:

- (A) 1-chloropropane (B) 2-chloropropane
(C) 1,2-dichloropropane (D) propan-2-ol

3 Easy 0.5 min NV

C15-T3

Find the number of π electrons in naphthalene.

Your answer (numerical value):

4

Easy

⌚ 0.5 min

Speed

C15-T4

But-2-yne is converted into *trans*-but-2-ene by:

- (A) H_2 , Lindlar catalyst (B) Na in liquid NH_3
 (C) H_2/Ni , excess (D) $\text{HgSO}_4/\text{H}_2\text{SO}_4$

5

Easy

⌚ 0.5 min

Speed

C15-T5

Which group deactivates the ring but directs ortho/para?

- (A) $-\text{CH}_3$ (B) $-\text{Cl}$ (C) $-\text{NO}_2$ (D) $-\text{NH}_2$

Answers & Solutions

C15 · Practice

- 1 3 2 (B) 3 (A) 4 (B) 5 (C) 6 (B) 7 (C) 8 (B) 9 (C) 10 (B)
 11 (B)

C15 · Chapter Test

- 1 (B) 2 (B) 3 10 4 (B) 5 (B)

C15 · PRACTICE

1

Answer: 3 Wurtz with mixed halides

Medium

⌚ 1 min

NV

C15-01

SOLUTION Ethane (CH_3-CH_3), propane ($\text{CH}_3-\text{C}_2\text{H}_5$) and butane ($\text{C}_2\text{H}_5-\text{C}_2\text{H}_5$).

COMMON TRAP Answering 1 (only the cross product).

2

Answer: (B) Peroxide effect

Easy

⌚ 0.5 min

Speed

C15-02

SOLUTION Free-radical addition: $\text{Br}\cdot$ adds to the terminal carbon, giving the more stable 2° radical.

3

Answer: (A) Ozonolysis

Easy

⌚ 0.75 min

Basic

C15-03

SOLUTION Cleaving $\text{CH}_3\text{CH}=\text{CHCH}_3$ at the double bond gives $2\text{CH}_3\text{CHO}$.

4

Answer: (B) Hydration of propyne

Easy

⌚ 0.75 min

Basic

C15-04

SOLUTION Markovnikov hydration gives an enol that tautomerises to the ketone.

COMMON TRAP Choosing propanal.

5

Answer: (C) Acidity of hydrocarbons

Easy

⌚ 0.5 min

Speed

C15-05

SOLUTION An sp carbon (50% s -character) holds the negative charge of the conjugate base best.

6

Answer: (B) Aromaticity

Medium

⌚ 1 min

Concept

C15-06

SOLUTION Planar, cyclic, conjugated, with 6π electrons ($4n + 2$ with $n = 1$). The cation (4π) and cyclobutadiene (4π) are antiaromatic. COT is non-planar.

7

Answer: (C) Meta director

Easy

⌚ 0.5 min

Speed

C15-07

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

8

Answer: (B) Ethane conformations

Easy

⌚ 0.5 min

Concept

C15-08

SOLUTION Torsional strain in the eclipsed form is about 12.5 kJ/mol (NCERT).

9

Answer: (C) Kolbe electrolysis

Easy

⌚ 0.75 min

Basic

C15-09

SOLUTION $2\text{CH}_3\text{CH}_2\text{COO}^- \rightarrow \text{CH}_3\text{CH}_2-\text{CH}_2\text{CH}_3 + 2\text{CO}_2 + 2\text{e}^-$.

SHORTCUT Product = 2R , where R = the alkyl group of the acid.

COMMON TRAP Losing the carbon as in decarboxylation (ethane).

10

Answer: (B) Chain initiation

Easy

⌚ 0.5 min

Speed

C15-10

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

11 Answer: (B) *Friedel–Crafts rearrangement* **Medium** ⌚ 1 min ⚡ Tricky

C15-11

SOLUTION The 1° carbocation rearranges by a hydride shift to the more stable 2° isopropyl cation.

COMMON TRAP Answering n-propylbenzene.

C15 · CHAPTER TEST

1 Answer: (B) *Baeyer's reagent* **Easy** ⌚ 0.5 min ⚡ Speed

C15-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (B) *Markovnikov product* **Easy** ⌚ 0.5 min ⚡ Speed

C15-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 10 *π electrons in naphthalene* **Easy** ⌚ 0.5 min NV

C15-T3

SOLUTION Five C=C bonds in the resonance structure, $4n + 2$ with $n = 2$.

4 Answer: (B) *Trans-alkene* **Easy** ⌚ 0.5 min ⚡ Speed

C15-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (B) *Deactivating o/p director* **Easy** ⌚ 0.5 min ⚡ Speed

C15-T5

SOLUTION Halogens: -I (deactivating) outweighs +R, but +R still directs ortho/para.

REVISION Revision tracker · Hydrocarbons

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III • CHEMISTRY • UNIT 16

Organic Compounds Containing Halogens

16

~1 per shift
TYPICAL QUESTIONSMedium
DIFFICULTYHigh
PRIORITYClass 12 • Ch 6
NCERT~6 hours
STUDY TIME

Important concepts

- **C–X bond:** polar ($C^{\delta+}-X^{\delta-}$). Bond length $C-F < C-Cl < C-Br < C-I$; bond strength runs the other way, so reactivity is $R-I > R-Br > R-Cl > R-F$.
- **Nucleophilic substitution** (S_N1/S_N2) and **β -elimination** compete. The substrate, nucleophile/base, solvent and temperature decide which wins.
- **Haloarenes** are far less reactive towards nucleophiles: resonance gives C–X partial double-bond character, the carbon is sp^2 , the phenyl cation is unstable, and electron-rich π clouds repel the nucleophile.

Preparation

Σ FORMULA From alcohols, alkenes and halogen exchange

Route	Reagent	Notes
$ROH \rightarrow RCl$	$HCl + \text{anhydrous } ZnCl_2; PCl_3; PCl_5; SOCl_2$	$SOCl_2$ is best: the byproducts SO_2 and HCl are gases
$ROH \rightarrow RBr / RI$	PBr_3 (red P + Br_2); red P + I_2	—
Alkene $\rightarrow RX$	HX (Markovnikov)	peroxide effect with HBr only
Finkelstein	$RCl/RBr + NaI$ in dry acetone $\rightarrow RI$	$NaCl/NaBr$ precipitate, which drives the reaction
Swarts	$RCl/RBr + AgF, Hg_2F_2, CoF_2$ or $SbF_3 \rightarrow RF$	route to alkyl fluorides
Haloarenes	$ArH + Cl_2$ ($FeCl_3$)	EAS
Sandmeyer	$ArN_2^+ + CuCl/HCl, CuBr/HBr, CuCN/KCN$	$\rightarrow ArCl, ArBr, ArCN$
Gattermann	$ArN_2^+ + Cu$ powder/ HX	$\rightarrow ArX$
Aryl iodide	$ArN_2^+ + KI$	no $Cu(I)$ needed

Physical properties: boiling point rises with halogen mass ($RI > RBr > RCl > RF$) and chain length, and **falls with branching**. The *para*-dihalobenzene has a higher melting point than the *ortho* or *meta* isomer (symmetric packing).

Nucleophilic substitution: S_N1 vs S_N2

	S_N2	S_N1
Steps / rate law	one step; $r = k[RX][Nu^-]$	two steps via a carbocation; $r = k[RX]$
Substrate reactivity	$CH_3X > 1^\circ > 2^\circ > 3^\circ$ (steric hindrance)	$3^\circ > 2^\circ > 1^\circ > CH_3X$ (carbocation stability)
Stereochemistry	inversion (Walden)	racemisation (with some inversion)
Favoured by	strong nucleophile, polar aprotic solvent	weak nucleophile, polar protic solvent
Special	—	allylic and benzylic halides react fast (resonance-stabilised cations)

REACTION MAP Reaction map • haloalkane R-X (1°, e.g. C₂H₅Br)

aq. KOH / NaOH	→ ROH (alcohol)
alc. KOH, heat	→ alkene (β-elimination, Saytzeff)
NaOR' (Williamson)	→ R-O-R' (ether)
KCN (ionic; attacks via C)	→ R-C≡N (nitrile), major
AgCN (covalent; attacks via N)	→ R-N≡C (isocyanide), major
KNO ₂	→ R-O-N=O (alkyl nitrite)
AgNO ₂	→ R-NO ₂ (nitroalkane)
NH ₃ (excess)	→ RNH ₂ (+ higher amines)
LiAlH ₄	→ RH
Mg, dry ether	→ RMgX (Grignard reagent)
Na, dry ether (Wurtz)	→ R-R

★ IMPORTANT Grignard reagents

RMgX is made in **dry ether**. It reacts with any compound that has an acidic H (water, alcohols, amines, terminal alkynes) to give the alkane: $\text{CH}_3\text{MgBr} + \text{H}_2\text{O} \longrightarrow \text{CH}_4 + \text{Mg}(\text{OH})\text{Br}$. That's why every trace of moisture must be excluded. Reactions with carbonyls are covered in Unit 17.

Σ FORMULA Wurtz family

Wurtz: $2\text{RX} + 2\text{Na} \longrightarrow \text{R-R}$. **Wurtz-Fittig:** $\text{ArX} + \text{RX} + 2\text{Na} \longrightarrow \text{Ar-R}$ (e.g. toluene from chlorobenzene + chloromethane). **Fittig:** $2\text{ArX} + 2\text{Na} \longrightarrow \text{Ar-Ar}$ (biphenyl).

Haloarenes: substitution only when activated

- Chlorobenzene → phenol needs **623 K, 300 atm** NaOH (Dow process).
- NO₂ groups at **ortho/para** positions activate the ring: *p*-nitrochlorobenzene reacts with NaOH at 443 K; the 2,4-dinitro compound at 368 K; the 2,4,6-trinitro compound with warm water.
- EAS** on haloarenes: ortho/para-directing but deactivating (halogenation, nitration, sulphonation, Friedel-Crafts).

Polyhalogen compounds: environmental effects (syllabus)

Compound	Uses	Environmental / health notes (NCERT)
CH ₂ Cl ₂	solvent, paint remover	affects the central nervous system
CHCl ₃	solvent (formerly an anaesthetic)	oxidised by air and light to poisonous phosgene (COCl ₂), so it's stored in dark bottles filled to the brim
CHI ₃ (iodoform)	antiseptic (releases I ₂)	objectionable smell; replaced by other antiseptics
CCl ₄	solvent, refrigerant feedstock	liver damage; depletes the ozone layer
Freons (CFCs)	refrigerants, aerosol propellants	reach the stratosphere, release Cl radicals, deplete ozone
DDT	insecticide (first chlorinated organic insecticide)	not biodegradable; accumulates in fatty tissue; banned in many countries

Common traps

MISTAKE ALERT Mistake alerts

- $\text{KCN} \rightarrow$ nitrile but $\text{AgCN} \rightarrow$ **isocyanide**. $\text{KNO}_2 \rightarrow$ nitrite but $\text{AgNO}_2 \rightarrow$ **nitro** compound.
- Aqueous KOH substitutes; **alcoholic** KOH eliminates.
- $\text{S}_{\text{N}}2$ on a chiral centre inverts the configuration. The R/S label may or may not change, depending on priorities.
- Vinyl and aryl halides don't undergo $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$ under normal conditions.

Question patterns

PYQ PATTERN Recurring structures

1. $\text{S}_{\text{N}}1$ vs $\text{S}_{\text{N}}2$ reactivity orders; stereochemical outcome.
2. Ambident nucleophiles (CN^- , NO_2^-) and their products.
3. Substitution vs elimination (reagent/conditions); Saytzeff product.
4. Named reactions: Finkelstein, Swarts, Sandmeyer, Gattermann, Wurtz, Wurtz-Fittig, Fittig.
5. Haloarene activation by nitro groups.
6. Environmental facts about CHCl_3 , CCl_4 , freons, DDT.

Practice questions

C16 · Practice

1 Easy ⌚ 0.5 min Speed C16-01

The best reagent for converting ethanol into pure chloroethane is:

- (A) HCl alone (B) SOCl_2 (C) Cl_2 (D) NaCl

2 Easy ⌚ 0.5 min Concept C16-02

The Finkelstein reaction converts an alkyl chloride or bromide into an alkyl iodide using:

- (A) KI in water (B) NaI in dry acetone
(C) $\text{I}_2/\text{red P}$ (D) HI

3 Easy ⌚ 0.75 min Concept C16-03

The correct order of $\text{S}_{\text{N}}2$ reactivity is:

- (A) $(\text{CH}_3)_3\text{CBr} > (\text{CH}_3)_2\text{CHBr} > \text{CH}_3\text{CH}_2\text{Br} > \text{CH}_3\text{Br}$ (B) $\text{CH}_3\text{Br} > \text{CH}_3\text{CH}_2\text{Br} > (\text{CH}_3)_2\text{CHBr} > (\text{CH}_3)_3\text{CBr}$
(C) $\text{CH}_3\text{CH}_2\text{Br} > \text{CH}_3\text{Br} > (\text{CH}_3)_3\text{CBr} > (\text{CH}_3)_2\text{CHBr}$ (D) all equal

4 Easy ⌚ 0.5 min Speed C16-04

Which halide undergoes $\text{S}_{\text{N}}1$ fastest?

- (A) CH_3Cl (B) $\text{CH}_3\text{CH}_2\text{Cl}$ (C) $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ (D) $\text{C}_6\text{H}_5\text{Cl}$

5 Easy ⌚ 0.5 min Speed C16-05

Ethyl bromide reacts with AgCN to give mainly:

- (A) ethyl cyanide (B) ethyl isocyanide (C) ethylamine (D) ethanol

6

Easy

⌚ 0.5 min

Speed

C16-06

CH_3MgBr reacts with water to give:

- (A) methanol (B) methane (C) ethane (D) formaldehyde

7

Easy

⌚ 0.75 min

Basic

C16-07

2-Bromobutane heated with alcoholic KOH gives mainly:

- (A) but-1-ene (B) but-2-ene (C) butan-2-ol (D) butane

8

Easy

⌚ 0.5 min

Concept

C16-08

Chloroform is stored in dark-coloured bottles filled to the brim because:

- (A) it is highly volatile
(B) in air and light it is oxidised to poisonous phosgene
(C) it reacts with glass
(D) it absorbs moisture

9

Medium

⌚ 1 min

Concept

C16-09

Which compound undergoes nucleophilic substitution with aqueous NaOH most easily?

- (A) chlorobenzene (B) 4-nitrochlorobenzene
(C) 2,4-dinitrochlorobenzene (D) 2,4,6-trinitrochlorobenzene

10

Medium

⌚ 1 min

Concept

C16-10

(*R*)-2-Bromobutane reacts with OH^- by a pure $\text{S}_{\text{N}}2$ mechanism. The product is:

- (A) (*R*)-butan-2-ol (B) (*S*)-butan-2-ol
(C) a racemic mixture (D) butan-1-ol

11

Medium

⌚ 1.5 min

NV

C16-11

Find the number of **structurally** distinct monochloro products from the free-radical chlorination of 2-methylbutane.

Your answer (numerical value):



C16 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

C16-T1

The correct order of boiling points is:

- (A) $\text{CH}_3\text{I} > \text{CH}_3\text{Br} > \text{CH}_3\text{Cl}$ (B) $\text{CH}_3\text{Cl} > \text{CH}_3\text{Br} > \text{CH}_3\text{I}$
(C) $\text{CH}_3\text{Br} > \text{CH}_3\text{I} > \text{CH}_3\text{Cl}$ (D) all equal

2

Easy

⌚ 0.5 min

Speed

C16-T2

Chlorobenzene + chloromethane + Na (dry ether) gives:

- (A) biphenyl (B) toluene (C) ethane only (D) benzene

3 Easy 0.5 min NV

C16-T3

How many stereoisomers does 2-chlorobutane have?

Your answer (numerical value):

4 Easy 0.5 min Concept

C16-T4

The main environmental concern with DDT is that it:

- (A) is a greenhouse gas
 (B) is not biodegradable and accumulates in fatty tissues
 (C) depletes ozone
 (D) forms phosgene

5 Medium 0.75 min Concept

C16-T5

Chlorobenzene is less reactive than chloroethane towards nucleophilic substitution mainly because:

- (A) Cl is larger in chlorobenzene
 (B) resonance gives the C-Cl bond partial double-bond character
 (C) chlorobenzene is non-polar
 (D) chlorobenzene forms a stable carbocation

Answers & Solutions

C16 · Practice

- 1 (B) 2 (B) 3 (B) 4 (C) 5 (B) 6 (B) 7 (B) 8 (B) 9 (D) 10 (B)
 11 4

C16 · Chapter Test

- 1 (A) 2 (B) 3 2 4 (B) 5 (B)

C16 · PRACTICE

1 Answer: (B) Best chlorinating agent Easy 0.5 min Speed

C16-01

SOLUTION The byproducts SO_2 and HCl escape as gases, leaving pure product.

2 Answer: (B) Finkelstein reaction Easy 0.5 min Concept

C16-02

SOLUTION NaCl/NaBr are insoluble in acetone and precipitate, pushing the equilibrium forward.3 Answer: (B) $\text{S}_{\text{N}}2$ reactivity order Easy 0.75 min Concept

C16-03

SOLUTION Backside attack is increasingly hindered as alkyl groups are added.

COMMON TRAP Using the $\text{S}_{\text{N}}1$ order.4 Answer: (C) $\text{S}_{\text{N}}1$ reactivity Easy 0.5 min Speed

C16-04

SOLUTION The benzyl carbocation is resonance-stabilised. Chlorobenzene (aryl) doesn't react.

5 Answer: (B) AgCN product Easy 0.5 min Speed

C16-05

SOLUTION AgCN is covalent. The nitrogen lone pair attacks, forming $\text{C}_2\text{H}_5\text{NC}$.

6 Answer: (B) Grignard + water Easy 0.5 min Speed

C16-06

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

7 Answer: (B) Saytzeff elimination Easy 0.75 min Basic

C16-07

SOLUTION Saytzeff: the more substituted alkene is the major product.

COMMON TRAP Choosing the substitution product (C), which needs aqueous KOH .

8 Answer: (B) Storing chloroform **Easy** ⌚ 0.5 min **Concept**

C16-08

SOLUTION $2\text{CHCl}_3 + \text{O}_2 \xrightarrow{\text{light}} 2\text{COCl}_2 + 2\text{HCl}$.

9 Answer: (D) Activated haloarene **Medium** ⌚ 1 min **Concept**

C16-09

SOLUTION Three $-\text{NO}_2$ groups at ortho/para positions stabilise the intermediate anion most. It reacts with warm water.

10 Answer: (B) Stereochemistry of $\text{S}_{\text{N}}2$ **Medium** ⌚ 1 min **Concept**

C16-10

SOLUTION Backside attack inverts the configuration. OH takes the priority position of Br, so the label changes from R to S.

COMMON TRAP Answering racemic ($\text{S}_{\text{N}}1$ thinking).

11 Answer: 4 Monochlorination isomers **Medium** ⌚ 1.5 min **NV**

C16-11

SOLUTION $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$ has 4 types of H: the two equivalent terminal methyls on C2, the C2-H, the CH_2 , and the far CH_3 .

SHORTCUT Count the distinct hydrogen environments.

COMMON TRAP Counting 5 by treating the two methyls on C2 as different.

C16 · CHAPTER TEST

1 Answer: (A) Boiling points **Easy** ⌚ 0.5 min **Speed**

C16-T1

SOLUTION van der Waals forces increase with the size of the halogen.

2 Answer: (B) Wurtz-Fittig **Easy** ⌚ 0.5 min **Speed**

C16-T2

SOLUTION Cross-coupling gives $\text{C}_6\text{H}_5-\text{CH}_3$ (biphenyl and ethane form as side products).

3 Answer: 2 Stereoisomers of 2-chlorobutane **Easy** ⌚ 0.5 min **NV**

C16-T3

SOLUTION One chiral centre: a pair of enantiomers.

4 Answer: (B) DDT **Easy** ⌚ 0.5 min **Concept**

C16-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (B) Unreactive aryl halides **Medium** ⌚ 0.75 min **Concept**

C16-T5

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

REVISION Revision tracker · Haloalkanes & Haloarenes

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 17

Organic Compounds Containing Oxygen

17

2–3 per shift
TYPICAL QUESTIONSMedium
DIFFICULTYMust-do (top-
weight organic
chapter)
PRIORITYClass 12 · Ch 7, 8
NCERT~14 hours
STUDY TIME PYQ PATTERN Weight evidence

Aldehydes, ketones & carboxylic acids alone: 35 questions across the 2026 Paper-1 shifts ($\approx 7.4\%$ of Chemistry)  THIRD-PARTY ANALYSIS [T-WT-CD]. Add alcohols, phenols and ethers, and this unit is the single largest block of organic marks.

Part A · Alcohols

 FORMULA Preparation

- **Alkenes:** acid-catalysed hydration (Markovnikov); **hydroboration–oxidation** (B_2H_6 , then $\text{H}_2\text{O}_2/\text{OH}^-$) gives **anti-Markovnikov** alcohols (propene $\rightarrow$ propan-1-ol).
- **Reduction of carbonyls:** H_2/Pd , NaBH_4 or LiAlH_4 . Aldehydes $\rightarrow 1^\circ$, ketones $\rightarrow 2^\circ$. Acids and esters need LiAlH_4 ($\rightarrow 1^\circ$).
- **Grignard:** $\text{HCHO} \rightarrow 1^\circ$; other aldehydes $\rightarrow 2^\circ$; ketones $\rightarrow 3^\circ$ (after hydrolysis).

 REACTION MAP Reaction map · ethanol $\text{CH}_3\text{CH}_2\text{OH}$

Na	$\rightarrow \text{CH}_3\text{CH}_2\text{ONa} + \frac{1}{2} \text{H}_2$ (acidic character)
RCOOH , conc. H_2SO_4 (Fischer)	$\rightarrow$ ester RCOOC_2H_5
$\text{HCl}/\text{ZnCl}_2 \cdot \text{PCl}_5 \cdot \text{SOCl}_2$	$\rightarrow \text{CH}_3\text{CH}_2\text{Cl}$
conc. H_2SO_4 , 443 K	$\rightarrow$ ethene (dehydration)
conc. H_2SO_4 , 413 K (excess alcohol)	$\rightarrow$ diethyl ether
PCC	$\rightarrow$ ethanal (stops at aldehyde)
acidified $\text{K}_2\text{Cr}_2\text{O}_7 / \text{KMnO}_4$	$\rightarrow$ ethanoic acid
Cu , 573 K	$\rightarrow$ ethanal (dehydrogenation)
I_2/NaOH	$\rightarrow$ iodoform (yellow): ethanol has the $\text{CH}_3\text{CH}(\text{OH})^-$ unit

 IMPORTANT Dehydration mechanism (syllabus-listed)

(1) Protonation of $-\text{OH} \rightarrow \text{R}-\text{OH}_2^+$; (2) loss of H_2O gives a **carbocation** (slow step); (3) loss of $\beta\text{-H}$ gives the alkene (Saytzeff). Ease: $3^\circ > 2^\circ > 1^\circ$, following carbocation stability. Rearrangements are possible.

Σ FORMULA Distinguishing 1°, 2° and 3° alcohols

Test	1°	2°	3°
Lucas (conc. HCl + anhydrous ZnCl ₂)	no turbidity at room temperature	turbidity in ~5 min	immediate turbidity
Oxidation (K ₂ Cr ₂ O ₇ /H ⁺)	aldehyde → acid	ketone	resistant
Cu at 573 K	aldehyde	ketone	alkene

Acidity (in solution): 1° > 2° > 3°. All are weaker acids than water, and much weaker than phenol.

Part B · Phenols

Σ FORMULA Preparation & acidity

Preparation: chlorobenzene (Dow: NaOH, 623 K, 300 atm); benzenesulphonic acid (fuse with NaOH, then acidify); **diazonium salt + warm water**; **cumene process** (cumene + O₂ → cumene hydroperoxide → dil. acid → phenol + **acetone**). **Acidity:** phenoxide is resonance-stabilised, so phenol (pK_a 10.0) is far more acidic than ethanol (15.9). It reacts with NaOH, but **not with** NaHCO₃ (picric acid is the exception).

Phenol	pK _a (NCERT)
<i>p</i> -nitrophenol	7.1
<i>o</i> -nitrophenol	7.2
<i>m</i> -nitrophenol	8.3
phenol	10.0
cresols (<i>o</i> , <i>m</i> , <i>p</i>)	~10.1–10.2
2,4,6-trinitrophenol (picric acid)	~0.4 (strongly acidic)

Electron-withdrawing groups at **ortho/para** raise acidity (–R and –I). Electron-donating groups lower it.

⚗ REACTION MAP Reaction map · phenol C₆H₅OH

dil. HNO ₃ , 298 K	→ <i>o</i>- + <i>p</i>-nitrophenol (separated by steam distillation: <i>o</i> is volatile)
conc. HNO ₃	→ 2,4,6-trinitrophenol (picric acid)
Br ₂ in CS ₂ , 273 K	→ <i>o</i>- + <i>p</i>-bromophenol
bromine water	→ 2,4,6-tribromophenol (white precipitate)
Kolbe: NaOH, then CO ₂ (400 K, 4–7 atm), then H ⁺	→ salicylic acid (<i>o</i>-hydroxybenzoic acid)
Reimer-Tiemann: CHCl ₃ + aq. NaOH, then H ⁺	→ salicylaldehyde (via dichlorocarbene : CCl₂)
Zn dust, heat	→ benzene
Na ₂ Cr ₂ O ₇ /H ₂ SO ₄	→ benzoquinone
neutral FeCl ₃	→ violet colour (test)

Part C · Ethers

Σ FORMULA Preparation & reactions

Dehydration of alcohols (conc. H_2SO_4 , 413 K): works for 1° alcohols only ($2^\circ/3^\circ$ give alkenes). **Williamson synthesis:** $\text{R-X} + \text{R}'\text{-ONa} \longrightarrow \text{R-O-R}' + \text{NaX}$ ($\text{S}_{\text{N}}2$). Use a **1° alkyl halide** and the bulky alkoxide: *tert*-butyl methyl ether = $(\text{CH}_3)_3\text{CONa} + \text{CH}_3\text{Br}$. The reverse combination ($(\text{CH}_3)_3\text{CBr} + \text{CH}_3\text{ONa}$) gives isobutylene by elimination. **Structure:** C–O–C angle $\approx 111.7^\circ$ (slightly above tetrahedral; bulky groups repel). **Cleavage by HI:** $\text{R-O-R}' + \text{HI} \longrightarrow \text{R-I} + \text{R}'\text{-OH}$. With a methyl alkyl ether (1°), I^- attacks the **methyl** carbon ($\text{S}_{\text{N}}2$), giving $\text{CH}_3\text{I} + \text{ROH}$. With a *tert*-alkyl group, the tertiary halide forms ($\text{S}_{\text{N}}1$). **Anisole + HI $\rightarrow$ phenol + CH_3I :** the O–aryl bond doesn't break. **Anisole EAS:** the $-\text{OCH}_3$ group is activating and o/p-directing (halogenation, nitration, Friedel–Crafts).

Part D · Aldehydes & Ketones

Σ FORMULA Preparation (named)

Reaction	Transformation
Rosenmund reduction	$\text{RCOCl} + \text{H}_2 \xrightarrow{\text{Pd/BaSO}_4} \text{RCHO}$
Stephen reduction	$\text{RCN} \xrightarrow{\text{SnCl}_2/\text{HCl}} \text{RCH=NH} \xrightarrow{\text{H}_3\text{O}^+} \text{RCHO}$
DIBAL-H	nitriles/esters $\rightarrow$ aldehydes
Etard reaction	toluene + CrO_2Cl_2 (chromyl chloride), then $\text{H}_3\text{O}^+ \rightarrow$ benzaldehyde
Side-chain chlorination	toluene $\rightarrow \text{C}_6\text{H}_5\text{CHCl}_2 \rightarrow$ hydrolysis $\rightarrow$ benzaldehyde
Gattermann–Koch	benzene + CO + HCl, anhydrous $\text{AlCl}_3/\text{CuCl} \rightarrow$ benzaldehyde
Ketones from nitriles	$\text{RCN} + \text{R}'\text{MgX}$, then $\text{H}_3\text{O}^+ \rightarrow \text{RCOR}'$
Ketones from acyl chlorides	$2 \text{RCOCl} + \text{R}'_2\text{Cd} \longrightarrow 2 \text{RCOR}'$
Friedel–Crafts acylation	benzene + $\text{RCOCl}/\text{AlCl}_3 \rightarrow$ aryl ketone
Ozonolysis; hydration of alkynes; oxidation of alcohols	see Units 15 and 17A

☆ IMPORTANT Reactivity of the carbonyl group

The carbonyl carbon is sp^2 , planar and electrophilic. **Nucleophilic addition** reactivity: $\text{HCHO} > \text{CH}_3\text{CHO} > \text{CH}_3\text{COCH}_3$. Alkyl groups donate electrons (+I) and add steric bulk. Aromatic carbonyls are less reactive (resonance with the ring).

REACTION MAP Reaction map · ethanal CH_3CHO (and ketones)

HCN (base)	→ cyanohydrin $\text{CH}_3\text{CH}(\text{OH})\text{CN}$
NaHSO_3	→ bisulphite adduct (crystalline; used for separation)
$\text{R}'\text{MgX}$, then H_3O^+	→ 2° alcohol (ketones → 3°)
ROH , dry HCl	→ hemiacetal → acetal (ketones + ethylene glycol → cyclic ketal)
NH_2OH	→ oxime
NH_2NH_2	→ hydrazone
2,4-DNP (Brady's reagent)	→ orange/yellow 2,4-DNP derivative (test for $\text{C}=\text{O}$)
semicarbazide	→ semicarbazone
NaBH_4 / LiAlH_4	→ ethanol
Clemmensen (Zn-Hg / conc. HCl)	→ ethane ($>\text{C}=\text{O} \rightarrow >\text{CH}_2$)
Wolff-Kishner (NH_2NH_2 , then KOH /ethylene glycol, heat)	→ ethane
Tollens' reagent	→ silver mirror (aldehydes only)
Fehling's solution	→ red Cu_2O (aliphatic aldehydes only)
I_2/NaOH (haloform)	→ iodoform + formate (all methyl carbonyls)
dil. NaOH (aldol)	→ 3-hydroxybutanal → (heat) but-2-enal

FORMULA Condensation & disproportionation

Aldol: aldehydes/ketones with $\alpha\text{-H}$ + dilute alkali $\rightarrow \beta$ -hydroxy carbonyl (aldol/ketol) $\rightarrow$ on heating, α,β -unsaturated carbonyl. **Cross-aldol** between two different carbonyls gives 4 products (fewer if one partner lacks $\alpha\text{-H}$). **Cannizzaro:** aldehydes without $\alpha\text{-H}$ (HCHO , $\text{C}_6\text{H}_5\text{CHO}$, $(\text{CH}_3)_3\text{CCHO}$) + conc. $\text{NaOH} \rightarrow$ alcohol + carboxylate salt (one molecule reduced, one oxidised). Cross-Cannizzaro with HCHO : HCHO is oxidised to formate. **Haloform:** compounds with CH_3CO^- or $\text{CH}_3\text{CH}(\text{OH})^-$ (ethanal, methyl ketones, ethanol, propan-2-ol) + $\text{X}_2/\text{NaOH} \rightarrow \text{CHX}_3$. Iodoform is a **yellow** precipitate. **EAS on benzaldehyde / acetophenone:** $-\text{CHO}$ and $-\text{COR}$ are **meta** directors.

Part E · Carboxylic Acids

FORMULA Preparation

Oxidation of 1° alcohols and aldehydes; **alkylbenzenes + alkaline $\text{KMnO}_4 \rightarrow$ benzoic acid** (whatever the chain length, provided there is a benzylic H); hydrolysis of nitriles and amides; **Grignard + CO_2 (dry ice)**, then $\text{H}_3\text{O}^+ \rightarrow \text{RCOOH}$; hydrolysis of acyl halides, anhydrides and esters.

FORMULA Acidity

The carboxylate ion has two **equivalent** resonance structures, so carboxylic acids are much more acidic than phenols and alcohols. They liberate CO_2 from NaHCO_3 (this is the test).

- EWG raise acidity: $\text{CF}_3\text{COOH} > \text{CCl}_3\text{COOH} > \text{CHCl}_2\text{COOH} > \text{ClCH}_2\text{COOH} > \text{HCOOH} > \text{CH}_3\text{COOH}$.
- α -substituent: $\text{F} > \text{Cl} > \text{Br} > \text{I}$. The effect falls off with distance: 2-chlorobutanoic $>$ 3-chloro $>$ 4-chloro.
- pK_a values: HCOOH 3.75 · $\text{C}_6\text{H}_5\text{COOH}$ 4.19 · CH_3COOH 4.76. *p*-Nitrobenzoic acid is stronger than benzoic acid; *p*-methoxybenzoic acid is weaker.

**REACTION MAP** Reaction map • ethanoic acid CH_3COOH

NaHCO_3	→ $\text{CH}_3\text{COONa} + \text{CO}_2$ (brisk effervescence)
$\text{PCl}_5 / \text{PCl}_3 / \text{SOCl}_2$	→ CH_3COCl (acetyl chloride)
P_2O_5 or heat	→ acetic anhydride
ROH / H^+	→ ester
NH_3 , then heat	→ ammonium acetate → acetamide
LiAlH_4 (not NaBH_4)	→ ethanol
$\text{NaOH} + \text{CaO}$ (soda lime), heat	→ methane (decarboxylation)
HVZ : Cl_2 / red P, then H_2O	→ chloroacetic acid (α-halogenation)

Named-reaction table (Unit 17)

Named reaction	Substrate → product	Key reagent
Reimer–Tiemann	phenol → salicylaldehyde	CHCl_3 , aq. NaOH
Kolbe (phenol)	sodium phenoxide → salicylic acid	CO_2 , 400 K, 4–7 atm
Cumene process	cumene → phenol + acetone	O_2 , then H_3O^+
Williamson	$\text{RX} + \text{R}'\text{ONa} \rightarrow \text{ether}$	1°RX
Hydroboration–oxidation	alkene → anti-Markovnikov alcohol	B_2H_6 ; $\text{H}_2\text{O}_2/\text{OH}^-$
Rosenmund	acyl chloride → aldehyde	H_2 , Pd/BaSO_4
Stephen	nitrile → aldehyde	SnCl_2/HCl , H_3O^+
Etard	toluene → benzaldehyde	CrO_2Cl_2
Gattermann–Koch	benzene → benzaldehyde	CO , HCl , $\text{AlCl}_3/\text{CuCl}$
Clemmensen	$\text{C}=\text{O} \rightarrow \text{CH}_2$	Zn-Hg , conc. HCl
Wolff–Kishner	$\text{C}=\text{O} \rightarrow \text{CH}_2$	NH_2NH_2 ; KOH , glycol, heat
Aldol	carbonyl with $\alpha\text{-H} \rightarrow \beta\text{-hydroxy carbonyl}$	dil. NaOH
Cannizzaro	aldehyde without $\alpha\text{-H} \rightarrow \text{alcohol} + \text{salt}$	conc. NaOH
Haloform	methyl ketone → $\text{CHX}_3 + \text{acid salt}$	X_2 , NaOH
HVZ	acid → $\alpha\text{-halo acid}$	X_2 , red P
Fischer esterification	acid + alcohol $\rightleftharpoons$ ester	conc. H_2SO_4

Distinguishing tests (very high frequency)

Pair	Test that separates them
aldehyde vs ketone	Tollens' (silver mirror), Fehling's (red precipitate, aliphatic aldehydes only)
benzaldehyde vs ethanal	Fehling's (ethanal +) and iodoform (ethanal +)
methyl ketone vs other ketones (pentan-2-one vs pentan-3-one)	iodoform
phenol vs alcohol	neutral FeCl_3 (violet); bromine water (white precipitate); NaOH solubility
phenol vs carboxylic acid	NaHCO_3 (acid effervesces)
$1^\circ/2^\circ/3^\circ$ alcohols	Lucas test
ethanol vs methanol	iodoform (ethanol +)

Common traps

MISTAKE ALERT Mistake alerts

- **Fehling's** is negative for aromatic aldehydes; Tollens' is positive for all aldehydes.
- Iodoform is positive for **ethanol** and **propan-2-ol** (they are oxidised to the methyl carbonyl in situ), but negative for methanol and propan-1-ol.
- Cannizzaro needs **no α -H** and **concentrated** alkali. Aldol needs α -H and **dilute** alkali.
- NaBH_4 doesn't reduce $-\text{COOH}$ or esters.
- Phenol does **not** react with NaHCO_3 , but picric acid does.
- In Williamson synthesis, never use a 3° alkyl halide.

Question patterns

PYQ PATTERN Recurring structures

1. Multi-step conversions ($A \rightarrow B \rightarrow C$) with named reagents; identify an intermediate or the final product.
2. Distinguishing tests (the table above).
3. Acidity orders (substituted phenols and carboxylic acids).
4. Aldol/Cannizzaro/haloform product prediction, including cross reactions.
5. Grignard products and the number of moles consumed.
6. Ether cleavage with HI; Williamson reagent choice.
7. Mechanism steps (alcohol dehydration, nucleophilic addition).

Practice questions

C17 · Practice

1 Easy ⌚ 0.5 min Speed

C17-01

Which alcohol gives immediate turbidity with Lucas reagent at room temperature?

- (A) butan-1-ol (B) butan-2-ol
(C) 2-methylpropan-2-ol (D) methanol

2 Easy ⌚ 0.5 min Speed

C17-02

Acetone treated with CH_3MgBr , then hydrolysed, gives:

- (A) propan-2-ol (B) 2-methylpropan-2-ol
(C) butan-2-ol (D) ethanol

3 Easy ⌚ 0.75 min Concept

C17-03

The correct order of acidity is:

- (A) *p*-nitrophenol > phenol > *p*-cresol (B) *p*-cresol > phenol > *p*-nitrophenol
(C) phenol > *p*-nitrophenol > *p*-cresol (D) all equal

4 Easy ⌚ 0.5 min Speed

C17-04

Phenol + chloroform + aqueous NaOH (then acidification) gives:

- (A) salicylic acid (B) salicylaldehyde (C) benzaldehyde (D) chlorobenzene

5 Medium 1 min Concept

C17-05

The best way to prepare *tert*-butyl methyl ether is:

- (A) $(\text{CH}_3)_3\text{CBr} + \text{CH}_3\text{ONa}$
(B) $(\text{CH}_3)_3\text{CONa} + \text{CH}_3\text{Br}$
(C) $(\text{CH}_3)_3\text{COH} + \text{CH}_3\text{OH}$ with conc. H_2SO_4 at 443 K
(D) $(\text{CH}_3)_3\text{CCl} + \text{CH}_3\text{OH}$

6 Easy 0.75 min Basic

C17-06

Anisole heated with HI gives:

- (A) iodobenzene + methanol
(B) phenol + iodomethane
(C) benzene + methanol
(D) iodobenzene + iodomethane

7 Easy 0.5 min Speed

C17-07

Benzaldehyde with concentrated NaOH gives:

- (A) benzoic acid only
(B) benzyl alcohol + sodium benzoate
(C) benzoin
(D) no reaction

8 Easy 0.5 min Speed

C17-08

Two molecules of ethanal in dilute NaOH give:

- (A) 3-hydroxybutanal
(B) butanal
(C) ethyl ethanoate
(D) 2-hydroxybutanal

9 Medium 1 min Concept

C17-09

Which compound does **not** give the iodoform test?

- (A) ethanol
(B) propan-2-ol
(C) acetone
(D) pentan-3-one

10 Easy 0.5 min Speed

C17-10

Acetophenone with Zn-Hg and conc. HCl gives:

- (A) 1-phenylethanol
(B) ethylbenzene
(C) styrene
(D) benzoic acid

11 Medium 1 min Concept

C17-11

The correct order of acid strength is:

- (A) $\text{Cl}_2\text{CHCOOH} > \text{ClCH}_2\text{COOH} > \text{CH}_3\text{COOH}$
(B) $\text{CH}_3\text{COOH} > \text{ClCH}_2\text{COOH} > \text{Cl}_2\text{CHCOOH}$
(C) $\text{ClCH}_2\text{COOH} > \text{Cl}_2\text{CHCOOH} > \text{CH}_3\text{COOH}$
(D) all equal

12 Easy 0.75 min Basic

C17-12

Propanoic acid with Cl_2 and red phosphorus gives:

- (A) 3-chloropropanoic acid
(B) 2-chloropropanoic acid
(C) propanoyl chloride
(D) chloroethane

13 Easy 0.75 min Concept

C17-13

Benzaldehyde:

- (A) gives both Tollens' and Fehling's tests
(B) gives Tollens' but not Fehling's test
(C) gives Fehling's but not Tollens' test
(D) gives neither

14 Medium 1 min NV

C17-T4

How many structurally isomeric **secondary** alcohols have the formula $C_5H_{12}O$?

Your answer (numerical value):

15 Easy 0.5 min Speed

C17-15

In the Stephen reduction, a nitrile is converted into an aldehyde using:

- (A) $LiAlH_4$ (B) $SnCl_2/HCl$, then hydrolysis
(C) $Pd/BaSO_4$ (D) $Zn-Hg/HCl$

C17 · Chapter Test

1 Easy 0.5 min Speed

C17-T1

Sodium phenoxide + CO_2 (400 K, 4–7 atm), then acidification, gives:

- (A) salicylaldehyde (B) salicylic acid (C) benzoic acid (D) phenyl acetate

2 Easy 0.5 min Speed

C17-T2

The catalyst used in Fischer esterification is:

- (A) anhydrous $AlCl_3$ (B) conc. H_2SO_4 (C) $NaOH$ (D) Pd

3 Medium 1 min NV

C17-T3

How many moles of CH_3MgBr does 1 mol of ethyl acetate consume in forming a tertiary alcohol (after hydrolysis)?

Your answer (numerical value):

4 Easy 0.5 min Speed

C17-T4

A carboxylic acid is distinguished from phenol by:

- (A) $FeCl_3$ (B) $NaHCO_3$ (C) $NaOH$ (D) bromine water

5 Easy 0.5 min Speed

C17-T5

Ethylbenzene heated with alkaline $KMnO_4$ and then acidified gives:

- (A) phenylacetic acid (B) benzoic acid
(C) acetophenone (D) benzaldehyde

6 Easy 0.5 min Speed

C17-T6

The cumene process produces:

- (A) phenol and acetone (B) phenol and formaldehyde
(C) benzoic acid and ethanol (D) benzene and propene

7 Easy 0.5 min Speed

C17-T7

Which compound is the most reactive towards nucleophilic addition?

- (A) $HCHO$ (B) CH_3CHO (C) CH_3COCH_3 (D) $C_6H_5COCH_3$

Answers & Solutions

C17 · Practice

1	(C)	2	(B)	3	(A)	4	(B)	5	(B)	6	(B)	7	(B)	8	(A)	9	(D)	10	(B)
11	(A)	12	(B)	13	(B)	14	3	15	(B)										

C17 · Chapter Test

1	(B)	2	(B)	3	2	4	(B)	5	(B)	6	(A)	7	(A)
---	-----	---	-----	---	---	---	-----	---	-----	---	-----	---	-----

C17 · PRACTICE

1 Answer: (C) Lucas test **Easy** ⌚ 0.5 min **Speed** C17-01

SOLUTION A 3° alcohol forms a stable carbocation at once, giving an insoluble chloride.

2 Answer: (B) Grignard + ketone **Easy** ⌚ 0.5 min **Speed** C17-02

SOLUTION Ketone + Grignard gives a 3° alcohol: $(\text{CH}_3)_3\text{COH}$.

3 Answer: (A) Acidity of phenols **Easy** ⌚ 0.75 min **Concept** C17-03

SOLUTION $-\text{NO}_2$ ($-\text{R}$, $-\text{I}$) stabilises phenoxide; $-\text{CH}_3$ ($+\text{I}$, hyperconjugation) destabilises it. $\text{p}K_a$ $7.1 < 10.0 < 10.2$.

4 Answer: (B) Reimer-Tiemann **Easy** ⌚ 0.5 min **Speed** C17-04

SOLUTION The electrophile is dichlorocarbene. It gives the *ortho*-CHO product.

COMMON TRAP Confusing it with the Kolbe product (salicylic acid).

5 Answer: (B) Williamson reagent choice **Medium** ⌚ 1 min **Concept** C17-05

SOLUTION $\text{S}_{\text{N}}2$ needs an unhindered halide (CH_3Br). In (A), the 3° halide undergoes elimination with the strong base.

SHORTCUT Put the halogen on the smaller group.

COMMON TRAP Choosing (A).

6 Answer: (B) Cleavage of anisole **Easy** ⌚ 0.75 min **Basic** C17-06

SOLUTION The O-aryl bond has partial double-bond character. I^- attacks the methyl carbon by $\text{S}_{\text{N}}2$.

7 Answer: (B) Cannizzaro **Easy** ⌚ 0.5 min **Speed** C17-07

SOLUTION There is no $\alpha\text{-H}$, so disproportionation occurs.

8 Answer: (A) Aldol product **Easy** ⌚ 0.5 min **Speed** C17-08

SOLUTION The enolate attacks a second ethanal: $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CHO}$.

9 Answer: (D) Iodoform test **Medium** ⌚ 1 min **Concept** C17-09

SOLUTION Pentan-3-one ($\text{C}_2\text{H}_5\text{COC}_2\text{H}_5$) has no CH_3CO^- group.

10 Answer: (B) Clemmensen reduction **Easy** ⌚ 0.5 min **Speed** C17-10

SOLUTION $\text{C}=\text{O}$ becomes CH_2 .

COMMON TRAP Choosing the alcohol (NaBH_4 product).

11 Answer: (A) Acidity of chloro acids **Medium** ⌚ 1 min **Concept** C17-11

SOLUTION More $-\text{I}$ groups stabilise the carboxylate more.

12 Answer: (B) HVZ reaction **Easy** ⌚ 0.75 min **Basic** C17-12

SOLUTION HVZ halogenates the α -carbon.

COMMON TRAP Choosing propanoyl chloride ($\text{PCl}_5/\text{SOCl}_2$ product).

13 Answer: (B) Tollens' vs Fehling's **Easy** ⌚ 0.75 min **Concept** C17-13

SOLUTION Aromatic aldehydes don't reduce Fehling's solution.

14 Answer: 3 Counting 2° alcohols **Medium** ⌚ 1 min **NV** C17-14

SOLUTION Pentan-2-ol, pentan-3-ol, 3-methylbutan-2-ol.

COMMON TRAP Including 2-methylbutan-2-ol (which is 3°).

15 Answer: (B) *Stephen reduction* **Easy** ⌚ 0.5 min **Speed**

C17-15

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

COMMON TRAP Confusing it with Rosenmund (Pd/BaSO₄, which works on acyl chlorides).

C17 · CHAPTER TEST

1 Answer: (B) *Kolbe's reaction* **Easy** ⌚ 0.5 min **Speed**

C17-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (B) *Fischer esterification catalyst* **Easy** ⌚ 0.5 min **Speed**

C17-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 2 *Grignard with an ester* **Medium** ⌚ 1 min **NV**

C17-T3

SOLUTION The first equivalent substitutes OC₂H₅ (giving acetone); the second adds to the ketone, giving 2-methylpropan-2-ol.

4 Answer: (B) *Test for carboxylic acids* **Easy** ⌚ 0.5 min **Speed**

C17-T4

SOLUTION Only the acid liberates CO₂ from bicarbonate.

5 Answer: (B) *Side-chain oxidation* **Easy** ⌚ 0.5 min **Speed**

C17-T5

SOLUTION Any alkyl chain with a benzylic H oxidises down to -COOH.

6 Answer: (A) *Cumene process* **Easy** ⌚ 0.5 min **Speed**

C17-T6

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

7 Answer: (A) *Most reactive carbonyl* **Easy** ⌚ 0.5 min **Speed**

C17-T7

SOLUTION It has the least steric hindrance and no electron-donating groups.

REVISION Revision tracker · Oxygen Compounds

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 18

Organic Compounds Containing Nitrogen

18

~1 per shift

TYPICAL QUESTIONS

Medium

DIFFICULTY

High

PRIORITY

Class 12 · Ch 9

NCERT

~6 hours

STUDY TIME

Amines: important concepts

- **Classification:** 1° RNH_2 , 2° R_2NH , 3° R_3N (and quaternary salts $\text{R}_4\text{N}^+\text{X}^-$). N is sp^3 and pyramidal ($\text{C-N-C} \approx 108^\circ$), with a lone pair that makes amines **bases and nucleophiles**.
- **Boiling points:** $1^\circ > 2^\circ > 3^\circ$ for isomers (N-H hydrogen bonding). Amines boil higher than alkanes but lower than alcohols of similar mass.
- **Basicity** is decided by three effects together: +I effect, **solvation** of the ammonium ion (H-bonding with water), and **steric** hindrance. Aromatic amines are much weaker bases (the lone pair is delocalised into the ring).

Preparation

Σ FORMULA Routes to amines

Method	Transformation	Note
Reduction of nitro compounds	$\text{ArNO}_2 \xrightarrow{\text{Fe/HCl or Sn/HCl or H}_2/\text{Pd}} \text{ArNH}_2$	Fe/HCl preferred (the FeCl_2 formed regenerates HCl)
Ammonolysis of alkyl halides	$\text{RX} + \text{NH}_3 \longrightarrow \text{RNH}_2$ (+ 2° , 3° , quaternary)	a mixture; excess NH_3 favours 1°
Reduction of nitriles	$\text{RCN} \xrightarrow{\text{LiAlH}_4 \text{ or } \text{H}_2/\text{Ni}} \text{RCH}_2\text{NH}_2$	adds one carbon (ascending series)
Reduction of amides	$\text{RCONH}_2 \xrightarrow{\text{LiAlH}_4} \text{RCH}_2\text{NH}_2$	same number of C
Gabriel phthalimide	phthalimide + KOH $\rightarrow$ + RX $\rightarrow$ hydrolysis $\rightarrow$ RNH_2	pure 1° aliphatic amines only (not aniline)
Hofmann bromamide degradation	$\text{RCONH}_2 + \text{Br}_2 + 4\text{NaOH} \longrightarrow \text{RNH}_2 + \text{Na}_2\text{CO}_3 + 2\text{NaBr} + 2\text{H}_2\text{O}$	amine has one carbon fewer

Basicity (NCERT $\text{p}K_b$ values; lower $\text{p}K_b$ = stronger base)

Amine	$\text{p}K_b$	Amine	$\text{p}K_b$
methanamine CH_3NH_2	3.38	ethanamine $\text{C}_2\text{H}_5\text{NH}_2$	3.29
N-methylmethanamine $(\text{CH}_3)_2\text{NH}$	3.27	N-ethylethanamine $(\text{C}_2\text{H}_5)_2\text{NH}$	3.00
N,N-dimethylmethanamine $(\text{CH}_3)_3\text{N}$	4.22	N,N-diethylethanamine $(\text{C}_2\text{H}_5)_3\text{N}$	3.25
ammonia	4.75	benzenamine (aniline)	9.38

☆ IMPORTANT Basicity orders to memorise

- **Methyl series (water):** $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N} > \text{NH}_3$.
- **Ethyl series (water):** $(\text{C}_2\text{H}_5)_2\text{NH} > (\text{C}_2\text{H}_5)_3\text{N} > \text{C}_2\text{H}_5\text{NH}_2 > \text{NH}_3$.
- **Gas phase** (inductive effect only): $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$.
- **Aromatic amines:** electron-donating groups raise basicity (*p*-toluidine > aniline), electron-withdrawing groups lower it (aniline > *p*-nitroaniline). Aniline $\ll$ NH_3 .

Reactions of amines

REACTION MAP Reaction map · aniline $C_6H_5NH_2$

HCl	→ anilinium chloride (salt)
CH_3COCl or $(CH_3CO)_2O$ / pyridine	→ acetanilide ($-NH_2$ protected; activation reduced)
C_6H_5COCl (benzoylation)	→ benzanilide
$CHCl_3$ + alc. KOH (carbylamine)	→ phenyl isocyanide (foul smell): 1° amines only
$NaNO_2$ + HCl, 273–278 K	→ benzenediazonium chloride
$C_6H_5SO_2Cl$ (Hinsberg)	→ N-phenylbenzenesulphonamide (soluble in alkali, as for all 1° amines)
bromine water	→ 2,4,6-tribromoaniline (white precipitate)
conc. HNO_3 + H_2SO_4 (unprotected)	→ tarry oxidation + <i>p</i> - (51%), <i>m</i> - (47%), <i>o</i> - (2%) nitroaniline
acetylate → nitrate → hydrolyse	→ mainly <i>p</i> -nitroaniline
conc. H_2SO_4 , 453–473 K	→ sulphanilic acid (exists as a zwitterion)
$AlCl_3$ (Friedel-Crafts)	→ no reaction: aniline complexes the Lewis acid

Σ FORMULA Hinsberg test (benzenesulphonyl chloride)

Amine	Product	Solubility in KOH
1°	$C_6H_5SO_2NHR$	soluble (acidic N–H)
2°	$C_6H_5SO_2NR_2$	insoluble
3°	no reaction	—

☆ IMPORTANT Nitrous acid

1° **aliphatic** amines + HNO_2 give an unstable diazonium salt, which decomposes to an alcohol + N_2 (quantitative N_2 evolution). 1° **aromatic** amines give **stable arenediazonium salts** at 273–278 K (stabilised by resonance with the ring).

Diazonium salts: importance in synthesis

Σ FORMULA Replacement of N_2 ($Ar-N_2^+X^- \rightarrow$)

Reagent	Product	Name
$CuCl/HCl$	$ArCl$	Sandmeyer
$CuBr/HBr$	$ArBr$	Sandmeyer
$CuCN/KCN$	$ArCN$	Sandmeyer
Cu powder + HX	ArX	Gattermann
KI	ArI	—
HBf_4 , then heat	ArF	Balz–Schiemann
H_3PO_2 or CH_3CH_2OH	ArH	(reductive removal of $-NH_2$)
H_2O , warm	$ArOH$	—
$NaNO_2/Cu$ (via fluoroborate)	$ArNO_2$	—

Σ FORMULA Coupling (N₂ retained): azo dyes

- Phenol in **alkaline** medium + $\text{C}_6\text{H}_5\text{N}_2^+ \rightarrow p\text{-hydroxyazobenzene}$ (**orange dye**).
- Aniline in **mildly acidic** medium + $\text{C}_6\text{H}_5\text{N}_2^+ \rightarrow p\text{-aminoazobenzene}$ (**aniline yellow**; also listed in Practical Chemistry).
- Coupling is an **electrophilic** substitution: the diazonium ion is a weak electrophile, so it needs a strongly activated ring.

⚡ SHORTCUT Synthesis logic with diazonium salts

To put a group on a benzene ring where direct substitution fails or gives the wrong orientation, go **via the amine**: nitrate $\rightarrow$ reduce $\rightarrow$ (use $-\text{NH}_2$ to direct/activate) $\rightarrow$ diazotise $\rightarrow$ replace. Classic: **1,3,5-tribromobenzene** from aniline (brominate to 2,4,6-tribromoaniline, then diazotise and remove N_2 with H_3PO_2).

Common traps**⚠ MISTAKE ALERT Mistake alerts**

- Gabriel synthesis **cannot** make aniline (aryl halides don't undergo $\text{S}_{\text{N}}2$).
- Hofmann bromamide **loses one carbon**; nitrile reduction **gains one carbon**.
- The carbylamine test works for 1° amines (aliphatic and aromatic) only.
- Direct nitration of aniline gives a lot of the **meta** product, because the anilinium ion ($-\text{NH}_3^+$) forms in acid.
- Order of basicity in water isn't simply $3^\circ > 2^\circ > 1^\circ$. Solvation and sterics matter.

Question patterns**▮ PYQ PATTERN Recurring structures**

- Basicity orders (aliphatic in water/gas; substituted anilines).
- Conversions using diazonium salts (Sandmeyer, Balz–Schiemann, H_3PO_2 , coupling).
- Distinguishing $1^\circ/2^\circ/3^\circ$ amines (Hinsberg, carbylamine, HNO_2).
- Hofmann bromamide / Gabriel / nitrile reduction: product and carbon count.
- EAS of aniline: bromination, nitration with and without protection, sulphanilic acid.

Practice questions**C18 · Practice****1** Medium 1 min Concept

C18-01

The correct order of basic strength in aqueous solution is:

- (A) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > \text{NH}_3$ (B) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N} > \text{NH}_3$
 (C) $\text{NH}_3 > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N}$ (D) $\text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N} > \text{NH}_3$

2 Easy 0.5 min Concept

C18-02

The Gabriel phthalimide synthesis can't be used to prepare:

- (A) methylamine (B) ethylamine (C) aniline (D) benzylamine

3 Easy 0.5 min Speed

C18-03

Acetamide with Br_2 and aqueous NaOH gives:

- (A) ethylamine (B) methylamine (C) acetonitrile (D) acetic acid

4 Easy 0.5 min Concept

C18-04

Which amine gives the carbylamine (isocyanide) test?

- (A) N-methylaniline (B) N,N-dimethylaniline
(C) aniline (D) triethylamine

5 Easy 0.75 min Concept

C18-05

In the Hinsberg test, a 2° amine gives:

- (A) no reaction (B) a sulphonamide soluble in KOH
(C) a sulphonamide insoluble in KOH (D) an isocyanide

6 Easy 0.5 min Speed

C18-06

Aniline with bromine water at room temperature gives:

- (A) *p*-bromoaniline (B) *o*-bromoaniline
(C) 2,4,6-tribromoaniline (D) *m*-bromoaniline

7 Medium 1 min Concept

C18-07

Direct nitration of aniline gives a large amount of *m*-nitroaniline because:

- (A) $-\text{NH}_2$ is a meta director
(B) in strongly acidic medium aniline is protonated to anilinium ion, whose $-\text{NH}_3^+$ group is meta-directing
(C) the para position is blocked
(D) nitration proceeds by a free-radical mechanism

8 Easy 0.5 min Speed

C18-08

Benzenediazonium chloride treated with CuCN/KCN gives:

- (A) phenyl isocyanide (B) benzonitrile
(C) benzoic acid (D) aniline

9 Easy 0.5 min Speed

C18-09

Benzenediazonium chloride couples with phenol in alkaline medium to give:

- (A) *p*-hydroxyazobenzene (orange) (B) *p*-aminoazobenzene (yellow)
(C) phenol + N_2 (D) benzene

10 Easy 0.75 min Concept

C18-10

Which is the weakest base?

- (A) NH_3 (B) aniline (C) *p*-toluidine (D) *p*-nitroaniline

11 Medium 1 min NV

C18-11

How many structurally isomeric **primary** amines have the formula $\text{C}_4\text{H}_{11}\text{N}$?

Your answer (numerical value):

12 Easy 0.5 min Speed

C18-12

Ethanenitrile (CH_3CN) reduced with LiAlH_4 gives:

- (A) methylamine (B) ethylamine (C) ethanamide (D) ethanal

C18 · Chapter Test

1 Easy ⌚ 0.5 min Speed C18-T1

Arenediazonium salts are prepared at:

(A) 273–278 K (B) 373 K (C) 323 K (D) 200 K

2 Easy ⌚ 0.5 min Concept C18-T2

Aniline doesn't undergo Friedel–Crafts reactions because:

(A) it is too reactive
(B) its $-\text{NH}_2$ group forms a salt/complex with AlCl_3
(C) it is meta-directing
(D) it is insoluble

3 Easy ⌚ 0.5 min NV C18-T3

The NCERT value of $\text{p}K_b$ for aniline is 9.38. Give it to the nearest integer.

Your answer (numerical value):

4 Easy ⌚ 0.5 min Speed C18-T4

Fluorobenzene is prepared from benzenediazonium chloride by:

(A) CuF/HF (B) HBF_4 , then heating
(C) KF (D) F_2 gas

5 Easy ⌚ 0.5 min Concept C18-T5

Aniline is acetylated before nitration to:

(A) increase the activating effect of $-\text{NH}_2$
(B) reduce the activating effect and protect $-\text{NH}_2$ from oxidation, giving mainly the para product
(C) make it meta-directing
(D) increase its solubility

Answers & Solutions

C18 · Practice

1 (B) 2 (C) 3 (B) 4 (C) 5 (C) 6 (C) 7 (B) 8 (B) 9 (A) 10 (D)

11 4 12 (B)

C18 · Chapter Test

1 (A) 2 (B) 3 9 4 (B) 5 (B)

C18 · PRACTICE

1 Answer: (B) Basicity in water Medium ⌚ 1 min Concept C18-01

SOLUTION +I effect, solvation and steric factors combine. The 3° amine's cation is poorly solvated (NCERT $\text{p}K_b$: 3.27, 3.38, 4.22, 4.75).

COMMON TRAP Using the gas-phase order (A).

2 Answer: (C) Gabriel synthesis limitation Easy ⌚ 0.5 min Concept C18-02

SOLUTION Aryl halides don't undergo nucleophilic substitution with phthalimide anion. Benzyl halides do.

- 3 Answer: (B)** *Hofmann bromamide* **Easy** ⌚ 0.5 min **Speed** C18-03
SOLUTION One carbon is lost as carbonate: $\text{CH}_3\text{CONH}_2 \longrightarrow \text{CH}_3\text{NH}_2$.
- 4 Answer: (C)** *Carbylamine test* **Easy** ⌚ 0.5 min **Concept** C18-04
SOLUTION Only primary amines react.
- 5 Answer: (C)** *Hinsberg test* **Easy** ⌚ 0.75 min **Concept** C18-05
SOLUTION The product $\text{C}_6\text{H}_5\text{SO}_2\text{NR}_2$ has no acidic N-H left.
- 6 Answer: (C)** *Bromination of aniline* **Easy** ⌚ 0.5 min **Speed** C18-06
SOLUTION $-\text{NH}_2$ strongly activates all the ortho and para positions.
- 7 Answer: (B)** *Meta product in aniline nitration* **Medium** ⌚ 1 min **Concept** C18-07
SOLUTION Direct recall: see the concept summary and formula box of this chapter.
- 8 Answer: (B)** *Sandmeyer with CuCN* **Easy** ⌚ 0.5 min **Speed** C18-08
SOLUTION Direct recall: see the concept summary and formula box of this chapter.
- 9 Answer: (A)** *Azo coupling* **Easy** ⌚ 0.5 min **Speed** C18-09
SOLUTION Direct recall: see the concept summary and formula box of this chapter.
COMMON TRAP Confusing it with coupling to aniline (B).
- 10 Answer: (D)** *Weakest base* **Easy** ⌚ 0.75 min **Concept** C18-10
SOLUTION $-\text{NO}_2$ pulls the lone pair further into the ring by resonance.
- 11 Answer: 4** *Primary amine isomers* **Medium** ⌚ 1 min **NV** C18-11
SOLUTION Butan-1-amine, butan-2-amine, 2-methylpropan-1-amine, 2-methylpropan-2-amine.
COMMON TRAP Including 2° and 3° amines (8 in total).
- 12 Answer: (B)** *Reduction of a nitrile* **Easy** ⌚ 0.5 min **Speed** C18-12
SOLUTION $\text{CH}_3\text{CN} \longrightarrow \text{CH}_3\text{CH}_2\text{NH}_2$ (the nitrile C becomes CH_2).

C18 · CHAPTER TEST

- 1 Answer: (A)** *Diazotisation temperature* **Easy** ⌚ 0.5 min **Speed** C18-T1
SOLUTION They decompose at higher temperatures.
- 2 Answer: (B)** *Friedel-Crafts and aniline* **Easy** ⌚ 0.5 min **Concept** C18-T2
SOLUTION The resulting positive N strongly deactivates the ring.
- 3 Answer: 9** *pK_b of aniline* **Easy** ⌚ 0.5 min **NV** C18-T3
SOLUTION 9.38 rounds to 9 (compared with NH_3 4.75: aniline is a much weaker base).
- 4 Answer: (B)** *Balz-Schiemann* **Easy** ⌚ 0.5 min **Speed** C18-T4
SOLUTION Direct recall: see the concept summary and formula box of this chapter.
- 5 Answer: (B)** *Why acetylate aniline?* **Easy** ⌚ 0.5 min **Concept** C18-T5
SOLUTION Direct recall: see the concept summary and formula box of this chapter.

 REVISION Revision tracker · Nitrogen Compounds

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 19

Biomolecules

19

~1 per shift
TYPICAL QUESTIONS

Easy (pure NCERT
recall)
DIFFICULTY

Must-do (fast
marks)
PRIORITY

Class 12 · Ch 10
NCERT

~4 hours
STUDY TIME

This is an NCERT-fact chapter. Learn the tables below until you can reproduce them on a blank sheet.

Carbohydrates

Σ FORMULA Classification

Class	Examples	Reducing?
Monosaccharides	glucose (aldohexose), fructose (ketohexose), ribose	yes
Disaccharides	sucrose (glucose + fructose)	no
	maltose (2 α-D-glucose), lactose (galactose + glucose)	yes
Polysaccharides	starch, cellulose, glycogen	no

Reducing sugars have a free aldehyde/ketone group (or a free hemiacetal) and reduce Tollens'/Fehling's reagents.

☆ IMPORTANT Structure of glucose: the NCERT evidence

Reaction	Result	Shows
Prolonged heating with HI	n-hexane	six carbons in a straight chain
NH ₂ OH / HCN	oxime / cyanohydrin	a carbonyl group
Bromine water	gluconic acid	the carbonyl is an aldehyde
Acetic anhydride	glucose pentaacetate	five –OH groups
Nitric acid	saccharic (glucaric) acid	one 1° alcohol (–CH ₂ OH)

The open chain fails to explain why glucose gives no Schiff's test, forms no NaHSO₃ adduct, and why its pentaacetate doesn't react with NH₂OH. Hence the **cyclic hemiacetal** (pyranose, C1–C5 ring) with α and β anomers at C1. **Mutarotation** is the interconversion of the anomers in solution.

Σ FORMULA Disaccharides & polysaccharides

Sugar	Units & linkage	Notes
Sucrose	α -D-glucose C1 – β -D-fructose C2	non-reducing; dextrorotatory; hydrolysis gives laevorotatory invert sugar
Maltose	α -D-glucose C1 – C4 α -D-glucose	reducing
Lactose	β -D-galactose C1 – C4 β -D-glucose	reducing; milk sugar
Starch: amylose (15–20%)	α -D-glucose, C1–C4	water-soluble, unbranched
Starch: amylopectin (80–85%)	α -D-glucose, C1–C4 + C1–C6 branches	insoluble, branched
Cellulose	β -D-glucose, C1–C4	straight chains; plant cell walls; not digested by humans
Glycogen	α -D-glucose (like amylopectin, more branched)	"animal starch": liver, muscles, brain

Fructose forms a five-membered (**furanose**) ring (C2–C5) and is laevorotatory: D-(–)-fructose.

Proteins

Σ FORMULA Amino acids & structure

- α -Amino acids** exist as **zwitterions** ($\text{H}_3\text{N}^+ - \text{CHR} - \text{COO}^-$). They are amphoteric, with an isoelectric point. **Glycine** is the only common achiral amino acid. Naturally occurring amino acids have the L-configuration.
- Essential amino acids** (must be in the diet): valine, leucine, isoleucine, arginine, lysine, threonine, methionine, phenylalanine, tryptophan, histidine.
- Peptide bond**: $-\text{CO}-\text{NH}-$ (amide). An n -peptide has $(n - 1)$ peptide bonds.

Level	Description	Held by
Primary	amino-acid sequence	covalent (peptide) bonds
Secondary	α -helix, β -pleated sheet	hydrogen bonds between C=O and N–H
Tertiary	overall 3D folding	H-bonds, disulphide bonds, van der Waals, electrostatic forces
Quaternary	arrangement of subunits (e.g. haemoglobin)	same as tertiary

Fibrous proteins (keratin, myosin) vs **globular** proteins (insulin, albumins). **Denaturation** (heat, pH change) destroys secondary and tertiary structure; the **primary structure stays intact**. Examples: coagulation of egg white, curdling of milk. **Enzymes**: mostly globular proteins, highly specific biocatalysts (e.g. maltase, invertase, pepsin).

Vitamins

Vitamin	Solubility	Deficiency disease
A	fat	xerophthalmia (hardening of the cornea), night blindness
B ₁ (thiamine)	water	beriberi (loss of appetite, retarded growth)
B ₂ (riboflavin)	water	cheilosis (fissuring at the corners of the mouth), digestive disorders, burning sensation of the skin
B ₆ (pyridoxine)	water	convulsions
B ₁₂ (cyanocobalamin)	water	pernicious anaemia
C (ascorbic acid)	water	scurvy (bleeding gums)

Vitamin	Solubility	Deficiency disease
D	fat	rickets (children), osteomalacia (adults)
E	fat	increased fragility of RBCs, muscular weakness
K	fat	increased blood-clotting time

Fat-soluble vitamins (A, D, E, K) are stored in the liver and adipose tissue. Water-soluble ones (B group, C) must be taken regularly, **except B₁₂**, which is stored in the liver.

Nucleic acids

	DNA	RNA
Sugar	2-deoxy-D-ribose	D-ribose
Bases	A, G, C, T	A, G, C, U
Structure	double helix	single strand
Types	—	mRNA, rRNA, tRNA
Function	heredity (carries genetic information)	protein synthesis

Nucleoside = base + sugar. **Nucleotide** = nucleoside + phosphate. Nucleotides are joined by **phosphodiester** bonds (5'–3'). Base pairing: **A=T (2 H-bonds)**, **G≡C (3 H-bonds)**.

Hormones (general introduction): peptide hormones (insulin, glucagon), amino-acid derivatives (thyroxine, adrenaline), and steroids (oestrogens, progesterone, testosterone). They are chemical messengers.

Common traps

MISTAKE ALERT Mistake alerts

- Sucrose is **non-reducing**; maltose and lactose are reducing.
- Cellulose has **β**-links; starch and glycogen have **α**-links.
- B₁₂ is water-soluble **but** stored in the liver.
- Denaturation leaves the primary structure intact.
- Thymine is found only in DNA; uracil only in RNA.

Question patterns

PYQ PATTERN Recurring structures

1. Glucose structure evidence (which reaction proves what).
2. Glycosidic linkages and reducing nature of disaccharides.
3. Vitamin ↔ deficiency matching; fat/water solubility.
4. DNA vs RNA (sugar, bases, pairing, H-bonds).
5. Protein structure levels and denaturation; essential amino acids.

Practice questions

C19 · Practice

1

Easy

⌚ 0.5 min

Speed

C19-01

On prolonged heating with HI, glucose gives n-hexane. This shows that glucose has:

- (A) an aldehyde group (B) five –OH groups
(C) six carbon atoms in a straight chain (D) a ring structure

2

Easy

⌚ 0.5 min

Speed

C19-02

Which sugar is non-reducing?

- (A) maltose (B) lactose (C) sucrose (D) glucose

3

Easy

⌚ 0.5 min

Speed

C19-03

Lactose is made of:

- (A) α -D-glucose + α -D-glucose (C1-C4) (B) β -D-galactose + β -D-glucose (C1-C4)
(C) α -D-glucose + β -D-fructose (C1-C2) (D) β -D-glucose + β -D-glucose (C1-C6)

4

Easy

⌚ 0.5 min

Speed

C19-04

Deficiency of vitamin B₁₂ causes:

- (A) scurvy (B) pernicious anaemia
(C) beriberi (D) rickets

5

Easy

⌚ 0.5 min

Speed

C19-05

Which vitamin is fat-soluble?

- (A) C (B) B₁ (C) K (D) B₆

6

Easy

⌚ 0.5 min

Speed

C19-06

Which base is present in RNA but not in DNA?

- (A) adenine (B) guanine (C) thymine (D) uracil

7

Easy

⌚ 0.5 min

Concept

C19-07

When a protein is denatured:

- (A) its primary structure is destroyed
(B) its secondary and tertiary structures are destroyed but the primary structure remains
(C) all peptide bonds are hydrolysed
(D) it becomes an amino acid

8

Easy

⌚ 0.5 min

NV

C19-08

Glucose forms a pentaacetate with acetic anhydride. How many -OH groups does glucose have?

Your answer (numerical value):

9

Easy

⌚ 0.5 min

Concept

C19-09

Hydrolysis of sucrose gives an equimolar mixture called invert sugar because:

- (A) the mixture is dextrorotatory
(B) the sign of optical rotation changes from (+) to (-)
(C) it doesn't rotate plane-polarised light
(D) it reverses the reducing nature

10

Easy

⌚ 0.5 min

Speed

C19-10

Amylopectin differs from amylose in having:

- (A) β -glycosidic links (B) branching through C1-C6 linkages
(C) fructose units (D) solubility in cold water

11

Easy

⌚ 0.5 min

NV

C19-11

How many peptide bonds are present in a pentapeptide?

Your answer (numerical value):

12

Easy

⌚ 0.5 min

Speed

C19-12

Which amino acid is essential?

- (A) glycine (B) alanine (C) lysine (D) glutamic acid



C19 · Chapter Test

1

Easy

⌚ 0.5 min

Speed

C19-T1

Cellulose is a polymer of:

- (A) α -D-glucose (B) β -D-glucose (C) fructose (D) galactose

2

Easy

⌚ 0.5 min

Speed

C19-T2

In aqueous solution, amino acids exist mainly as:

- (A) cations only (B) anions only
(C) zwitterions (dipolar ions) (D) neutral molecules without charge

3

Easy

⌚ 0.5 min

NV

C19-T3

How many hydrogen bonds form between guanine and cytosine in DNA?

Your answer (numerical value):

4

Easy

⌚ 0.5 min

Speed

C19-T4

Deficiency of vitamin D causes:

- (A) night blindness (B) rickets (C) scurvy (D) beriberi

5

Easy

⌚ 0.5 min

Concept

C19-T5

Which observation is evidence that glucose exists in a cyclic form?

- (A) it gives an oxime (B) it doesn't give Schiff's test
(C) it gives gluconic acid with bromine water (D) it forms a pentaacetate

Answers & Solutions

C19 · Practice

- 1 (C) 2 (C) 3 (B) 4 (B) 5 (C) 6 (D) 7 (B) 8 5 9 (B) 10 (B)
11 4 12 (C)

C19 · Chapter Test

- 1 (B) 2 (C) 3 3 4 (B) 5 (B)

C19 · PRACTICE

1 Answer: (C) *Glucose + HI* **Easy** ⌚ 0.5 min **Speed** C19-01

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (C) *Non-reducing sugar* **Easy** ⌚ 0.5 min **Speed** C19-02

SOLUTION Both anomeric carbons (glucose C1 and fructose C2) are tied up in the glycosidic bond.

3 Answer: (B) *Lactose linkage* **Easy** ⌚ 0.5 min **Speed** C19-03

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

COMMON TRAP Choosing (A), which is maltose.

4 Answer: (B) *Vitamin B₁₂ deficiency* **Easy** ⌚ 0.5 min **Speed** C19-04

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (C) *Fat-soluble vitamin* **Easy** ⌚ 0.5 min **Speed** C19-05

SOLUTION A, D, E and K are fat-soluble.

6 Answer: (D) *RNA-only base* **Easy** ⌚ 0.5 min **Speed** C19-06

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

7 Answer: (B) *Denaturation* **Easy** ⌚ 0.5 min **Concept** C19-07

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

8 Answer: 5 *Hydroxyl groups in glucose* **Easy** ⌚ 0.5 min **NV** C19-08

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

9 Answer: (B) *Invert sugar* **Easy** ⌚ 0.5 min **Concept** C19-09

SOLUTION Fructose's laevorotation (−92.4°) outweighs glucose's dextrorotation (+52.5°).

10 Answer: (B) *Amylopectin* **Easy** ⌚ 0.5 min **Speed** C19-10

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

11 Answer: 4 *Peptide bonds* **Easy** ⌚ 0.5 min **NV** C19-11

SOLUTION $n - 1 = 4$.

COMMON TRAP Answering 5.

12 Answer: (C) *Essential amino acid* **Easy** ⌚ 0.5 min **Speed** C19-12

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

C19 · CHAPTER TEST

1 Answer: (B) *Cellulose* **Easy** ⌚ 0.5 min **Speed** C19-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (C) *Zwitterions* **Easy** ⌚ 0.5 min **Speed** C19-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 3 *G–C hydrogen bonds* **Easy** ⌚ 0.5 min **NV** C19-T3

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

COMMON TRAP Answering 2 (that's A–T).

4 Answer: (B) *Vitamin D deficiency* **Easy** ⌚ 0.5 min **Speed** C19-T4

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (B) *Evidence for ring structure* **Easy** ⌚ 0.5 min **Concept** C19-T5

SOLUTION The open-chain aldehyde would restore Schiff's reagent's colour. The absence of that colour points to the hemiacetal ring.

**REVISION** Revision tracker · Biomolecules

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week

PART III · CHEMISTRY · UNIT 20

Principles Related to Practical Chemistry

20

~1 per shift

TYPICAL QUESTIONS

Easy (Lab-manual recall)

DIFFICULTY

High

PRIORITY

Laboratory Manual
Chemistry, Class 11 & 12
 NCERT

~5 hours

STUDY TIME

**VERIFIED FACT Syllabus scope (Unit 20)**

Detection of N, S and halogens; functional-group tests (alcoholic and phenolic –OH, aldehyde/ketone, –COOH, –NH₂); preparation of Mohr's salt, potash alum, acetanilide, *p*-nitroacetanilide, aniline yellow and iodoform; acid–base titrations with indicators, oxalic acid vs KMnO₄ and Mohr's salt vs KMnO₄; salt analysis for the listed cations and anions (insoluble salts excluded); and four physical-chemistry experiments [NTA-SYL]. 2026 analyses noted questions taken **directly from NCERT lab manuals** [THIRD-PARTY ANALYSIS](#) [T-ANALYSIS26].

Functional-group tests

Group	Test	Positive result
Alcoholic –OH	ceric ammonium nitrate	red colour
	sodium metal	effervescence (H ₂)
Phenolic –OH	neutral FeCl ₃	violet/blue/green colour
	bromine water	white precipitate (2,4,6-tribromophenol)
Carbonyl (any)	2,4-DNP (Brady's)	orange-yellow precipitate
Aldehyde	Tollens' reagent	silver mirror
	Fehling's solution	red precipitate (Cu ₂ O); not for aromatic aldehydes
	Schiff's reagent	pink/magenta colour restored
Ketone	sodium nitroprusside + NaOH	red/wine-red colour (methyl ketones)
–COOH	NaHCO ₃	brisk effervescence (CO ₂)
	ethanol + conc. H ₂ SO ₄	fruity smell (ester)
–NH ₂ (1°)	carbylamine (CHCl ₃ + KOH)	offensive smell (isocyanide)
Aromatic –NH ₂ (1°)	azo-dye test: NaNO ₂ /HCl (0–5 °C), then alkaline β-naphthol	orange-red dye

Preparations

Compound	How (principle)	Appearance
Mohr's salt FeSO ₄ · (NH ₄) ₂ SO ₄ · 6 H ₂ O	equimolar FeSO ₄ + (NH ₄) ₂ SO ₄ in water with dil. H ₂ SO ₄ (prevents hydrolysis of Fe ²⁺); crystallise	pale green crystals
Potash alum K ₂ SO ₄ · Al ₂ (SO ₄) ₃ · 24 H ₂ O	equimolar K ₂ SO ₄ + Al ₂ (SO ₄) ₃ with dil. H ₂ SO ₄ ; crystallise	colourless octahedral crystals
Acetanilide	aniline + acetic anhydride (glacial acetic acid)	white crystals
<i>p</i>-Nitroacetanilide	acetanilide + conc. HNO ₃ /conc. H ₂ SO ₄ , kept cold	pale yellow crystals

Compound	How (principle)	Appearance
Aniline yellow (<i>p</i> -aminoazobenzene)	diazotise aniline (0–5 °C), couple with aniline	yellow/orange solid
Iodoform	ethanol or acetone + I ₂ + alkali (e.g. KOH/Na ₂ CO ₃), warm	pale yellow crystals, characteristic smell

Titrimetric exercises

Σ FORMULA Indicators & redox titrations

Titration	Suitable indicator
strong acid vs strong base	phenolphthalein or methyl orange
weak acid vs strong base (e.g. CH ₃ COOH/NaOH)	phenolphthalein (pH 8.3–10; colourless → pink)
strong acid vs weak base (e.g. HCl/NH ₃)	methyl orange (pH 3.1–4.4; red → yellow)

Oxalic acid vs KMnO₄: $2 \text{MnO}_4^- + 5 \text{C}_2\text{O}_4^{2-} + 16 \text{H}^+ \longrightarrow 2 \text{Mn}^{2+} + 10 \text{CO}_2 + 8 \text{H}_2\text{O}$. Heat to **60–70 °C**; acidify with dil. H₂SO₄. KMnO₄ is a **self-indicator**: the end point is a permanent light pink. Mn²⁺ formed catalyses the reaction (autocatalysis). **Mohr's salt vs KMnO₄:** $\text{MnO}_4^- + 5 \text{Fe}^{2+} + 8 \text{H}^+ \longrightarrow \text{Mn}^{2+} + 5 \text{Fe}^{3+} + 4 \text{H}_2\text{O}$. **No heating** (air would oxidise Fe²⁺); dil. H₂SO₄ is added. Mole ratios: KMnO₄ : oxalate = 2 : 5; KMnO₄ : Fe²⁺ = 1 : 5.

Qualitative salt analysis

Σ FORMULA Cation group scheme (syllabus cations)

Group	Group reagent	Cations (precipitate)	Confirmatory tests
0	NaOH, heat	NH ₄ ⁺ (NH ₃ gas)	Nessler's reagent → brown precipitate
I	dil. HCl	Pb ²⁺ (PbCl ₂ white, soluble in hot water)	KI → yellow PbI ₂ ; K ₂ CrO ₄ → yellow PbCrO ₄
II	H ₂ S in dil. HCl	Cu ²⁺ (CuS black); Pb ²⁺ (PbS black)	Cu ²⁺ : NH ₄ OH → deep blue ; K ₄ [Fe(CN) ₆] → chocolate-brown
III	NH ₄ Cl + NH ₄ OH	Al ³⁺ (white gelatinous Al(OH) ₃), Fe ³⁺ (reddish-brown Fe(OH) ₃)	Fe ³⁺ : K ₄ [Fe(CN) ₆] → Prussian blue ; KSCN → blood red . Al ³⁺ : blue lake test
IV	H ₂ S in ammoniacal medium	Zn ²⁺ (ZnS white), Ni ²⁺ (NiS black)	Ni ²⁺ : dimethylglyoxime → rosy-red ; Zn ²⁺ : K ₄ [Fe(CN) ₆] → bluish-white
V	(NH ₄) ₂ CO ₃ + NH ₄ Cl/NH ₄ OH	Ba ²⁺ , Ca ²⁺ (white carbonates)	Ba ²⁺ : K ₂ CrO ₄ → yellow; flame apple green . Ca ²⁺ : ammonium oxalate → white; flame brick red
VI	Na ₂ HPO ₄ + NH ₄ OH	Mg ²⁺ (white crystalline MgNH ₄ PO ₄)	—

Why NH₄Cl in Group III? The common-ion effect suppresses NH₄OH ionisation, keeping [OH[−]] low enough that only the Group III hydroxides (very low *K_{sp}*) precipitate.

Σ FORMULA Anion tests (syllabus anions)

Anion	Preliminary test	Confirmatory test
CO_3^{2-}	dil. $\text{H}_2\text{SO}_4 \rightarrow$ effervescence	CO_2 turns lime water milky
S^{2-}	dil. $\text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}$ (rotten eggs)	lead acetate paper turns black; sodium nitroprusside $\rightarrow$ violet
NO_2^-	dil. $\text{H}_2\text{SO}_4 \rightarrow$ brown fumes	$\text{KI} + \text{starch} \rightarrow$ blue
Cl^-	conc. $\text{H}_2\text{SO}_4 \rightarrow \text{HCl}$ fumes	$\text{AgNO}_3 \rightarrow$ white precipitate, soluble in NH_4OH ; chromyl chloride test (orange-red CrO_2Cl_2 vapours $\rightarrow$ yellow PbCrO_4)
Br^-	conc. $\text{H}_2\text{SO}_4 \rightarrow$ brown Br_2 vapours	$\text{AgNO}_3 \rightarrow$ pale-yellow precipitate (partly soluble in NH_4OH); layer test $\rightarrow$ orange-brown
I^-	conc. $\text{H}_2\text{SO}_4 \rightarrow$ violet I_2 vapours	$\text{AgNO}_3 \rightarrow$ yellow precipitate (insoluble); layer test $\rightarrow$ violet
NO_3^-	conc. $\text{H}_2\text{SO}_4 + \text{Cu} \rightarrow$ brown fumes	brown ring test: $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$
SO_4^{2-}	(no gas)	$\text{BaCl}_2 \rightarrow$ white BaSO_4 , insoluble in HCl

Physical-chemistry experiments (syllabus)

- Enthalpy of solution of CuSO_4 :** measure the temperature change in a calorimeter; $\Delta H = -\frac{(mc + W)\Delta T}{n}$. Anhydrous CuSO_4 dissolves exothermically.
- Enthalpy of neutralisation** (strong acid + strong base): about -57.1 kJ per mol of water.
- Lyophilic sols** (starch, gum, gelatin: warm with water directly) vs **lyophobic sols** ($\text{Fe}(\text{OH})_3$ by hydrolysis of FeCl_3 in boiling water; As_2S_3 by double decomposition).
- Kinetics of $\text{I}^- + \text{H}_2\text{O}_2$:** $\text{H}_2\text{O}_2 + 2\text{I}^- + 2\text{H}^+ \rightarrow \text{I}_2 + 2\text{H}_2\text{O}$. The time for the blue starch-iodine colour to appear (thiosulphate clock) measures the rate at different concentrations.

Common traps

⚠ MISTAKE ALERT Mistake alerts

- Don't heat in the Mohr's salt- KMnO_4 titration. Do heat ($60-70^\circ\text{C}$) for oxalic acid.
- HCl is never used** to acidify KMnO_4 titrations (it's oxidised to Cl_2).
- The chromyl chloride test is given by **chlorides only**.
- Nessler's reagent (alkaline K_2HgI_4) $\rightarrow$ **brown** precipitate with $\text{NH}_4^+/\text{NH}_3$.
- DMG gives a rosy-red precipitate with Ni^{2+} , not with Zn^{2+} .

Question patterns

📊 PYQ PATTERN Recurring structures

- Group reagent / confirmatory test / colour matching.
- Indicator choice; conditions for KMnO_4 titrations; titration stoichiometry (NV).
- Reagents and appearance in the listed preparations.
- Functional-group tests for a given compound.
- Principles of the four physical-chemistry experiments.

Practice questions

C20 · Practice

1 Easy 0.5 min Speed

C20-01

A test that distinguishes an aldehyde from a ketone is:

- (A) 2,4-DNP test (B) Tollens' test
(C) sodium bicarbonate test (D) Lucas test

2 Easy 0.5 min Speed

C20-02

The best indicator for titrating acetic acid with NaOH is:

- (A) methyl orange (B) phenolphthalein (C) litmus only (D) starch

3 Easy 0.5 min Concept

C20-03

In the titration of oxalic acid with KMnO_4 :

- (A) no indicator is needed, and the acid solution is warmed to 60–70 °C
(B) phenolphthalein is used at room temperature
(C) HCl is used to acidify
(D) starch is the indicator

4 Easy 0.5 min Concept

C20-04

Dilute H_2SO_4 is added while preparing Mohr's salt to:

- (A) increase its solubility (B) prevent hydrolysis of Fe^{2+}
(C) oxidise Fe^{2+} (D) act as an indicator

5 Easy 0.5 min Speed

C20-05

The brown ring in the test for nitrate is due to:

- (A) $\text{Fe}_2(\text{SO}_4)_3$ (B) $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$
(C) NO_2 (D) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$

6 Easy 0.5 min Speed

C20-06

The group reagent for Group II cations is:

- (A) dil. HCl (B) H_2S in the presence of dil. HCl
(C) $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$ (D) $(\text{NH}_4)_2\text{CO}_3$

7 Easy 0.5 min Speed

C20-07

Ni^{2+} is confirmed by:

- (A) potassium ferrocyanide
(B) dimethylglyoxime in ammoniacal solution, giving a rosy-red precipitate
(C) Nessler's reagent
(D) the flame test

8 Easy 0.5 min Concept

C20-08

The chromyl chloride test is used to confirm:

- (A) bromide (B) iodide (C) chloride (D) sulphate

9 Easy 0.5 min Speed

C20-09

Barium salts impart which flame colour?

- (A) brick red (B) apple (grassy) green
(C) crimson red (D) golden yellow

10 Medium 1.5 min NV

C20-10

Find the volume (in mL) of 0.02 M KMnO_4 needed to titrate 20 mL of 0.1 M oxalic acid in acidic medium.

Your answer (numerical value):

11 Easy 0.5 min Concept

C20-11

The azo-dye test (diazotisation followed by alkaline β -naphthol) is characteristic of:

- (A) 1° aliphatic amines (B) 1° aromatic amines
(C) 2° amines (D) phenols

12 Easy 0.5 min Speed

C20-12

A lyophobic sol of ferric hydroxide is prepared by:

- (A) dissolving $\text{Fe}(\text{OH})_3$ in cold water
(B) adding FeCl_3 solution to boiling water (hydrolysis)
(C) adding NH_4OH to FeCl_3 at room temperature
(D) dissolving iron in HCl

C20 · Chapter Test

1 Easy 0.5 min Speed

C20-T1

The cations precipitated by NH_4OH in the presence of NH_4Cl (from the syllabus list) are:

- (A) Zn^{2+} , Ni^{2+} (B) Fe^{3+} , Al^{3+} (C) Ca^{2+} , Ba^{2+} (D) Cu^{2+} , Pb^{2+}

2 Easy 0.5 min Speed

C20-T2

Nessler's reagent gives a brown precipitate with:

- (A) NH_4^+ (ammonia) (B) Ni^{2+} (C) Cl^- (D) Fe^{3+}

3 Easy 1 min NV

C20-T3

Find the volume (in mL) of 0.1 M KMnO_4 needed to oxidise 50 mL of 0.1 M Fe^{2+} (Mohr's salt) in acidic medium.

Your answer (numerical value):

4 Easy 0.5 min Speed

C20-T4

Lead chloride precipitated in Group I:

- (A) is insoluble in hot water (B) dissolves in hot water
(C) is black (D) doesn't form

5 Easy 0.5 min Speed

C20-T5

Acetanilide is prepared from aniline using:

- (A) acetic anhydride (or acetyl chloride) (B) acetaldehyde
(C) ethanol (D) CHCl_3/KOH

Answers & Solutions

C20 · Practice

- 1 (B) 2 (B) 3 (A) 4 (B) 5 (B) 6 (B) 7 (B) 8 (C) 9 (B) 10 40
11 (B) 12 (B)

C20 · Chapter Test

- 1 (B) 2 (A) 3 10 4 (B) 5 (A)

C20 · PRACTICE

1 Answer: (B) Aldehyde vs ketone Easy 0.5 min Speed

C20-01

SOLUTION 2,4-DNP is positive for both.

COMMON TRAP Choosing 2,4-DNP.

2 Answer: (B) Indicator for weak acid–strong base Easy 0.5 min Speed

C20-02

SOLUTION The equivalence point is basic ($\text{pH} \approx 8-9$), which falls in phenolphthalein's range.

3 Answer: (A) Oxalic acid titration Easy 0.5 min Concept

C20-03

SOLUTION KMnO_4 is a self-indicator. Warming speeds up the slow initial reaction.

4 Answer: (B) Mohr's salt preparation Easy 0.5 min Concept

C20-04

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

5 Answer: (B) Brown ring Easy 0.5 min Speed

C20-05

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

6 Answer: (B) Group II reagent Easy 0.5 min Speed

C20-06

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

7 Answer: (B) Confirming nickel Easy 0.5 min Speed

C20-07

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

8 Answer: (C) Chromyl chloride test Easy 0.5 min Concept

C20-08

SOLUTION Only chlorides give volatile orange-red CrO_2Cl_2 .

9 Answer: (B) Flame test for barium Easy 0.5 min Speed

C20-09

SOLUTION Ca is brick red, Sr crimson, Na golden yellow.

10 Answer: 40 Permanganate–oxalate titration Medium 1.5 min NV

C20-10

SOLUTION Moles of oxalic acid = 2×10^{-3} . KMnO_4 needed = $\frac{2}{5} \times 2 \times 10^{-3} = 8 \times 10^{-4}$ mol. $V = 8 \times 10^{-4} / 0.02 = 0.040 \text{ L} = 40 \text{ mL}$.

SHORTCUT Equivalents: $5 \times 0.02 \times V = 2 \times 0.1 \times 20$.

COMMON TRAP Using a 1 : 1 ratio (100 mL).

11 Answer: (B) Azo-dye test Easy 0.5 min Concept

C20-11

SOLUTION Only aromatic diazonium salts are stable enough to couple.

12 Answer: (B) *Lyophobic sol* **Easy** ⌚ 0.5 min **Speed**

C20-12

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

COMMON TRAP (C) gives a precipitate, not a sol.

C20 · CHAPTER TEST

1 Answer: (B) *Group III* **Easy** ⌚ 0.5 min **Speed**

C20-T1

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

2 Answer: (A) *Nessler's reagent* **Easy** ⌚ 0.5 min **Speed**

C20-T2

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

3 Answer: 10 *Mohr's salt titration* **Easy** ⌚ 1 min **NV**

C20-T3

SOLUTION $\text{MnO}_4^- : \text{Fe}^{2+} = 1 : 5$, so $\text{KMnO}_4 = 5 \times 10^{-3} / 5 = 10^{-3} \text{ mol} = 10 \text{ mL}$.

4 Answer: (B) *Group I* **Easy** ⌚ 0.5 min **Speed**

C20-T4

SOLUTION That's why some Pb^{2+} also appears in Group II.

5 Answer: (A) *Acetanilide* **Easy** ⌚ 0.5 min **Speed**

C20-T5

SOLUTION Direct recall: see the concept summary and formula box of this chapter.

REVISION Revision tracker · Practical Chemistry

☐ Same day ☐ Day 2 ☐ Day 7 ☐ Day 14 ☐ Day 30 ☐ Final week